

1 INTRODUCTION

The finite element method (FEM) is a powerful numerical technique designed to approximate solutions to **boundary value problems**, which are prevalent in numerous fields of physics, such as solid mechanics, heat conduction, potential flows, and electrostatics, among others. A boundary value problem is defined by a partial differential equation paired with a set of constraints, termed boundary conditions, specified along the boundaries of the problem domain.

In the FEM, the domain is divided into smaller, manageable sub-regions known as finite elements. This subdivision enables the method to approximate complex governing equations over the entire domain. Through this process, the problem is transformed into a system of linear equations. The FEM can be implemented through various mathematical frameworks, including the direct stiffness method, integral formulations, and variational calculus.

Most engineering problems lack analytical solutions due to their complexity, rendering numerical methods like the FEM essential. Effective application of the FEM relies on computational tools to handle the resulting system of equations. **Figure 1.1** shows an example of a finite element mesh and its corresponding stress field.

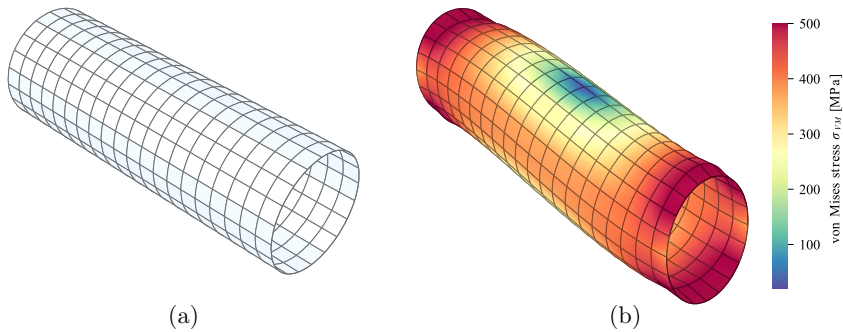


Figure 1.1: Application of the finite element mesh in a mechanical analysis of a steel tube clamped at both ends: (a) finite element mesh composed by shell elements; (b) von Mises stress field.

1.1 Brief history

The following timeline highlights key milestones in the history of FEM, showcasing the foundational ideas, pivotal publications, and technological advances that have defined its progress over the decades. These milestones are summarized in **Figure 1.2**.

- 1943: Richard Courant applied the Rayleigh-Ritz method to the torsion problem by subdividing the domain into triangular subregions, an important precursor of FEM [1]
- Early 1950s: M. J. Turner and collaborators at Boeing developed the Direct Stiffness Method for aerospace structural analysis, helping set the stage for modern FEM.
- 1950s and 1960s: J. H. Argyris and R. W. Clough played major roles in extending and disseminating matrix and finite element methods.
- 1956: Turner, Clough, Martin and Topp (TCMT) published a paper widely recognized as the foundation of modern finite element method technology [2]
- 1960: Clough coined the term **finite element** [3]
- 1963: Melosh showed that conforming displacement models can be interpreted as a Rayleigh-Ritz method based on the principle of minimum potential energy, strengthening the variational foundation of FEM [4]

- 1965: Zienkiewicz and Y. K. Cheung showed that FEM could be extended beyond structural mechanics to more general field problems [5]
- 1967: Zienkiewicz and Y. K. Cheung published **The Finite Element Method in Structural and Continuum Mechanics**, one of the first books dedicated to FEM [6]
- 1968: Irons and Zienkiewicz formalized the isoparametric concept, enabling curved elements and more accurate geometry representation [7]
- Late 1960s and 1970s: General-purpose FEM programs emerged, including NASA's NAS-TRAN, Berkeley's SAP series, and ANSYS, helping establish computational structural analysis in engineering practice [8] [9] [10] [11]
- Late 1960s and 1970s: FEM expanded beyond solid mechanics into heat transfer, seepage, electromagnetics, and fluid-related field problems.
- 1980s: Adaptive meshing, error estimation, and improved nonlinear formulations matured, while FEM became increasingly important in computational fluid dynamics.
- 1990s: Parallel computing and CAD/CAE integration enabled larger three-dimensional industrial simulations and broader multiphysics applications.
- 2000s to present: Growth of multiphysics FEM (coupled thermal-structural, fluid-structure interaction, etc.). FEM becomes widely used in biomechanics, geotechnics, materials science, and additive manufacturing.

1.2 Definitions

The following definitions introduce the main concepts used in the finite element method, covering problem definition, discretization, mathematical formulation, and analysis type.

Domain: The physical region or geometric space over which the problem is defined and analyzed, such as a solid structure, fluid volume, or heat transfer area.

Boundary condition: Constraints or prescribed values (e.g., fixed displacement, applied force) imposed on the boundaries of the domain to define the problem.

Initial condition: The state of the system at the initial time (e.g., initial displacement, temperature, or velocity) for time-dependent problems.

Constitutive model: A mathematical relationship describing the material behavior, linking quantities like stress and strain (e.g., Hooke's law for elasticity, von Mises model for metals, etc.).

Mesh: A discretization of the domain into smaller, simpler geometric shapes (elements), used to approximate the solution over the entire domain.

Finite element: An individual geometric subunit of the mesh, such as a triangle, quadrilateral, or tetrahedron, over which the governing equations are approximated.

Node: A specific point in the mesh, typically at the vertices or along the edges of elements, where solution variables (e.g., displacements, temperatures) are computed.

Degree of freedom: The independent variables at a node (e.g., displacement in x , y , or z directions) that describe the system's behavior and are solved for in the analysis.

Integration point: Specific locations within a finite element where numerical integration is performed to evaluate quantities like stiffness or stress.

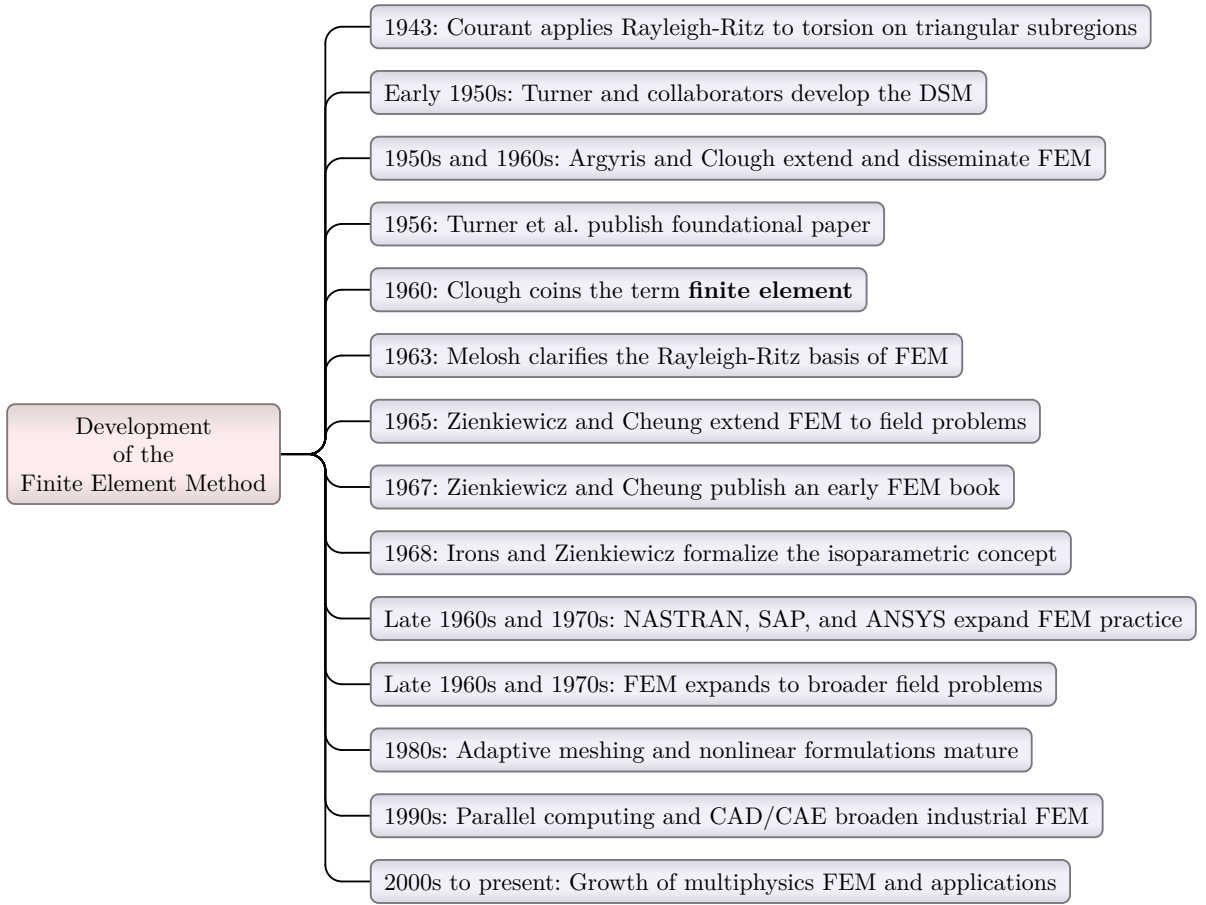


Figure 1.2: Timeline of key milestones in the development of the finite element method.

Strong form: The original differential equations of the problem, expressed in their exact, local form, typically requiring higher continuity of the solution. For example, for the static equilibrium problem, the strong form is given by:

$$\begin{cases} \nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0} & \text{in } \Omega \\ \mathbf{u} = \mathbf{u}_D & \text{on } \Gamma_D \text{ for essential boundary conditions} \\ \boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{t} & \text{on } \Gamma_N \text{ for natural boundary conditions} \end{cases} \quad (1.1)$$

where

- $\boldsymbol{\sigma}$ represents the stress tensor.
- $\mathbf{b}$ is the body force vector.
- $\mathbf{u}$ is the displacement vector.
- $\mathbf{u}_D$ is the prescribed displacement on the essential boundary.
- $\mathbf{n}$ is the outward unit normal vector to the boundary.
- $\mathbf{t}$ is the traction vector on the natural boundary.
- Ω denotes the problem domain.

- Γ_D represents the essential boundary of the domain.
- Γ_N represents the natural boundary of the domain.

Weak form: A reformulation of the governing differential equations into an integral form, reducing the continuity requirements on the solution and enabling numerical approximation. The weak form for the static equilibrium problem can be expressed as:

$$\int_{\Omega} \boldsymbol{\sigma} : \nabla \mathbf{u} \, d\Omega = \int_{\Omega} \mathbf{b} \cdot \mathbf{v} \, d\Omega + \int_{\Gamma_N} \mathbf{t} \cdot \mathbf{v} \, d\Gamma \quad (1.2)$$

where $\mathbf{v}$ is the virtual displacement vector (test function).

Finite element form: The finite element discretization of the weak form, applied to a single element, leads to the following equation:

$$\int_{\Omega} \mathbf{B}^T \mathbf{D} \mathbf{B} \, d\Omega \mathbf{U} = \int_{\Gamma_N} \mathbf{N}^T \mathbf{t} \, d\Gamma_N + \int_{\Omega} \mathbf{N}^T \mathbf{b} \, d\Omega \quad (1.3)$$

where:

- $\mathbf{B}$ is the strain-displacement matrix, relating displacements to strains.
- $\mathbf{D}$ is the constitutive matrix, representing the stress-strain relationship.
- $\mathbf{U}$ is the displacement vector of an element.
- $\mathbf{N}$ is the shape function matrix, used for interpolating within the element.
- $\mathbf{t}$ is the traction vector, representing surface forces.

Linear analysis: An analysis in which the governing equations, material behavior, and geometric response are assumed linear, so the principle of superposition applies.

Nonlinear analysis: An analysis in which the response depends nonlinearly on the unknowns due to material nonlinearity, large deformations, contact, or other nonlinear effects.

Steady-state analysis: An analysis assuming the system has reached equilibrium, where variables (e.g., temperature, stress) no longer change with time.

Transient analysis: A time-dependent analysis that captures how the system evolves over time under changing conditions (e.g., dynamic response or heat diffusion).

Figure 1.3 summarizes how these concepts are organized within a typical finite element formulation.

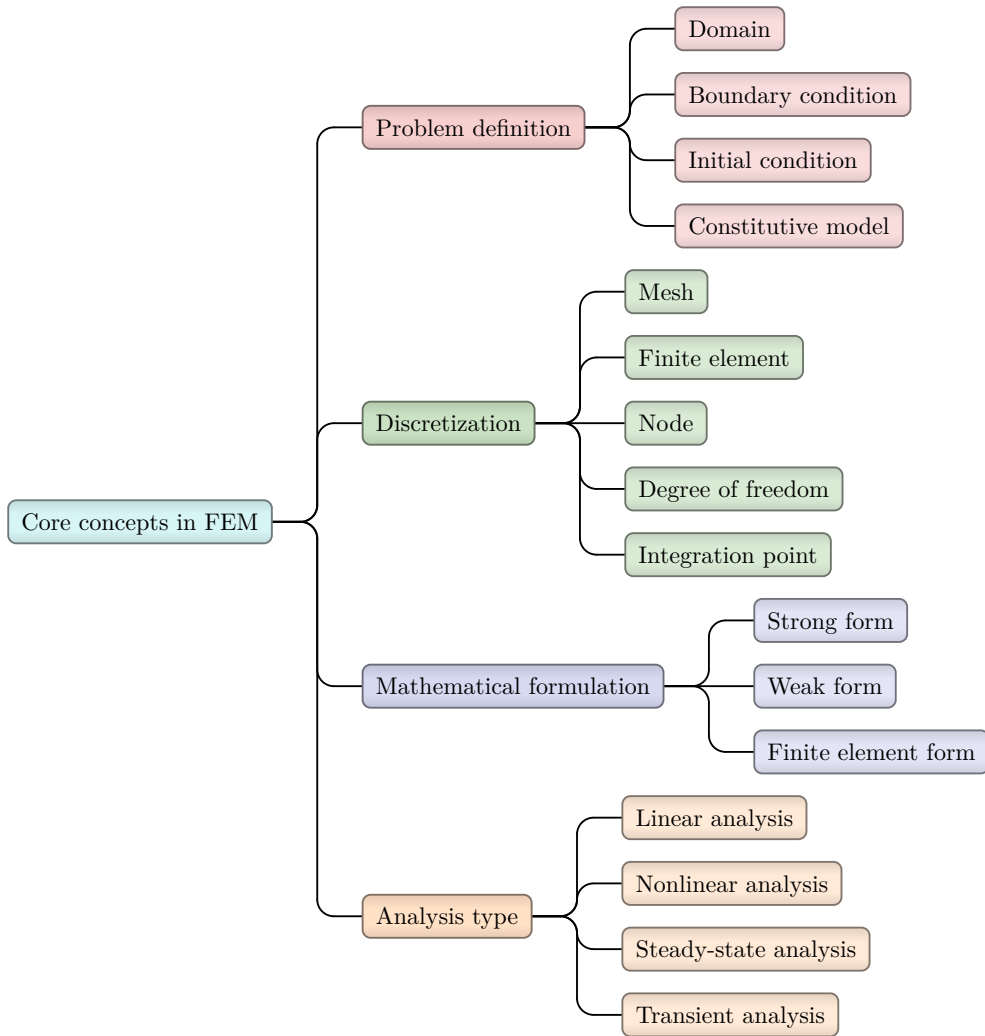


Figure 1.3: Hierarchy of core concepts used in the finite element method.

1.3 Main features and applications of the FEM

The finite element method (FEM) is a versatile numerical technique for solving a broad range of engineering and physics problems. It is especially useful when analytical solutions are difficult or impossible to obtain due to complex geometries, heterogeneous materials, or intricate boundary conditions and versatility. Compared with other numerical methods, FEM offers several important advantages, which are described in more detail below and summarized in Figure 1.4.

Geometrical flexibility: FEM can represent irregular and complex geometries by discretizing the domain into smaller and simpler elements, such as triangles, quadrilaterals, or tetrahedra.

Applicability to linear and nonlinear problems: FEM can be used for both linear and nonlinear analyses, enabling the study of material nonlinearity, large deformations, contact, and other complex behaviors.

Stability and convergence: When based on a well-posed formulation, FEM exhibits desirable numerical properties. Stability helps prevent spurious solutions, while convergence means that the numerical solution approaches the exact solution as the mesh is refined.

Ease of imposing boundary conditions: FEM provides a convenient framework for prescribing essential and natural boundary conditions, including fixed displacements, applied loads, fluxes, and mixed conditions.

Local refinement capability: FEM allows selective mesh refinement in regions where the solution varies rapidly, improving accuracy without unnecessarily increasing the computational cost over the entire domain.

Interdisciplinary applications: Although FEM originated in structural mechanics, it is now widely used in many areas, including biomechanics, geophysics, environmental modeling, materials science, and finance.

A common application in structural engineering is the static and dynamic analysis of frames used in the design of reinforced concrete and steel structures.

Despite these advantages, FEM can be **computationally demanding**, especially for fine meshes, transient problems, nonlinear analyses, and large-scale three-dimensional models. In such cases, efficient numerical solvers and high-performance computing resources are often required to make simulations practical.

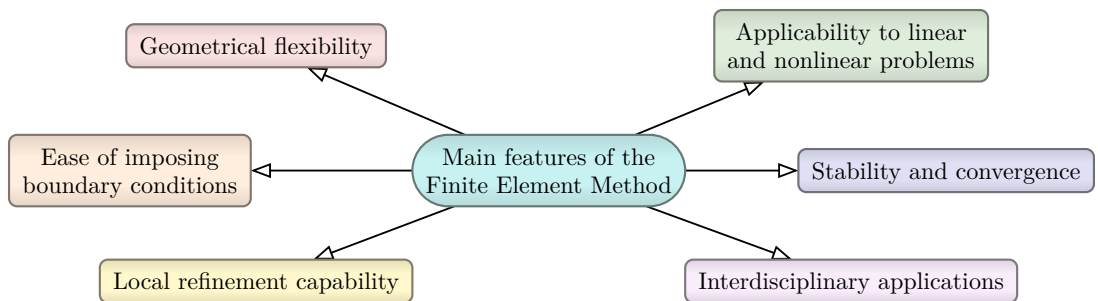


Figure 1.4: Main features of the finite element method when compared to other numerical methods.

1.4 General application procedure

Applying the finite element method involves a sequence of interconnected steps. The process begins with a clear definition of the physical problem, followed by the choice of an appropriate idealization, the formulation of the mathematical model, the discretization of the domain, the assembly and solution of the finite element equations, and, finally, the interpretation of the results during postprocessing. In practice, this workflow is rarely strictly linear: the results may indicate the need to refine the mesh, revise the modeling assumptions, or adjust the selected material model. The main stages of this procedure are outlined in Figure 1.5.

1.4.1 The physical problem

The first step in any finite element analysis is to define the physical problem clearly. This includes identifying the system to be studied, the loads acting on it, the relevant constraints, and the quantities of interest. As an example, consider the following statement:

Problem: Determine the stress and displacement fields within the holed steel plate depicted in Figure 1.6 when subjected to a uniform tensile traction of 50 MPa applied to the top and bottom edges.

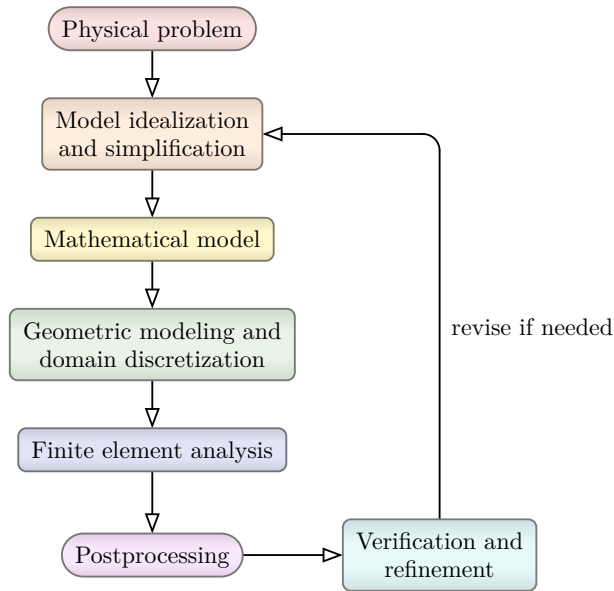


Figure 1.5: Main steps in a typical finite element application procedure.

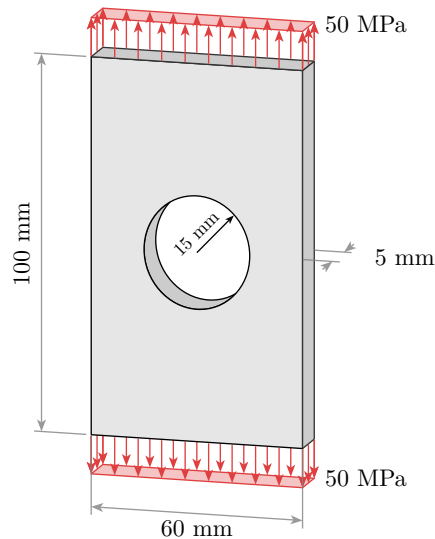


Figure 1.6: Holed steel plate under tensile traction.

1.4.2 Model idealization and simplification

Before writing the governing equations, it is often advantageous to replace the full physical problem with an idealized model that captures the dominant mechanical behavior while remaining computationally efficient. This stage is optional: some problems must be analyzed in their full geometric and physical complexity, whereas others benefit greatly from symmetry reductions, dimensional simplifications, or assumptions such as plane stress or plane strain.

In the case of the holed plate, symmetry allows us to analyze only one quarter of the domain (Figure 1.7a), which substantially reduces the computational cost. This idealization requires

boundary conditions that reproduce the behavior of the complete plate correctly. The model can be simplified further by adopting a plane stress formulation (Figure 1.7b).

When moving from a three-dimensional model to a two-dimensional one, the displacement constraints must be reformulated so that they remain consistent with the reduced kinematics. In addition, the plate thickness must still be taken into account when defining loads and constitutive relations, so that the simplified model remains physically equivalent to the original problem.

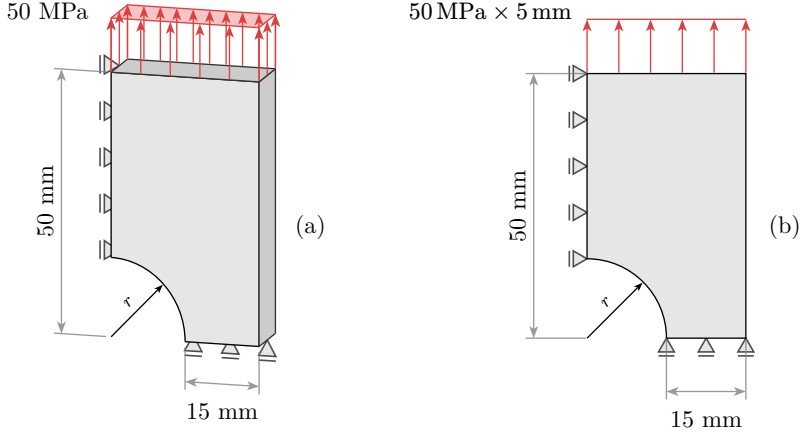


Figure 1.7: Model idealization: (a) Reduced model considering one quarter of the whole domain; (b) further reduction by considering plane stress conditions.

1.4.3 Mathematical model

For this example, we adopt the static equilibrium formulation. The mathematical model can be expressed as the following boundary value problem (BVP):

$$\text{BVP: } \begin{cases} \nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0} & \text{in } \Omega \\ \text{Boundary conditions on } \partial\Omega \end{cases} \quad (1.4)$$

where $\partial\Omega$ represents the boundary of the domain.

The differential equation $\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0}$ enforces equilibrium within the domain. The boundary conditions, defined on $\partial\Omega$, prescribe the displacement constraints and the applied tractions.

Next, a constitutive model must be selected. In this example, the material is assumed to be isotropic and linearly elastic. For steel, we adopt the material parameters Young's modulus $E = 200 \text{ GPa}$ and Poisson's ratio $\nu = 0.3$.

Starting from the strong form of the boundary value problem, we derive a weak form and then approximate the unknown field over each finite element. This discretization transforms the continuous problem into a system of algebraic equations. For a subregion of the domain, referred to as an element, the finite element equilibrium equation can be written as:

$$\underbrace{\int_{\Omega} \mathbf{B}^T \mathbf{D} \mathbf{B} d\Omega}_{\mathbf{K}_e} \mathbf{U}_e = \underbrace{\int_{\Gamma_N} \mathbf{N}^T \mathbf{t} d\Gamma_N + \int_{\Omega} \mathbf{N}^T \mathbf{b} d\Omega}_{\mathbf{F}_e} \quad (1.5)$$

where:

- $\mathbf{U}_e$ is the displacement vector of the element.
- $\mathbf{K}_e$ is the element stiffness matrix.
- $\mathbf{F}_e$ is the element force vector.

By assembling the contributions from all elements and enforcing the boundary conditions, we obtain the global system of equations:

$$\bar{\mathbf{K}}\mathbf{U}_u = \bar{\mathbf{F}} \quad (1.6)$$

where:

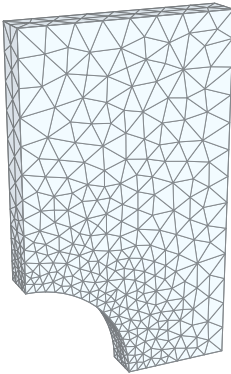
- $\bar{\mathbf{K}}$ is the reduced global stiffness matrix.
- $\bar{\mathbf{F}}$ is the modified global force vector.
- $\mathbf{U}_u$ is the vector containing the unknown nodal displacements.

1.4.4 Geometric modeling and domain discretization

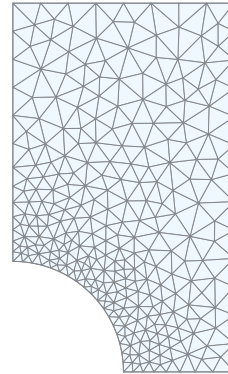
Domain discretization is the process of transforming a continuous medium into a discrete representation. In the finite element method, this is done by subdividing the physical domain into smaller subregions called finite elements. The resulting collection of elements, connected at nodes, is called a mesh. **Figure 1.8** illustrates examples of both two- and three-dimensional meshes.

Finite element meshes are commonly classified as either **structured** or **unstructured**. Many software packages and libraries are specifically designed for mesh generation, and some finite element programs include this capability directly.

The choice of element type depends on the nature of the analysis and the desired level of accuracy. Common examples include truss and beam elements, as well as triangular, quadrilateral, tetrahedral, and hexahedral continuum elements.



3D mesh using tetrahedral elements



2D mesh using triangular elements

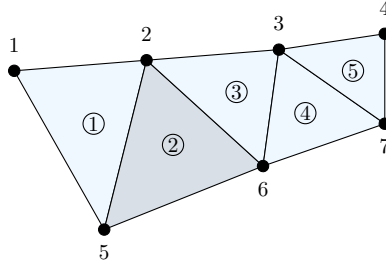
Figure 1.8: Unstructured finite element meshes.

1.4.5 Finite element analysis

The finite element analysis stage consists of assembling the element contributions into a global system, applying the prescribed boundary conditions, and solving the resulting algebraic equations (e.g. **Eq. (1.6)**). In structural problems, the solution provides the unknown nodal quantities, such as displacements. In practice, these operations are performed by a computer program.

To organize the assembly process, all nodes and elements in the model are assigned unique identifiers. The numbering of nodes within each element is referred to as local numbering. In two-dimensional elements, local numbering is often defined in a counterclockwise sequence. For instance, as illustrated in [Figure 1.9](#), the local numbering for the highlighted element is $\langle 5, 6, 2 \rangle$.

Once the nodal solution has been obtained, additional quantities such as strains, stresses, reaction forces, and other derived field variables can be recovered. These results are then examined during postprocessing.



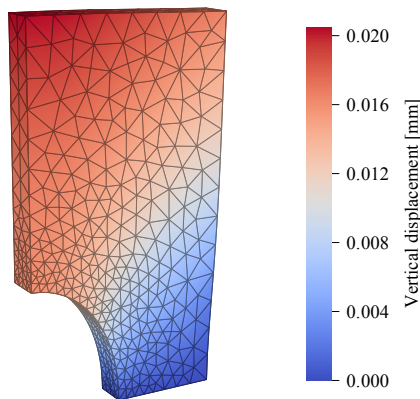
[Figure 1.9](#): A chunk of elements displaying nodal and element numbering.

1.4.6 Postprocessing

Following the finite element analysis, the next stage is [postprocessing](#), in which the output data are extracted, organized, and presented in a clear and informative way. This stage is essential for interpreting the results of the analysis. Typical outputs include force-displacement curves, strain and stress field plots, and visualizations of the deformed geometry.

In general, postprocessing can present results in many formats, such as scalar and vector field distributions, colormaps, isolines, isosurfaces, animations, deformed shapes, and charts. These visualizations help reveal how the analyzed system behaves. For example, [Figure 1.10](#) shows the deformed three-dimensional configuration and the vertical displacement field, [Figure 1.11](#) illustrates the displacement fields, and [Figure 1.12](#) shows the stress distributions.

Although many finite element packages offer basic postprocessing capabilities, more advanced analysis and visualization often require specialized software.



[Figure 1.10](#): Deformed 3D model showing the vertical displacement field (scale = 200).

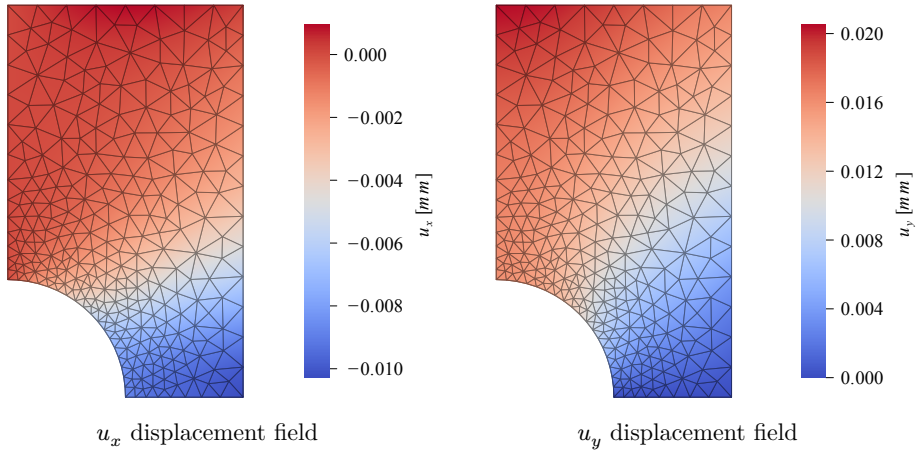


Figure 1.11: Horizontal and vertical displacement fields for the 2D model.

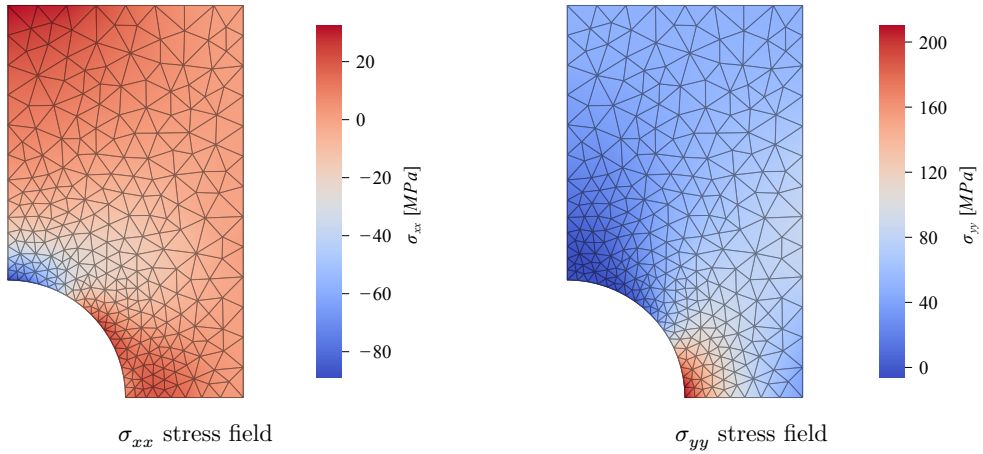


Figure 1.12: Horizontal and vertical stresses fields for the 2D model.

1.4.7 Verification and refinement

Postprocessing should not be viewed solely as a visualization stage. It is also the stage at which the analyst evaluates whether the model is producing physically meaningful and numerically reliable results. Typical verification tasks include checking that boundary conditions and reaction forces are consistent, assessing whether the stress and displacement fields are plausible, and examining how strongly the solution changes under mesh refinement.

If these checks reveal deficiencies, the analysis cycle should be repeated. The analyst may refine the mesh, improve the geometric idealization, modify the material model, or revise the loading and boundary conditions. In this sense, finite element analysis is usually an iterative modeling process rather than a single linear sequence of steps.

1.5 Methodological variants of the FEM

Over time, the classical finite element method has been extended to address broader classes of problems and improve modeling capabilities. The variants listed below differ in formulation, field coupling, treatment of discontinuities, and representation of geometry or uncertainty; [Figure 1.13](#) summarizes them according to their main purpose and modeling strategy.

Classical Finite Element Method (Standard FEM): Assumes small strains, small displacements, and linear material behavior. Commonly applied in structural analysis and heat conduction.

Nonlinear Finite Element Method: Addresses geometric nonlinearity (large displacements/rotations), material nonlinearity (e.g., plasticity, damage, hyperelasticity), and contact mechanics. Used in metal forming, buckling, and crash simulations.

Mixed/Coupled Finite Element Method: Solves multiple field variables simultaneously (e.g., displacements and temperature). Ideal for thermoelasticity, poroelasticity, and multiphysics problems.

Extended Finite Element Method (XFEM/GFEM): Enriches solution spaces to model discontinuities (e.g., cracks, voids, phase interfaces) without remeshing for evolving features.

Cohesive Zone Modeling (CZM) in FEM: Simulates fracture initiation and propagation using interface (cohesive) elements between continuum elements. These follow a traction–separation law governing material resistance and failure.

Meshless/Meshfree Methods: Uses nodes without fixed connectivity (e.g., Element-Free Galerkin, Smoothed Particle Hydrodynamics). Ideal for problems with large deformations, fragmentation or moving boundaries, where meshing is challenging.

Isogeometric Analysis (IGA): Employs NURBS or B-splines from CAD models as shape functions, representing geometry exactly rather than approximating it, unlike classical FEM.

Hybrid FEM: Combines FEM with other numerical methods (e.g., boundary element method, finite volume method) to leverage their respective strengths. Used in problems like fluid–structure interaction or electromagnetic simulations.

Multiscale FEM: Captures fine-scale effects at macro scales through homogenization or nested models. Includes FE^2 methods, where local FEM analyses inform the global behavior.

Stochastic Finite Element Method (SFEM): Incorporates random fields or variables to model uncertainty, supporting reliability analysis and probabilistic design.

Adaptive / hp-FEM: Improves accuracy by refining the mesh size (h), increasing the interpolation order (p), or combining both strategies adaptively according to the solution behavior.

Discontinuous Galerkin FEM (DG-FEM): Uses discontinuous interpolation between elements, offering flexibility for convection-dominated problems, wave propagation, and high-order formulations.

Stabilized FEM: Introduces stabilization terms into the weak formulation to improve numerical behavior in problems such as incompressible flow or convection-dominated transport.

Spectral Element Method: Combines the geometric flexibility of FEM with high-order polynomial interpolation, providing very high accuracy in problems such as structural dynamics, wave propagation, and fluid mechanics.

The following mind map groups these methodological variants according to their main purpose and modeling strategy.

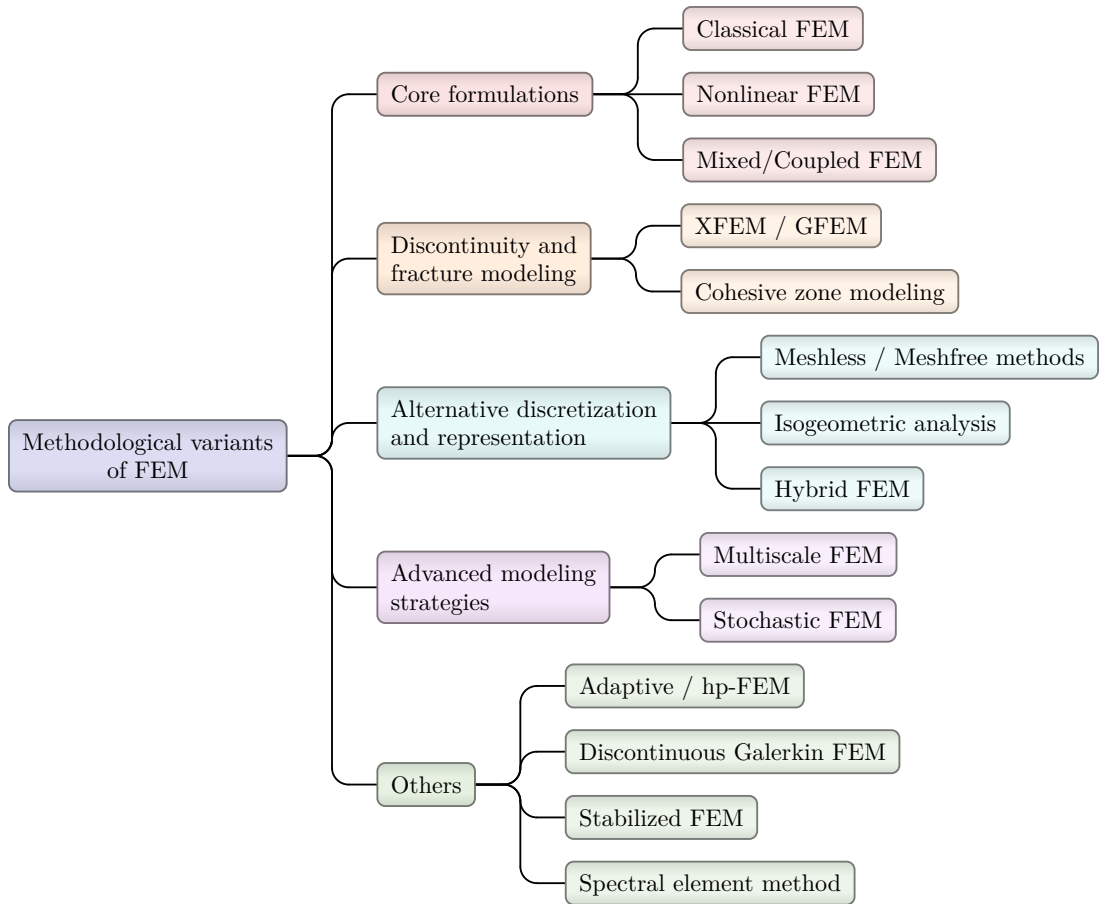


Figure 1.13: Mind map of the main methodological variants of the FEM.

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2 DIRECT STIFFNESS METHOD

The Direct Stiffness Method (DSM), also called the **Displacement Method** or the Matrix Method, is a method of structural analysis to find internal forces and displacements. This method is commonly used to solve trusses, frames and spring systems. The analysis of an structure using the DSM is usually divided in three major steps: breakdown, assembly and solution.

In the breakdown step, the structure is divided into elements and the stiffness matrix for each element is computed. The stiffness matrix compiles together material and geometric properties of the element and is derived using concepts such as **equilibrium and elasticity**. For each element we have:

$$K^e U^e = F^e \quad (2.1)$$

where K^e is the element's stiffness matrix, U^e is the element's displacements vector and F^e is the element's forces vector. In the assembly step the equilibrium equations from all elements are compiled into one global system of linear equations using the **superposition principle**:

$$K^g U^g = F^g \quad (2.2)$$

The final step is solving the system to find unknown displacements and reaction forces.

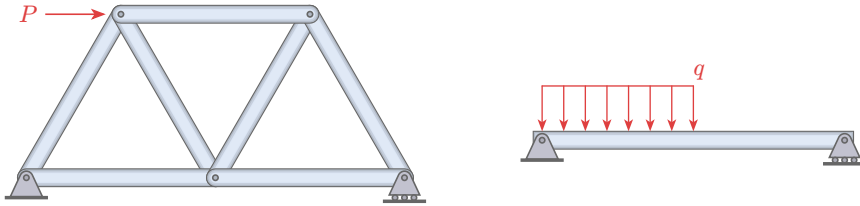


Figure 2.1: Truss and beam structures; examples of structures that can be analyzed using the direct stiffness method

2.1 Trusses

A truss is a structural system composed by three or more bar elements. It is assumed that each bar only supports **axial loads** and the bar ends converge at points called **nodes**. Also, external and reaction forces are considered to act at nodes.

Given an axial elongation $\Delta\ell$ in a bar element, the axial strain is given by:

$$\varepsilon = \frac{\Delta\ell}{\ell} = \frac{du}{dx} \quad (2.3)$$

where ℓ is the bar length, u is the displacement and x is the longitudinal coordinate.

Using Hooke's law, the axial stress is given by:

$$\sigma = \frac{f}{A} = E\varepsilon \quad (2.4)$$

where F is the axial force, A is the section area and E is the Young's modulus. From the above relations, when an axial force is applied, the elongation $\Delta\ell$ is found as:

$$\Delta\ell = \frac{f\ell}{AE} \quad (2.5)$$

Once $\Delta\ell$ is known, the strain and stress can be easily calculated.

2.1.1 1D-truss element

The 1D truss element is a fundamental finite element consisting of two nodes. Each node has a single degree of freedom, corresponding to the axial displacement along the element's longitudinal axis. The displacements at these two nodes are denoted by u_1 and u_2 , respectively. For each displacement, there is a corresponding axial force, denoted by f_1 and f_2 , respectively. Figure 2.2 illustrates the nodal displacements and forces in a 1D-truss element.

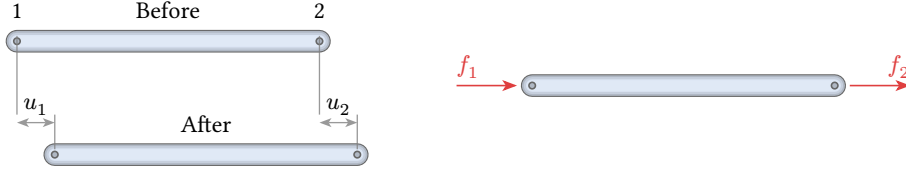


Figure 2.2: Nodal displacements and forces in a 1D-truss element

Considering the change in length $\Delta\ell = u_2 - u_1$, the internal axial force can be expressed as:

$$f = \frac{EA}{\ell}(u_2 - u_1) \quad (2.6)$$

From nodal equilibrium conditions, we know that $f = -f_1 = f_2$. Therefore, the nodal forces are:

$$\begin{aligned} f_1 &= \frac{EA}{\ell}(u_1 - u_2) \\ f_2 &= \frac{EA}{\ell}(u_2 - u_1) \end{aligned} \quad (2.7)$$

These relations can be assembled into matrix form as:

$$\underbrace{\frac{EA}{\ell} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}}_{\mathbf{K}^e} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} \quad (2.8)$$

or, more compactly, as:

$$\mathbf{K}^e \mathbf{U}^e = \mathbf{F}^e \quad (2.9)$$

where $\mathbf{K}^e$ represents the element stiffness matrix, $\mathbf{U}^e$ is the vector of nodal displacements, and $\mathbf{F}^e$ is the vector of nodal forces. It is important to note that the element stiffness matrix $\mathbf{K}^e$ is both **symmetric** and **singular**.

2.1.2 2D-truss element

In two dimensions, each node has two degrees of freedom, corresponding to the displacements u_x and u_y in the global x and y directions, respectively, as shown in Figure 2.3. Consequently, each node is subjected to two force components, f_x and f_y . To evaluate the internal axial force, we must determine the axial elongation $\Delta\ell$. Under the assumption of small deformations, $\Delta\ell$ is obtained by projecting the nodal displacements onto the element's longitudinal axis.

For convenience, the stiffness matrix of a plane truss element can be formulated using the stiffness matrix of a 1D-truss element along with a coordinate transformation. Using trigonometric relations, the local axial displacements, denoted by $\bar{u}_1$ and $\bar{u}_2$, can be calculated from the global displacement degrees of freedom as:

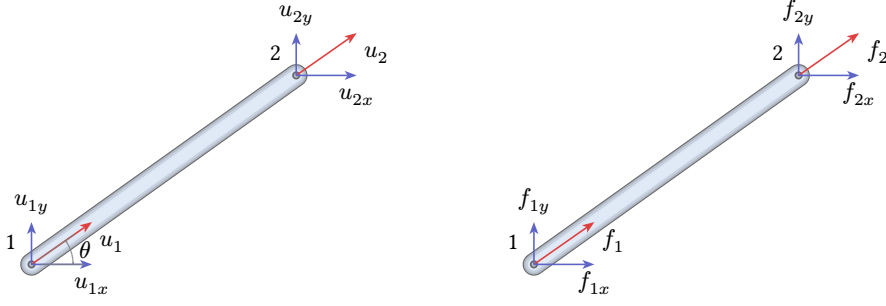


Figure 2.3: Nodal displacements and forces in a 2D-truss element

$$\begin{pmatrix} \bar{u}_1 \\ \bar{u}_2 \end{pmatrix} = \begin{pmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 \\ 0 & 0 & \cos(\theta) & \sin(\theta) \end{pmatrix} \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \end{pmatrix} \quad (2.10)$$

or, in short:

$$\bar{U}^e = T U^e \quad (2.11)$$

where T is the transformation matrix that projects the global displacements onto the local element axis. Note that overbars are used to denote quantities in the local coordinate system of the truss element.

The local internal forces $\bar{F}^e$ are related to the local displacements by the 1D-truss stiffness matrix $\bar{K}^e$:

$$\bar{F}^e = \bar{K}^e \bar{U}^e = \bar{K}^e T U^e \quad (2.12)$$

Likewise, the global nodal forces are related to the local nodal forces by

$$F^e = T^T \bar{F}^e \quad (2.13)$$

This relation follows from the principle of virtual work. Since the local and global displacements are related by $\bar{U}^e = T U^e$, the corresponding virtual displacements satisfy

$$\delta \bar{U}^e = T \delta U^e \quad (2.14)$$

The virtual work must be identical whether evaluated in the local or global coordinate system:

$$\delta U^{eT} F^e = \delta \bar{U}^{eT} \bar{F}^e \quad (2.15)$$

Substituting the virtual displacement transformation into this expression gives

$$\delta U^{eT} F^e = (T \delta U^e)^T \bar{F}^e = \delta U^{eT} T^T \bar{F}^e \quad (2.16)$$

Since this equality must hold for any virtual displacement δU^e , equation Eq. (2.13) follows.

Substituting Eq. (2.13) into Eq. (2.12) yields

$$\underbrace{T^T \bar{K}^e T}_{K^e} U^e = F^e \quad (2.17)$$

Therefore, the global nodal displacements and forces are related through the element stiffness matrix

$$K^e = T^T \bar{K}^e T \quad (2.18)$$

Expanding this equation, the equilibrium equations of a 2D-truss element are expressed in matrix form as:

$$\frac{EA}{\ell} \begin{pmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{pmatrix} \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \end{pmatrix} = \begin{pmatrix} f_{1x} \\ f_{1y} \\ f_{2x} \\ f_{2y} \end{pmatrix} \quad (2.19)$$

where $c = \cos(\theta)$ and $s = \sin(\theta)$.

Example 2.1: Calculate the stiffness matrix for a 2D rod element with endpoints at coordinates (2, 3) and (3, 4) (in meters). The element has a Young's modulus of $E = 200 \times 10^6$ kPa and a cross-sectional area of $A = 0.00707$ m².

Solution:

First, we determine the element's length ℓ and orientation θ . The length is calculated using the distance formula which leads to $\ell \approx 1.414$.

Taking the first node as reference, the angle with the horizontal axis is:

$$\theta = \text{atan}\left(\frac{4-3}{3-2}\right) = 45^\circ$$

The stiffness matrix in the element's local coordinate system, $\bar{\mathbf{K}}^e$, is given by:

$$\bar{\mathbf{K}}^e = \frac{EA}{\ell} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

The transformation matrix, $\mathbf{T}$, which rotates the global displacements to the local system, is:

$$\mathbf{T} = \begin{pmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 \\ 0 & 0 & \cos(\theta) & \sin(\theta) \end{pmatrix}$$

Substituting $\theta = 45^\circ$, we get $\cos(\theta) = \sin(\theta) \approx 0.707$, so:

$$\mathbf{T} = \begin{pmatrix} 0.707 & 0.707 & 0 & 0 \\ 0 & 0 & 0.707 & 0.707 \end{pmatrix}$$

The global stiffness matrix $\mathbf{K}^e$ is found by transforming the local stiffness matrix to the global coordinate system:

$$\mathbf{K}^e = \mathbf{T}^T \bar{\mathbf{K}}^e \mathbf{T}$$

Substituting the known values and matrices yields:

$$\mathbf{K}^e = \begin{pmatrix} 5 \times 10^5 & 5 \times 10^5 & -5 \times 10^5 & -5 \times 10^5 \\ 5 \times 10^5 & 5 \times 10^5 & -5 \times 10^5 & -5 \times 10^5 \\ -5 \times 10^5 & -5 \times 10^5 & 5 \times 10^5 & 5 \times 10^5 \\ -5 \times 10^5 & -5 \times 10^5 & 5 \times 10^5 & 5 \times 10^5 \end{pmatrix} \frac{\text{kN}}{\text{m}}$$

2.1.3 3D-truss element

The formulation of the 3D-truss element follows the same rationale as the 2D-truss element, extended to three dimensions. Each node now has three degrees of freedom, corresponding to the translations in the global x , y , and z directions. Figure 2.4 illustrates the nodal displacements for a 3D-truss element.

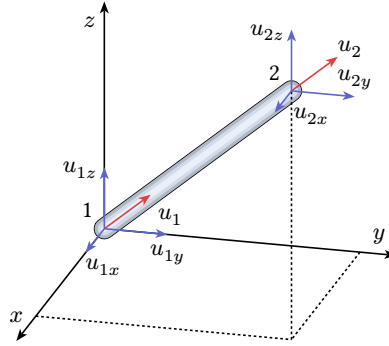


Figure 2.4: Nodal displacements in a 3D-truss element

The orientation of the truss element in 3D space is defined by a directional unit vector, which can be calculated from the nodal coordinate vectors $\mathbf{x}_1$ and $\mathbf{x}_2$:

$$\hat{\mathbf{n}} = \frac{\mathbf{x}_2 - \mathbf{x}_1}{\|\mathbf{x}_2 - \mathbf{x}_1\|} = (l, m, n) \quad (2.20)$$

where $l = \cos(\theta_x)$, $m = \cos(\theta_y)$, and $n = \cos(\theta_z)$ are the direction cosines with respect to the global axes.

Using these direction cosines, the local axial displacements u_1 and u_2 are related to the global nodal displacements through the transformation matrix $\mathbf{T}$:

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \underbrace{\begin{pmatrix} l & m & n & 0 & 0 & 0 \\ 0 & 0 & 0 & l & m & n \end{pmatrix}}_{\mathbf{T}} \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{1z} \\ u_{2x} \\ u_{2y} \\ u_{2z} \end{pmatrix} \quad (2.21)$$

As in the 2D case, the global stiffness matrix of the 3D-truss element is obtained by transforming the 1D-truss stiffness matrix:

$$\mathbf{K}^e = \mathbf{T}^T \bar{\mathbf{K}}^e \mathbf{T} \quad (2.22)$$

Expanding this matrix multiplication yields the explicit form of the 3D-truss element stiffness matrix:

$$\mathbf{K}^e = \frac{EA}{\ell} \begin{pmatrix} l^2 & lm & ln & -l^2 & -lm & -ln \\ lm & m^2 & mn & -lm & -m^2 & -mn \\ ln & mn & n^2 & -ln & -mn & -n^2 \\ -l^2 & -lm & -ln & l^2 & lm & ln \\ -lm & -m^2 & -mn & lm & m^2 & mn \\ -ln & -mn & -n^2 & ln & mn & n^2 \end{pmatrix} \quad (2.23)$$

Also, the global displacement and force vectors are given by:

$$U^e = \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{1z} \\ u_{2x} \\ u_{2y} \\ u_{2z} \end{pmatrix} \quad \text{and} \quad F^e = \begin{pmatrix} f_{1x} \\ f_{1y} \\ f_{1z} \\ f_{2x} \\ f_{2y} \\ f_{2z} \end{pmatrix} \quad (2.24)$$

2.2 Assembly and solution of the global system

Once the stiffness matrices for all elements have been determined, the global system of linear equations is assembled using the **superposition principle**:

$$KU = F \quad (2.25)$$

where K is the global stiffness matrix, and U and F are the global displacement and force vectors, respectively.

The global displacement vector U contains all the displacement degrees of freedom of the system, including both unknown and prescribed displacements (such as support conditions).

The global stiffness matrix is typically **sparse**, meaning that most of its entries are zero. Proper node numbering strategies can significantly reduce the matrix bandwidth, making computations more efficient. For large-scale finite element analyses, specialized sparse linear solvers are essential.

2.2.1 System solution

Once the global stiffness matrix and global vectors are assembled, we can solve for the unknown nodal values: **displacements and reaction forces**.

First, it is convenient to partition the system $KU = F$ to separate the unknown displacements from the prescribed ones. The system can be rewritten in block matrix form as:

$$\begin{pmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{pmatrix} \begin{pmatrix} U_1 \\ U_2 \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} \quad (2.26)$$

where:

- K_{11} to K_{22} are stiffness submatrices
- U_1 contains the unknown displacements
- U_2 contains prescribed displacements
- F_2 contains the unknown reactions
- F_1 contains prescribed forces

This partitioned system can be expanded into two matrix equations:

$$\begin{aligned} K_{11}U_1 + K_{12}U_2 &= F_1 \\ K_{21}U_1 + K_{22}U_2 &= F_2 \end{aligned} \quad (2.27)$$

The first equation leads to the following system which is used to find the unknown displacements U_1 :

$$K_{11}U_1 = F_1 - K_{12}U_2 \quad (2.28)$$

thus, the unknown displacements are found as:

$$\mathbf{U}_1 = (\mathbf{K}_{11})^{-1}(\mathbf{F}_1 - \mathbf{K}_{12}\mathbf{U}_2). \quad (2.29)$$

Although matrix inversion is shown here conceptually, direct or iterative linear solvers are used in practice for computational efficiency.

Once the unknown displacements $\mathbf{U}_1$ are found, the unknown reaction forces $\mathbf{F}_2$ are computed using the second equation:

$$\mathbf{F}_2 = \mathbf{K}_{21}\mathbf{U}_1 + \mathbf{K}_{22}\mathbf{U}_2 \quad (2.30)$$

2.2.2 Direct imposition of boundary conditions

Direct imposition of boundary conditions is a technique used to enforce prescribed displacements directly within the global system of equations. This approach is exact and applies to both zero (homogeneous) and nonzero (non-homogeneous) prescribed displacements.

For instance, consider the following system with six degrees of freedom:

$$\begin{pmatrix} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ k_{31} & k_{32} & k_{33} & k_{34} & k_{35} & k_{36} \\ k_{41} & k_{42} & k_{43} & k_{44} & k_{45} & k_{46} \\ k_{51} & k_{52} & k_{53} & k_{54} & k_{55} & k_{56} \\ k_{61} & k_{62} & k_{63} & k_{64} & k_{65} & k_{66} \end{pmatrix} \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \end{pmatrix} = \begin{pmatrix} f_{1x} \\ f_{1y} \\ f_{2x} \\ f_{2y} \\ f_{3x} \\ f_{3y} \end{pmatrix} \quad (2.31)$$

If the displacements u_{2x} , u_{3x} , and u_{3y} are prescribed as $\bar{u}_{2x}$, $\bar{u}_{3x}$, and $\bar{u}_{3y}$ respectively, we replace the corresponding rows in the stiffness matrix and the force vector with the constraint equations. The modified system becomes:

$$\begin{pmatrix} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ 0 & 0 & 1 & 0 & 0 & 0 \\ k_{41} & k_{42} & k_{43} & k_{44} & k_{45} & k_{46} \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \end{pmatrix} = \begin{pmatrix} f_{1x} \\ f_{1y} \\ \bar{u}_{2x} \\ f_{2y} \\ \bar{u}_{3x} \\ \bar{u}_{3y} \end{pmatrix} \quad (2.32)$$

To preserve the symmetry of the stiffness matrix, the columns corresponding to the prescribed displacements are also zeroed out (except for the diagonal entries), and their known contributions are subtracted from the load vector on the right-hand side. The resulting symmetric system is:

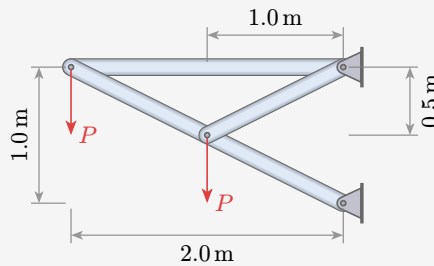
$$\begin{pmatrix} k_{11} & k_{12} & 0 & k_{14} & 0 & 0 \\ k_{21} & k_{22} & 0 & k_{24} & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ k_{41} & k_{42} & 0 & k_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \end{pmatrix} = \begin{pmatrix} f_{1x} - k_{13}\bar{u}_{2x} - k_{15}\bar{u}_{3x} - k_{16}\bar{u}_{3y} \\ f_{1y} - k_{23}\bar{u}_{2x} - k_{25}\bar{u}_{3x} - k_{26}\bar{u}_{3y} \\ \bar{u}_{2x} \\ f_{2y} - k_{43}\bar{u}_{2x} - k_{45}\bar{u}_{3x} - k_{46}\bar{u}_{3y} \\ \bar{u}_{3x} \\ \bar{u}_{3y} \end{pmatrix} \quad (2.33)$$

When the prescribed displacements are exactly zero, the right-hand side modifications vanish, and the system simplifies to:

$$\begin{pmatrix} k_{11} & k_{12} & 0 & k_{14} & 0 & 0 \\ k_{21} & k_{22} & 0 & k_{24} & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ k_{41} & k_{42} & 0 & k_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \end{pmatrix} = \begin{pmatrix} f_{1x} \\ f_{1y} \\ 0 \\ f_{2y} \\ 0 \\ 0 \end{pmatrix} \quad (2.34)$$

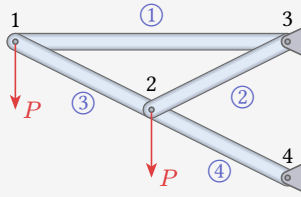
Example 2.2:

Given the truss in the figure, find all nodal displacements and reaction forces. Consider $E = 200 \times 10^3$ MPa and $A = 0.005 \text{ m}^2$ and $P = 1 \text{ MN}$.



Solution

First, we define the node and element numbering. The nodal numbering sequence can be arbitrary, however it is suggested to start from one side and continue gradually in direction to the opposite side to **reduce the stiffness matrix bandwidth**.



According to the element numbering, the stiffness matrices for each element are

$$\mathbf{K}^{(1)} = \begin{pmatrix} 500 & 0 & -500 & 0 \\ 0 & 0 & 0 & 0 \\ -500 & 0 & 500 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \text{ MN/m}$$

$$\mathbf{K}^{(2)} = \begin{pmatrix} 715.542 & 357.771 & -715.542 & -357.771 \\ 357.771 & 178.885 & -357.771 & -178.885 \\ -715.542 & -357.771 & 715.542 & 357.771 \\ -357.771 & -178.885 & 357.771 & 178.885 \end{pmatrix} \text{ MN/m}$$

$$\mathbf{K}^{(3)} = \mathbf{K}^{(4)} = \begin{pmatrix} 715.54 & -357.77 & -715.54 & 357.77 \\ -357.77 & 178.88 & 357.77 & -178.88 \\ -715.54 & 357.77 & 715.54 & -357.77 \\ 357.77 & -178.88 & -357.77 & 178.88 \end{pmatrix} \text{ MN/m}$$

Note that elements 3 and 4 have the same length and orientation, thus, the corresponding stiffness matrices are the same.

The assembly of the linear equations leads to the global system

$$\begin{matrix} f_{1x} \\ f_{1y} \\ f_{2x} \\ f_{2y} \\ f_{3x} \\ f_{3y} \\ f_{4x} \\ f_{4y} \end{matrix} \begin{pmatrix} u_{1x} & u_{1y} & u_{2x} & u_{2y} & u_{3x} & u_{3y} & u_{4x} & u_{4y} \\ 1215.542 & -357.771 & -715.542 & 357.771 & -500.000 & 0.000 & 0.000 & 0.000 \\ -357.771 & 178.885 & 357.771 & -178.885 & 0.000 & 0.000 & 0.000 & 0.000 \\ -715.542 & 357.771 & 2146.625 & -357.771 & -715.542 & -357.771 & -715.542 & 357.771 \\ 357.771 & -178.885 & -357.771 & 536.656 & -357.771 & -178.885 & 357.771 & -178.885 \\ -500.000 & 0.000 & -715.542 & -357.771 & 1215.542 & 357.771 & 0.000 & 0.000 \\ 0.000 & 0.000 & -357.771 & -178.885 & 357.771 & 178.885 & 0.000 & 0.000 \\ 0.000 & 0.000 & -715.542 & 357.771 & 0.000 & 0.000 & 715.542 & -357.771 \\ 0.000 & 0.000 & 357.771 & -178.885 & 0.000 & 0.000 & -357.771 & 178.885 \end{pmatrix} \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 0 \\ -1 \\ f_{3x} \\ f_{3y} \\ f_{4x} \\ f_{4y} \end{pmatrix}$$

Solving for the unknown displacements we have

$$\begin{aligned} K_{11}U_1 &= F_1 - K_{12}U_2 \\ U_1 &= \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \end{pmatrix} = \begin{pmatrix} -0.004 \\ -0.0192 \\ 0.0014 \\ -0.0028 \end{pmatrix} \text{ m} \end{aligned}$$

Finally, the reaction forces are found as

$$\begin{aligned} F_2 &= K_{21}U_1 + K_{22}U_2 \\ F_2 &= \begin{pmatrix} f_{3x} \\ f_{3y} \\ f_{4x} \\ f_{4y} \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ -2 \\ 1 \end{pmatrix} \text{ MN} \end{aligned}$$

The final step in the solution process is the **postprocessing**. For example, we can compute the axial strain and stress for each bar. For this purpose, we need to compute the nodal displacements in the local reference.

$$\bar{U}^e = TU^e$$

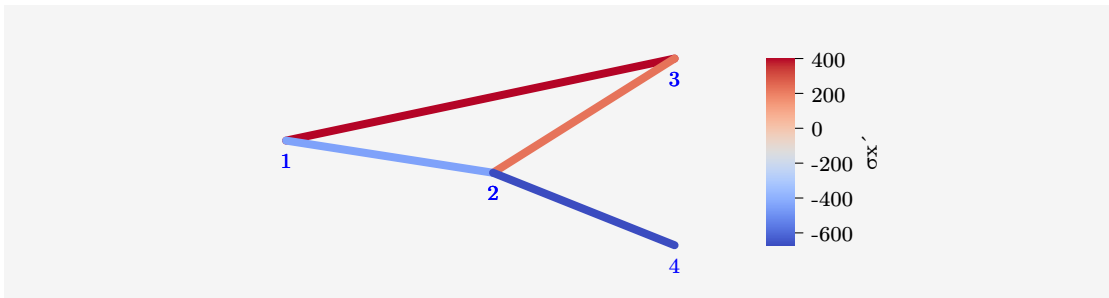
For instance, for the first element we compute

$$\bar{U}^1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} -0.004 \\ -0.0192 \\ 0 \\ 0 \end{pmatrix} \text{ m} = \begin{pmatrix} -0.004 \\ 0 \end{pmatrix} \text{ m}$$

Then, the axial strain can be computed as

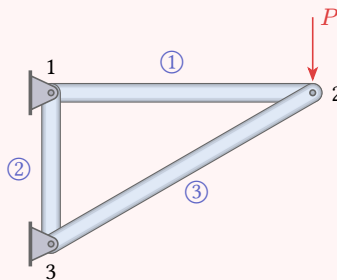
$$\varepsilon = \frac{\bar{U}_2^1 - \bar{U}_1^1}{\ell} = \frac{0.004}{2} = 0.002$$

Finally, the stress is computed as $\sigma = \varepsilon E = 400 \text{ MPa}$. The figure below shows the axial stress distribution in the truss structure.

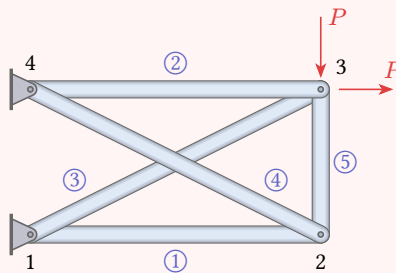


Exercises 2.1:

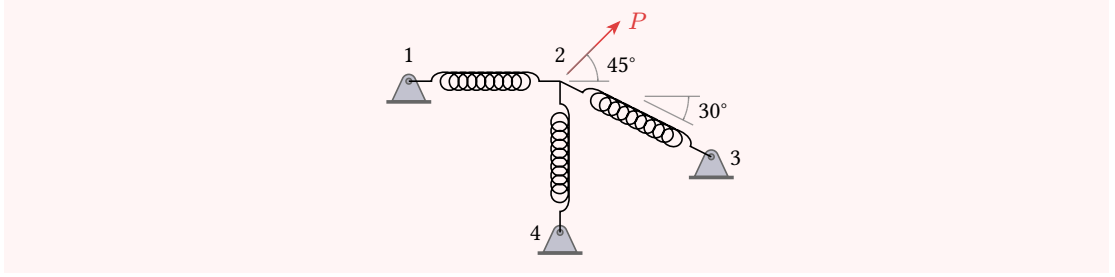
1. Determine the nodal displacements and reaction forces for the truss system shown below. Calculate the axial force and axial stress in each element. Solve the linear system by partitioning the unknown displacements from the prescribed ones. Assume $P = 10 \text{ kN}$, $E = 200 \text{ GPa}$, and $A = 0.001 \text{ m}^2$. The lengths of elements 2 and 3 are 1 m and 2 m, respectively.



2. Determine the nodal displacements of the given truss. Assume an applied load $P = 10 \text{ kN}$ and an axial stiffness $EA = 2000 \text{ kN}$. The lengths of elements 1 and 5 are 1.6 m and 1.2 m, respectively.



3. Derive the finite element equations for a linear spring element in both 2D and 3D space.
4. Calculate the displacements of node 2 in the spring system shown below. Each spring has a stiffness of $k = 1000 \text{ kN/m}$, and the applied load is $P = 7.07 \text{ kN}$.



2.3 Classical beam element

A beam element is used to model the structural behavior of beams. A classical beam element is composed of two nodes, with each node having two degrees of freedom: transverse displacement and rotation. In the convention adopted here, upward displacements and counterclockwise rotations are considered positive.

The nodal displacements are collected into a vector, and the corresponding vector of nodal forces is given by:

$$\mathbf{U}^e = \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix} \quad \mathbf{F}^e = \begin{pmatrix} V_1 \\ M_1 \\ V_2 \\ M_2 \end{pmatrix} \quad (2.35)$$

The stiffness matrix relates the nodal displacements to the nodal forces. In classical beam theory, the stiffness matrix is formulated in terms of the Young's modulus E , the moment of inertia of the cross-section I , and the element length ℓ .

2.3.1 Euler-Bernoulli beam theory

The Euler-Bernoulli beam theory is a simplification of the linear theory of elasticity. This theory states that the transverse deflection v along a beam is related to the distributed load $q(x)$ through the following differential equation:

$$\frac{d^2}{dx^2} \left(EI \frac{d^2 v}{dx^2} \right) = q(x) \quad (2.36)$$

For a uniform beam section where E and I are constant, this equation simplifies to:

$$EI \frac{d^4 v}{dx^4} = q(x) \quad (2.37)$$

The stiffness matrix can be obtained by successive integrations of this differential equation, establishing the relations between nodal displacements and forces.

2.3.2 Beam with no distributed load



Figure 2.11: Nodal forces in a beam element.

For the case of no distributed load ($q(x) = 0$), successive integrations of the differential equation yield expressions for the shear force, bending moment, rotation, and deflection:

$$\begin{aligned}
 V(x) &= EI \frac{d^3v}{dx^3} = c_1 \\
 M(x) &= EI \frac{d^2v}{dx^2} = c_1x + c_2 \\
 \theta(x) &= \frac{dv}{dx} = \frac{1}{EI} \left(c_1 \frac{x^2}{2} + c_2x + c_3 \right) \\
 v(x) &= \frac{1}{EI} \left(c_1 \frac{x^3}{6} + c_2 \frac{x^2}{2} + c_3x + c_4 \right)
 \end{aligned} \tag{2.38}$$

where $c_1, \dots, c_4$ are integration constants.

Evaluating these expressions at the element endpoints, we have:

$$\begin{aligned}
 v_1 &= v(0) = \frac{1}{EI} c_4 \\
 \theta_1 &= \theta(0) = \frac{1}{EI} c_3 \\
 v_2 &= v(\ell) = \frac{1}{EI} \left(c_1 \frac{\ell^3}{6} + c_2 \frac{\ell^2}{2} + c_3\ell + c_4 \right) \\
 \theta_2 &= \theta(\ell) = \frac{1}{EI} \left(c_1 \frac{\ell^2}{2} + c_2\ell + c_3 \right)
 \end{aligned} \tag{2.39}$$

These expressions can be rearranged into matrix form:

$$\begin{pmatrix} 0 & 0 & 0 & \frac{1}{EI} \\ 0 & 0 & \frac{1}{EI} & 0 \\ \frac{\ell^3}{6EI} & \frac{\ell^2}{2EI} & \frac{\ell}{EI} & \frac{1}{EI} \\ \frac{\ell^2}{2EI} & \frac{\ell}{EI} & \frac{1}{EI} & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix} \tag{2.40}$$

which allows us to express the integration constants as functions of the nodal displacements:

$$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} \frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} & -\frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} \\ -\frac{6EI}{\ell^2} & -\frac{4EI}{\ell} & \frac{6EI}{\ell^2} & -\frac{2EI}{\ell} \\ 0 & EI & 0 & 0 \\ EI & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix} \tag{2.41}$$

On the other hand, the nodal forces can be defined as:

$$\begin{aligned}
 V_1 &= V(0) = c_1 \\
 M_1 &= -M(0) = -c_2 \\
 V_2 &= -V(\ell) = -c_1 \\
 M_2 &= M(\ell) = c_1\ell + c_2
 \end{aligned} \tag{2.42}$$

Substituting the expressions for c_1 and c_2 into these force equations, we obtain:

$$\begin{aligned}
 V_1 &= \frac{12EI}{\ell^3}v_1 + \frac{6EI}{\ell^2}\theta_1 - \frac{12EI}{\ell^3}v_2 + \frac{6EI}{\ell^2}\theta_2 \\
 M_1 &= \frac{6EI}{\ell^2}v_1 + \frac{4EI}{\ell}\theta_1 - \frac{6EI}{\ell^2}v_2 + \frac{2EI}{\ell}\theta_2 \\
 V_2 &= -\frac{12EI}{\ell^3}v_1 - \frac{6EI}{\ell^2}\theta_1 + \frac{12EI}{\ell^3}v_2 - \frac{6EI}{\ell^2}\theta_2 \\
 M_2 &= \left(\frac{12EI}{\ell^3}v_1 + \frac{6EI}{\ell^2}\theta_1 - \frac{12EI}{\ell^3}v_2 + \frac{6EI}{\ell^2}\theta_2 \right) \ell + M_1 \\
 &= \frac{6EI}{\ell^2}v_1 + \frac{2EI}{\ell}\theta_1 - \frac{6EI}{\ell^2}v_2 + \frac{4EI}{\ell}\theta_2
 \end{aligned} \tag{2.43}$$

Finally, organizing these equations in matrix form yields the finite element stiffness system:

$$\frac{EI}{\ell^3} \begin{pmatrix} 12 & 6\ell & -12 & 6\ell \\ 6\ell & 4\ell^2 & -6\ell & 2\ell^2 \\ -12 & -6\ell & 12 & -6\ell \\ 6\ell & 2\ell^2 & -6\ell & 4\ell^2 \end{pmatrix} \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} V_1 \\ M_1 \\ V_2 \\ M_2 \end{pmatrix} \tag{2.44}$$

2.3.3 Constant load

For a constant distributed load we have

$$q(x) = q \tag{2.45}$$

where q is a constant.

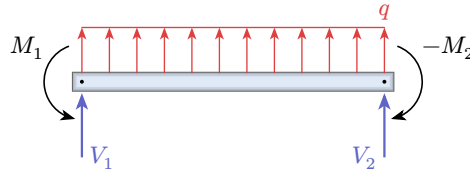


Figure 2.12: Nodal forces in a beam element with constant load.

Proceeding with successive integrations as before, we obtain the following equations:

$$\begin{aligned}
 V(x) &= EI \frac{d^3v}{dx^3} = qx + c_1 \\
 M(x) &= EI \frac{d^2v}{dx^2} = q \frac{x^2}{2} + c_1x + c_2 \\
 \theta(x) &= \frac{dv}{dx} = \frac{1}{EI} \left(q \frac{x^3}{6} + c_1 \frac{x^2}{2} + c_2x + c_3 \right) \\
 v(x) &= \frac{1}{EI} \left(q \frac{x^4}{24} + c_1 \frac{x^3}{6} + c_2 \frac{x^2}{2} + c_3x + c_4 \right)
 \end{aligned} \tag{2.46}$$

Evaluating the displacements at both ends gives:

$$\begin{aligned}
 v_1 &= v(0) = \frac{1}{EI}c_4 \\
 \theta_1 &= \theta(0) = \frac{1}{EI}c_3 \\
 v_2 &= v(\ell) = \frac{1}{EI} \left(q\frac{\ell^4}{24} + c_1\frac{\ell^3}{6} + c_2\frac{\ell^2}{2} + c_3\ell + c_4 \right) \\
 \theta_2 &= \theta(\ell) = \frac{1}{EI} \left(q\frac{\ell^3}{6} + c_1\frac{\ell^2}{2} + c_2\ell + c_3 \right)
 \end{aligned} \tag{2.47}$$

Writing these equations in matrix form we have

$$\begin{pmatrix} 0 & 0 & 0 & \frac{1}{EI} \\ 0 & 0 & \frac{1}{EI} & 0 \\ \frac{\ell^3}{6EI} & \frac{\ell^2}{2EI} & \frac{\ell}{EI} & \frac{1}{EI} \\ \frac{\ell^2}{2EI} & \frac{\ell}{EI} & \frac{1}{EI} & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 - \frac{q\ell^4}{24EI} \\ \theta_2 - \frac{q\ell^3}{6EI} \end{pmatrix} \tag{2.48}$$

Solving for the integration constants yields:

$$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} \frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} & -\frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} \\ -\frac{6EI}{\ell^2} & -\frac{4EI}{\ell} & \frac{6EI}{\ell^2} & -\frac{2EI}{\ell} \\ 0 & EI & 0 & 0 \\ EI & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 - \frac{q\ell^4}{24EI} \\ \theta_2 - \frac{q\ell^3}{6EI} \end{pmatrix} \tag{2.49}$$

Subsequently, defining the nodal forces as $V_1 = V(0)$, $M_1 = -M(0)$, $V_2 = -V(\ell)$, and $M_2 = M(\ell)$, we can express them as functions of the integration constants:

$$\begin{aligned}
 V_1 &= c_1 \\
 M_1 &= -c_2 \\
 V_2 &= -q\ell - c_1 \\
 M_2 &= q\frac{\ell^2}{2} + c_1\ell + c_2
 \end{aligned} \tag{2.50}$$

Substituting the expressions for c_1 and c_2 and simplifying gives:

$$\begin{aligned}
 V_1 &= \frac{12EI}{\ell^3}v_1 + \frac{6EI}{\ell^2}\theta_1 - \frac{12EI}{\ell^3}v_2 + \frac{6EI}{\ell^2}\theta_2 - \frac{q\ell}{2} \\
 M_1 &= \frac{6EI}{\ell^2}v_1 + \frac{4EI}{\ell}\theta_1 - \frac{6EI}{\ell^2}v_2 + \frac{2EI}{\ell}\theta_2 - \frac{q\ell^2}{12} \\
 V_2 &= -\frac{12EI}{\ell^3}v_1 - \frac{6EI}{\ell^2}\theta_1 + \frac{12EI}{\ell^3}v_2 - \frac{6EI}{\ell^2}\theta_2 - \frac{q\ell}{2} \\
 M_2 &= \frac{6EI}{\ell^2}v_1 + \frac{2EI}{\ell}\theta_1 - \frac{6EI}{\ell^2}v_2 + \frac{4EI}{\ell}\theta_2 + \frac{q\ell^2}{12}
 \end{aligned} \tag{2.51}$$

Organizing the above equations in matrix form we finally get the finite element system

$$\frac{EI}{\ell^3} \begin{pmatrix} 12 & 6\ell & -12 & 6\ell \\ 6\ell & 4\ell^2 & -6\ell & 2\ell^2 \\ -12 & -6\ell & 12 & -6\ell \\ 6\ell & 2\ell^2 & -6\ell & 4\ell^2 \end{pmatrix} \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} V_1 \\ M_1 \\ V_2 \\ M_2 \end{pmatrix} + \frac{q\ell}{12} \begin{pmatrix} 6 \\ \ell \\ 6 \\ -\ell \end{pmatrix} \quad (2.52)$$

2.3.4 Trapezoidal load

For a trapezoidal distributed load we can consider

$$q(x) = q_1 + \frac{q_2 - q_1}{\ell} x \quad (2.53)$$

where q_1 and q_2 are the values of the distributed load at the left and right ends, respectively.

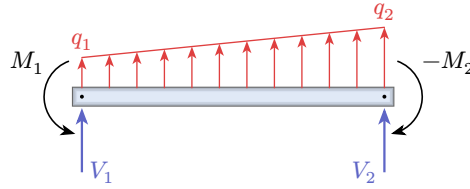


Figure 2.13: Nodal forces in a beam element with trapezoidal load.

Again, successive integrations of the load function yield:

$$\begin{aligned} V(x) &= EI \frac{d^3v}{dx^3} = q_1 x + \frac{q_2 - q_1}{\ell} \frac{x^2}{2} + c_1 \\ M(x) &= EI \frac{d^2v}{dx^2} = q_1 \frac{x^2}{2} + \frac{q_2 - q_1}{\ell} \frac{x^3}{6} + c_1 x + c_2 \\ \theta(x) &= \frac{dv}{dx} = \frac{1}{EI} \left(q_1 \frac{x^3}{6} + \frac{q_2 - q_1}{\ell} \frac{x^4}{24} + c_1 \frac{x^2}{2} + c_2 x + c_3 \right) \\ v(x) &= \frac{1}{EI} \left(q_1 \frac{x^4}{24} + \frac{q_2 - q_1}{\ell} \frac{x^5}{120} + c_1 \frac{x^3}{6} + c_2 \frac{x^2}{2} + c_3 x + c_4 \right) \end{aligned} \quad (2.54)$$

Considering the boundary displacements $v_1 = v(0)$, $\theta_1 = \theta(0)$, $v_2 = v(\ell)$, and $\theta_2 = \theta(\ell)$, the integration constants can be expressed as:

$$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} \frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} & -\frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} \\ -\frac{6EI}{\ell^2} & -\frac{4EI}{\ell} & \frac{6EI}{\ell^2} & -\frac{2EI}{\ell} \\ 0 & EI & 0 & 0 \\ EI & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 - \frac{q_1 \ell^4}{24EI} - \frac{(q_2 - q_1) \ell^4}{120EI} \\ \theta_2 - \frac{q_1 \ell^3}{6EI} - \frac{(q_2 - q_1) \ell^3}{24EI} \end{pmatrix} \quad (2.55)$$

Defining the nodal forces as $V_1 = V(0)$, $M_1 = -M(0)$, $V_2 = -V(\ell)$, and $M_2 = M(\ell)$, we obtain:

$$\begin{aligned} V_1 &= c_1 \\ M_1 &= -c_2 \\ V_2 &= -q_1 \ell - \frac{(q_2 - q_1) \ell}{2} - c_1 \\ M_2 &= q_1 \frac{\ell^2}{2} + \frac{(q_2 - q_1) \ell^2}{6} + c_1 \ell + c_2 \end{aligned} \quad (2.56)$$

Finally, substituting the expressions for c_1 and c_2 and organizing the results into matrix form yields the finite element system:

$$\frac{EI}{\ell^3} \begin{pmatrix} 12 & 6\ell & -12 & 6\ell \\ 6\ell & 4\ell^2 & -6\ell & 2\ell^2 \\ -12 & -6\ell & 12 & -6\ell \\ 6\ell & 2\ell^2 & -6\ell & 4\ell^2 \end{pmatrix} \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} V_1 \\ M_1 \\ V_2 \\ M_2 \end{pmatrix} + \begin{pmatrix} \frac{7q_1\ell}{20} + \frac{3q_2\ell}{20} \\ \frac{q_1\ell^2}{20} + \frac{q_2\ell^2}{30} \\ \frac{3q_1\ell}{20} + \frac{7q_2\ell}{20} \\ -\frac{q_1\ell^2}{30} - \frac{q_2\ell^2}{20} \end{pmatrix} \quad (2.57)$$

2.3.5 Plotting beam diagrams

Beam diagrams are obtained by evaluating the relevant field variables at several points along the element, with $0 \leq x \leq \ell$. For a beam element subjected to a constant distributed load, the exact shear force, bending moment, rotation, and deflection fields are:

$$\begin{aligned} V(x) &= qx + c_1 \\ M(x) &= q\frac{x^2}{2} + c_1x + c_2 \\ \theta(x) &= \frac{1}{EI} \left(q\frac{x^3}{6} + c_1\frac{x^2}{2} + c_2x + c_3 \right) \\ v(x) &= \frac{1}{EI} \left(q\frac{x^4}{24} + c_1\frac{x^3}{6} + c_2\frac{x^2}{2} + c_3x + c_4 \right) \end{aligned} \quad (2.58)$$

where c_1 , c_2 , c_3 , and c_4 are obtained from Eq. (2.49).

Hermite interpolation functions

The analytical expressions presented above depend on the applied load and on the integration constants. As a result, the corresponding field expressions must be derived separately for each loading case. In contrast, a general finite element approximation can be written solely in terms of the nodal degrees of freedom, without explicit dependence on the applied load. This is achieved by using cubic Hermite interpolation functions.

For a two-node beam element, these functions are given by

$$\begin{aligned} N_1(x) &= 1 - \frac{3x^2}{\ell^2} + \frac{2x^3}{\ell^3} \\ N_2(x) &= x - \frac{2x^2}{\ell} + \frac{x^3}{\ell^2} \\ N_3(x) &= \frac{3x^2}{\ell^2} - \frac{2x^3}{\ell^3} \\ N_4(x) &= \frac{x^3}{\ell^2} - \frac{x^2}{\ell} \end{aligned} \quad (2.59)$$

and are illustrated in Figure 2.14. Using the nodal degrees of freedom v_1 , θ_1 , v_2 , and θ_2 , and adopting the Hermite interpolation functions as basis functions, the transverse deflection is approximated by

$$v(x) = v_1N_1(x) + \theta_1N_2(x) + v_2N_3(x) + \theta_2N_4(x) \quad (2.60)$$

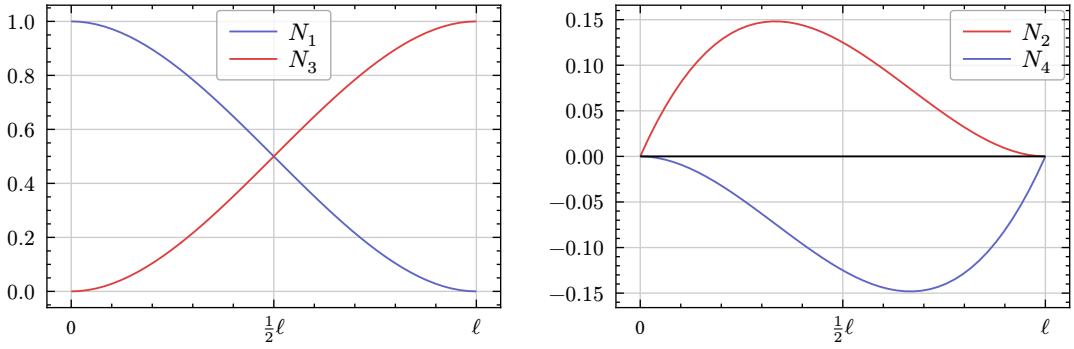


Figure 2.14: Cubic Hermite interpolation functions for a beam element.

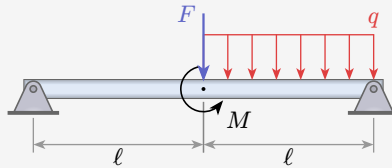
Using the Euler-Bernoulli relations $\theta(x) = \frac{dv(x)}{dx}$, $M(x) = EI \frac{d^2v(x)}{dx^2}$, and $V(x) = EI \frac{d^3v(x)}{dx^3}$, the corresponding finite element approximations for the rotation, bending moment, and shear force are:

$$\begin{aligned} \theta(x) &= v_1 \frac{dN_1(x)}{dx} + \theta_1 \frac{dN_2(x)}{dx} + v_2 \frac{dN_3(x)}{dx} + \theta_2 \frac{dN_4(x)}{dx} \\ M(x) &= EI \left(v_1 \frac{d^2N_1(x)}{dx^2} + \theta_1 \frac{d^2N_2(x)}{dx^2} + v_2 \frac{d^2N_3(x)}{dx^2} + \theta_2 \frac{d^2N_4(x)}{dx^2} \right) \\ V(x) &= EI \left(v_1 \frac{d^3N_1(x)}{dx^3} + \theta_1 \frac{d^3N_2(x)}{dx^3} + v_2 \frac{d^3N_3(x)}{dx^3} + \theta_2 \frac{d^3N_4(x)}{dx^3} \right) \end{aligned} \quad (2.61)$$

Because the interpolated deflection $v(x)$ is cubic within each element, the rotation is quadratic, the bending moment is linear, and the shear force is constant. Therefore, when internal forces are recovered from interpolation alone, a single beam element cannot reproduce the exact quadratic bending-moment diagram or the exact linear shear-force diagram associated with a uniformly distributed load.

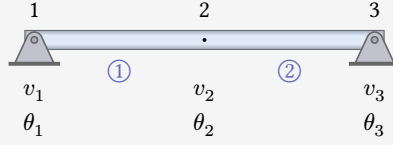
Example 2.3:

Given the beam in the figure, find all nodal displacements and plot the shear, bending moment, and deflection diagrams. Consider $\ell = 1$ m, $F = 1$ kN, $M = 1$ kNm, $q = 12 \frac{\text{kN}}{\text{m}}$ and $EI = 10$ kN m².



Solution

Two beam elements can be defined. The figure below shows the node and element numbering. The figure also lists the degrees of freedom per node. Note that θ_1 , v_2 , θ_2 , and θ_3 are unknown displacements, while v_1 and v_3 are equal to zero.



The stiffness matrices for each element are computed from the material and geometry properties. Then, each row and column in these matrices can be related to global degrees of freedom. Thus

$$\mathbf{K}^{(1)} = \begin{pmatrix} v_1 & \theta_1 & v_2 & \theta_2 \\ 120 & 60 & -120 & 60 \\ 60 & 40 & -60 & 20 \\ -120 & -60 & 120 & -60 \\ 60 & 20 & -60 & 40 \end{pmatrix} \begin{matrix} V_1 \\ M_1 \\ V_2 \\ M_2 \end{matrix} \quad \text{and} \quad \mathbf{K}^{(2)} = \begin{pmatrix} v_2 & \theta_2 & v_3 & \theta_3 \\ 120 & 60 & -120 & 60 \\ 60 & 40 & -60 & 20 \\ -120 & -60 & 120 & -60 \\ 60 & 20 & -60 & 40 \end{pmatrix} \begin{matrix} V_2 \\ M_2 \\ V_3 \\ M_3 \end{matrix}$$

Assembling the global stiffness matrix, we get the following global system

$$\begin{matrix} V_1 \\ M_1 \\ V_2 \\ M_2 \\ V_3 \\ M_3 \end{matrix} \begin{pmatrix} v_1 & \theta_1 & v_2 & \theta_2 & v_3 & \theta_3 \\ 120 & 60 & -120 & 60 & 0 & 0 \\ 60 & 40 & -60 & 20 & 0 & 0 \\ -120 & -60 & 240 & 0 & -120 & 60 \\ 60 & 20 & 0 & 80 & -60 & 20 \\ 0 & 0 & -120 & -60 & 120 & -60 \\ 0 & 0 & 60 & 20 & -60 & 40 \end{pmatrix} \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{pmatrix} = \begin{pmatrix} V_1 \\ 0 \\ -1 \\ 1 \\ V_3 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -6 \\ -1 \\ -6 \\ 1 \end{pmatrix}$$

The system above can be reorganized to place the unknown displacements, θ_1 , v_2 , θ_2 and θ_3 , at the beginning of the displacement vector

$$\begin{matrix} M_1 \\ V_2 \\ M_2 \\ M_3 \end{matrix} \left(\begin{array}{cccc|cc} \theta_1 & v_2 & \theta_2 & \theta_3 & v_1 & v_3 \\ 40 & -60 & 20 & 0 & 60 & 0 \\ -60 & 240 & 0 & 60 & -120 & -120 \\ 20 & 0 & 80 & 20 & 60 & -60 \\ 0 & 60 & 20 & 40 & 0 & -60 \end{array} \right) \begin{pmatrix} \theta_1 \\ v_2 \\ \theta_2 \\ \theta_3 \end{pmatrix} = \begin{pmatrix} 0 \\ -7 \\ 0 \\ 1 \end{pmatrix}$$

Therein $V'_3 = V_3 - 6$.

Next, the unknown displacements are found from the system below

$$\begin{pmatrix} 40 & -60 & 20 & 0 \\ -60 & 240 & 0 & 60 \\ 20 & 0 & 80 & 20 \\ 0 & 60 & 20 & 40 \end{pmatrix} \begin{pmatrix} \theta_1 \\ v_2 \\ \theta_2 \\ \theta_3 \end{pmatrix} = \begin{pmatrix} 0 \\ -7 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 600 \\ -120 & -120 \\ 60 & -60 \\ 0 & -60 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The obtained displacements are $\theta_1 = -\frac{5}{24}$ rad, $v_2 = -\frac{17}{120}$ m, $\theta_2 = -\frac{1}{120}$ rad, and $\theta_3 = \frac{29}{120}$ rad. Next, the unknown reaction forces are computed as:

$$\begin{pmatrix} V_1 \\ V_3' \end{pmatrix} = \begin{pmatrix} 60 & -120 & 60 & 0 \\ 0 & -120 & -60 & -60 \end{pmatrix} \begin{pmatrix} \theta_1 \\ v_2 \\ \theta_2 \\ \theta_3 \end{pmatrix} + \begin{pmatrix} 120 & 0 \\ 0 & 120 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The resulting values are $V_1 = 4$ kN and $V_3' = 3$ kN. Finally, $V_3 = V_3' + 6 = 9$ kN.

Using direct imposition for the zero-displacement constraints, the initial system is modified as

$$\begin{matrix} & v_1 & \theta_1 & v_2 & \theta_2 & v_3 & \theta_3 \\ \begin{matrix} V_1 \\ M_1 \\ V_2 \\ M_2 \\ V_3 \\ M_3 \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 40 & -60 & 20 & 0 & 0 \\ 0 & -60 & 240 & 0 & 0 & 60 \\ 0 & 20 & 0 & 80 & 0 & 20 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 60 & 20 & 0 & 40 \end{pmatrix} & \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{pmatrix} & = & \begin{pmatrix} 0 \text{ m} \\ 0 \text{ kNm} \\ -7 \text{ kN} \\ 1 \text{ kNm} \\ 0 \text{ m} \\ 0 \text{ kNm} \end{pmatrix} \end{matrix}$$

Solving this system, the displacement vector is given by

$$U = \begin{pmatrix} 0 \text{ m} \\ -\frac{5}{24} \text{ rad} \\ -\frac{17}{120} \text{ m} \\ -\frac{1}{120} \text{ rad} \\ 0 \text{ m} \\ \frac{29}{120} \text{ rad} \end{pmatrix}$$

Once the nodal displacements are found we can proceed to plot the diagrams for deflection, bending moment and shear force according to the functions

$$v(x) = \frac{1}{EI} \left(q \frac{x^4}{24} + c_1 \frac{x^3}{6} + c_2 \frac{x^2}{2} + c_3 x + c_4 \right)$$

$$\theta(x) = \frac{1}{EI} \left(q \frac{x^3}{6} + c_1 \frac{x^2}{2} + c_2 x + c_3 \right)$$

$$M(x) = q \frac{x^2}{2} + c_1 x + c_2$$

$$V(x) = qx + c_1$$

where the integration constants are computed from

$$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} \frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} & -\frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} \\ -\frac{6EI}{\ell^2} & -\frac{4EI}{\ell} & \frac{6EI}{\ell^2} & -\frac{2EI}{\ell} \\ 0 & EI & 0 & 0 \\ EI & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 - \frac{q\ell^4}{24EI} \\ \theta_2 - \frac{q\ell^3}{6EI} \end{pmatrix}$$

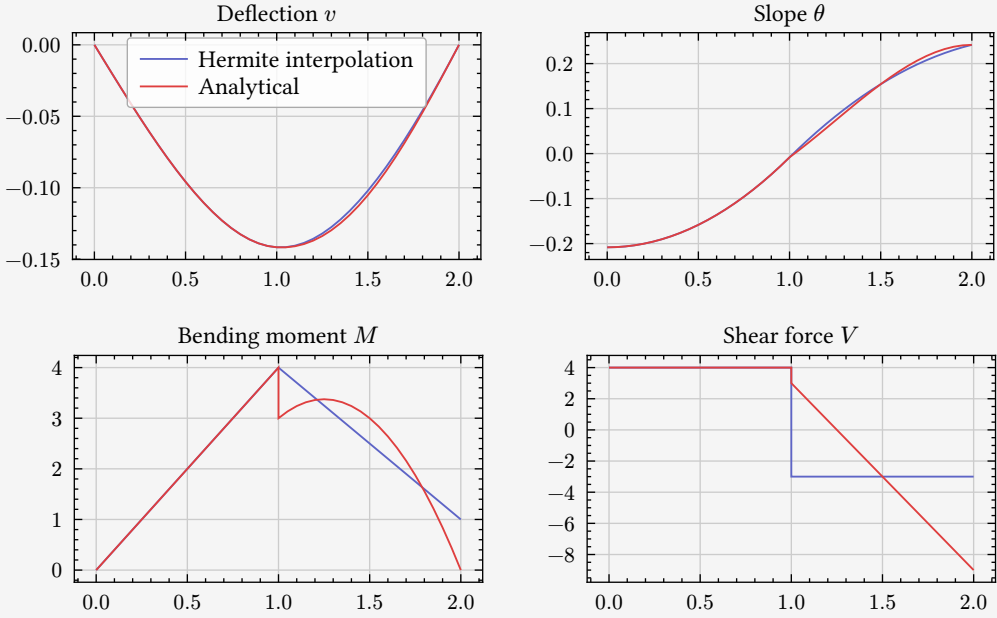
On the other hand, using Hermite interpolation functions, the diagrams can be plotted from the following expressions:

$$v(x) = v_1 N_1(x) + \theta_1 N_2(x) + v_2 N_3(x) + \theta_2 N_4(x)$$

$$\theta(x) = v_1 \frac{dN_1(x)}{dx} + \theta_1 \frac{dN_2(x)}{dx} + v_2 \frac{dN_3(x)}{dx} + \theta_2 \frac{dN_4(x)}{dx}$$

$$M(x) = EI \left(v_1 \frac{d^2 N_1(x)}{dx^2} + \theta_1 \frac{d^2 N_2(x)}{dx^2} + v_2 \frac{d^2 N_3(x)}{dx^2} + \theta_2 \frac{d^2 N_4(x)}{dx^2} \right)$$

$$V(x) = EI \left(v_1 \frac{d^3 N_1(x)}{dx^3} + \theta_1 \frac{d^3 N_2(x)}{dx^3} + v_2 \frac{d^3 N_3(x)}{dx^3} + \theta_2 \frac{d^3 N_4(x)}{dx^3} \right)$$



2.4 Frame element

By applying the **superposition principle**, the finite element equations of a beam element can be combined with those of a truss element to simultaneously account for bending and axial deformations.

2.4.1 2D frame element

In two dimensions, the result is the so-called 2D frame element, which possesses three degrees of freedom per node: two translations and one rotation (**Figure 2.18**). The local finite element equilibrium equations for a 2D frame element are expressed as the following linear system:

$$\begin{pmatrix} \frac{EA}{\ell} & 0 & 0 & -\frac{EA}{\ell} & 0 & 0 \\ 0 & \frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} & 0 & -\frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} \\ 0 & \frac{6EI}{\ell^2} & \frac{4EI}{\ell} & 0 & -\frac{6EI}{\ell^2} & \frac{2EI}{\ell} \\ -\frac{EA}{\ell} & 0 & 0 & \frac{EA}{\ell} & 0 & 0 \\ 0 & -\frac{12EI}{\ell^3} & -\frac{6EI}{\ell^2} & 0 & \frac{12EI}{\ell^3} & -\frac{6EI}{\ell^2} \\ 0 & \frac{6EI}{\ell^2} & \frac{2EI}{\ell} & 0 & -\frac{6EI}{\ell^2} & \frac{4EI}{\ell} \end{pmatrix} \begin{pmatrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} f_1 \\ V_1 \\ M_1 \\ f_2 \\ V_2 \\ M_2 \end{pmatrix} \quad (2.62)$$

where u represents axial displacements and v represents normal displacements.

It is important to note that in linear frame elements, axial deformation and bending are **uncoupled**. This means that axial forces do not induce bending, and bending moments do not cause axial elongation.

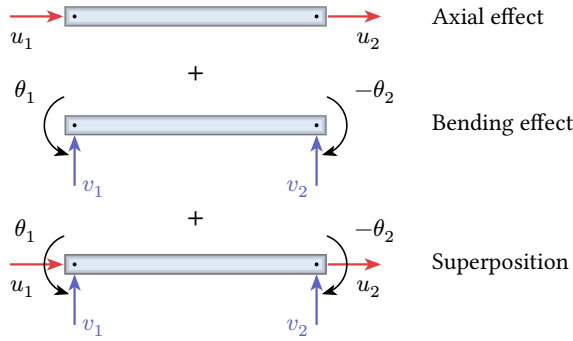


Figure 2.18: Frame element obtained by superposition.

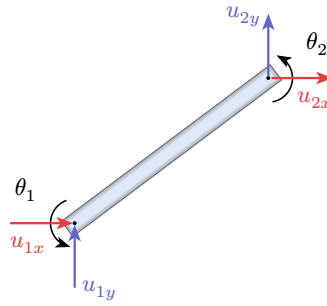


Figure 2.19: Tilted frame element in the xy reference.

The stiffness matrix for a frame element inclined at an angle θ relative to the global xy coordinate system is obtained using a transformation matrix. This matrix relates the global nodal displacements to the local ones as follows:

$$\begin{pmatrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} c & s & 0 & 0 & 0 & 0 \\ -s & c & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & c & s & 0 \\ 0 & 0 & 0 & -s & c & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_{1x} \\ u_{1y} \\ \theta_1 \\ u_{2x} \\ u_{2y} \\ \theta_2 \end{pmatrix} \quad (2.63)$$

where $c = \cos(\theta)$ and $s = \sin(\theta)$.

This equation can be written compactly as:

$$\bar{\mathbf{U}}^e = \mathbf{T} \mathbf{U}^e \quad (2.64)$$

Likewise, global nodal forces are related to local nodal forces by the transpose of the transformation matrix:

$$\mathbf{F}^e = \mathbf{T}^T \bar{\mathbf{F}}^e \quad (2.65)$$

Combining these kinematic and static transformations with the local equilibrium equation, $\bar{\mathbf{K}}^e \bar{\mathbf{U}}^e = \bar{\mathbf{F}}^e$, yields the global element stiffness matrix:

$$\mathbf{K}^e = \mathbf{T}^T \bar{\mathbf{K}}^e \mathbf{T} \quad (2.66)$$

2.4.2 3D frame element

Often referred to as a 3D beam element, the 3D frame element extends the 2D frame formulation into three dimensions. Each node possesses six degrees of freedom: three translations and three rotations. The formulation of the 3D frame element superposes four distinct, uncoupled structural behaviors: axial deformation, torsional deformation, and bending in two mutually orthogonal planes.

The axial stiffness along the local x' axis is identical to the 1D-truss element:

$$\mathbf{K}^{\text{axial}} = \frac{EA}{\ell} \begin{pmatrix} u_{1x'} & u_{2x'} \\ 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} f_{1x'} \\ f_{2x'} \end{pmatrix} \quad (2.67)$$

The torsional stiffness around the local x' axis relates twisting moments to twist angles, where G is the shear modulus and J is the torsional constant of the cross-section:

$$\mathbf{K}^{\text{torsion}} = \frac{GJ}{\ell} \begin{pmatrix} \theta_{1x'} & \theta_{2x'} \\ 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} M_{1x'} \\ M_{2x'} \end{pmatrix} \quad (2.68)$$

The bending stiffness in the local $x'y'$ plane involves transverse displacements along y' and rotations around the z' axis (where I_z is the moment of inertia about the local z' axis):

$$\mathbf{K}_{x'y'}^{\text{bend}} = \frac{EI_z}{\ell^3} \begin{pmatrix} u_{1y'} & \theta_{1z'} & u_{2y'} & \theta_{2z'} \\ 12 & 6\ell & -12 & 6\ell \\ 6\ell & 4\ell^2 & -6\ell & 2\ell^2 \\ -12 & -6\ell & 12 & -6\ell \\ 6\ell & 2\ell^2 & -6\ell & 4\ell^2 \end{pmatrix} \begin{pmatrix} f_{1y'} \\ M_{1z'} \\ f_{2y'} \\ M_{2z'} \end{pmatrix} \quad (2.69)$$

Similarly, the bending stiffness in the local $x'z'$ plane involves transverse displacements along z' and rotations around the y' axis (where I_y is the moment of inertia about the local y' axis). Note the sign differences in the off-diagonal terms due to the right-hand rule convention:

$$\mathbf{K}_{x'z'}^{\text{bend}} = \frac{EI_y}{\ell^3} \begin{pmatrix} u_{1z'} & \theta_{1y'} & u_{2z'} & \theta_{2y'} \\ 12 & -6\ell & -12 & -6\ell \\ -6\ell & 4\ell^2 & 6\ell & 2\ell^2 \\ -12 & 6\ell & 12 & 6\ell \\ -6\ell & 2\ell^2 & 6\ell & 4\ell^2 \end{pmatrix} \begin{pmatrix} f_{1z'} \\ M_{1y'} \\ f_{2z'} \\ M_{2y'} \end{pmatrix} \quad (2.70)$$

Superposing these independent behaviors assembles the complete 12×12 local stiffness matrix $\bar{\mathbf{K}}^e$:

$$\bar{\mathbf{K}}^e = \begin{pmatrix} u_{1x'} & u_{1y'} & u_{1z'} & \theta_{1x'} & \theta_{1y'} & \theta_{1z'} & u_{2x'} & u_{2y'} & u_{2z'} & \theta_{2x'} & \theta_{2y'} & \theta_{2z'} \\ \frac{EA}{\ell} & 0 & 0 & 0 & 0 & 0 & -\frac{EA}{\ell} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{12EI_z}{\ell^3} & 0 & 0 & 0 & -\frac{6EI_z}{\ell^2} & 0 & -\frac{12EI_z}{\ell^3} & 0 & 0 & 0 & \frac{6EI_z}{\ell^2} \\ 0 & 0 & \frac{12EI_y}{\ell^3} & 0 & -\frac{6EI_y}{\ell^2} & 0 & 0 & 0 & -\frac{12EI_y}{\ell^3} & 0 & -\frac{6EI_y}{\ell^2} & 0 \\ 0 & 0 & 0 & \frac{GJ}{\ell} & 0 & 0 & 0 & 0 & 0 & -\frac{GJ}{\ell} & 0 & 0 \\ 0 & 0 & -\frac{6EI_y}{\ell^2} & 0 & \frac{4EI_y}{\ell} & 0 & 0 & 0 & \frac{6EI_y}{\ell^2} & 0 & \frac{2EI_y}{\ell} & 0 \\ 0 & \frac{6EI_z}{\ell^2} & 0 & 0 & 0 & \frac{4EI_z}{\ell} & 0 & -\frac{6EI_z}{\ell^2} & 0 & 0 & 0 & \frac{2EI_z}{\ell} \\ -\frac{EA}{\ell} & 0 & 0 & 0 & 0 & 0 & \frac{EA}{\ell} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{12EI_z}{\ell^3} & 0 & 0 & 0 & -\frac{6EI_z}{\ell^2} & 0 & \frac{12EI_z}{\ell^3} & 0 & 0 & 0 & -\frac{6EI_z}{\ell^2} \\ 0 & 0 & -\frac{12EI_y}{\ell^3} & 0 & \frac{6EI_y}{\ell^2} & 0 & 0 & 0 & \frac{12EI_y}{\ell^3} & 0 & \frac{6EI_y}{\ell^2} & 0 \\ 0 & 0 & 0 & -\frac{GJ}{\ell} & 0 & 0 & 0 & 0 & 0 & \frac{GJ}{\ell} & 0 & 0 \\ 0 & 0 & -\frac{6EI_y}{\ell^2} & 0 & \frac{2EI_y}{\ell} & 0 & 0 & 0 & \frac{6EI_y}{\ell^2} & 0 & \frac{4EI_y}{\ell} & 0 \\ 0 & \frac{6EI_z}{\ell^2} & 0 & 0 & 0 & \frac{2EI_z}{\ell} & 0 & -\frac{6EI_z}{\ell^2} & 0 & 0 & 0 & \frac{4EI_z}{\ell} \end{pmatrix} \begin{pmatrix} f_{1x'} \\ f_{1y'} \\ f_{1z'} \\ M_{1x'} \\ M_{1y'} \\ M_{1z'} \\ f_{2x'} \\ f_{2y'} \\ f_{2z'} \\ M_{2x'} \\ M_{2y'} \\ M_{2z'} \end{pmatrix} \quad (2.71)$$

To transform this local stiffness matrix into the global coordinate system, a 3×3 rotation matrix $\mathbf{R}$ is established. This matrix is composed of the direction cosines of the local x' , y' , and z' axes with respect to the global x , y , and z axes:

$$\mathbf{R} = \begin{pmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{pmatrix} \quad (2.72)$$

where (l_1, m_1, n_1) are the direction cosines for the local x' axis, (l_2, m_2, n_2) for the local y' axis, and (l_3, m_3, n_3) for the local z' axis.

The full 12×12 block-diagonal transformation matrix $\mathbf{T}$ is then constructed as:

$$\mathbf{T} = \begin{pmatrix} \mathbf{R} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{R} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{R} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{R} \end{pmatrix} \quad (2.73)$$

Finally, the global element stiffness matrix is obtained through the standard transformation:

$$\mathbf{K}^e = \mathbf{T}^T \bar{\mathbf{K}}^e \mathbf{T} \quad (2.74)$$

2.5 Timoshenko beam element

The Euler-Bernoulli beam theory neglects transverse shear deformation. For short or deep beams this assumption may be too restrictive, so the **Timoshenko beam theory** is preferred. The Timoshenko beam theory uses two coupled differential equations to account for both bending and shear deformation:

$$\underbrace{\frac{d}{dx} \left(kGA \left(\frac{dv}{dx} - \theta \right) \right)}_V = q(x) \quad (2.75)$$

$$\frac{d}{dx} \left(EI \frac{d\theta}{dx} \right) + kGA \left(\frac{dv}{dx} - \theta \right) = 0$$

where G is the shear modulus and k is the shear correction factor.

For a two-node beam element without axial deformation, each node has two degrees of freedom: the transverse displacement v and the cross-section rotation θ . Thus,

$$\mathbf{U}^e = \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix} \quad \mathbf{F}^e = \begin{pmatrix} V_1 \\ M_1 \\ V_2 \\ M_2 \end{pmatrix} \quad (2.76)$$

In the Timoshenko theory, the rotation of the cross section is independent of the slope of the deflection curve. The bending curvature and the shear strain are

$$\kappa_b = \frac{d\theta}{dx} \quad (2.77)$$

$$\gamma = \frac{dv}{dx} - \theta$$

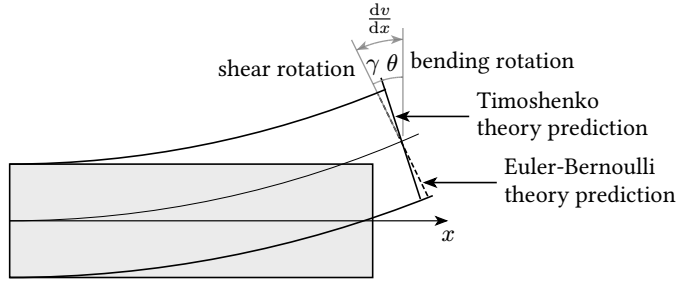


Figure 2.20: Kinematics of a Timoshenko beam section.

and the constitutive relations are

$$\begin{aligned} M &= EI\kappa_b = EI \frac{d\theta}{dx} \\ V &= -\kappa GA\gamma = \kappa GA \left(\theta - \frac{dv}{dx} \right) \end{aligned} \quad (2.78)$$

The shear correction factor κ is introduced because the transverse shear stress is not uniformly distributed over the cross section. Por exemplo, para uma seção retangular, a distribuição de tensão de cisalhamento é parabólica como apresentado na [Figure 2.21](#). In the Timoshenko beam model, the shear strain is represented by an average value over the section, which would overestimate the shear energy if the factor κ were not included. Therefore, κGA represents an **effective shear stiffness**. For a rectangular section, a common approximation is $\kappa = \frac{5}{6}$, while for a solid circular section a common approximation is $\kappa = \frac{9}{10}$.

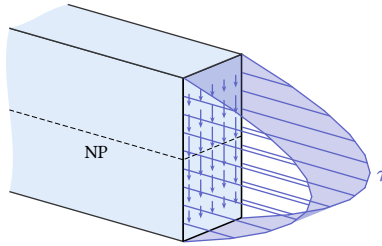


Figure 2.21: Parabolic shear stress distribution in a rectangular section.

For a beam element with no distributed load, equilibrium gives

$$\begin{aligned} \frac{dV}{dx} &= 0 \\ \frac{dM}{dx} - V &= 0 \end{aligned} \quad (2.79)$$

Hence,

$$\begin{aligned} V(x) &= c_1 \\ M(x) &= c_1 x + c_2 \end{aligned} \quad (2.80)$$

Substituting these expressions into the constitutive equations, we obtain

$$\frac{d\theta}{dx} = \frac{M}{EI} = \frac{1}{EI}(c_1 x + c_2) \quad (2.81)$$

and

$$\frac{dv}{dx} = \theta - \frac{V}{\kappa GA} \quad (2.82)$$

After integration,

$$\begin{aligned} \theta(x) &= \frac{c_1}{2EI}x^2 + \frac{c_2}{EI}x + c_3 \\ v(x) &= \frac{c_1}{6EI}x^3 + \frac{c_2}{2EI}x^2 + c_3x - \frac{c_1}{\kappa GA}x + c_4 \end{aligned} \quad (2.83)$$

Evaluating these functions at the element nodes gives

$$\begin{aligned} v_1 &= v(0) = c_4 \\ \theta_1 &= \theta(0) = c_3 \\ v_2 &= c_1 \frac{\ell^3}{6EI} + c_2 \frac{\ell^2}{2EI} + c_3\ell - c_1 \frac{\ell}{\kappa GA} + c_4 \\ \theta_2 &= c_1 \frac{\ell^2}{2EI} + c_2 \frac{\ell}{EI} + c_3 \end{aligned} \quad (2.84)$$

From the last two equations,

$$\begin{aligned} \theta_2 - \theta_1 &= c_1 \frac{\ell^2}{2EI} + c_2 \frac{\ell}{EI} \\ v_2 - v_1 - \ell\theta_1 &= c_1 \frac{\ell^3}{6EI} + c_2 \frac{\ell^2}{2EI} - c_1 \frac{\ell}{\kappa GA} \end{aligned} \quad (2.85)$$

Replacing c_2 from the first equation into the second, we obtain

$$c_1 = 12E \frac{I}{\ell^3(1 + \varphi)} \left(v_1 - v_2 + \frac{\ell}{2}(\theta_1 + \theta_2) \right) \quad (2.86)$$

where

$$\varphi = \frac{12EI}{\kappa GA \ell^2} \quad (2.87)$$

The nodal forces are related to the internal forces by

$$\begin{aligned} V_1 &= V(0) = c_1 \\ M_1 &= -M(0) = -c_2 \\ V_2 &= -V(\ell) = -c_1 \\ M_2 &= M(\ell) = c_1\ell + c_2 \end{aligned} \quad (2.88)$$

Using the expression for c_1 and

$$c_2 = E \frac{I}{\ell} (\theta_2 - \theta_1) - c_1 \frac{\ell}{2} \quad (2.89)$$

the nodal forces can be written in terms of the nodal displacements as

$$\begin{aligned}
 V_1 &= \frac{12EI}{\ell^3(1+\varphi)}v_1 + \frac{6EI}{\ell^2(1+\varphi)}\theta_1 - \frac{12EI}{\ell^3(1+\varphi)}v_2 + \frac{6EI}{\ell^2(1+\varphi)}\theta_2 \\
 M_1 &= \frac{6EI}{\ell^2(1+\varphi)}v_1 + \frac{(4+\varphi)EI}{\ell(1+\varphi)}\theta_1 - \frac{6EI}{\ell^2(1+\varphi)}v_2 + \frac{(2-\varphi)EI}{\ell(1+\varphi)}\theta_2 \\
 V_2 &= -\frac{12EI}{\ell^3(1+\varphi)}v_1 - \frac{6EI}{\ell^2(1+\varphi)}\theta_1 + \frac{12EI}{\ell^3(1+\varphi)}v_2 - \frac{6EI}{\ell^2(1+\varphi)}\theta_2 \\
 M_2 &= \frac{6EI}{\ell^2(1+\varphi)}v_1 + \frac{(2-\varphi)EI}{\ell(1+\varphi)}\theta_1 - \frac{6EI}{\ell^2(1+\varphi)}v_2 + \frac{(4+\varphi)EI}{\ell(1+\varphi)}\theta_2
 \end{aligned} \tag{2.90}$$

Therefore, the local stiffness matrix is

$$\mathbf{K}^e = \frac{1}{1+\varphi} \begin{pmatrix} \frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} & -\frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} \\ \frac{6EI}{\ell^2} & \frac{(4+\varphi)EI}{\ell} & -\frac{6EI}{\ell^2} & \frac{(2-\varphi)EI}{\ell} \\ -\frac{12EI}{\ell^3} & -\frac{6EI}{\ell^2} & \frac{12EI}{\ell^3} & -\frac{6EI}{\ell^2} \\ \frac{6EI}{\ell^2} & \frac{(2-\varphi)EI}{\ell} & -\frac{6EI}{\ell^2} & \frac{(4+\varphi)EI}{\ell} \end{pmatrix} \tag{2.91}$$

Therefore, the finite element equations for the Timoshenko beam element are

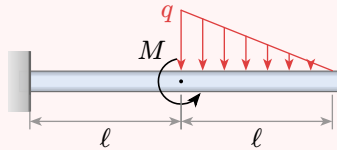
$$\mathbf{K}^e \mathbf{U}^e = \mathbf{F}^e \tag{2.92}$$

When the shear stiffness becomes very large, that is, $\kappa GA \rightarrow \infty$, we obtain $\varphi \rightarrow 0$ and the stiffness matrix reduces to the Euler-Bernoulli beam element matrix.

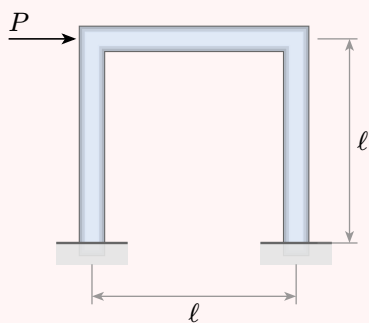
In practice, low-order Timoshenko beam elements may suffer from **shear locking** when the beam is slender. A common remedy is to use reduced or selective numerical integration for the shear term. For slender beams, another practical alternative is to use the Euler-Bernoulli theory, since transverse shear deformation becomes negligible in that limit.

Exercises 2.2:

1. Compute the nodal displacements and reaction forces for the cantilever beam shown below. Discretize the beam using two finite elements and assume a flexural rigidity of $EI = 100 \text{ kN m}^2$, segment length $\ell = 1 \text{ m}$, concentrated moment $M = 20 \text{ kNm}$, and distributed load $q = 120 \text{ kN/m}$. Plot the corresponding shear force, bending moment, and deflection diagrams.



2. Compute the nodal displacements and reaction forces for the one-story portal frame. Assume an applied load of $P = 10 \text{ kN}$, flexural rigidity $EI = 10 \text{ kN m}^2$, segment length $\ell = 1 \text{ m}$, and axial stiffness $EA = 100 \text{ kN}$. Plot the resulting shear force and bending moment diagrams.



3. Demonstrate that the Bernoulli beam differential equation can be derived from the Timoshenko beam equations by taking the limit as $\frac{dv}{dx} \rightarrow \theta$.

3 ISOPARAMETRIC FORMULATION

Isoparametric elements are finite elements in which the same **shape functions** are used to interpolate both the element geometry and the field variables, such as displacement or temperature. In most practical elements, these functions are low-order polynomials written in terms of **natural** or **parametric** coordinates.

The central idea is to define a simple **reference element** in a parametric space, such as a reference triangle or a reference square. The coordinates of the physical element are then obtained by a mapping from this reference element, using the nodal coordinates and the same shape functions adopted for the field approximation.

This approach provides a unified way to formulate many element types. Once the interpolation is defined in the reference element, derivatives and integrals in the physical domain can be obtained through the mapping, usually by means of the Jacobian matrix.

The isoparametric formulation is therefore one of the main reasons finite elements can represent irregular geometries while retaining a systematic interpolation and numerical integration procedure.

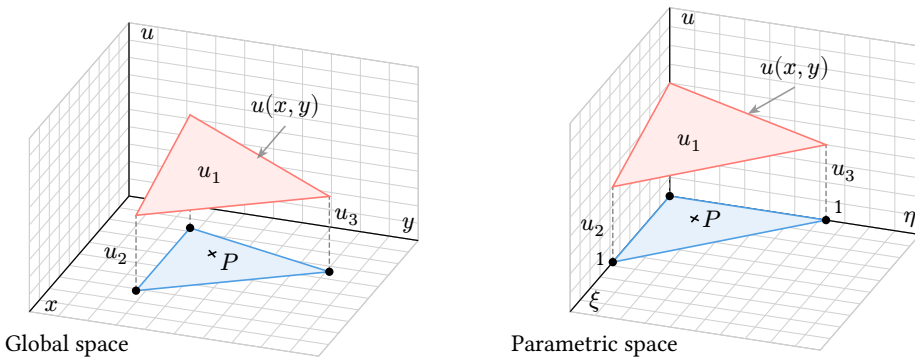


Figure 3.1: Parametric mapping and interpolation

3.1 Coordinate systems and interpolation

3.1.1 Coordinate systems

The location of a point in an element can be described by different coordinate systems, each one convenient for a specific purpose.

Global coordinate system: Global coordinates describe the position of points in the physical domain.

Nodal coordinates of the mesh are therefore usually given in this system.

Local coordinate system: A local coordinate system is attached to the element and is used to describe points with respect to that element. For a 1D element, it is common to use a coordinate that varies from one end node to the other, for example $x \in [0, \ell]$.

Parametric coordinate system: Parametric coordinates are dimensionless coordinates defined on the reference element. They are also commonly called **natural coordinates**. For a two-node 1D element, the natural coordinate is $\xi \in [-1, 1]$. In many texts, natural and local coordinates are treated interchangeably; here they are distinguished because the natural system is the one used to define isoparametric interpolation and mapping.

Figure 3.2 illustrates these three coordinate systems for a two-node 1D element.

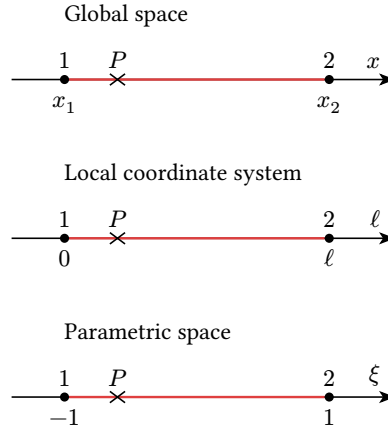


Figure 3.2: Coordinate systems for a two-node 1D element

3.1.2 Interpolation functions

In the finite element method, a continuous field is approximated inside each element from values stored at a finite number of nodes. For example, the displacement at a point of a two-node 1D element can be interpolated in terms of the local coordinate x as

$$u = N_1(x)u_1 + N_2(x)u_2 \quad (3.1)$$

where $x \in [0, \ell]$ and $N_i(x)$ is the interpolation function associated with node i .

In isoparametric elements, the interpolation is more conveniently written in terms of the natural coordinate ξ . In this context, the interpolation functions are usually called **shape functions**. The displacement approximation becomes

$$u = N_1(\xi)u_1 + N_2(\xi)u_2 \quad (3.2)$$

where $\xi \in [-1, 1]$.

For isoparametric interpolation, the same functions are also used to interpolate the geometry. In general, shape functions satisfy the partition-of-unity property

$$1 = \sum_{i=1}^n N_i \quad (3.3)$$

and the interpolations

$$\begin{aligned} x &= \sum_{i=1}^n N_i x_i \\ u &= \sum_{i=1}^n N_i u_i \end{aligned} \quad (3.4)$$

For multidimensional elements, the geometric interpolation is written componentwise in the same manner.

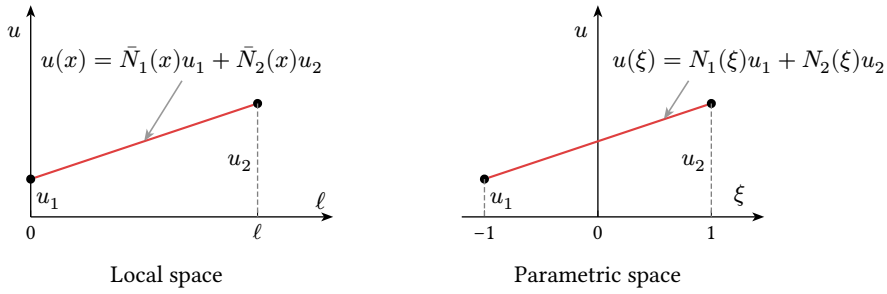


Figure 3.3: Interpolation in local and natural coordinates

3.1.3 Element mapping

The mapping process relates points of the reference element to points of the physical element. Once the natural coordinates of a point are known, its global coordinates are obtained by interpolation of the nodal coordinates. For example, if a point P has natural coordinate ξ in a three-node 1D element, its global coordinate is

$$x = N_1(\xi)x_1 + N_2(\xi)x_2 + N_3(\xi)x_3 \quad (3.5)$$

If the same element is embedded in a 2D space, both global coordinates are interpolated with the same shape functions:

$$\begin{pmatrix} x & y \end{pmatrix} = \begin{pmatrix} N_1(\xi) & N_2(\xi) & N_3(\xi) \end{pmatrix} \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \end{pmatrix} \quad (3.6)$$

or

$$\mathbf{x}^T = \mathbf{N}^T \mathbf{X} \quad (3.7)$$

where $\mathbf{x} = (x, y)$, $\mathbf{N} = (N_1, N_2, N_3)$, and $\mathbf{X}$ is the matrix collecting the nodal coordinates of the element.

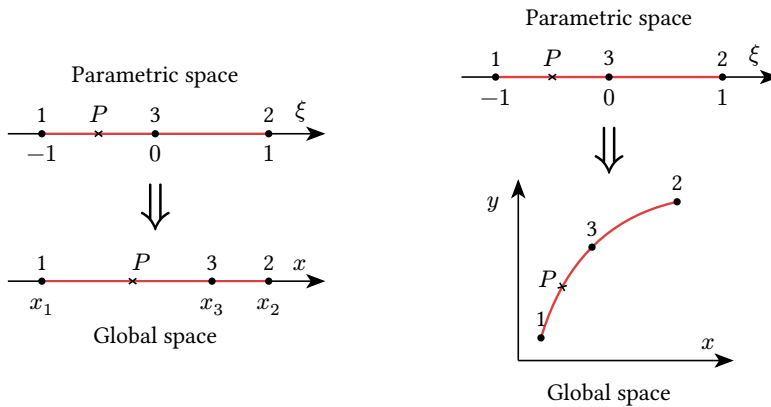


Figure 3.4: Isoparametric mapping of a three-node line element

3.1.4 Two-node 1D element

For a two-node 1D element, two linear interpolation functions are required to interpolate the field and the geometry. For this reason, this element is also called a **linear** element. The general form of these functions is

$$\begin{aligned} N_1(\xi) &= a_1 + b_1\xi \\ N_2(\xi) &= a_2 + b_2\xi \end{aligned} \quad (3.8)$$

where a_i and b_i are coefficients. These coefficients are obtained from the nodal interpolation conditions

$$\begin{aligned} N_1(-1) &= 1, & N_1(1) &= 0 \\ N_2(-1) &= 0, & N_2(1) &= 1 \end{aligned} \quad (3.9)$$

Substituting these conditions into the expressions above yields

$$\begin{aligned} a_1 + b_1(-1) &= 1 \\ a_1 + b_1(+1) &= 0 \\ a_2 + b_2(-1) &= 0 \\ a_2 + b_2(+1) &= 1 \end{aligned} \quad (3.10)$$

which leads to the systems

$$\begin{aligned} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a_2 \\ b_2 \end{pmatrix} &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{aligned} \quad (3.11)$$

Solving them leads to the coefficients $a_1 = \frac{1}{2}$, $b_1 = -\frac{1}{2}$, $a_2 = \frac{1}{2}$, and $b_2 = \frac{1}{2}$. Thus, the resulting shape functions are

$$\begin{aligned} N_1(\xi) &= \frac{1}{2} - \frac{\xi}{2} \\ N_2(\xi) &= \frac{1}{2} + \frac{\xi}{2} \end{aligned} \quad (3.12)$$

The derivatives of the shape functions with respect to the natural coordinate ξ are frequently used in the finite element formulation. For this element, they are constant and given by

$$\frac{\partial N_1}{\partial \xi} = -\frac{1}{2}, \quad \frac{\partial N_2}{\partial \xi} = \frac{1}{2} \quad (3.13)$$

3.1.5 Three-node 1D element

For a three-node 1D element, the interpolation is quadratic. The shape functions may therefore be assumed in the form

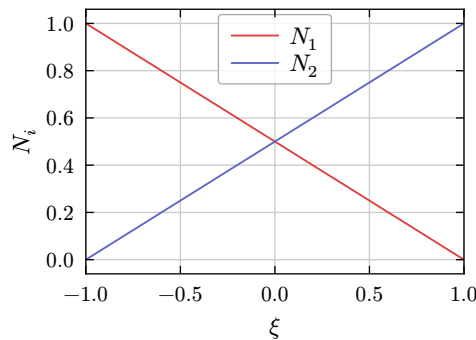


Figure 3.5: Shape functions of a two-node line element

$$N_i(\xi) = a_i + b_i\xi + c_i\xi^2 \quad (3.14)$$

The coefficients are obtained by imposing the nodal conditions as

$$\begin{aligned} N_1(-1) &= 1, & N_1(+1) &= 0, & N_1(0) &= 0 \\ N_2(-1) &= 0, & N_2(+1) &= 1, & N_2(0) &= 0 \\ N_3(-1) &= 0, & N_3(+1) &= 0, & N_3(0) &= 1 \end{aligned} \quad (3.15)$$

This produces three linear systems with the same coefficient matrix:

$$\begin{aligned} \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix} &= \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_2 \\ b_2 \\ c_2 \end{pmatrix} &= \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_3 \\ b_3 \\ c_3 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \end{aligned} \quad (3.16)$$

These systems can be condensed into a single matrix equation:

$$\begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (3.17)$$

where the coefficient matrix is obtained by inversion:

$$\begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}^{-1} \quad (3.18)$$

Once the coefficients are obtained, the shape functions are written as

$$\begin{aligned} N_1(\xi) &= \frac{1}{2}\xi^2 - \frac{1}{2}\xi \\ N_2(\xi) &= \frac{1}{2}\xi^2 + \frac{1}{2}\xi \\ N_3(\xi) &= 1 - \xi^2 \end{aligned} \quad (3.19)$$

These shape functions are plotted in [Figure 3.6](#).

In this element, the derivatives of the shape functions with respect to the natural coordinate are

$$\frac{\partial N_1}{\partial \xi} = \xi - \frac{1}{2}, \quad \frac{\partial N_2}{\partial \xi} = \xi + \frac{1}{2}, \quad \frac{\partial N_3}{\partial \xi} = -2\xi \quad (3.20)$$

3.1.6 1D Lagrangian shape functions

For an n -node 1D element, the shape functions can be constructed directly from the Lagrange interpolation polynomials defined on the interval $[-1, 1]$. For a set of nodal natural coordinates $\xi_1, \xi_2, \dots, \xi_n$, the i -th shape function is

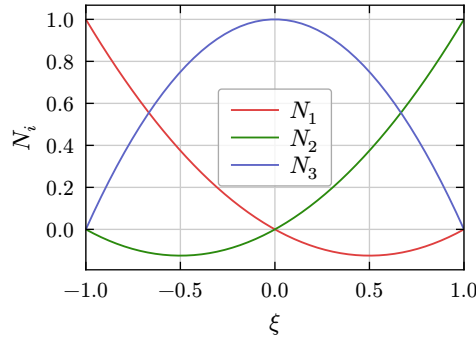


Figure 3.6: Shape functions of a three-node line element

$$N_i(\xi) = \prod_{\substack{j=1 \\ j \neq i}}^n \frac{\xi - \xi_j}{\xi_i - \xi_j} \quad (3.21)$$

These functions satisfy the interpolation property

$$N_i(\xi_j) = \delta_{ij} \quad (3.22)$$

so each function is equal to 1 at its own node and 0 at the remaining nodes.

For example, for a two-node element with nodal natural coordinates $\xi_1 = -1$ and $\xi_2 = 1$, we obtain

$$\begin{aligned} N_1(\xi) &= \frac{\xi - 1}{-1 - 1} = \frac{1}{2} - \frac{\xi}{2} \\ N_2(\xi) &= \frac{\xi - (-1)}{1 - (-1)} = \frac{1}{2} + \frac{\xi}{2} \end{aligned} \quad (3.23)$$

and for a three-node element with nodal coordinates $\xi_1 = -1$, $\xi_2 = 1$, and $\xi_3 = 0$, the result is

$$\begin{aligned} N_1(\xi) &= \frac{(\xi - 1)(\xi - 0)}{(-1 - 1)(-1 - 0)} = \frac{1}{2}\xi^2 - \frac{1}{2}\xi \\ N_2(\xi) &= \frac{(\xi - (-1))(\xi - 0)}{(1 - (-1))(1 - 0)} = \frac{1}{2}\xi^2 + \frac{1}{2}\xi \\ N_3(\xi) &= \frac{(\xi - (-1))(\xi - 1)}{(0 - (-1))(0 - 1)} = 1 - \xi^2 \end{aligned} \quad (3.24)$$

These functions are the same as those obtained by solving the linear systems.

The same procedure can be applied to higher-order elements. For a four-node line element with nodal natural coordinates $\xi_1 = -1$, $\xi_2 = 1$, $\xi_3 = -\frac{1}{3}$, and $\xi_4 = \frac{1}{3}$, the corresponding shape functions are shown in Figure 3.7.

Example 3.1: Consider a 4-node 1D Lagrangian element with the following nodal displacement values

$$\mathbf{U}^e = (0.1, 0.3, 0.2, 0.15)$$

Use the corresponding shape functions to interpolate the displacement at $\xi = -0.5$ and $\xi = 0.5$.

Solution

Using the nodal natural coordinates $\xi_1 = -1$, $\xi_2 = 1$, $\xi_3 = -\frac{1}{3}$, and $\xi_4 = \frac{1}{3}$, the shape functions are

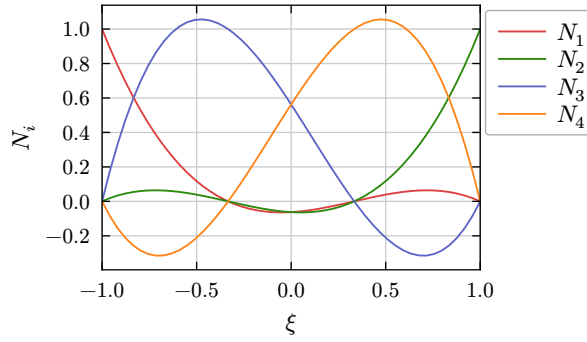


Figure 3.7: Shape functions of a four-node line element

$$\begin{aligned}
 N_1(\xi) &= \frac{1}{16}(-9\xi^3 + 9\xi^2 + \xi - 1) \\
 N_2(\xi) &= \frac{1}{16}(9\xi^3 + 9\xi^2 - \xi - 1) \\
 N_3(\xi) &= \frac{1}{16}(27\xi^3 - 9\xi^2 - 27\xi + 9) \\
 N_4(\xi) &= \frac{1}{16}(-27\xi^3 - 9\xi^2 + 27\xi + 9)
 \end{aligned}$$

The displacement at any point can be found as:

$$\begin{aligned}
 u &= \sum_{i=1}^n N_i(\xi)u_i \\
 &= N_1(\xi)u_1 + N_2(\xi)u_2 + N_3(\xi)u_3 + N_4(\xi)u_4
 \end{aligned}$$

where u_i are the nodal displacements and n is the number of nodes.

Substituting $\xi = -0.5$ and $\xi = 0.5$ into this expression, the following displacement values are obtained:

$$\begin{aligned}
 u(-0.5) &= 0.2027 \\
 u(0.5) &= 0.155
 \end{aligned}$$

Example 3.2: Write a routine that interpolates the value of displacement at any point within a 4-node 1D element.

Solution:

This routine can be implemented in any programming language. The code snippet below is written in Julia and uses the shape functions for a 4-node element to compute the displacement at a given ξ coordinate.

```

function lin4_shape_func(xi)
    return [
        1.0/16.0*( -9.0*xi^3 + 9.0*xi*xi + xi - 1.0)
        1.0/16.0*( 9.0*xi^3 + 9.0*xi*xi - xi - 1.0)
        1.0/16.0*( 27.0*xi^3 - 9.0*xi*xi - 27.0*xi + 9.0)
        1.0/16.0*(-27.0*xi^3 - 9.0*xi*xi + 27.0*xi + 9.0)
    ]
end

U = [ 0.1, 0.3, 0.2, 0.15 ]

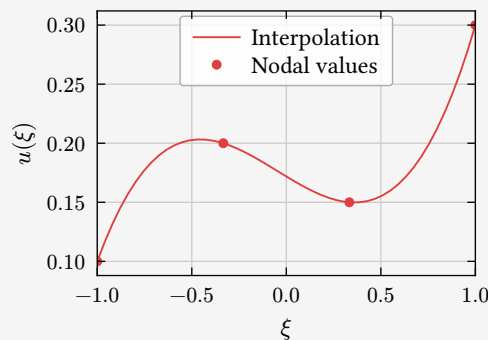
function interp_lin4(U, xi)
    N = lin4_shape_func(xi)
    return U'*N
end

println("Displacement value at xi = -0.5: ", interp_lin4(U, -0.5))
println("Displacement value at xi = 0.5: ", interp_lin4(U, 0.5))

Displacement value at xi = -0.5: 0.202734375
Displacement value at xi = 0.5: 0.15507812499999998

```

This function `interp_lin4` can be used to plot the displacement field over the element as shown below.



Exercises 3.1:

1. What are the global, local and parametric coordinate systems? How are they related to each other?
2. What does it mean for an element to be isoparametric?
3. Why are parametric coordinates useful in finite element interpolation?
4. State the nodal interpolation property and the partition-of-unity property of shape functions.
5. Explain the difference between a linear two-node 1D element and a quadratic three-node 1D element.
6. The nodal displacements at the three nodes of a 1D element are shown in the table. Find the displacement at $\xi = 0.5$. Explain why the result of the interpolation is negative even with all positive nodal values?

u_1	u_2	u_3
0.99	0.11	0.05

7. Determine the shape functions for a 1D isoparametric element with four nodes by calculating the polynomial coefficients using a system of equations. Later, compare the results with the shape functions obtained using Lagrange polynomials. Plot the obtained shape functions.
8. The values of a nodal variable u at the four nodes of a 1D element are shown in the table below. Plot the u field along the element by using shape functions.

u_1	u_2	u_3	u_4
0.0	0.22	-0.15	0.12

3.1.7 Three-node triangular element

A three-node triangle is the simplest 2D solid finite element. It is a linear element and, in solid mechanics, is frequently referred to as the **constant strain triangle** (CST), because its strain field is constant inside the element.

Conventionally, 2D elements have their nodes numbered counterclockwise, as shown in the figure. In the reference element, that is, in the parametric space, the nodal coordinates are $(\xi_1, \eta_1) = (0, 0)$, $(\xi_2, \eta_2) = (1, 0)$, and $(\xi_3, \eta_3) = (0, 1)$.

For this element, the natural coordinates are conveniently interpreted as **barycentric** or **area** coordinates. Defining

$$L_1 = 1 - \xi - \eta, \quad L_2 = \xi, \quad L_3 = \eta \quad (3.25)$$

gives

$$L_1 + L_2 + L_3 = 1 \quad (3.26)$$

and each coordinate is equal to 1 at one node and 0 at the other two nodes. Therefore, the linear triangle shape functions are simply

$$N_1 = L_1, \quad N_2 = L_2, \quad N_3 = L_3 \quad (3.27)$$

or, equivalently,

$$N_1(\xi, \eta) = 1 - \xi - \eta, \quad N_2(\xi, \eta) = \xi, \quad N_3(\xi, \eta) = \eta \quad (3.28)$$

The same result can also be obtained by assuming the linear polynomial form

$$N_i(\xi, \eta) = a_i + b_i\xi + c_i\eta \quad (3.29)$$

where ξ and η are natural coordinates.

Imposing the nodal interpolation conditions $N_i(\xi_j, \eta_j) = \delta_{ij}$ leads to the matrix equation

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (3.30)$$

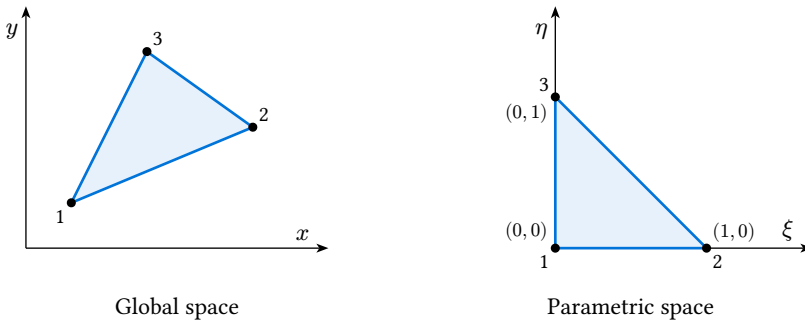


Figure 3.9: Three-node triangular element in global and reference spaces

This matrix system can be written in condensed form as

$$AC = I \quad (3.31)$$

where $\mathbf{A}$ contains the nodal-coordinate data of the reference element and the columns of $\mathbf{C}$ contain the coefficients of the shape functions associated with each node, and $\mathbf{I}$ is the identity matrix. Therefore,

$$\mathbf{C} = \mathbf{A}^{-1} \quad (3.32)$$

Extracting the coefficients a_i , b_i , and c_i from $\mathbf{C}$ gives the same shape functions:

$$\begin{aligned} N_1(\xi, \eta) &= 1 - \xi - \eta \\ N_2(\xi, \eta) &= \xi \\ N_3(\xi, \eta) &= \eta \end{aligned} \quad (3.33)$$

Figure 3.10 shows the surfaces of the shape functions for nodes 1 and 3. All shape functions are linear over the reference triangle and satisfy

$$N_1(\xi, \eta) + N_2(\xi, \eta) + N_3(\xi, \eta) = 1 \quad (3.34)$$

In other words, the three-node triangle shape functions are exactly the barycentric coordinates of the reference triangle.

Their derivatives with respect to the natural coordinates are constant and can be arranged as

$$\begin{pmatrix} \frac{\partial N_1}{\partial \xi} & \frac{\partial N_1}{\partial \eta} \\ \frac{\partial N_2}{\partial \xi} & \frac{\partial N_2}{\partial \eta} \\ \frac{\partial N_3}{\partial \xi} & \frac{\partial N_3}{\partial \eta} \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (3.35)$$

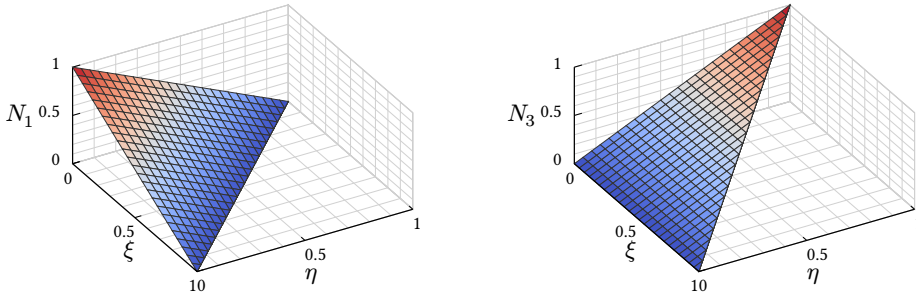


Figure 3.10: Shape function surfaces for nodes 1 and 3 of a three-node triangle element

3.1.8 Shape function requirements

For an element to be suitable for standard displacement-based finite element formulations, its shape functions should satisfy a few basic requirements:

Interpolation and completeness: The shape function associated with node i must satisfy $N_i = 1$ at node i and $N_i = 0$ at all other nodes of the same element. In addition, the shape functions must satisfy the partition-of-unity condition

$$\sum_{i=1}^n N_i = 1 \quad (3.36)$$

The interpolation must also reproduce polynomial fields up to the order required by the element. In particular, it should represent rigid-body motion exactly and reproduce the corresponding constant-strain state for linear elements.

Local support: For nodal elements, a shape function N_i vanishes on sides that do not contain node i .

Interelement compatibility: Along the boundary shared by adjacent elements, the interpolation must provide the continuity required by the formulation. In standard displacement-based elements, this is usually C^0 continuity of the primary variable. Elements that satisfy this condition are called **conforming** or **compatible**. Figure 3.11 shows examples of conforming and non-conforming elements.

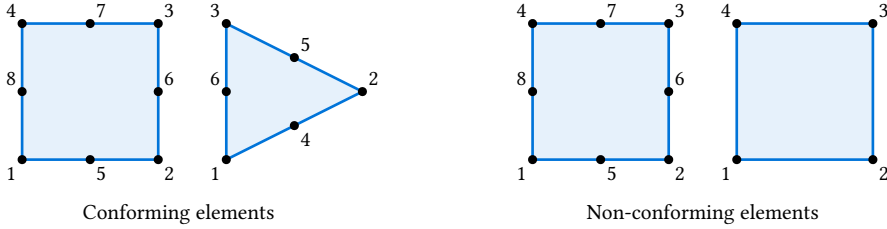


Figure 3.11: Examples of conforming and non-conforming elements

To construct shape functions with the required completeness, the polynomial basis must contain all terms up to the appropriate order. For example, a four-node 1D element uses a cubic polynomial basis, while a six-node triangle uses a complete quadratic basis. The required polynomial terms can be identified from Pascal's triangle shown in Figure 3.12. For triangle elements, the terms are selected in triangular patterns; for Lagrangian quadrilateral elements, they are selected in diamond patterns. In Serendipity elements, some terms are omitted while preserving symmetry.

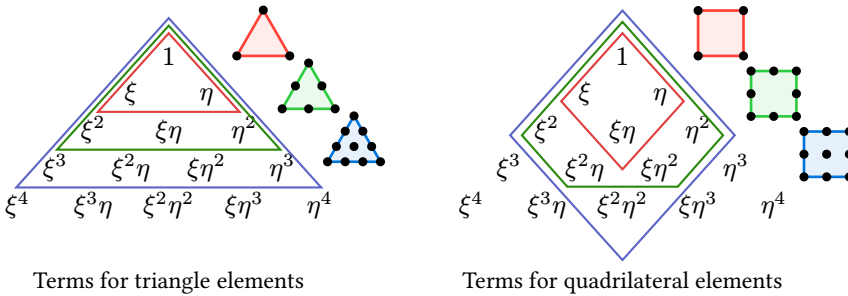


Figure 3.12: Pascal's triangles for triangle and quadrilateral elements

3.1.9 Quadratic triangular element

The six-node triangle is the quadratic counterpart of the three-node triangle. Because both the geometry and the field variables are interpolated quadratically, this element can represent curved boundaries more accurately than the linear triangle. For this reason, it is widely used in 2D finite element analyses, especially in unstructured meshes. In displacement-based mechanical formulations, quadratic displacement interpolation leads to strain fields that vary linearly inside the element.

The conventional nodal numbering and the natural coordinates of the reference element are shown in Figure 3.13.

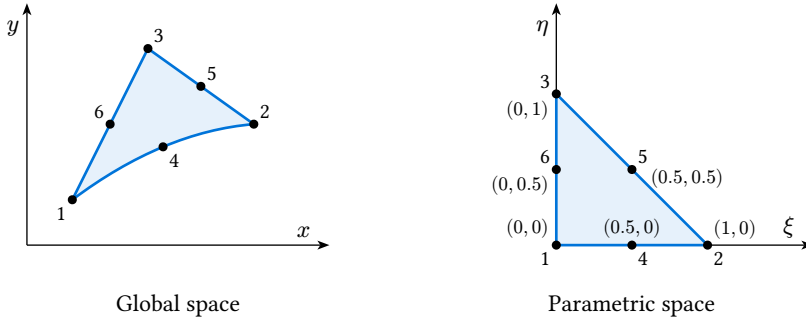


Figure 3.13: Six-node triangular element in global and reference spaces

As in the three-node triangle, it is convenient to use the barycentric, or area, coordinates

$$L_1 = 1 - \xi - \eta, \quad L_2 = \xi, \quad L_3 = \eta \quad (3.37)$$

The six-node triangle shape functions are obtained from the complete quadratic combinations of L_1 , L_2 , and L_3 :

$$\begin{aligned} N_1 &= L_1(2L_1 - 1) \\ N_2 &= L_2(2L_2 - 1) \\ N_3 &= L_3(2L_3 - 1) \\ N_4 &= 4L_1L_2 \\ N_5 &= 4L_2L_3 \\ N_6 &= 4L_3L_1 \end{aligned} \quad (3.38)$$

The first three functions are associated with the corner nodes, while N_4 , N_5 , and N_6 are associated with the midside nodes. Expanding the expressions above in terms of ξ and η gives

$$\begin{aligned} N_1(\xi, \eta) &= 1 - 3(\xi + \eta) + 2(\xi + \eta)^2 \\ N_2(\xi, \eta) &= 2\xi^2 - \xi \\ N_3(\xi, \eta) &= 2\eta^2 - \eta \\ N_4(\xi, \eta) &= 4\xi - 4\xi(\xi + \eta) \\ N_5(\xi, \eta) &= 4\xi\eta \\ N_6(\xi, \eta) &= 4\eta - 4\eta(\xi + \eta) \end{aligned} \quad (3.39)$$

Equivalently, they can be interpreted as the result of imposing the nodal interpolation conditions on the complete second-order polynomial form

$$N_i(\xi, \eta) = a_i + b_i\xi + c_i\eta + d_i\xi^2 + e_i\eta^2 + f_i\xi\eta \quad (3.40)$$

for $i = 1, 2, \dots, 6$. Figure 3.14 shows the surfaces of the shape functions for nodes 1 and 6.

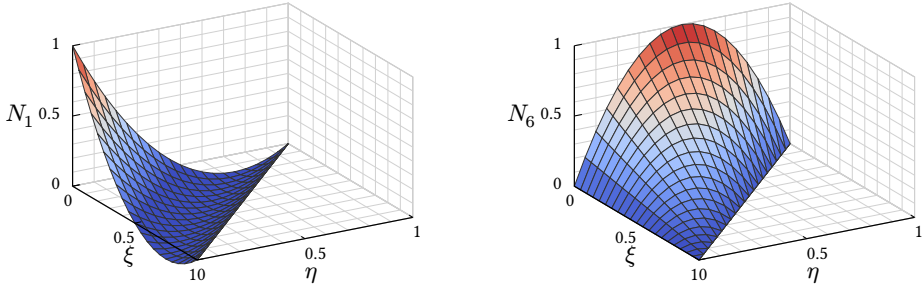


Figure 3.14: Shape function surfaces for nodes 1 and 6 of a six-node triangle element

The corresponding derivatives are

$$\frac{\partial N}{\partial \xi} = \begin{pmatrix} -3 + 4\xi + 4\eta & -3 + 4\xi + 4\eta \\ -1 + 4\xi & 0 \\ 0 & -1 + 4\eta \\ 4 - 8\xi - 4\eta & -4\xi \\ 4\eta & 4\xi \\ -4\eta & 4 - 4\xi - 8\eta \end{pmatrix} \quad (3.41)$$

where N is a vector with the shape function expressions and $\xi = (\xi, \eta)$ is a vector with the parametric coordinates.

3.1.10 Four-node quadrilateral element

The four-node quadrilateral is the simplest quadrilateral isoparametric element. It is usually called a **bilinear quadrilateral element**, because each shape function is linear in ξ for fixed η , linear in η for fixed ξ , and contains the bilinear term $\xi\eta$.

The reference element is the square $[-1, 1] \times [-1, 1]$, with counterclockwise nodal numbering as shown in Figure 3.15. The interpolation is written in the bilinear form

$$N_i(\xi, \eta) = a_i + b_i\xi + c_i\eta + d_i\xi\eta \quad (3.42)$$

for $i = 1, 2, 3, 4$.

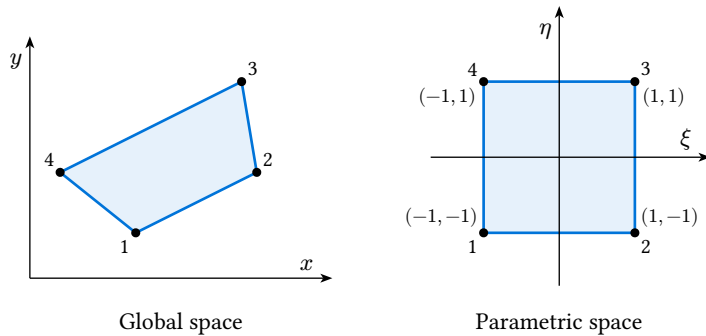


Figure 3.15: Four-node quadrilateral element in global and reference spaces

The shape functions can be easily derived by using Lagrange polynomials. The resulting bilinear shape functions are

$$\begin{aligned}
 N_1(\xi, \eta) &= \frac{1}{4}(1 - \xi)(1 - \eta) \\
 N_2(\xi, \eta) &= \frac{1}{4}(1 + \xi)(1 - \eta) \\
 N_3(\xi, \eta) &= \frac{1}{4}(1 + \xi)(1 + \eta) \\
 N_4(\xi, \eta) &= \frac{1}{4}(1 - \xi)(1 + \eta)
 \end{aligned} \tag{3.43}$$

A compact expression is

$$N_i(\xi, \eta) = \frac{1}{4}(1 + \xi_i \xi)(1 + \eta_i \eta) \tag{3.44}$$

where (ξ_i, η_i) are the nodal natural coordinates of node i , with $i = 1, 2, 3, 4$. Figure 3.16 shows the surfaces of the shape functions for nodes 1 and 3.

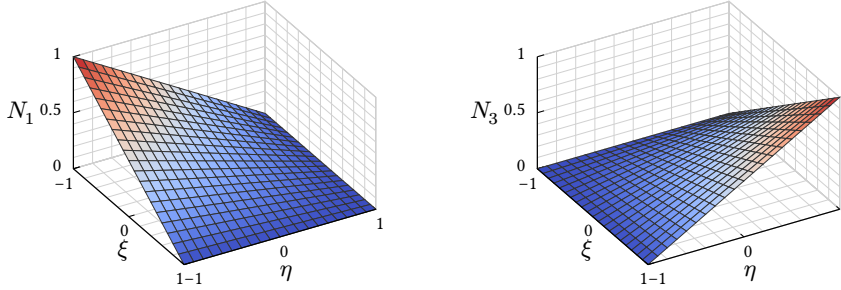


Figure 3.16: Shape function surfaces for nodes 1 and 3 of a four-node quadrilateral element

Their derivatives with respect to the natural coordinates can be written in matrix form as

$$\frac{\partial \mathbf{N}}{\partial \xi} = \begin{pmatrix} -\frac{1}{4}(1 - \eta) & -\frac{1}{4}(1 - \xi) \\ \frac{1}{4}(1 - \eta) & -\frac{1}{4}(1 + \xi) \\ \frac{1}{4}(1 + \eta) & \frac{1}{4}(1 + \xi) \\ -\frac{1}{4}(1 + \eta) & \frac{1}{4}(1 - \xi) \end{pmatrix} \tag{3.45}$$

3.1.11 Nine-node quadrilateral element

The nine-node quadrilateral is the quadratic counterpart of the four-node quadrilateral. It is a **biquadratic** Lagrangian element, since the interpolation is quadratic in ξ , quadratic in η , and includes the term $\xi^2 \eta^2$.

The reference element is the square $[-1, 1] \times [-1, 1]$, with the nodal numbering shown in Figure 3.17. Because this is a tensor-product Lagrangian element, its shape functions can be obtained from the quadratic 1D Lagrange polynomials associated with the nodal coordinates -1 , 1 , and 0 .

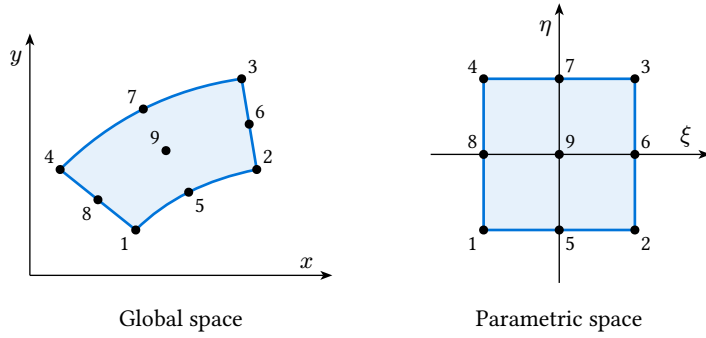


Figure 3.17: Nine-node quadrilateral element in global and reference spaces

Using the 1D shape functions

$$\begin{aligned}
 L_1(s) &= \frac{1}{2}s(s-1) \\
 L_2(s) &= \frac{1}{2}s(s+1) \\
 L_3(s) &= 1-s^2
 \end{aligned} \tag{3.46}$$

the 2D shape functions are formed by products in the ξ and η directions. For the numbering shown in Figure 3.17, they are

$$\begin{aligned}
 N_1(\xi, \eta) &= \frac{1}{4}\xi\eta(1-\xi)(1-\eta) \\
 N_2(\xi, \eta) &= -\frac{1}{4}\xi\eta(1+\xi)(1-\eta) \\
 N_3(\xi, \eta) &= \frac{1}{4}\xi\eta(1+\xi)(1+\eta) \\
 N_4(\xi, \eta) &= -\frac{1}{4}\xi\eta(1-\xi)(1+\eta) \\
 N_5(\xi, \eta) &= \frac{1}{2}\eta(\eta-1)(1-\xi^2) \\
 N_6(\xi, \eta) &= \frac{1}{2}\xi(\xi+1)(1-\eta^2) \\
 N_7(\xi, \eta) &= \frac{1}{2}\eta(\eta+1)(1-\xi^2) \\
 N_8(\xi, \eta) &= \frac{1}{2}\xi(\xi-1)(1-\eta^2) \\
 N_9(\xi, \eta) &= (1-\xi^2)(1-\eta^2)
 \end{aligned} \tag{3.47}$$

The corner-node functions are associated with nodes 1 to 4, the midside functions with nodes 5 to 8, and N_9 with the internal node.

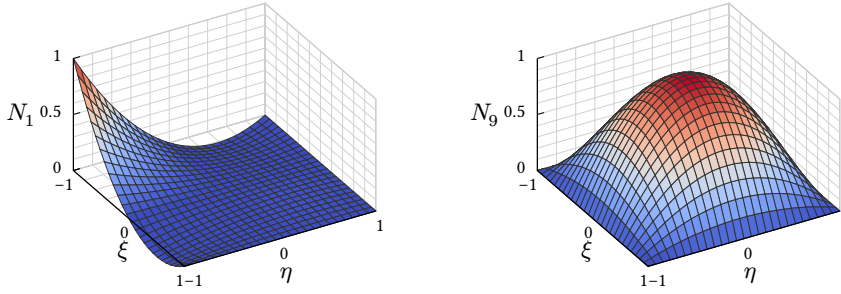


Figure 3.18: Shape function surfaces for nodes 1 and 9 of a nine-node quadrilateral element

3.1.12 Eight-node quadrilateral element

This quadratic quadrilateral element contains 8 nodes as shown in Figure 3.19. It is one of the most widely used elements in 2D analyses and belongs to the family of **Serendipity elements**, in which nodes are placed only at the vertices and along the edges, without an internal node.

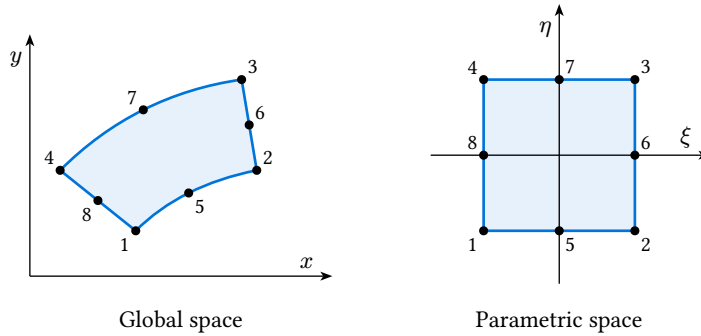


Figure 3.19: Eight-node Serendipity quadrilateral element

Because the center node is omitted, this element does not use the complete biquadratic Lagrangian interpolation of the nine-node quadrilateral. Instead, its shape functions are chosen so that the nodal interpolation conditions are satisfied while preserving quadratic variation along the edges.

The shape functions associated with the corner nodes are

$$N_i(\xi, \eta) = \frac{1}{4}(1 + \xi\xi_i)(1 + \eta\eta_i)(\xi\xi_i + \eta\eta_i - 1) \quad \text{for } i = 1 \dots 4 \quad (3.48)$$

which leads to

$$\begin{aligned} N_1(\xi, \eta) &= \frac{1}{4}(1 - \xi)(1 - \eta)(-1 - \xi - \eta) \\ N_2(\xi, \eta) &= \frac{1}{4}(1 + \xi)(1 - \eta)(-1 + \xi - \eta) \\ N_3(\xi, \eta) &= \frac{1}{4}(1 + \xi)(1 + \eta)(-1 + \xi + \eta) \\ N_4(\xi, \eta) &= \frac{1}{4}(1 - \xi)(1 + \eta)(-1 - \xi + \eta) \end{aligned} \quad (3.49)$$

Likewise, the midside-node shape functions are

$$\begin{aligned} N_i(\xi, \eta) &= \frac{1}{2}(1 + \eta\eta_i)(1 - \xi^2) \quad \text{for } i = 5, 7 \\ N_i(\xi, \eta) &= \frac{1}{2}(1 + \xi\xi_i)(1 - \eta^2) \quad \text{for } i = 6, 8 \end{aligned} \quad (3.50)$$

thus,

$$\begin{aligned} N_5(\xi, \eta) &= \frac{1}{2}(1 - \eta)(1 - \xi^2) \\ N_6(\xi, \eta) &= \frac{1}{2}(1 + \xi)(1 - \eta^2) \\ N_7(\xi, \eta) &= \frac{1}{2}(1 + \eta)(1 - \xi^2) \\ N_8(\xi, \eta) &= \frac{1}{2}(1 - \xi)(1 - \eta^2) \end{aligned} \quad (3.51)$$

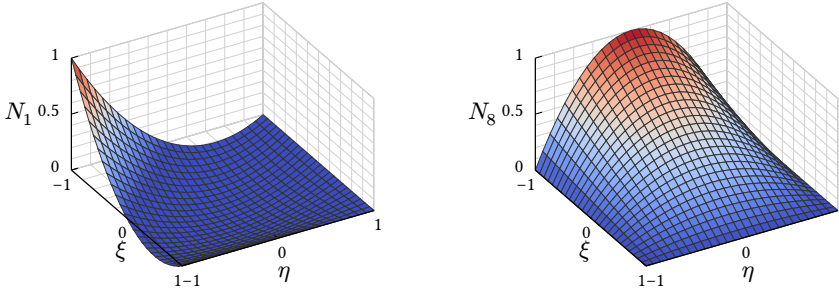


Figure 3.20: Shape function surfaces for nodes 1 and 8 of an eight-node Serendipity quadrilateral element

3.1.13 Lagrangian and serendipity families

For higher-order elements, two important families appear repeatedly: **Lagrangian** elements and **serendipity** elements. The distinction is best understood by comparing the nine-node and eight-node quadrilaterals (Figure 3.21).

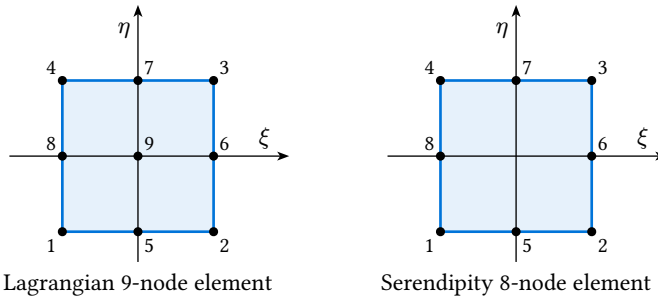


Figure 3.21: The 9-node Lagrangian quadrilateral element and its serendipity counterpart.

A **Lagrangian** element contains the nodes required to reproduce the complete polynomial interpolation of the chosen order. In practice, this usually means that, besides corner and midside nodes, interior nodes are also included whenever they are required by the polynomial basis. The nine-node quadrilateral is therefore a complete quadratic element, since its interpolation contains all terms generated by the quadratic basis in ξ and η .

A **serendipity** element reduces the number of nodes by omitting some interior nodes while preserving the nodal interpolation conditions and the desired approximation order along the element

boundaries. The eight-node quadrilateral is the quadratic serendipity counterpart of the nine-node quadrilateral: it still interpolates quadratically along each edge, but it does not contain the complete biquadratic polynomial basis over the interior.

Note 3.1: The origin of the term “serendipity” is often linked to the somewhat accidental discovery of these elements while seeking more efficient higher-order formulations. The word derives from “Serendip,” an old name for Sri Lanka, popularized in the 18th century to describe fortunate, unexpected discoveries in a fairy tale.

In practice, Lagrangian elements are more systematic and provide a complete interpolation of the corresponding order, whereas serendipity elements are more economical because they use fewer degrees of freedom. Even so, serendipity elements often deliver an approximation that is very close to that of the corresponding Lagrangian element, especially when the element distortion is moderate and the solution does not vary too sharply within the element.

Other examples of 2D Lagrangian elements are the 3-node triangle, the 4-node quadrilateral, the 6-node triangle, and the 9-node quadrilateral, whereas 2D serendipity elements include the 8-node quadrilateral and the 12-node quadrilateral. In three dimensions, examples of Lagrangian elements include the 4-node tetrahedron, the 10-node tetrahedron, and the 27-node hexahedron, while examples of serendipity elements include the 20-node hexahedron and the 32-node hexahedron.

3.1.14 Four-node tetrahedron element

The four-node tetrahedron is the simplest 3D solid isoparametric element (Figure 3.22). It is a linear simplex element and is widely used in unstructured meshes.

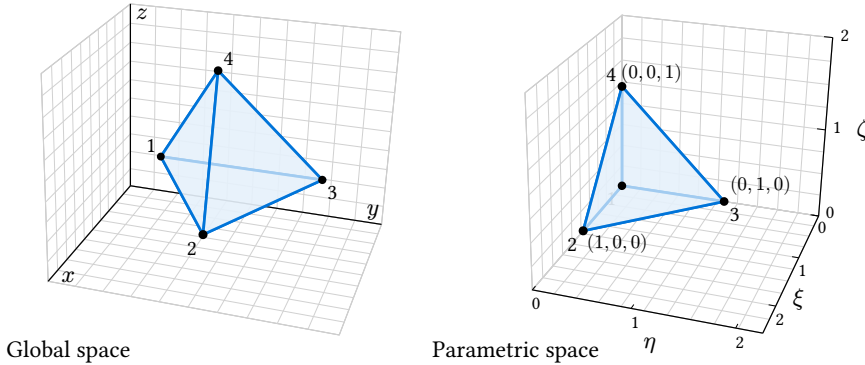


Figure 3.22: Four-node tetrahedron element

The reference tetrahedron has nodal natural coordinates $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$. As in the triangular element, these natural coordinates can be interpreted as barycentric, or volume, coordinates. Defining

$$L_1 = 1 - \xi - \eta - \zeta, \quad L_2 = \xi, \quad L_3 = \eta, \quad L_4 = \zeta \quad (3.52)$$

gives

$$L_1 + L_2 + L_3 + L_4 = 1 \quad (3.53)$$

and each coordinate is equal to 1 at one node and 0 at the other three nodes. Therefore, the shape functions are

$$\begin{aligned}
 N_1(\xi, \eta, \zeta) &= 1 - \xi - \eta - \zeta \\
 N_2(\xi, \eta, \zeta) &= \xi \\
 N_3(\xi, \eta, \zeta) &= \eta \\
 N_4(\xi, \eta, \zeta) &= \zeta
 \end{aligned} \tag{3.54}$$

or, equivalently,

$$N_1 = L_1, \quad N_2 = L_2, \quad N_3 = L_3, \quad N_4 = L_4 \tag{3.55}$$

All shape functions are linear; therefore, this is a linear element. Their derivatives with respect to the natural coordinates are constant:

$$\frac{\partial \mathbf{N}}{\partial \xi} = \begin{pmatrix} -1 & -1 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{3.56}$$

3.1.15 Eight-node hexahedron element

The eight-node hexahedron is the simplest member of the hexahedron family. It is the three-dimensional counterpart of the four-node quadrilateral.

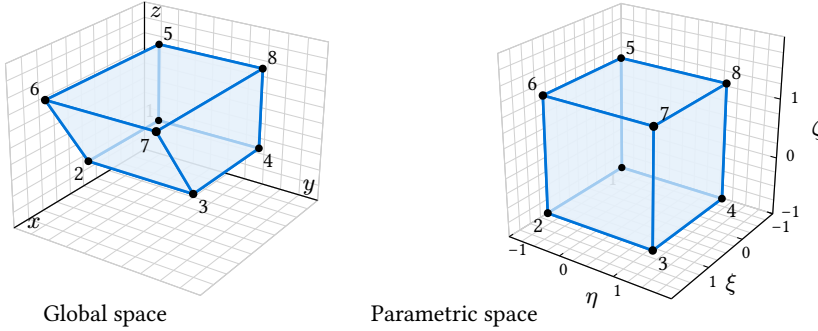


Figure 3.23: 8-node hexahedron element

The reference element is the cube $[-1, 1] \times [-1, 1] \times [-1, 1]$, with the nodal numbering shown in the figure. Its shape functions can be written in compact form as

$$N_i(\xi, \eta, \zeta) = \frac{1}{8}(1 + \xi_i \xi)(1 + \eta_i \eta)(1 + \zeta_i \zeta) \quad \text{for } i = 1 \dots 8 \tag{3.57}$$

where (ξ_i, η_i, ζ_i) are the natural coordinates of node i . Expanding this expression gives

$$\begin{aligned}
 N_1(\xi, \eta, \zeta) &= \frac{1}{8}(1 - \xi)(1 - \eta)(1 - \zeta) \\
 N_2(\xi, \eta, \zeta) &= \frac{1}{8}(1 + \xi)(1 - \eta)(1 - \zeta) \\
 N_3(\xi, \eta, \zeta) &= \frac{1}{8}(1 + \xi)(1 + \eta)(1 - \zeta) \\
 N_4(\xi, \eta, \zeta) &= \frac{1}{8}(1 - \xi)(1 + \eta)(1 - \zeta) \\
 N_5(\xi, \eta, \zeta) &= \frac{1}{8}(1 - \xi)(1 - \eta)(1 + \zeta) \\
 N_6(\xi, \eta, \zeta) &= \frac{1}{8}(1 + \xi)(1 - \eta)(1 + \zeta) \\
 N_7(\xi, \eta, \zeta) &= \frac{1}{8}(1 + \xi)(1 + \eta)(1 + \zeta) \\
 N_8(\xi, \eta, \zeta) &= \frac{1}{8}(1 - \xi)(1 + \eta)(1 + \zeta)
 \end{aligned} \tag{3.58}$$

This element is also called a **brick element**. More precisely, it is a **trilinear hexahedron**, because each shape function is linear in each natural coordinate when the other two are held fixed. Its highest-order term is proportional to $\xi\eta\zeta$, so describing it simply as a linear element is imprecise.

3.1.16 Twenty-node hexahedron element

The twenty-node hexahedron is the quadratic serendipity counterpart of the eight-node hexahedron. It is also known as a **quadratic brick element**. In addition to the eight corner nodes, it has twelve midside nodes, one on each edge, but no internal node. Consequently, it provides quadratic interpolation along the edges and represents curved boundaries more accurately than the eight-node hexahedron, while still using fewer degrees of freedom than the 27-node Lagrangian hexahedron.

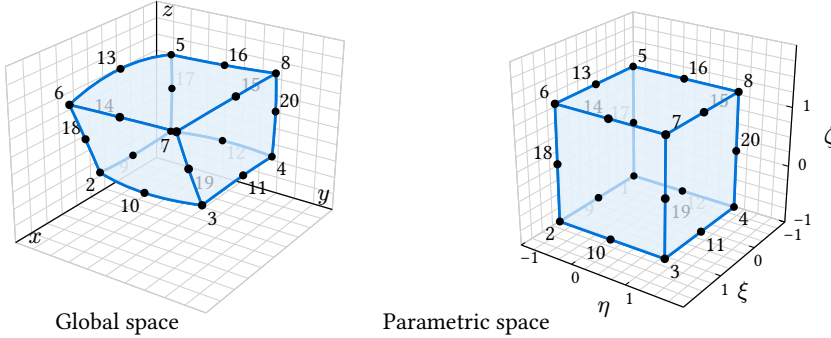


Figure 3.24: 20-node hexahedron element.

The reference domain is again the cube $[-1, 1] \times [-1, 1] \times [-1, 1]$. Because midside nodes are present on all edges, the interpolation is quadratic along each edge of the element. However, since no center node is included, the element does not form a complete quadratic Lagrangian interpolation in three dimensions and therefore belongs to the **serendipity** family.

3.1.17 Other elements

Besides the most common isoparametric elements discussed so far, many other reference elements are also used in practice. They are introduced to improve geometric flexibility, to provide higher-order interpolation, or to connect regions meshed with different element topologies.

Figures 3.25 to 3.28 summarize some representative higher-order 2D and 3D elements in parametric space. The purpose here is not to derive all of their shape functions, but to show their nodal arrangement, their natural-coordinate domains, and the variety of element families used in finite element meshes. Finally, Figure 3.29 summarizes the most common 2D and 3D isoparametric element families, along with their most commonly used members.

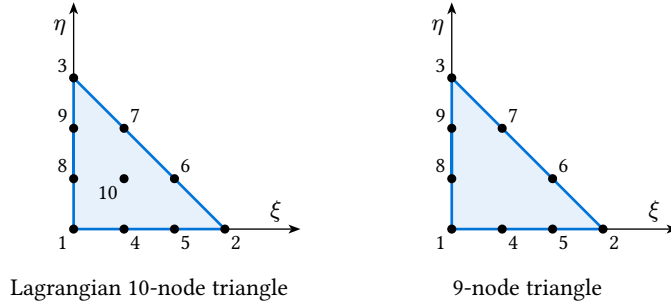


Figure 3.25: The 10-node Lagrangian triangle and its 9-node counterpart.

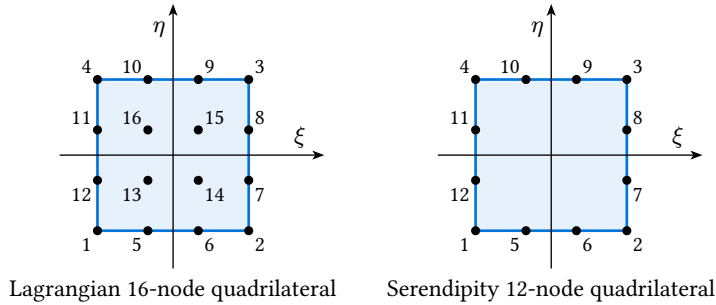


Figure 3.26: The 16-node Lagrangian quadrilateral and its serendipity counterpart.

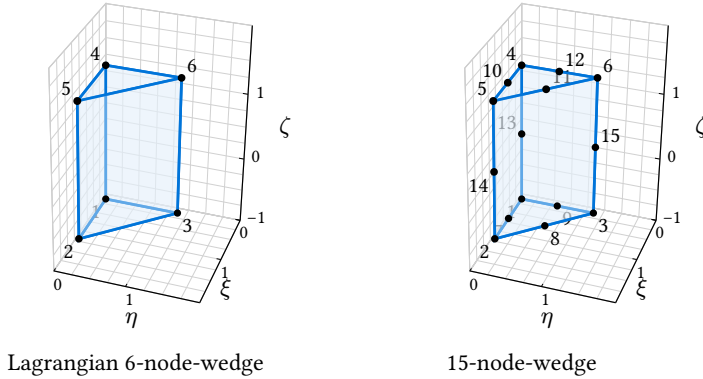


Figure 3.27: The 6-node wedge and its 15-node counterpart.

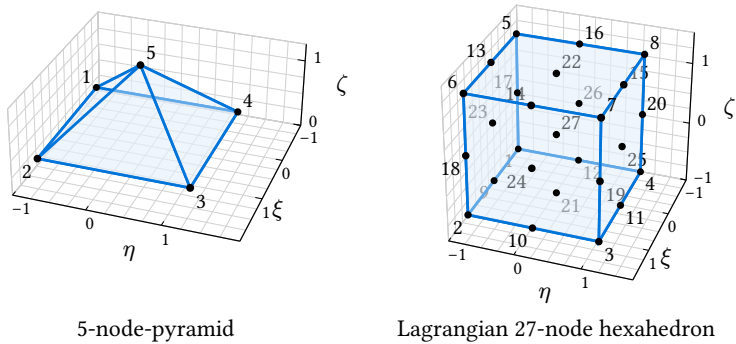


Figure 3.28: Other 3D elements in parametric space.

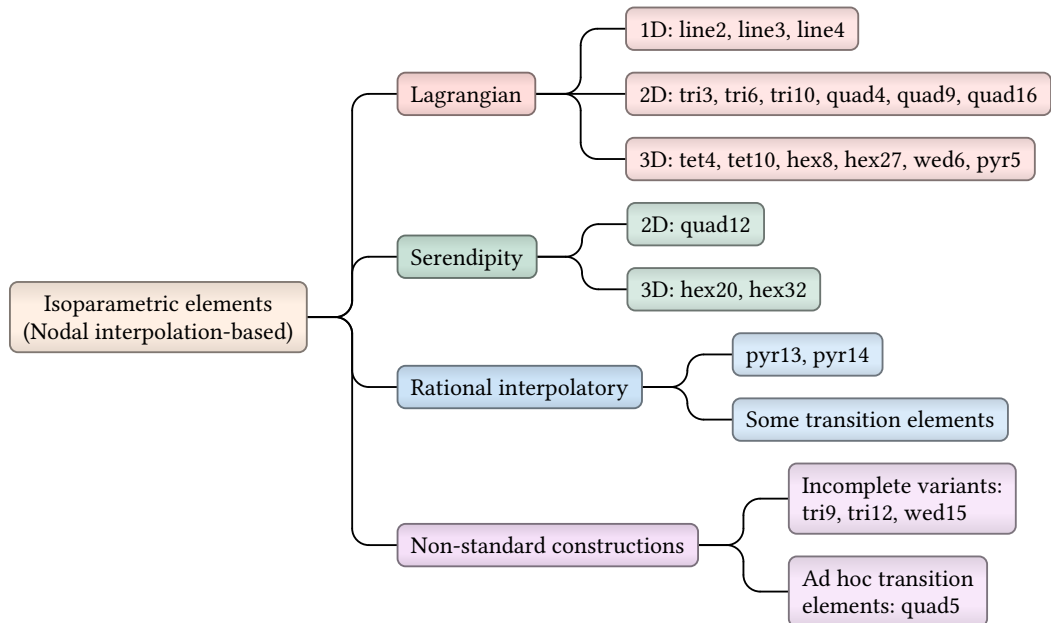
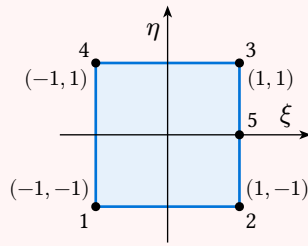


Figure 3.29: Classification of nodal interpolation-based isoparametric element families.

Exercises 3.2:

1. What are the differences between Lagrangian and serendipity elements?
2. Why is the 4-node quadrilateral called a bilinear element rather than simply a linear element?
3. Why does the 20-node hexahedron belong to the serendipity family, whereas the 27-node hexahedron belongs to the Lagrangian family?
4. Why are pyramid elements useful in three-dimensional meshes?
5. List all terms needed in the shape functions of a serendipity quadrilateral element with 12 nodes.
6. Find the shape functions of nodes 1, 4 and 10 of a 10-node 2D Lagrangian triangular element.
7. Find all shape functions for the transition element shown below. Plot the shape functions for nodes 1 and 5.



3.2 Coordinates transformation

In the isoparametric formulation, interpolation functions are defined in the reference element, whereas the governing equations and boundary conditions are written in the physical domain. For this reason, it is necessary to transform coordinates, derivatives, and integrals from the natural-coordinate system to the global-coordinate system.

This transformation is performed through the element mapping introduced in the previous section. The key quantity in this process is the Jacobian matrix, which relates differential changes between both coordinate systems and allows derivatives and integrals to be expressed consistently in either space.

3.2.1 Jacobian matrix

In multivariable calculus, the Jacobian matrix is the matrix of all first-order partial derivatives of a vector-valued function.

For a mapping $\mathbf{f}: \mathbb{R}^m \rightarrow \mathbb{R}^n$ written as

$$\mathbf{f} = (f_1, f_2, \dots, f_n)^T \quad (3.1)$$

with independent variables $(x_1, x_2, \dots, x_m)$, the Jacobian matrix is defined by

$$\mathbf{J} = \frac{\partial(f_1, f_2, \dots, f_n)}{\partial(x_1, x_2, \dots, x_m)} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_m} \end{pmatrix} \quad (3.2)$$

For example, if

$$\mathbf{f}(x, y) = \begin{pmatrix} f_1(x, y) \\ f_2(x, y) \end{pmatrix} \quad (3.3)$$

then

$$\mathbf{J} = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} \quad (3.4)$$

When $m = n$, the Jacobian matrix is square and its determinant

$$J = \det(\mathbf{J}) \quad (3.5)$$

is called the **Jacobian determinant**.

The Jacobian matrix is used to relate differential changes between coordinate systems, for example between global coordinates (x, y, z) and natural coordinates (ξ, η, ζ) . In change-of-variables formulas, the Jacobian determinant gives the local ratio between measures in both systems. Thus,

$$\begin{aligned} dx &= |J| d\xi \\ dA &= |J| d\xi d\eta \\ dV &= |J| d\xi d\eta d\zeta \end{aligned} \quad (3.6)$$

and the corresponding integrals can be written as

$$\begin{aligned}
 \int_L f(x) dx &= \int_{\ell'} f(\xi) |J| d\xi \\
 \int_A f(x, y) dA &= \int_{A'} f(\xi, \eta) |J| d\xi d\eta \\
 \int_V f(x, y, z) dV &= \int_{V'} f(\xi, \eta, \zeta) |J| d\xi d\eta d\zeta
 \end{aligned}
 \tag{3.7}$$

Example 3.1: Using the Jacobian in one-dimensional integration, evaluate

$$\int_0^1 \sqrt{2x+1} dx$$

Solution:

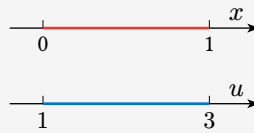
Making the substitution

$$u = 2x + 1$$

we get

$$x = \frac{u}{2} - \frac{1}{2}$$

The relation between the x -space and the u -space is illustrated below:



The corresponding Jacobian is

$$J = \frac{dx}{du} = \frac{1}{2}$$

Therefore,

$$dx = \frac{1}{2} du$$

and the integral becomes

$$\int_1^3 \sqrt{u} \frac{1}{2} du$$

Hence,

$$\begin{aligned}
 \int_0^1 \sqrt{2x+1} dx &= \frac{1}{2} \int_1^3 \sqrt{u} du \\
 &= \sqrt{3} - \frac{1}{3}
 \end{aligned}$$

Example 3.2: Using the Jacobian in two-dimensional integration, evaluate

$$I = \iint_R (4x^2 - y^2)^4 \, dR$$

Solution:

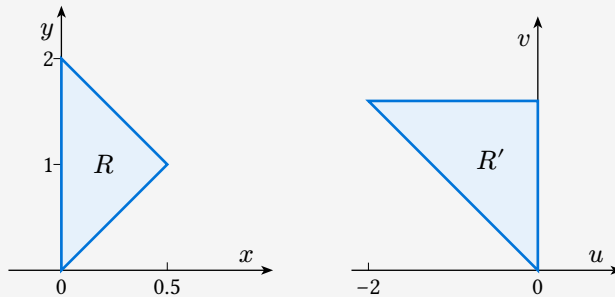
This integral can be rewritten as:

$$I = \iint_R (2x - y)^4 (2x + y)^4 \, dx \, dy$$

Consider the two-dimensional change of variables

$$u = 2x - y, \quad v = 2x + y$$

Under this transformation, the original region R is mapped into the region R' in the uv -plane, as shown below:



The Jacobian matrix of the transformation is

$$J = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 2 & 1 \end{pmatrix}$$

and its determinant is $J = 4$. Hence,

$$dx \, dy = \frac{1}{4} \, du \, dv$$

Therefore, the integral can be rewritten as

$$I = \int_0^2 \int_{-v}^0 u^4 v^4 \frac{1}{4} \, du \, dv$$

and the final result is

$$\begin{aligned} I &= \frac{1}{4} \int_0^2 v^4 \int_{-v}^0 u^4 \, du \, dv \\ &= \frac{1}{20} \int_0^2 v^9 \, dv \\ &= \frac{128}{25} \end{aligned}$$

3.2.2 Jacobian in finite elements

In finite elements, the Jacobian matrix describes the local change of variables between the parametric coordinates and the physical coordinates of the element. It is therefore used to map derivatives, lengths, areas, and volumes from the reference domain to the actual element geometry. For two-dimensional elements, this role can be interpreted as a local area-scaling relation, as illustrated in [Figure 3.1](#).

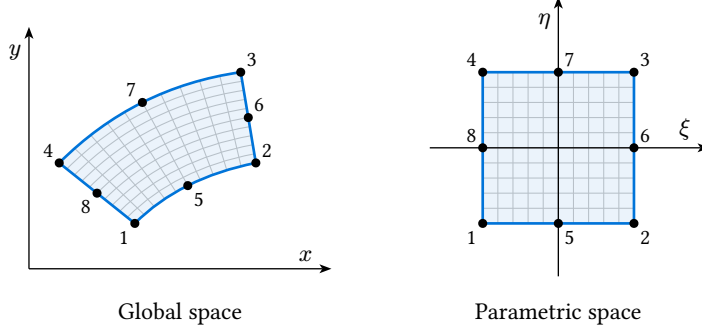


Figure 3.1: Area mapping described by the Jacobian in a 8-node quadrilateral element

In some finite element formulations, the Jacobian matrix is defined as the transpose of its standard mathematical form. As a result, both conventions can be found in the literature.

Because the determinant of a matrix is equal to the determinant of its transpose, the area and volume relations remain unchanged. Thus, most integral formulations that depend only on the Jacobian determinant are unaffected by this choice of convention. However, the convention does affect the explicit construction of the Jacobian matrix.

In this text, the Jacobian matrix is defined according to the standard mathematical convention, and all subsequent derivations follow this definition.

3.2.3 Differential length transformation

Consider first a one-dimensional mapping between a physical coordinate x and a natural coordinate ξ , written as

$$x = x(\xi) \quad (3.8)$$

Its total differential is

$$dx = \frac{\partial x}{\partial \xi} d\xi \quad (3.9)$$

Thus, the Jacobian matrix reduces to the scalar

$$J(\xi) = \frac{\partial x}{\partial \xi} \quad (3.10)$$

Therefore, the differential length in physical space is related to the differential length in natural space by

$$dx = |J(\xi)| d\xi \quad (3.11)$$

In one dimension, the Jacobian determinant is the local scaling factor between both coordinate systems.

3.2.4 Differential area transformation

Consider now two two-dimensional coordinate systems, for example the global system (x, y) and the natural system (ξ, η) , with

$$x = x(\xi, \eta), \quad y = y(\xi, \eta) \quad (3.12)$$

The total differentials

$$\begin{aligned} dx &= \frac{\partial x}{\partial \xi} d\xi + \frac{\partial x}{\partial \eta} d\eta \\ dy &= \frac{\partial y}{\partial \xi} d\xi + \frac{\partial y}{\partial \eta} d\eta \end{aligned} \quad (3.13)$$

lead to the matrix equation

$$\begin{pmatrix} dx \\ dy \end{pmatrix} = \begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \end{pmatrix} \begin{pmatrix} d\xi \\ d\eta \end{pmatrix} \quad (3.14)$$

which defines the Jacobian matrix

$$\mathbf{J}(\xi, \eta) = \begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \end{pmatrix} \quad (3.15)$$

The differential area in global coordinates is related to the differential area in natural coordinates by the determinant of the Jacobian matrix:

$$dA = |\mathbf{J}(\xi, \eta)| d\xi d\eta \quad (3.16)$$

This result can be interpreted geometrically by introducing the differential vectors

$$\begin{aligned} \mathbf{a} &= \begin{pmatrix} \frac{\partial x}{\partial \xi} d\xi \\ \frac{\partial y}{\partial \xi} d\xi \end{pmatrix} \\ \mathbf{b} &= \begin{pmatrix} \frac{\partial x}{\partial \eta} d\eta \\ \frac{\partial y}{\partial \eta} d\eta \end{pmatrix} \end{aligned} \quad (3.17)$$

as shown in [Figure 3.2](#).

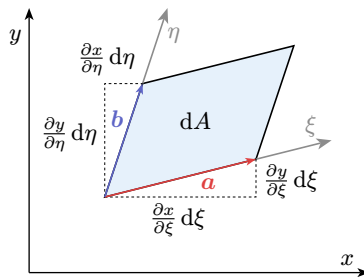


Figure 3.2: Differential parametric area in global space.

The image of the differential rectangle $d\xi \times d\eta$ is a parallelogram in physical space, and its area is

$$dA = \|\mathbf{a} \times \mathbf{b}\| \quad (3.18)$$

which leads directly to the expression above.

3.2.5 Differential volume transformation

For the three-dimensional systems (x, y, z) and (ξ, η, ζ) , let

$$x = x(\xi, \eta, \zeta), \quad y = y(\xi, \eta, \zeta), \quad z = z(\xi, \eta, \zeta) \quad (3.19)$$

Then the total differentials lead to

$$\begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} = \begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} & \frac{\partial x}{\partial \zeta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} & \frac{\partial y}{\partial \zeta} \\ \frac{\partial z}{\partial \xi} & \frac{\partial z}{\partial \eta} & \frac{\partial z}{\partial \zeta} \end{pmatrix} \begin{pmatrix} d\xi \\ d\eta \\ d\zeta \end{pmatrix} \quad (3.20)$$

which defines the Jacobian matrix

$$\mathbf{J}(\xi, \eta, \zeta) = \begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} & \frac{\partial x}{\partial \zeta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} & \frac{\partial y}{\partial \zeta} \\ \frac{\partial z}{\partial \xi} & \frac{\partial z}{\partial \eta} & \frac{\partial z}{\partial \zeta} \end{pmatrix} \quad (3.21)$$

The differential volume in global coordinates is related to the differential volume in natural coordinates by

$$dV = |\mathbf{J}(\xi, \eta, \zeta)| d\xi d\eta d\zeta \quad (3.22)$$

This relation follows from the scalar triple product of the auxiliary vectors

$$\mathbf{a} = \begin{pmatrix} \frac{\partial x}{\partial \xi} d\xi \\ \frac{\partial y}{\partial \xi} d\xi \\ \frac{\partial z}{\partial \xi} d\xi \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} \frac{\partial x}{\partial \eta} d\eta \\ \frac{\partial y}{\partial \eta} d\eta \\ \frac{\partial z}{\partial \eta} d\eta \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} \frac{\partial x}{\partial \zeta} d\zeta \\ \frac{\partial y}{\partial \zeta} d\zeta \\ \frac{\partial z}{\partial \zeta} d\zeta \end{pmatrix} \quad (3.23)$$

shown in [Figure 3.3](#).

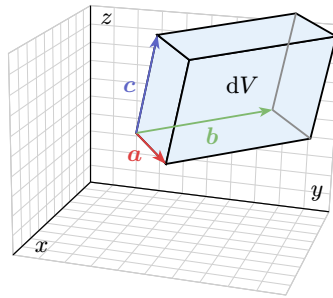


Figure 3.3: Differential parametric volume in global space.

The image of the differential box $d\xi \times d\eta \times d\zeta$ is a parallelepiped in physical space, whose volume is

$$dV = \|(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}\| \quad (3.24)$$

and this yields the expression above.

3.2.6 Inverse of the Jacobian matrix

For two coordinate systems, say (x, y, z) and (ξ, η, ζ) , the direct differential relations are

$$\begin{aligned}
 dx &= \frac{\partial x}{\partial \xi} d\xi + \frac{\partial x}{\partial \eta} d\eta + \frac{\partial x}{\partial \zeta} d\zeta \\
 dy &= \frac{\partial y}{\partial \xi} d\xi + \frac{\partial y}{\partial \eta} d\eta + \frac{\partial y}{\partial \zeta} d\zeta \\
 dz &= \frac{\partial z}{\partial \xi} d\xi + \frac{\partial z}{\partial \eta} d\eta + \frac{\partial z}{\partial \zeta} d\zeta
 \end{aligned} \tag{3.25}$$

The inverse differential relations are

$$\begin{aligned}
 d\xi &= \frac{\partial \xi}{\partial x} dx + \frac{\partial \xi}{\partial y} dy + \frac{\partial \xi}{\partial z} dz \\
 d\eta &= \frac{\partial \eta}{\partial x} dx + \frac{\partial \eta}{\partial y} dy + \frac{\partial \eta}{\partial z} dz \\
 d\zeta &= \frac{\partial \zeta}{\partial x} dx + \frac{\partial \zeta}{\partial y} dy + \frac{\partial \zeta}{\partial z} dz
 \end{aligned} \tag{3.26}$$

If the first set of equations is used to define a Jacobian matrix $\mathbf{J}$, then the second set defines the inverse Jacobian matrix and is written as

$$\begin{pmatrix} d\xi \\ d\eta \\ d\zeta \end{pmatrix} = \begin{pmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial \xi}{\partial y} & \frac{\partial \xi}{\partial z} \\ \frac{\partial \eta}{\partial x} & \frac{\partial \eta}{\partial y} & \frac{\partial \eta}{\partial z} \\ \frac{\partial \zeta}{\partial x} & \frac{\partial \zeta}{\partial y} & \frac{\partial \zeta}{\partial z} \end{pmatrix} \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} \tag{3.27}$$

Whenever $\mathbf{J}$ is nonsingular, the inverse Jacobian matrix is obtained directly by matrix inversion.

The Jacobian matrix is singular when the mapping between coordinate systems is not locally invertible, for example when two or more points in global coordinates are mapped to the same point in natural coordinates. In such cases, the Jacobian determinant is zero and the inverse Jacobian matrix does not exist.

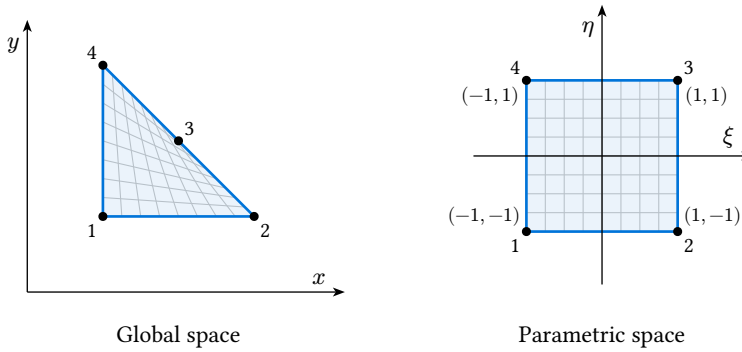


Figure 3.4: Element distortion that leads to a singular Jacobian matrix.

3.2.7 Jacobian matrix for isoparametric elements

The Jacobian matrix is composed by the partial derivatives as $\frac{\partial x}{\partial \xi}$, $\frac{\partial x}{\partial \eta}$, $\frac{\partial y}{\partial \xi}$, etc. For instance, to compute $\frac{\partial x}{\partial \xi}$, the coordinate x is written as

$$x = \sum_{i=1}^n N_i(\xi, \eta, \zeta) x_i \tag{3.28}$$

and therefore, the derivative becomes

$$\begin{aligned}
 \frac{\partial x}{\partial \xi} &= \frac{\partial}{\partial \xi} \left(\sum_{i=1}^n N_i(\xi, \eta, \zeta) x_i \right) \\
 &= \sum_{i=1}^n \frac{\partial N_i}{\partial \xi} x_i \\
 &= (x_1 \ x_2 \ \dots \ x_n) \begin{pmatrix} \frac{\partial N_1}{\partial \xi} & \frac{\partial N_2}{\partial \xi} & \dots & \frac{\partial N_n}{\partial \xi} \end{pmatrix}^T
 \end{aligned} \tag{3.29}$$

The same procedure is applied to all other partial derivatives required to form the Jacobian matrix. For example,

$$\begin{aligned}
 \frac{\partial x}{\partial \eta} &= \sum_{i=1}^n \frac{\partial N_i}{\partial \eta} x_i \\
 \frac{\partial x}{\partial \zeta} &= \sum_{i=1}^n \frac{\partial N_i}{\partial \zeta} x_i \\
 \frac{\partial y}{\partial \xi} &= \sum_{i=1}^n \frac{\partial N_i}{\partial \xi} y_i \\
 \frac{\partial y}{\partial \eta} &= \sum_{i=1}^n \frac{\partial N_i}{\partial \eta} y_i
 \end{aligned} \tag{3.30}$$

Then, the Jacobian matrix is assembled as

$$\mathbf{J}(\xi, \eta, \zeta) = \begin{pmatrix} \sum_{i=1}^n \frac{\partial N_i}{\partial \xi} x_i & \sum_{i=1}^n \frac{\partial N_i}{\partial \eta} x_i & \sum_{i=1}^n \frac{\partial N_i}{\partial \zeta} x_i \\ \sum_{i=1}^n \frac{\partial N_i}{\partial \xi} y_i & \sum_{i=1}^n \frac{\partial N_i}{\partial \eta} y_i & \sum_{i=1}^n \frac{\partial N_i}{\partial \zeta} y_i \\ \sum_{i=1}^n \frac{\partial N_i}{\partial \xi} z_i & \sum_{i=1}^n \frac{\partial N_i}{\partial \eta} z_i & \sum_{i=1}^n \frac{\partial N_i}{\partial \zeta} z_i \end{pmatrix} \tag{3.31}$$

For convenience, if the derivatives of the shape functions are arranged in matrix form as

$$\frac{\partial \mathbf{N}}{\partial \boldsymbol{\xi}} = \begin{pmatrix} \frac{\partial N_1}{\partial \xi} & \frac{\partial N_1}{\partial \eta} & \frac{\partial N_1}{\partial \zeta} \\ \frac{\partial N_2}{\partial \xi} & \frac{\partial N_2}{\partial \eta} & \frac{\partial N_2}{\partial \zeta} \\ \vdots & \vdots & \vdots \\ \frac{\partial N_n}{\partial \xi} & \frac{\partial N_n}{\partial \eta} & \frac{\partial N_n}{\partial \zeta} \end{pmatrix} \tag{3.32}$$

and the nodal coordinates are arranged in matrix form as

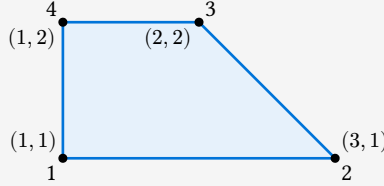
$$\mathbf{X} = \begin{pmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ \vdots & \vdots & \vdots \\ x_n & y_n & z_n \end{pmatrix} \tag{3.33}$$

then the Jacobian matrix can be written compactly as

$$\mathbf{J} = \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \boldsymbol{\xi}} \tag{3.34}$$

This expression is also valid for 2D and 1D elements, with the corresponding reduced number of coordinates and shape function derivatives.

Example 3.3: The figure below shows a quadrilateral element in global coordinates. Find the Jacobian matrix at points $(0, 0)$, $(1, 0)$, $(-1, 1)$ and $(-1, -1)$ in natural coordinates.



Solution:

The Jacobian matrix can be computed by the formula $\mathbf{J} = \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \xi}$. For this element, the matrix coordinates can be written as

$$\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 3 & 1 \\ 2 & 2 \\ 1 & 2 \end{pmatrix}$$

Also, the derivatives of the shape functions are

$$\frac{\partial \mathbf{N}}{\partial \xi} = \begin{pmatrix} -\frac{1}{4}(1-\eta) & -\frac{1}{4}(1-\xi) \\ \frac{1}{4}(1-\eta) & -\frac{1}{4}(1+\xi) \\ \frac{1}{4}(1+\eta) & \frac{1}{4}(1+\xi) \\ -\frac{1}{4}(1+\eta) & \frac{1}{4}(1-\xi) \end{pmatrix}$$

Then, for the natural coordinate $(\xi, \eta) = (0, 0)$ we get

$$\frac{\partial \mathbf{N}}{\partial \xi} = \begin{pmatrix} -\frac{1}{4} & -\frac{1}{4} \\ \frac{1}{4} & -\frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} \end{pmatrix}$$

Then the Jacobian matrix is

$$\begin{aligned} \mathbf{J}(0,0) &= \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \xi} \\ &= \begin{pmatrix} 1 & 3 & 2 & 1 \\ 1 & 1 & 2 & 2 \end{pmatrix} \begin{pmatrix} -\frac{1}{4} & -\frac{1}{4} \\ \frac{1}{4} & -\frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} \end{pmatrix} \\ &= \begin{pmatrix} 0.75 & -0.25 \\ 0.0 & 0.5 \end{pmatrix} \end{aligned}$$

Repeating this procedure for the other points, we get

$$\begin{aligned} \mathbf{J}(1,0) &= \begin{pmatrix} 0.75 & 0.5 \\ 0.0 & 0.5 \end{pmatrix} \\ \mathbf{J}(-1,1) &= \begin{pmatrix} 0.75 & 0.0 \\ 0.0 & 0.5 \end{pmatrix} \\ \mathbf{J}(-1,-1) &= \begin{pmatrix} 0.75 & -0.5 \\ 0.0 & 0.5 \end{pmatrix} \end{aligned}$$

Example 3.4: Write a code to compute the Jacobian matrix for quadrilateral elements and print the values at different locations in parametric coordinates.

Solution:

The code below computes the corresponding Jacobian matrix and print the values at different locations in parametric coordinates.

```
# function to compute the derivatives of a quad4 element
function quad4_derivs(xi, eta)
    return [
        0.25*(-1.0 + eta)  0.25*(-1.0 + xi)
        0.25*(+1.0 - eta)  0.25*(-1.0 - xi)
        0.25*(+1.0 + eta)  0.25*(+1.0 + xi)
        0.25*(-1.0 - eta)  0.25*(+1.0 - xi)
    ]
end

# coordinates matrix
X = [ 1 1; 3 1; 2 2; 1 2 ]

# function to compute the Jacobian matrix
function quad4_jacobian(X, xi, eta)
    dNdξ = quad4_derivs(xi, eta)
    return X'*dNdξ
end

println("Jacobian at (0,0): ", quad4_jacobian(X, 0, 0))
println("Jacobian at (1,0): ", quad4_jacobian(X, 1, 0))
println("Jacobian at (-1,1): ", quad4_jacobian(X, -1, 1))
println("Jacobian at (-1,-1): ", quad4_jacobian(X, -1, -1))

Jacobian at (0,0): [0.75 -0.25; 0.0 0.5]
Jacobian at (1,0): [0.75 -0.5; 0.0 0.5]
Jacobian at (-1,1): [0.5 0.0; 0.0 0.5]
Jacobian at (-1,-1): [1.0 0.0; 0.0 0.5]
```

Exercises 3.1:

1. What is the geometric meaning of the Jacobian determinant in a change of variables?
2. Explain why the Jacobian matrix is essential in isoparametric finite elements.
3. Under what condition does the Jacobian matrix admit an inverse?
4. Find the Jacobian determinants for the transformation from Cartesian coordinates to polar coordinates

$$x = r \cos(\theta), \quad y = r \sin(\theta)$$

and from Cartesian coordinates to spherical coordinates

$$x = \rho \sin(\varphi) \cos(\theta), \quad y = \rho \sin(\varphi) \sin(\theta), \quad z = \rho \cos(\varphi)$$

A. $J = r, J = \rho^2 \sin(\varphi)$

5. Calculate the integral below and show the Jacobian matrix used.

$$I = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{-x^2-y^2} dx \, dy$$

A. $I = \pi$

6. Nodal coordinates of a 1D element are: $x_1 = 2.0$, $x_2 = 10.0$ and $x_3 = 9.0$. Plot the Jacobian J on the natural coordinates space and explain the behavior between nodes 2 and 3. **A.** $J = 4 - 6\xi$

7. If the area of a triangular element is equal to A , show that the Jacobian determinant is equal to $2A$. Assume nodal coordinates (x_1, y_1) , (x_2, y_2) and (x_3, y_3) .

3.2.8 Jacobian limitations

Inside each element in a mesh, the determinant $J = \det(\mathbf{J})$ must be positive. This requirement is called **Jacobian positiveness**.

To comply the requirement, $J > 0$, all nodes must be properly positioned and numbered. This is called **convexity condition**. For example, in the convex four-node element shown at right J is positive over whole element.

In the example, the scalar field defined by the Jacobian is linear showing that the area ratio between local and global coordinates is variable and positive for the element domain. Finite element codes should check node numbering and Jacobian positiveness for all elements prior analysis.

There are some elements that always show constant jacobian determinat over its domain. This is the case of 2-node line elements, 3-node triangle elements and 4-node tetrahedrons, for example. Also, any element can have constant Jacobian if its shape and orientation in global coordinates are the same as in natural coordinates.

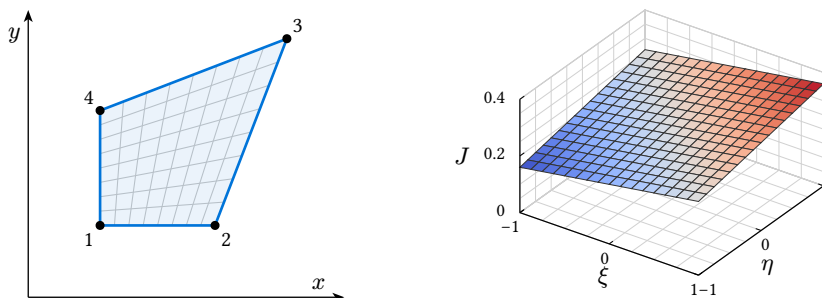


Figure 3.5: Convex 4-node quadrilateral element and its Jacobian determinant field.

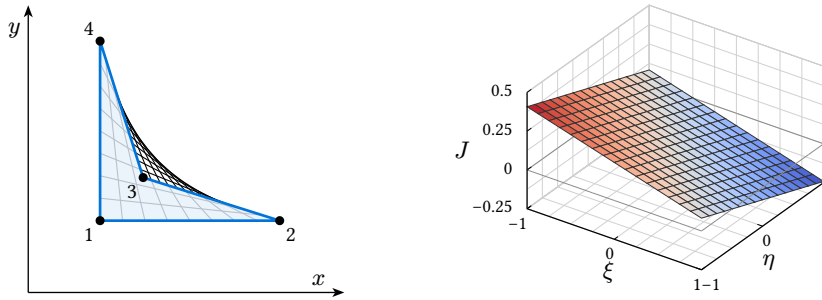


Figure 3.6: Concave 4-node quadrilateral element and its Jacobian determinant field.

For 2D elements with more than three nodes, distortions may render regions with negative Jacobian. For example, the figure at right shows a quadrilateral element where node 3 was dislocated inwards rendering the shape concave.

The Jacobian along the element is variable as show in the second figure; however, near the node 3 there is a region where the Jacobian is negative. In this case Jacobian positiveness is not complied, thus the element is invalid for finite element analysis.

In general, zero or negative Jacobians render the element unacceptable for finite element analyses. Locations where Jacobian determinant is zero imply that there is not an inverse Jacobian relation.

In locations with negative Jacobians may not be unique correspondence between points in natural and global coordinates. This means that some points in natural coordinates can be mapped to the global space but not all of them can be mapped back.

The example below shows the effect of gradually moving a node of a quadrilateral element. The top left figure shows the initial geometry while the other figures show the distortions obtained after moving node 4.

The shape of the top right figure is a triangle. Note that the Jacobian determinant at node 4 is zero.

3.2.9 Restriction on mid-side nodes

For higher order elements, proper location of corner nodes is not enough. Mid-side and interior nodes must be placed sufficiently close to their natural locations in order to render positive Jacobians.

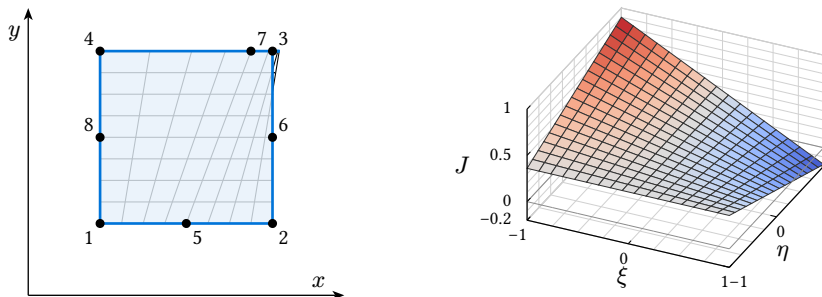
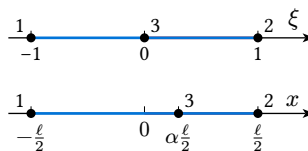


Figure 3.7: Midside-node 8-node quadrilateral element and its Jacobian determinant field.



For instance, consider a three-node bar element with the following nodal coordinates: $x_1 = -\frac{L}{2}$, $x_2 = \frac{\ell}{2}$ and $x_3 = \alpha\ell$, as shown in the figure. In this case the Jacobian determinant is given by

$$J(\xi) = \left(\frac{1}{2} - 2\xi\alpha \right) \ell \quad (3.35)$$

where the expression $\left(\frac{1}{2} - 2\xi\alpha \right)$ must be greater than zero. Considering the element domain in natural coordinates $\xi \in [-1, 1]$, it can be shown that α can vary along the interval. That means that for three-node bars the mid-node must be located along the central region with length $\frac{\ell}{2}$.

This result can be extended for the case of mid-side nodes on 2D and 3D elements, where their locations must be within the central part of the side with length $\frac{\ell}{2}$, where, in this case, ℓ represents the side length.

3.2.10 Non-square Jacobian matrices

So far, the presented Jacobian matrices have been square because the natural coordinate space and the physical space had the same dimension. Non-square Jacobians arise when a lower-dimensional element is embedded in a higher-dimensional domain. These Jacobians are especially important for line and surface integrations, such as those appearing in reinforcement elements, joint elements, and distributed tractions applied along edges or over faces.

1D non-square Jacobian

One important case of a non-square Jacobian arises when a one-dimensional element is embedded in two- or three-dimensional space. Such Jacobians appear in reinforcement, cable, truss, beam, and joint elements, and they are also needed in line integration for distributed loads, edge tractions, and other quantities defined along embedded curves.

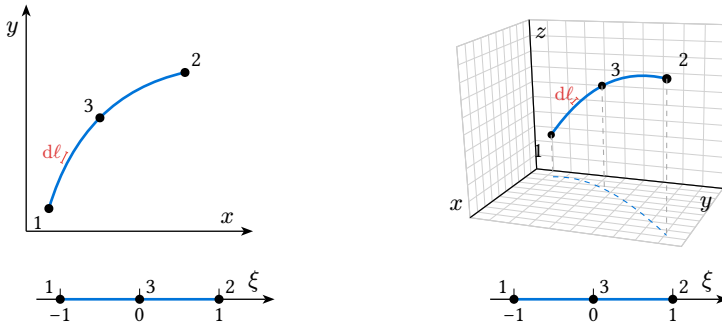


Figure 3.9: 1D elements embedded in 2D and 3D spaces.

For a one-dimensional element embedded in two-dimensional space, the Jacobian is a vector:

$$\mathbf{J}(\xi) = \begin{pmatrix} \frac{\partial x}{\partial \xi} \\ \frac{\partial y}{\partial \xi} \end{pmatrix} \quad (3.36)$$

Similarly, for a one-dimensional element embedded in three-dimensional space, we have:

$$\mathbf{J}(\xi) = \begin{pmatrix} \frac{\partial x}{\partial \xi} \\ \frac{\partial y}{\partial \xi} \\ \frac{\partial z}{\partial \xi} \end{pmatrix} \quad (3.37)$$

Using the coordinate interpolation, the Jacobian vector can be written as

$$\mathbf{J}(\xi) = \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \xi} \quad (3.38)$$

where $\mathbf{X}$ is the coordinate matrix in the global space and $\frac{\partial \mathbf{N}}{\partial \xi}$ is the vector of shape function derivatives with respect to the natural coordinate ξ .

The differential length in global coordinates is related to the differential in natural coordinates by

$$d\ell = \|\mathbf{J}(\xi)\| d\xi \quad (3.39)$$

where $\|\mathbf{J}(\xi)\| = J(\xi)$ is the Euclidean norm of the Jacobian vector, ℓ is the curvilinear coordinate along the element, and

$$d\ell = \sqrt{dx^2 + dy^2 + dz^2} \quad (3.40)$$

is the corresponding differential arc length. Therefore,

$$J(\xi) = \frac{d\ell}{d\xi} \quad (3.41)$$

Application to curved 1D elements orientation

For 1D elements in 2D or 3D space, the orientation at a point is often needed. This orientation can be described by a unit tangent vector $\mathbf{r}$, as shown in the figure. For a given coordinate ξ , it is defined by

$$\mathbf{r}(\xi) = \mathbf{r}(x, y, z) = \begin{pmatrix} \frac{\partial x}{\partial \ell} \\ \frac{\partial y}{\partial \ell} \\ \frac{\partial z}{\partial \ell} \end{pmatrix} = \frac{\partial \mathbf{x}}{\partial \ell} \quad (3.42)$$

Using the chain rule,

$$\mathbf{r}(\xi) = \frac{\partial \mathbf{x}}{\partial \xi} \frac{\partial \xi}{\partial \ell} \quad (3.43)$$

In this expression, $\frac{\partial \mathbf{x}}{\partial \xi}$ is precisely the Jacobian vector $\mathbf{J}$, while

$$\frac{\partial \xi}{\partial \ell} = \frac{1}{\|\mathbf{J}\|} \quad (3.44)$$

is the inverse Jacobian in the curvilinear coordinate. Therefore, the orientation vector can be written as

$$\mathbf{r}(\xi) = \mathbf{J} \frac{1}{\|\mathbf{J}\|} = \hat{\mathbf{J}} \quad (3.45)$$

This result shows that, for isoparametric bar elements, $\mathbf{J}$ is parallel to the element orientation at the point ξ , and the unit vector $\hat{\mathbf{J}}$ gives the local orientation of the element. This local tangent direction is used, for example, in the definition of element local axes, in the transformation of vectors and forces between local and global coordinates, and in the evaluation of loads or constitutive relations that depend on the element orientation.

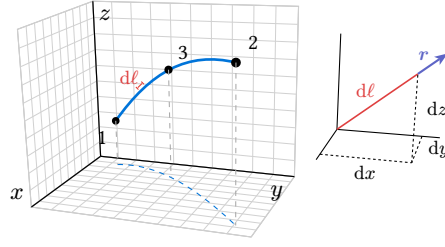


Figure 3.10: Local orientation vector $\mathbf{r}$ in a 1D element embedded in 3D space.

Surface mapping

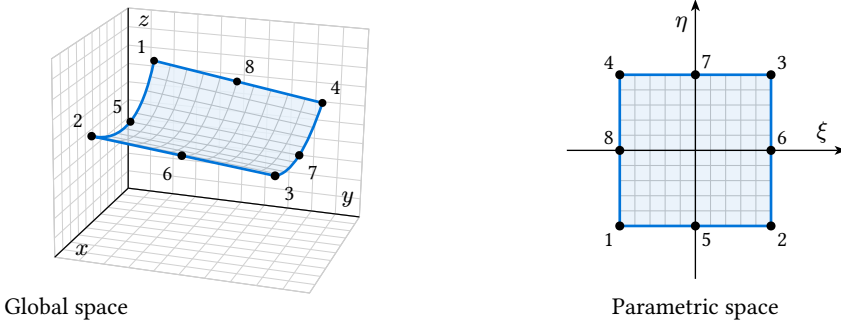


Figure 3.11: Area mapping described by the Jacobian in a 8-node quadrilateral element

Surface elements embedded in three-dimensional space also lead to non-square Jacobians. This situation appears, for example, in joint elements and in the integration of traction forces over the faces of 3D elements.

When a two-dimensional natural domain (ξ, η) is mapped to a surface in 3D space, the Jacobian matrix is rectangular:

$$\mathbf{J} = \begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \\ \frac{\partial z}{\partial \xi} & \frac{\partial z}{\partial \eta} \end{pmatrix} \quad (3.46)$$

so $\mathbf{J}$ is a 3×2 matrix.

Using the interpolation of the coordinates, we obtain

$$\begin{aligned} \mathbf{J}(\xi, \eta) &= \begin{pmatrix} \sum \frac{\partial N_i}{\partial \xi} x_i & \sum \frac{\partial N_i}{\partial \eta} x_i \\ \sum \frac{\partial N_i}{\partial \xi} y_i & \sum \frac{\partial N_i}{\partial \eta} y_i \\ \sum \frac{\partial N_i}{\partial \xi} z_i & \sum \frac{\partial N_i}{\partial \eta} z_i \end{pmatrix} \\ &= \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \xi} \end{aligned} \quad (3.47)$$

For surface elements, the corresponding differential area relation is

$$dA = J d\xi d\eta \quad (3.48)$$

where J is the scalar surface Jacobian associated with the matrix $\mathbf{J}$. To obtain it, define the two tangent vectors

$$\mathbf{a} = \begin{pmatrix} \frac{dx}{d\xi} d\xi \\ \frac{dy}{d\xi} d\xi \\ \frac{dz}{d\xi} d\xi \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} \frac{dx}{d\eta} d\eta \\ \frac{dy}{d\eta} d\eta \\ \frac{dz}{d\eta} d\eta \end{pmatrix} \quad (3.49)$$

Then the differential area is given by

$$dA = \|\mathbf{a} \times \mathbf{b}\| \quad (3.50)$$

which leads to the following expression for the scalar surface Jacobian:

$$J = \sqrt{\begin{vmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{vmatrix}^2 + \begin{vmatrix} J_{21} & J_{22} \\ J_{31} & J_{32} \end{vmatrix}^2 + \begin{vmatrix} J_{31} & J_{32} \\ J_{11} & J_{12} \end{vmatrix}^2} \quad (3.51)$$

where J_{ij} are the components of the matrix $\mathbf{J}$.

Thus, the scalar Jacobian J acts as the area-scaling factor between the natural surface and the physical surface, and it is the quantity used in surface integration over curved geometries.

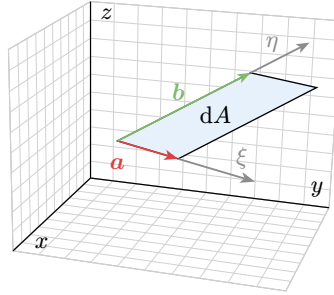


Figure 3.12: Differential parametric embedded in 3D global space.

Applications to surface normals and local bases

The columns of $\mathbf{J}$, denoted by $\mathbf{J}_1$ and $\mathbf{J}_2$, are tangent to the surface because they describe how the position vector varies with respect to ξ and η . Their cross product therefore gives a normal vector:

$$\mathbf{n} = \mathbf{J}_1 \times \mathbf{J}_2 \quad (3.52)$$

In general, $\mathbf{J}_1$ and $\mathbf{J}_2$ are not orthogonal.

This construction is used in shell, joint, and interface elements, in surface integration over the faces of 3D elements, and in the evaluation of pressure, traction, and contact contributions, all of which require a consistent surface normal.

When an orthonormal basis $\mathcal{B}$ attached to the surface is needed, one convenient choice is to introduce the tangent vectors $\mathbf{l}$ and $\mathbf{m}$. The first is defined directly as

$$\mathbf{l} = \mathbf{J}_1 \quad (3.53)$$

The second tangent vector $\mathbf{m}$ is defined as

$$\mathbf{m} = \mathbf{n} \times \mathbf{l} \quad (3.54)$$

Normalizing the three vectors then gives the orthonormal basis

$$\mathcal{B} = (\hat{l}, \hat{m}, \hat{n}) \quad (3.55)$$

Such a basis is needed when quantities must be resolved into local tangential and normal directions, as in shell and interface formulations, anisotropic surface models, and contact or traction laws that distinguish between normal and tangential components.

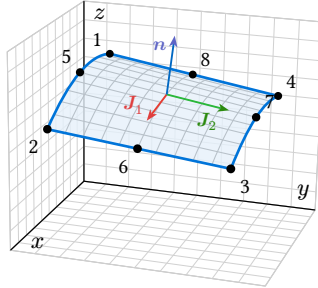
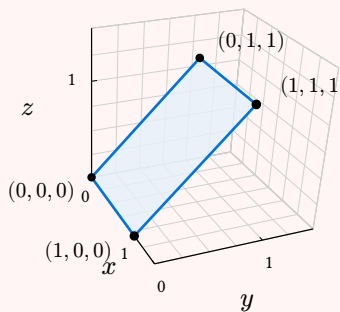


Figure 3.13: Area mapping described by the Jacobian in a 8-node quadrilateral element

Exercises 3.2:

1. What does Jacobian positiveness mean, and why is it important in finite element analysis?
2. What is the meaning of a negative Jacobian determinant? What are the implications for finite element analysis?
3. Why can the location of midside nodes affect the sign or magnitude of the Jacobian determinant?
4. In what sense can the Jacobian norm be used in line and surface elements embedded in higher-dimensional spaces?
5. Give examples of elements that have constant Jacobian determinant over the whole domain.
6. For a three-node bar element with coordinates $(x_1, y_1) = (1, 1)$, $(x_2, y_2) = (3, 1)$ and $(x_3, y_3) = (2, 2)$, determine an expression for its orientation as a function of the local coordinate ξ . Find the orientation vector at node 3. **A.** $\mathbf{r} = \langle 1, 0 \rangle$.
7. Using integration, calculate the length of the element in the last exercise. **A.** $\ell = \frac{\ln \sqrt{5} + 2}{2} + \sqrt{5}$
8. By inspection determine the Jacobian norm of the four-node element shown below. Then, compare the result with the one obtained by using the Jacobian matrix. **A.** $J = \frac{\sqrt{2}}{4}$



9. Compute the Jacobian norm as function of ξ and η of the element in the last exercise changing the coordinates of node 3 to $x_3 = (1, 1, 0)$. Later calculate the surface area by integration. **A.** $A = 1.281$

3.3 Numerical integration

In numerical analysis, a quadrature rule approximates a definite integral by a weighted sum of function values evaluated at prescribed points in the integration domain:

$$\int_a^b f(x) dx \approx \sum_{i=1}^n f(x_i) w_i \quad (3.56)$$

where x_i are the quadrature points and w_i are the corresponding weights.

For partition-based rules, each term $f(x_i)w_i$ can be interpreted as the contribution of a small subinterval, which is closely related to a Riemann-sum approximation (Figure 3.15). In this case, the quadrature rule can be expressed as:

$$\int_a^b f(x) dx \approx \sum_{i=1}^n A_i \quad (3.57)$$

Many quadrature rules are constructed by approximating $f(x)$ with an interpolating polynomial and then integrating that polynomial exactly. Two widely used families of quadrature rules are the Newton-Cotes and Gauss-Legendre formulae.

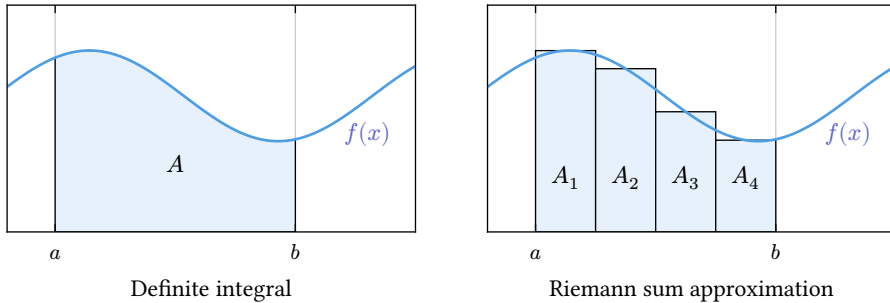


Figure 3.15: Interpretação geométrica de uma integral definida e sua aproximação por uma soma de Riemann.

In finite elements, numerical integration is used to evaluate integrals over element lines, surfaces, and volumes. Typical applications include the computation of element lengths, areas, volumes, stiffness matrices, equivalent nodal loads, and internal force vectors. For instance, the area of a two-dimensional element can be computed in natural coordinates as

$$A = \iint_{A'} |J(\xi, \eta)| d\xi d\eta \approx \sum_{i=1}^m |J(\xi_i, \eta_i)| w_i \quad (3.58)$$

where A' is the element domain in natural coordinates and (ξ_i, η_i) are the integration points.

3.3.1 Newton-Cotes formulae

Newton-Cotes formulae are quadrature rules obtained by interpolating the integrand at equally spaced points and integrating the interpolating polynomial exactly over the interval.

Newton-Cotes formulae are classified as **closed** when the interval endpoints are included among the sampling points, and as **open** when the endpoints are excluded. In what follows, we consider closed Newton-Cotes rules on a single panel $[x_1, x_n]$.

For a closed Newton-Cotes rule with n equally spaced sampling points, the integral over the panel is approximated as

$$\int_{x_1}^{x_n} f(x) dx \approx \sum_{i=1}^n w_i f(x_i) \quad (3.59)$$

where

$$x_i = x_1 + (i-1)h, \quad h = \frac{x_n - x_1}{n-1}, \quad i = 1, \dots, n \quad (3.60)$$

The weights are obtained by integrating the Lagrange basis polynomials associated with the nodes x_i . If

$$p_{n-1}(x) = \sum_{i=1}^n f(x_i) L_i(x) \quad (3.61)$$

is the interpolating polynomial of degree $n-1$, then

$$\begin{aligned} \int_{x_1}^{x_n} f(x) dx &\approx \int_{x_1}^{x_n} p_{n-1}(x) dx \\ &= \int_{x_1}^{x_n} \sum_{i=1}^n f(x_i) L_i(x) dx \\ &= \sum_{i=1}^n f(x_i) \underbrace{\int_{x_1}^{x_n} L_i(x) dx}_{w_i} \end{aligned} \quad (3.62)$$

The closed two-point Newton-Cotes rule, also called the trapezoidal rule, approximates the integral as

$$\begin{aligned} \int_{x_1}^{x_2} f(x) dx &\approx \frac{x_2 - x_1}{2} (f(x_1) + f(x_2)) \\ &= \frac{h}{2} f(x_1) + \frac{h}{2} f(x_2) \end{aligned} \quad (3.63)$$

where $h = x_2 - x_1$.

The closed three-point Newton-Cotes rule, also called Simpson's $1/3$ rule, is given by

$$\begin{aligned} \int_{x_1}^{x_3} f(x) dx &\approx \frac{x_3 - x_1}{6} (f(x_1) + 4f(x_2) + f(x_3)) \\ &\approx \frac{h}{3} f(x_1) + \frac{4h}{3} f(x_2) + \frac{h}{3} f(x_3) \end{aligned} \quad (3.64)$$

where $h = \frac{x_3 - x_1}{2}$.

Applying these single-panel rules repeatedly over a partition of the interval leads to the composite trapezoidal and composite Simpson rules ([Figure 3.16](#)). In the trapezoidal rule, the integrand is approximated by piecewise linear segments; in Simpson's rule, it is approximated by piecewise quadratic

polynomials over pairs of subintervals. Table 3.1 lists the first four closed Newton-Cotes rules together with the degree of the interpolating polynomial and the corresponding normalized weights.

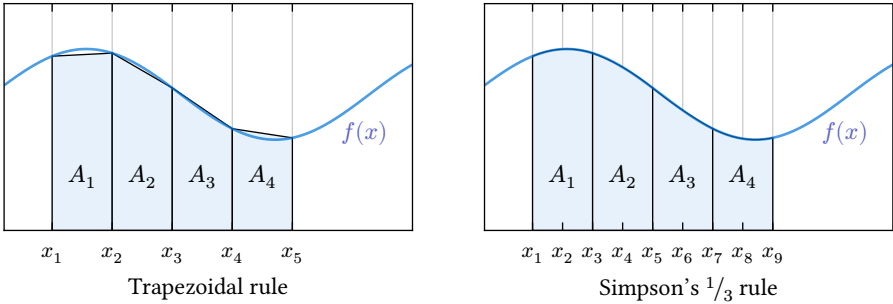


Figure 3.16: Composite trapezoidal and composite Simpson approximations.

Table 3.1: Normalized weights of the first four closed Newton-Cotes rules.

Rule	n	Interp. degree	Normalized weights w_i/h
Trapezoid	2	1	$\frac{1}{2}, \frac{1}{2}$
Simpson's $1/3$	3	2	$\frac{1}{3}, \frac{4}{3}, \frac{1}{3}$
Simpson's $3/8$	4	3	$\frac{3}{8}, \frac{9}{8}, \frac{9}{8}, \frac{3}{8}$
Boole's	5	4	$\frac{14}{45}, \frac{64}{45}, \frac{8}{45}, \frac{64}{45}, \frac{14}{45}$

3.3.2 Gauss-Legendre quadrature

Gauss-Legendre quadrature is a family of quadrature rules defined on the reference interval $[-1, 1]$. For a rule with n quadrature points, the approximation has the form

$$\int_{-1}^{+1} f(\xi) \, d\xi \approx \sum_{i=1}^n w_i f(\xi_i) \tag{3.65}$$

where ξ_i are the quadrature points and w_i are the corresponding weights.

The points and weights are chosen so that the rule integrates exactly every polynomial of degree up to $2n - 1$. This number is called the **degree of exactness** of the rule. For a general smooth integrand, the rule is accurate when the function is well approximated by a polynomial over the interval of integration.

In Gauss-Legendre quadrature, the quadrature points ξ_i are the roots of the Legendre polynomial $P_n(\xi)$. Because the rules are tabulated on the reference interval $[-1, 1]$, an integral over an arbitrary interval $[a, b]$ is first mapped to this range by the linear transformation

$$x = \frac{b-a}{2}\xi + \frac{a+b}{2} \tag{3.66}$$

for which

$$dx = \frac{b-a}{2} d\xi \tag{3.67}$$

Therefore,

$$\begin{aligned}\int_a^b f(x) dx &= \frac{b-a}{2} \int_{-1}^{+1} f\left(\frac{b-a}{2}\xi + \frac{a+b}{2}\right) d\xi \\ &\approx \frac{b-a}{2} \sum_{i=1}^n w_i f\left(\frac{b-a}{2}\xi_i + \frac{a+b}{2}\right)\end{aligned}\tag{3.68}$$

Example 3.5: Find the quadrature points and weights for the two-point Gauss-Legendre rule on the interval $[-1, 1]$.

Solution:

A two-point Gauss-Legendre rule has degree of exactness $2n - 1 = 3$. Therefore, it must integrate exactly the monomials $1, \xi, \xi^2$, and ξ^3 , which gives the system

$$\begin{aligned}w_1 + w_2 &= \int_{-1}^{+1} 1 d\xi = 2 \\ w_1\xi_1 + w_2\xi_2 &= \int_{-1}^{+1} \xi d\xi = 0 \\ w_1(\xi_1)^2 + w_2(\xi_2)^2 &= \int_{-1}^{+1} \xi^2 d\xi = \frac{2}{3} \\ w_1(\xi_1)^3 + w_2(\xi_2)^3 &= \int_{-1}^{+1} \xi^3 d\xi = 0\end{aligned}$$

Solving this system gives the quadrature points

$$\xi_1 = -\sqrt{\frac{1}{3}} \quad \text{and} \quad \xi_2 = +\sqrt{\frac{1}{3}}$$

with corresponding weights

$$w_1 = 1 \quad \text{and} \quad w_2 = 1$$

The first four Gauss-Legendre rules on $[-1, 1]$ are listed in [Table 3.2](#), where $p = 2n - 1$ denotes the degree of exactness. [Figure 3.17](#) illustrates the two- and three-point rules geometrically. In this representation, each rectangle has area $w_i f(\xi_i)$, so its width represents the weight w_i and its height represents the sampled value $f(\xi_i)$.

Table 3.2: Quadrature points and weights of the first four Gauss-Legendre rules on $[-1, 1]$.

n	Degree of exactness	ξ_i	w_i
1	1	0	2
2	3	$-\sqrt{\frac{1}{3}}$	1
		$+\sqrt{\frac{1}{3}}$	1
3	5	0	$\frac{8}{9}$
		$\pm\sqrt{\frac{3}{5}}$	$\frac{5}{9}$
4	7	± 0.3399810436	0.6521451549
		± 0.8611363116	0.3478548451

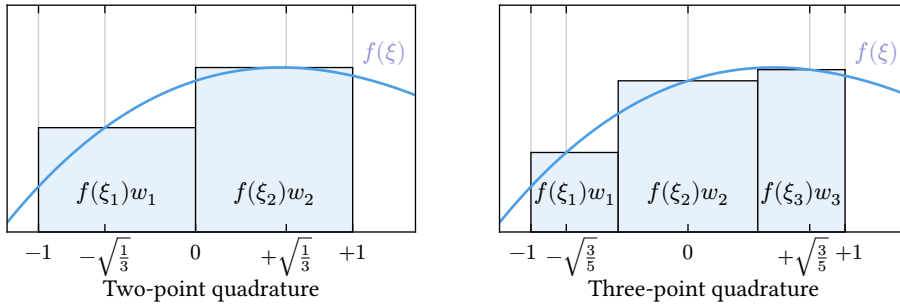


Figure 3.17: Geometric interpretation of the two- and three-point Gauss-Legendre rules on $[-1, 1]$.

3.3.3 Multiple integrals

For multiple integrals over tensor-product domains, Gauss rules are constructed as **product rules** by applying one-dimensional Gauss rules to each independent variable. For example, over the reference square $[-1, 1] \times [-1, 1]$,

$$\int_{-1}^{+1} \int_{-1}^{+1} f(\xi, \eta) \, d\xi \, d\eta \approx \sum_{i=1}^{n_\xi} \sum_{j=1}^{n_\eta} w_i^\xi w_j^\eta f(\xi_i, \eta_j) \quad (3.69)$$

where n_ξ and n_η are the numbers of Gauss points in the ξ and η directions, respectively.

If the tensor-product points are renumbered with a single index $k = 1, \dots, n$, where $n = n_\xi n_\eta$, the same rule can be written as

$$\int_{-1}^{+1} \int_{-1}^{+1} f(\xi, \eta) \, d\xi \, d\eta \approx \sum_{k=1}^n w_k f(\xi_k, \eta_k) \quad (3.70)$$

where each w_k is the product of the corresponding one-dimensional weights.

3.3.4 Quadrature rules for finite elements

For one-dimensional finite elements, the quadrature points and weights are those of the one-dimensional Gauss-Legendre rules on the reference interval $[-1, 1]$, listed in Table 3.2.

For two-dimensional elements, quadrilateral elements use product Gauss rules on the reference square $[-1, 1] \times [-1, 1]$, whereas triangular elements use simplex rules such as the Dunavant rules on the reference triangle. For three-dimensional elements, hexahedral elements use product Gauss rules on the reference cube $[-1, 1] \times [-1, 1] \times [-1, 1]$, whereas tetrahedral elements use simplex rules such as the Keast rules on the reference tetrahedron.

Figure 3.18 and Figure 3.19 show the locations of the quadrature points for selected rules on triangular and quadrilateral reference elements. Table 3.3 through Table 3.6 list quadrature points and weights for triangular, quadrilateral, tetrahedral, and hexahedral elements.

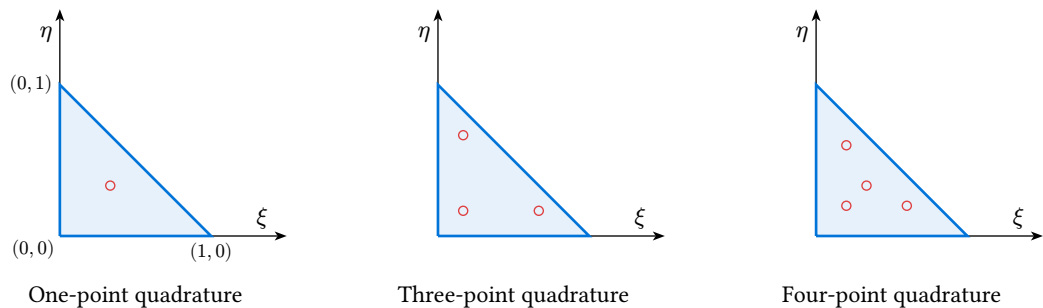


Figure 3.18: Quadrature points for selected rules on the reference triangle.

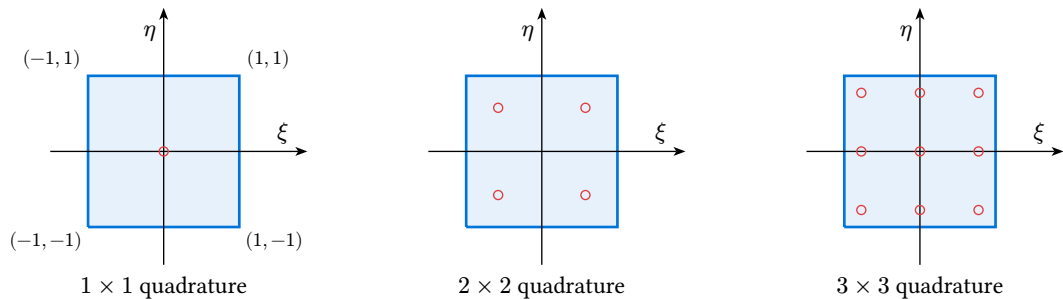


Figure 3.19: Quadrature points for selected rules on the reference square.

Table 3.3: Quadrature points and weights for triangular elements.

n	Degree of exactness	i	ξ_i	η_i	w_i
1	1	1	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{2}$
3	2	1	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
		2	$\frac{2}{3}$	$\frac{1}{6}$	$\frac{1}{6}$
		3	$\frac{1}{6}$	$\frac{2}{3}$	$\frac{1}{6}$
4	3	1	$\frac{1}{3}$	$\frac{1}{3}$	$-\frac{9}{32}$
		2	$\frac{3}{5}$	$\frac{1}{5}$	$\frac{25}{96}$
		3	$\frac{1}{5}$	$\frac{3}{5}$	$\frac{25}{96}$
		4	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{25}{96}$

Table 3.4: Quadrature points and weights for quadrilateral elements.

n	Degree of exactness	i	ξ_i	η_i	w_i
1	1	1	0	0	4
4	3	1	$-\sqrt{\frac{1}{3}}$	$-\sqrt{\frac{1}{3}}$	1
		2	$\sqrt{\frac{1}{3}}$	$-\sqrt{\frac{1}{3}}$	1
		3	$-\sqrt{\frac{1}{3}}$	$\sqrt{\frac{1}{3}}$	1
		4	$\sqrt{\frac{1}{3}}$	$\sqrt{\frac{1}{3}}$	1

9	5	1	$-\sqrt{\frac{3}{5}}$	$-\sqrt{\frac{3}{5}}$	$\frac{25}{81}$
		2	0	$-\sqrt{\frac{3}{5}}$	$\frac{40}{81}$
		3	$\sqrt{\frac{3}{5}}$	$-\sqrt{\frac{3}{5}}$	$\frac{25}{81}$
		4	$-\sqrt{\frac{3}{5}}$	0	$\frac{40}{81}$
		5	0	0	$\frac{64}{81}$
		6	$\sqrt{\frac{3}{5}}$	0	$\frac{40}{81}$
		7	$-\sqrt{\frac{3}{5}}$	$\sqrt{\frac{3}{5}}$	$\frac{25}{81}$
		8	0	$\sqrt{\frac{3}{5}}$	$\frac{40}{81}$
		9	$\sqrt{\frac{3}{5}}$	$\sqrt{\frac{3}{5}}$	$\frac{25}{81}$

Table 3.5: Quadrature points and weights for tetrahedral elements.

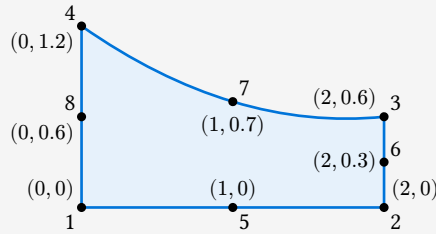
n	Degree of exactness	i	ξ_i	η_i	ζ_i	w_i
1	1	1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{6}$
4	1	1	a	b	b	$\frac{1}{24}$
		2	b	a	b	$\frac{1}{24}$
		3	b	b	a	$\frac{1}{24}$
		4	b	b	b	$\frac{1}{24}$
5	3	1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$-\frac{2}{15}$
		2	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{3}{40}$
		3	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{3}{40}$
		4	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{40}$
		5	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{3}{40}$
$a = \frac{5+\sqrt{5}}{20} = 0.5854101966249685$						
$b = \frac{5-\sqrt{5}}{20} = 0.1381966011250105$						

Table 3.6: Quadrature points and weights for hexahedral elements.

n	Degree of exactness	i	ξ_i	η_i	ζ_i	w_i
1	1	1	0	0	0	8
8	3	1	$-\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	1
		2	$\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	1
		3	$-\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	1
		$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
		8	$\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$	1

27	5	1	$-\sqrt{\frac{3}{5}}$	$-\sqrt{\frac{3}{5}}$	$-\sqrt{\frac{3}{5}}$	$\frac{125}{243}$
		2	0	$-\sqrt{\frac{3}{5}}$	$-\sqrt{\frac{3}{5}}$	$\frac{200}{243}$
		3	$\sqrt{\frac{3}{5}}$	$-\sqrt{\frac{3}{5}}$	$-\sqrt{\frac{3}{5}}$	$\frac{125}{243}$
		4	$-\sqrt{\frac{3}{5}}$	0	$-\sqrt{\frac{3}{5}}$	$\frac{200}{243}$
		$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
		27	$\sqrt{\frac{3}{5}}$	$\sqrt{\frac{3}{5}}$	$\sqrt{\frac{3}{5}}$	$\frac{125}{243}$

Example 3.6: Consider the 8-node quadrilateral element below. Compute its area using a 2×2 Gauss-Legendre rule.



Solution:

The element area is given by

$$A = \iint_A dA$$

which, in natural coordinates, becomes

$$A = \int_{-1}^1 \int_{-1}^1 J(\xi, \eta) d\xi d\eta$$

where $J(\xi, \eta) = \det \mathbf{J}(\xi, \eta)$ is the Jacobian determinant.

Using numerical integration, this integral is approximated by

$$A \approx \sum_{i=1}^m J_i w_i$$

where $J_i = J(\xi_i, \eta_i)$. For the 2×2 rule, $m = 4$ and all weights are equal to 1.

This requires computing the Jacobian matrix at each integration point. It is obtained from

$$\mathbf{J} = \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \xi}$$

The coordinates matrix of the element is given by

$$\mathbf{X} = \begin{pmatrix} 0 & 0 \\ 2 & 0 \\ 2 & 0.6 \\ 0 & 1.2 \\ 1 & 0 \\ 2 & 0.3 \\ 1 & 0.7 \\ 0 & 0.6 \end{pmatrix},$$

and the derivatives matrix is

$$\frac{\partial N}{\partial \xi} = \begin{pmatrix} \frac{1}{4}(1-\eta)(2\xi+\eta) & \frac{1}{4}(1-\xi)(\xi+2\eta) \\ \frac{1}{4}(1-\eta)(2\xi-\eta) & \frac{1}{4}(1+\xi)(2\eta-\xi) \\ \frac{1}{4}(1+\eta)(2\xi+\eta) & \frac{1}{4}(1+\xi)(\xi+2\eta) \\ \frac{1}{4}(1+\eta)(2\xi-\eta) & \frac{1}{4}(1-\xi)(2\eta-\xi) \\ -\xi(1-\eta) & -\frac{1}{2}(1-\xi^2) \\ \frac{1}{2}(1-\eta^2) & -\eta(1+\xi) \\ -\xi(1+\eta) & \frac{1}{2}(1-\xi^2) \\ -\frac{1}{2}(1-\eta^2) & -\eta(1-\xi) \end{pmatrix}$$

Evaluating the derivatives at the first integration point, $(\xi_1, \eta_1) = \left(-\sqrt{\frac{1}{3}}, -\sqrt{\frac{1}{3}}\right)$, gives

$$\frac{\partial N}{\partial \xi} = \begin{pmatrix} -0.6830127018922196 & -0.6830127018922196 \\ -0.2276709006307399 & -0.0610042339640731 \\ -0.1830127018922193 & -0.1830127018922193 \\ -0.0610042339640731 & -0.2276709006307399 \\ 0.9106836025229595 & -0.3333333333333326 \\ 0.3333333333333326 & 0.2440169358562924 \\ 0.2440169358562924 & 0.3333333333333326 \\ -0.3333333333333326 & 0.9106836025229595 \end{pmatrix}$$

The Jacobian matrix at this point is therefore

$$\mathbf{J}_1 = \begin{pmatrix} 1 & 0 \\ -0.1122008468 & 0.4699358737 \end{pmatrix}$$

and its determinant is

$$J_1 = \det \mathbf{J}_1 = 0.4699358737$$

Repeating the calculation at the remaining integration points gives

$$J_1 = 0.4699358737$$

$$J_2 = 0.296730793$$

$$J_3 = 0.4699358737$$

$$J_4 = 0.296730793$$

Finally, the area is computed as

$$\begin{aligned}
 A &= \sum_{i=1}^4 J_i w_i \\
 &= 0.4699358737 \times 1.0 + 0.296730793 \times 1.0 \\
 &\quad + 0.4699358737 \times 1.0 + 0.296730793 \times 1.0 \\
 &= 1.5333333333999999
 \end{aligned}$$

Example 3.7: Write code to compute the area of an 8-node quadrilateral element using Gauss quadrature with 1, 4, and 9 integration points.

Solution:

First, define the derivatives of the shape functions with respect to the natural coordinates. For the 8-node quadrilateral element, they are computed by the following function:

```

using LinearAlgebra

function quad8_deriv(xi, eta)
    dN = zeros(8,2)

    dN[1,1] = 0.25*(1.0 - eta)*(2.0*xi + eta)
    dN[2,1] = 0.25*(1.0 - eta)*(2.0*xi - eta)
    dN[3,1] = 0.25*(1.0 + eta)*(2.0*xi + eta)
    dN[4,1] = 0.25*(1.0 + eta)*(2.0*xi - eta)
    dN[5,1] = -xi*(1.0 - eta)
    dN[6,1] = 0.50*(1.0 - eta*eta)
    dN[7,1] = -xi*(1.0 + eta)
    dN[8,1] = -0.5*(1.0 - eta*eta)

    dN[1,2] = 0.25*(1.0 - xi)*(xi + 2.0*eta)
    dN[2,2] = 0.25*(1.0 + xi)*(2.0*eta - xi)
    dN[3,2] = 0.25*(1.0 + xi)*(xi + 2.0*eta)
    dN[4,2] = 0.25*(1.0 - xi)*(2.0*eta - xi)
    dN[5,2] = -0.50*(1.0 - xi*xi)
    dN[6,2] = -eta*(1.0 + xi)
    dN[7,2] = 0.50*(1.0 - xi*xi)
    dN[8,2] = -eta*(1.0 - xi)

    return dN
end

```

Next, define the quadrature rules for 1, 4, and 9 integration points:

```

# 1-point quadrature rule
quadrature_1_pts = [ 0.0  0.0  4.0 ]

# 4-point quadrature rule
quadrature_4_pts = [
    -0.577350269189626  -0.577350269189626  1.0
     0.577350269189626  -0.577350269189626  1.0
    -0.577350269189626   0.577350269189626  1.0
     0.577350269189626   0.577350269189626  1.0 ]

# 9-point quadrature rule
quadrature_9_pts = [
    -0.774596669241483  -0.774596669241483  0.3086419753086419
      0.0                -0.774596669241483  0.4938271604938271
     0.774596669241483  -0.774596669241483  0.3086419753086419
    -0.774596669241483   0.0                0.4938271604938271
      0.0                0.0                0.7901234567901234
     0.774596669241483   0.0                0.4938271604938271
    -0.774596669241483   0.774596669241483  0.3086419753086419
      0.0                0.774596669241483  0.4938271604938271
     0.774596669241483   0.774596669241483  0.3086419753086419 ]

```

Finally, define a function that evaluates the Jacobian determinant at each integration point and accumulates its weighted contribution to the area:

```

function area(X, quadrature)
    npts = size(quadrature, 1)
    A = 0.0
    for i in 1:npts
        xi = quadrature[i, 1]
        eta = quadrature[i, 2]
        w = quadrature[i, 3]
        dN = quad8_deriv(xi, eta)
        J = X'*dN
        A = A + abs(det(J))*w
    end
    return A
end

# element coordinates
X = [0 0; 2 0; 2 0.6; 0 1.2; 1 0; 2 0.3; 1 0.7; 0 0.6]
# area computation
A1pts = area(X, quadrature_1_pts)
A4pts = area(X, quadrature_4_pts)
A9pts = area(X, quadrature_9_pts)
println("Area using 1-point quadrature: ", A1pts)
println("Area using 4-point quadrature: ", A4pts)
println("Area using 9-point quadrature: ", A9pts)

```

```

Area using 1-point quadrature: 1.4
Area using 4-point quadrature: 1.5333333333333332
Area using 9-point quadrature: 1.5333333333333332

```

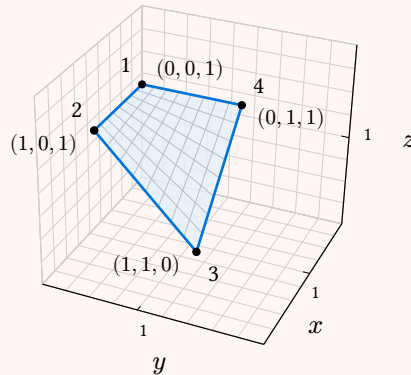
For this element, the areas computed with 4 and 9 integration points are identical. Therefore, the 2×2 Gauss rule is sufficient to evaluate the area in this example.

Exercises 3.3:

1. What is the main idea of numerical integration in the finite element method?
2. Why is Gauss quadrature usually more efficient than Newton-Cotes rules?
3. Why does the required number of integration points depend on the polynomial order of the integrand?
4. Evaluate the integral $\int_0^1 \frac{x}{\sqrt{1+x^2}} dx$ using the Trapezoid and Simpson's rules with 4 subdivisions of the integration limits. Later use the Gauss quadrature rule with 4 integration points. Compare the results.

A. $I_T = 0.4108$, $I_S = 0.4143$, $I_G = 0.4142$

1. The nodal coordinates of a 3-node bar element are $(x_1, y_1) = (-1, -1)$, $(x_2, y_2) = (1, 1)$ and $(x_3, y_3) = (0, 1)$. Determine the element length analytically; then use the Gauss quadrature testing different numbers of integration points. How many integration points are required to find the element length accurately?
2. The nodal coordinates of a 4-node quadrilateral element are $(x_1, y_1) = (1, 1)$, $(x_2, y_2) = (5, 2)$, $(x_3, y_3) = (6, 5)$ and $(x_4, y_4) = (2, 4)$. Determine the element area using the Gauss quadrature with 1, 4, and 9 integration points. Compare the results.
3. Compute the surface area shown at right. How many integration points are required to find the area with precision of 0.01? **A.** $n = 4$



4 MECHANICAL EQUILIBRIUM

Given a solid subjected to body forces, surface loads, and prescribed displacements, the central goal of a static mechanical analysis is to determine the displacement, strain, and stress fields that satisfy kinematics, constitutive behavior, and equilibrium throughout the body.

This chapter develops that boundary-value problem from its continuum form to its finite element discretization. It first explains how deformations and internal forces are described in a continuum, then shows how the governing equilibrium equations are converted into a finite element model, and finally discusses how the resulting system is assembled, solved, and checked in practice.

4.1 Problem statement and boundary conditions

We consider a continuum body occupying a domain Ω and subjected to body forces and mechanical boundary conditions. In a static analysis, the goal is to determine the displacement, strain, and stress fields that satisfy kinematics, constitutive behavior, and equilibrium.

The boundary Γ is split into two complementary parts:

$$\Gamma = \Gamma_u \cup \Gamma_t, \quad \Gamma_u \cap \Gamma_t = \emptyset \quad (4.1)$$

Displacements are prescribed on Γ_u , whereas surface tractions are prescribed on Γ_t :

$$\mathbf{u} = \bar{\mathbf{u}} \quad \text{on } \Gamma_u \quad (4.2)$$

$$\mathbf{t} = \bar{\mathbf{t}} \quad \text{on } \Gamma_t \quad (4.3)$$

The traction vector $\mathbf{t}$ represents the force per unit area acting on the boundary. Its relation to the stress tensor will be introduced in the stress section and then used to write the equilibrium equations.

This distinction between prescribed displacements and prescribed tractions is fundamental. In the weak formulation, displacement conditions become essential boundary conditions, whereas traction conditions appear naturally as load terms.

4.2 Strain kinematics

Kinematics describes how the body deforms, independently of the constitutive law that will later relate strain to stress. Let $\mathbf{X}$ denote the position of a material point in the reference configuration and let

$$\mathbf{x} = \mathbf{X} + \mathbf{u}(\mathbf{X}) \quad (4.4)$$

be its position after deformation, where $\mathbf{u}$ is the displacement field.

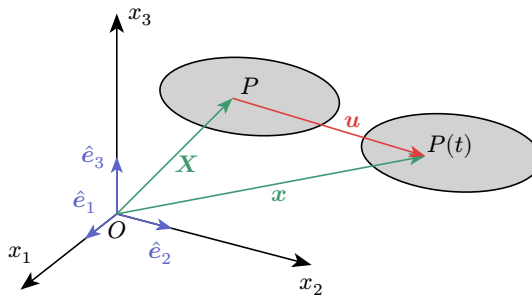


Figure 4.1: Displacement vector.

The deformation gradient is

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} = \mathbf{I} + \nabla_{\mathbf{X}} \mathbf{u} \quad (4.5)$$

and measures the local change of lengths, angles, and orientation induced by the motion.

For a general deformation, the strain measure naturally associated with the reference configuration is the Green-Lagrange strain tensor:

$$\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I}) \quad (4.6)$$

Substituting $\mathbf{F} = \mathbf{I} + \nabla_{\mathbf{X}} \mathbf{u}$ gives

$$\mathbf{E} = \frac{1}{2}(\nabla_{\mathbf{X}} \mathbf{u} + \nabla_{\mathbf{X}} \mathbf{u}^T + \nabla_{\mathbf{X}} \mathbf{u}^T \nabla_{\mathbf{X}} \mathbf{u}) \quad (4.7)$$

This expression shows that strain contains both a linear part and a quadratic part in the displacement gradient.

In this chapter we restrict attention to small deformations and small displacement gradients. Under that assumption, the quadratic term can be neglected and

$$\mathbf{E} \approx \boldsymbol{\varepsilon} = \frac{1}{2}(\nabla_{\mathbf{X}} \mathbf{u} + \nabla_{\mathbf{X}} \mathbf{u}^T) \quad (4.8)$$

where $\boldsymbol{\varepsilon}$ is the infinitesimal strain tensor.

This result is important because it separates true deformation from rigid-body motion. In small-strain kinematics, the displacement gradient can be decomposed into a symmetric part associated with deformation and a skew-symmetric part associated with rigid-body rotation:

$$\nabla_{\mathbf{X}} \mathbf{u} = \boldsymbol{\varepsilon} + \boldsymbol{\omega} \quad (4.9)$$

with

$$\boldsymbol{\omega} = \frac{1}{2}(\nabla_{\mathbf{X}} \mathbf{u} - \nabla_{\mathbf{X}} \mathbf{u}^T) \quad (4.10)$$

The components of the infinitesimal strain tensor are therefore

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial X_j} + \frac{\partial u_j}{\partial X_i} \right) \quad (4.11)$$

or, in tensor form,

$$\begin{aligned}
 \boldsymbol{\varepsilon} &= \begin{pmatrix} \varepsilon_{11} & \varepsilon_{12} & \varepsilon_{13} \\ \varepsilon_{21} & \varepsilon_{22} & \varepsilon_{23} \\ \varepsilon_{31} & \varepsilon_{32} & \varepsilon_{33} \end{pmatrix} = \begin{pmatrix} \varepsilon_{xx} & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_{yy} & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_{zz} \end{pmatrix} \\
 &= \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) & \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \\ \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) & \frac{\partial v}{\partial y} & \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \\ \frac{1}{2} \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) & \frac{1}{2} \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right) & \frac{\partial w}{\partial z} \end{pmatrix} \\
 &= \begin{pmatrix} \varepsilon_{xx} & \frac{1}{2} \gamma_{xy} & \frac{1}{2} \gamma_{xz} \\ \frac{1}{2} \gamma_{xy} & \varepsilon_{yy} & \frac{1}{2} \gamma_{yz} \\ \frac{1}{2} \gamma_{xz} & \frac{1}{2} \gamma_{yz} & \varepsilon_{zz} \end{pmatrix}
 \end{aligned} \tag{4.12}$$

For small deformations, $\boldsymbol{\varepsilon}$ is symmetric, so $\varepsilon_{ij} = \varepsilon_{ji}$. As a result, only six independent components remain.

The diagonal terms represent normal strains:

$$\varepsilon_{xx} = \frac{\partial u}{\partial x}, \quad \varepsilon_{yy} = \frac{\partial v}{\partial y}, \quad \varepsilon_{zz} = \frac{\partial w}{\partial z} \tag{4.13}$$

They measure extension or contraction along the coordinate directions.

The off-diagonal terms represent tensorial shear strains:

$$\begin{aligned}
 \varepsilon_{xy} &= \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \\
 \varepsilon_{yz} &= \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \\
 \varepsilon_{xz} &= \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)
 \end{aligned} \tag{4.14}$$

In engineering notation, it is common to use the engineering shear strains

$$\gamma_{xy} = 2\varepsilon_{xy}, \quad \gamma_{yz} = 2\varepsilon_{yz}, \quad \gamma_{xz} = 2\varepsilon_{xz} \tag{4.15}$$

Engineering shear strains are convenient because they correspond directly to the change of the right angle between material line elements. For example,

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \tag{4.16}$$

Figure 4.2 shows the components of the infinitesimal strain tensor.

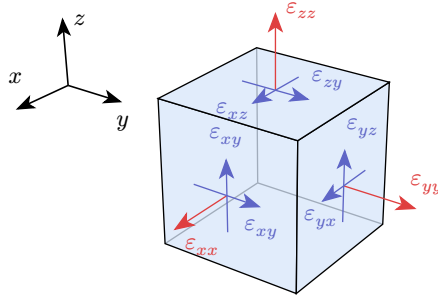


Figure 4.2: Components of the infinitely small strain tensor.

The infinitesimal rotation tensor ω describes local rigid-body-like rotation. Because it is skew-symmetric, it does not contribute to the strain energy.

For matrix calculations, the strain tensor is often written in vector form. Using tensorial shear strains,

$$\boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{zz} \\ \epsilon_{xy} \\ \epsilon_{yz} \\ \epsilon_{xz} \end{pmatrix} \quad (4.17)$$

whereas, using engineering shear strains,

$$\boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{xz} \end{pmatrix} = \begin{pmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{zz} \\ 2\epsilon_{xy} \\ 2\epsilon_{yz} \\ 2\epsilon_{xz} \end{pmatrix} \quad (4.18)$$

The two vector conventions are not interchangeable and must be used consistently with the chosen constitutive equations.

4.3 Stress description

4.3.1 Traction vector

Stress describes the internal forces transmitted across imaginary surfaces inside a continuum. Consider a body and an internal surface patch ΔA passing through a point P , with unit normal vector $\mathbf{n}$.

The force acting on that patch is $\Delta \mathbf{F}$. The force per unit area at the limit defines the traction vector, also called the Cauchy stress vector:

$$\mathbf{t} = \lim_{\Delta A \rightarrow 0} \frac{\Delta \mathbf{F}}{\Delta A} = \frac{d\mathbf{F}}{dA} \quad (4.19)$$

Accordingly,

$$d\mathbf{F} = \mathbf{t} dA \quad (4.20)$$

The traction vector depends on the orientation of the plane, that is, on the normal vector $\mathbf{n}$. At the same material point, different cut orientations generally produce different traction vectors.

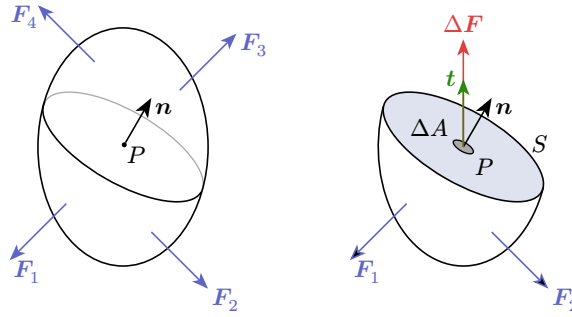


Figure 4.3: A solid under external forces and a cut showing the internal forces vector

4.3.2 Cauchy stress tensor

The traction vector varies with the orientation of the surface passing through P . Cauchy's theorem states that this dependence is linear, so there exists a second-order tensor σ such that

$$t = \sigma \cdot n \quad (4.21)$$

for any unit normal vector n .

If traction vectors are known on three mutually orthogonal planes, the tensor components are obtained by projecting those tractions onto the Cartesian basis vectors.

The stress components are given by the projections of the traction vectors onto the Cartesian basis vectors:

$$\sigma = \begin{pmatrix} t_x \cdot e_x & t_y \cdot e_x & t_z \cdot e_x \\ t_x \cdot e_y & t_y \cdot e_y & t_z \cdot e_y \\ t_x \cdot e_z & t_y \cdot e_z & t_z \cdot e_z \end{pmatrix} \quad (4.22)$$

In general, the components of the Cauchy stress tensor are defined by

$$\sigma_{ij} = \lim_{\Delta A_j \rightarrow 0} \frac{\Delta F_i}{\Delta A_j} = \frac{dF_i}{dA_j} \quad (4.23)$$

where ΔA_j is an area element whose normal is aligned with direction j , and ΔF_i is the i -component of the force acting on that area. Because this tensor is defined in the current configuration, it is called the Cauchy stress tensor.

In matrix form,

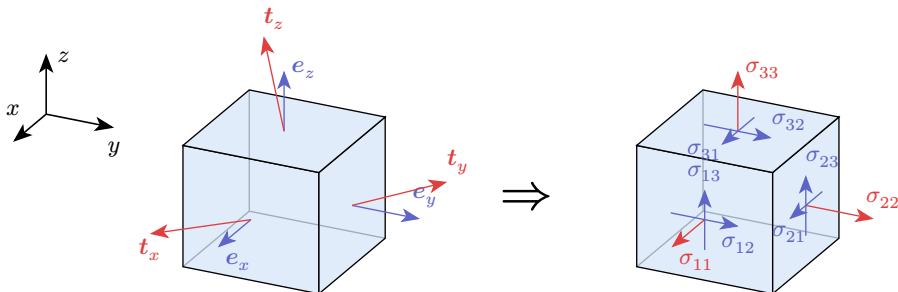


Figure 4.4: Decomposition of traction vectors into stress components.

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix} \quad (4.24)$$

4.3.3 Symmetry and engineering notation

In the absence of distributed body couples, moment equilibrium of an infinitesimal material element implies

$$\sigma_{ij} = \sigma_{ji} \quad (4.25)$$

so the Cauchy stress tensor is symmetric. Hence, only six independent stress components remain.

Using the conventional notation with normal and shear components,

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_{yy} & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_{zz} \end{pmatrix} \quad (4.26)$$

For matrix manipulations, the tensor is often written as a stress vector in engineering notation:

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \tau_{yz} \\ \tau_{xz} \\ \tau_{xy} \end{pmatrix} \quad (4.27)$$

4.3.4 Typical stress states

Several idealized stress states are used frequently in analysis and modeling.

Uniaxial stress: stress acts in only one direction. For example,

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{xx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (4.28)$$

More generally, if the loading acts in direction $\mathbf{n}$,

$$\boldsymbol{\sigma} = \sigma \mathbf{n} \otimes \mathbf{n} \quad (4.29)$$

Pure shear: only shear components are present in a given coordinate system. For example,

$$\boldsymbol{\sigma} = \begin{pmatrix} 0 & \sigma_{xy} & 0 \\ \sigma_{yx} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{or} \quad \boldsymbol{\sigma}' = \begin{pmatrix} \sigma & 0 & 0 \\ 0 & -\sigma & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (4.30)$$

where $\sigma = \sigma_{x'x'} = -\sigma_{y'y'}$.

Isotropic or hydrostatic stress: all normal stresses are equal and shear stresses vanish:

$$\boldsymbol{\sigma} = \begin{pmatrix} p & 0 & 0 \\ 0 & p & 0 \\ 0 & 0 & p \end{pmatrix} = p\mathbf{I} \quad (4.31)$$

Plane stress: stress is essentially confined to a plane, which is a common approximation for thin bodies:

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & 0 \\ \sigma_{yx} & \sigma_{yy} & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (4.32)$$

Plane strain: deformation is confined to a plane, but the out-of-plane stress may be nonzero:

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & 0 \\ \sigma_{yx} & \sigma_{yy} & 0 \\ 0 & 0 & \sigma_{zz} \end{pmatrix} \quad (4.33)$$

Axisymmetric stress: the geometry, loading, and response are invariant with respect to rotation about an axis. In cylindrical coordinates,

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{rr} & 0 & \sigma_{rz} \\ 0 & \sigma_{\theta\theta} & 0 \\ \sigma_{zr} & 0 & \sigma_{zz} \end{pmatrix} \quad (4.34)$$

4.4 Constitutive relations

4.4.1 General constitutive form

A constitutive model closes the mechanical problem by relating stress and strain. In the most general setting, this relation may be nonlinear, anisotropic, path-dependent, or time-dependent.

Taking a differential form, the constitutive relation can be expressed as

$$d\boldsymbol{\sigma} = \mathbf{D} d\boldsymbol{\varepsilon} \quad (4.35)$$

In component form, this equation can be written as

$$\begin{pmatrix} d\sigma_{xx} \\ d\sigma_{yy} \\ d\sigma_{zz} \\ d\tau_{yz} \\ d\tau_{xz} \\ d\tau_{xy} \end{pmatrix} = \begin{pmatrix} \frac{\partial \sigma_{xx}}{\partial \varepsilon_{xx}} & \frac{\partial \sigma_{xx}}{\partial \varepsilon_{yy}} & \frac{\partial \sigma_{xx}}{\partial \varepsilon_{zz}} & \frac{\partial \sigma_{xx}}{\partial \gamma_{yz}} & \frac{\partial \sigma_{xx}}{\partial \gamma_{xz}} & \frac{\partial \sigma_{xx}}{\partial \gamma_{xy}} \\ \frac{\partial \sigma_{yy}}{\partial \varepsilon_{xx}} & \frac{\partial \sigma_{yy}}{\partial \varepsilon_{yy}} & \frac{\partial \sigma_{yy}}{\partial \varepsilon_{zz}} & \frac{\partial \sigma_{yy}}{\partial \gamma_{yz}} & \frac{\partial \sigma_{yy}}{\partial \gamma_{xz}} & \frac{\partial \sigma_{yy}}{\partial \gamma_{xy}} \\ \frac{\partial \sigma_{zz}}{\partial \varepsilon_{xx}} & \frac{\partial \sigma_{zz}}{\partial \varepsilon_{yy}} & \frac{\partial \sigma_{zz}}{\partial \varepsilon_{zz}} & \frac{\partial \sigma_{zz}}{\partial \gamma_{yz}} & \frac{\partial \sigma_{zz}}{\partial \gamma_{xz}} & \frac{\partial \sigma_{zz}}{\partial \gamma_{xy}} \\ \frac{\partial \tau_{yz}}{\partial \varepsilon_{xx}} & \frac{\partial \tau_{yz}}{\partial \varepsilon_{yy}} & \frac{\partial \tau_{yz}}{\partial \varepsilon_{zz}} & \frac{\partial \tau_{yz}}{\partial \gamma_{yz}} & \frac{\partial \tau_{yz}}{\partial \gamma_{xz}} & \frac{\partial \tau_{yz}}{\partial \gamma_{xy}} \\ \frac{\partial \tau_{xz}}{\partial \varepsilon_{xx}} & \frac{\partial \tau_{xz}}{\partial \varepsilon_{yy}} & \frac{\partial \tau_{xz}}{\partial \varepsilon_{zz}} & \frac{\partial \tau_{xz}}{\partial \gamma_{yz}} & \frac{\partial \tau_{xz}}{\partial \gamma_{xz}} & \frac{\partial \tau_{xz}}{\partial \gamma_{xy}} \\ \frac{\partial \tau_{xy}}{\partial \varepsilon_{xx}} & \frac{\partial \tau_{xy}}{\partial \varepsilon_{yy}} & \frac{\partial \tau_{xy}}{\partial \varepsilon_{zz}} & \frac{\partial \tau_{xy}}{\partial \gamma_{yz}} & \frac{\partial \tau_{xy}}{\partial \gamma_{xz}} & \frac{\partial \tau_{xy}}{\partial \gamma_{xy}} \end{pmatrix} \begin{pmatrix} d\varepsilon_{xx} \\ d\varepsilon_{yy} \\ d\varepsilon_{zz} \\ d\gamma_{yz} \\ d\gamma_{xz} \\ d\gamma_{xy} \end{pmatrix} \quad (4.36)$$

Under the assumption of **small-strains** and **linear elasticity**, the stress depends linearly on the current strain, and unloading follows the same path as loading. In that setting,

$$\boldsymbol{\sigma} = \mathbf{D}\boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} = \mathbf{C}\boldsymbol{\sigma} \quad (4.37)$$

where $\mathbf{D}$ is the constitutive or stiffness matrix and $\mathbf{C}$ is the compliance matrix.

4.4.2 Elastic constants

For isotropic linear elasticity, only two independent elastic constants are needed. Common choices are:

- Young's modulus E
- Poisson's ratio ν
- shear modulus G
- bulk modulus K
- Lamé constants λ and μ

These constants are related by

$$\mu = G = \frac{E}{2(1 + \nu)} \quad (4.38)$$

$$\lambda = \frac{E\nu}{(1 + \nu)(1 - 2\nu)} \quad (4.39)$$

$$K = \frac{E}{3(1 - 2\nu)} \quad (4.40)$$

and therefore

$$\lambda = K - \frac{2}{3}G \quad (4.41)$$

Once any two of these constants are known, the others follow from these relations.

4.4.3 Hooke's law for isotropic linear elasticity

An isotropic material responds in the same way in every direction. Under the additional assumption of linear elasticity, the constitutive law reduces to the generalized Hooke's law.

In compliance form, for a three-dimensional solid,

$$\begin{aligned} \varepsilon_{xx} &= \frac{\sigma_{xx}}{E} - \nu \frac{\sigma_{yy}}{E} - \nu \frac{\sigma_{zz}}{E} \\ \varepsilon_{yy} &= \frac{\sigma_{yy}}{E} - \nu \frac{\sigma_{xx}}{E} - \nu \frac{\sigma_{zz}}{E} \\ \varepsilon_{zz} &= \frac{\sigma_{zz}}{E} - \nu \frac{\sigma_{xx}}{E} - \nu \frac{\sigma_{yy}}{E} \\ \gamma_{xy} &= \frac{\tau_{xy}}{G} \\ \gamma_{yz} &= \frac{\tau_{yz}}{G} \\ \gamma_{xz} &= \frac{\tau_{xz}}{G} \end{aligned} \quad (4.42)$$

In index notation,

$$\varepsilon_{ij} = \left(\frac{1 + \nu}{E} \right) \sigma_{ij} - \left(\frac{\nu}{E} \right) \sigma_{kk} \delta_{ij} \quad (4.43)$$

The inverse relation, in stiffness form, is

$$\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij} \quad (4.44)$$

where δ_{ij} is the Kronecker delta. This expression makes the physical split transparent: the term with λ depends on the volumetric strain, whereas the term with 2μ governs the distortional response.

The volumetric strain follows as

$$\varepsilon_v = \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz} = \left(\frac{1-2\nu}{E} \right) (\sigma_{xx} + \sigma_{yy} + \sigma_{zz}) = \frac{p}{K} \quad (4.45)$$

where $p = \frac{\sigma_{xx} + \sigma_{yy} + \sigma_{zz}}{3}$ is the mean stress.

4.4.4 Engineering notation

Hooke's law is usually written in vector-matrix form. Using engineering notation, the three-dimensional constitutive equation becomes

$$\begin{pmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{yz} \\ \sigma_{xz} \\ \sigma_{xy} \end{pmatrix} = \begin{pmatrix} \lambda + 2\mu & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda + 2\mu & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda + 2\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu \end{pmatrix} \begin{pmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{yz} \\ \gamma_{xz} \\ \gamma_{xy} \end{pmatrix} \quad (4.46)$$

Rewriting the coefficients matrix $\mathbf{D}$ in terms of E and ν , we have

$$\mathbf{D} = \frac{E}{(1+\nu)(1-2\nu)} \begin{pmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} \end{pmatrix} \quad (4.47)$$

Then, the corresponding compliance matrix is

$$\mathbf{C} = \begin{pmatrix} \frac{1}{E} & -\frac{\nu}{E} & -\frac{\nu}{E} & 0 & 0 & 0 \\ -\frac{\nu}{E} & \frac{1}{E} & -\frac{\nu}{E} & 0 & 0 & 0 \\ -\frac{\nu}{E} & -\frac{\nu}{E} & \frac{1}{E} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2(1+\nu)}{E} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2(1+\nu)}{E} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2(1+\nu)}{E} \end{pmatrix} \quad (4.48)$$

4.4.5 Reduced constitutive matrices

For plane stress, plane strain, and axisymmetric models, only the stress and strain components compatible with the kinematic assumption are retained. The constitutive matrix is then reduced accordingly.

4.4.5.1 Plane strain

In plane strain, the Hooke's law relates the in-plane stresses σ_{xx} , σ_{yy} and τ_{xy} to the in-plane strains ε_{xx} , ε_{yy} and γ_{xy} . The out-of-plane strain ε_{zz} is zero, but the out-of-plane stress σ_{zz} may be nonzero. The constitutive matrix becomes

$$D = \frac{E}{(1+\nu)(1-2\nu)} \begin{pmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{pmatrix} \quad (4.49)$$

4.4.5.2 Plane stress

Similarly to plane strain, in plane stress, the Hooke's law relates the in-plane stresses σ_{xx} , σ_{yy} and τ_{xy} to the in-plane strains ε_{xx} , ε_{yy} and γ_{xy} . However, in plane stress, the out-of-plane stress σ_{zz} is zero, but the out-of-plane strain ε_{zz} may be nonzero. The constitutive matrix becomes

$$D = \frac{E}{1-\nu^2} \begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{pmatrix} \quad (4.50)$$

4.4.5.3 Axisymmetric

For axisymmetric linear elasticity,

$$\gamma_{r\theta} = 0, \quad \gamma_{z\theta} = 0, \quad \varepsilon_{rr} \neq 0, \quad \varepsilon_{zz} \neq 0, \quad \varepsilon_{\theta\theta} \neq 0, \quad \gamma_{rz} \neq 0 \quad (4.51)$$

The constitutive matrix becomes

$$\begin{pmatrix} \sigma_{rr} \\ \sigma_{zz} \\ \sigma_{\theta\theta} \\ \tau_{rz} \end{pmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{pmatrix} 1-\nu & \nu & \nu & 0 \\ \nu & 1-\nu & \nu & 0 \\ \nu & \nu & 1-\nu & 0 \\ 0 & 0 & 0 & \frac{1-2\nu}{2} \end{pmatrix} \begin{pmatrix} \varepsilon_{rr} \\ \varepsilon_{zz} \\ \varepsilon_{\theta\theta} \\ \gamma_{rz} \end{pmatrix} \quad (4.52)$$

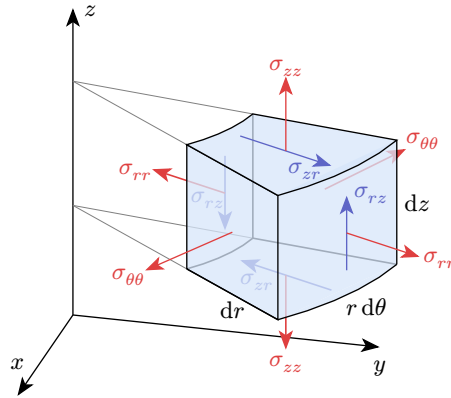


Figure 4.5: Stress components in an axisymmetric differential volume.

4.4.6 Orthotropic linear elasticity

An orthotropic material is a kind of anisotropic material, but with material axes aligned with three orthogonal material directions. In those material directions, linear elasticity is described by the compliance matrix

$$\mathbf{C} = \begin{pmatrix} \frac{1}{E_x} & -\frac{\nu_{xy}}{E_x} & -\frac{\nu_{xz}}{E_x} & 0 & 0 & 0 \\ -\frac{\nu_{yx}}{E_y} & \frac{1}{E_y} & -\frac{\nu_{yz}}{E_y} & 0 & 0 & 0 \\ -\frac{\nu_{zx}}{E_z} & -\frac{\nu_{zy}}{E_z} & \frac{1}{E_z} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{yz}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{xz}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{xy}} \end{pmatrix} \quad (4.53)$$

The reciprocal relations

$$\frac{\nu_{xy}}{E_x} = \frac{\nu_{yx}}{E_y}, \quad \frac{\nu_{xz}}{E_x} = \frac{\nu_{zx}}{E_z}, \quad \frac{\nu_{yz}}{E_y} = \frac{\nu_{zy}}{E_z} \quad (4.54)$$

ensure that $\mathbf{C}$ is symmetric. They arise from linear elasticity being derivable from a strain-energy density, which requires the constitutive matrix to satisfy the major symmetry relations.

Exercises 4.1:

1. In which cases the strain tensor is symmetric?
2. Why is the stress tensor always symmetric?
3. Why does the classical finite element formulation assume small deformations?
4. Give examples of typical structures corresponding to each stress state: plane stress, plane strain, and axisymmetric conditions. Provide at least three examples for each case.
5. A flat metallic plate with length 100 mm, width 40 mm, and thickness 5 mm is subjected to a uniform tensile load. When the axial force was measured as 20 kN the axial elongation was 0.1 mm and in-plane lateral contraction 0.008 mm. Determine the corresponding constitutive matrix $\mathbf{D}$ for a plane stress 2D linear elastic model. **A.** $\mathbf{D} = (1.04, 0.21, 0; 0.21, 1.04, 0; 0, 0, 0.42) \times 10^5$ MPa.
6. A cylindrical concrete sample with height 20 cm and diameter 10 cm is tested in uniaxial compression. At a point within the elastic range, the measured values are: axial stress 20 MPa, axial shortening 0.2 mm, and radial expansion 0.015 mm. Using these measurements, determine the elastic parameters E and ν , and construct the corresponding constitutive matrix $\mathbf{D}$ for a plane stress linear elastic model. **A.** $\mathbf{D} = (2.2, 0.66, 0; 0.66, 2.2, 0; 0, 0, 0.78) \times 10^4$ MPa.
7. A long thick-walled cylindrical pipe with inner radius $a = 50$ mm and outer radius $b = 100$ mm is subjected to a uniform internal pressure $p = 10$ MPa. The material is linear elastic and isotropic, with Young's modulus $E = 200$ GPa and Poisson's ratio $\nu = 0.2$. Assume that the pipe is sufficiently long so that the axial strain can be neglected, i.e. $\varepsilon_z = 0$. At a point located at $r = 75$ mm within the wall, the stress state is characterized by normal stresses only, with no shear components. Determine the stress components at this point and the corresponding strain components. **A.** $\sigma_{rr} = -2.6$ MPa, $\sigma_{\theta\theta} = 9.3$ MPa, $\sigma_z = 1.3$ MPa, $\varepsilon_{rr} = -2.4 \times 10^{-5}$ and $\varepsilon_{\theta\theta} = 4.8 \times 10^{-5}$.

4.5 Mechanical equilibrium

Mechanical equilibrium expresses the local balance of forces inside the continuum. Since the stress tensor is now available, we can write this balance in both traction and stress form.

Consider a small parallelepiped element of a continuum body with dimensions Δx , Δy , and Δz . The traction forces acting on its faces are shown in Figure 4.6.

According to Newton's second law of motion, the balance of forces on the element gives

$$\begin{aligned} \left(t_x + \frac{\partial t_x}{\partial x} \Delta x - t_x \right) \Delta y \Delta z + \left(t_y + \frac{\partial t_y}{\partial y} \Delta y - t_y \right) \Delta x \Delta z + \\ \left(t_z + \frac{\partial t_z}{\partial z} \Delta z - t_z \right) \Delta x \Delta y + b \Delta x \Delta y \Delta z = \rho a \Delta x \Delta y \Delta z \end{aligned} \quad (4.55)$$

where $\mathbf{b}$ is the body force vector (e.g. $\mathbf{b} = \gamma \hat{\mathbf{e}}_z$, with $\gamma = \rho g$ the material unit weight, ρ the material density and g the acceleration due to gravity) and $\mathbf{a}$ is the acceleration vector.

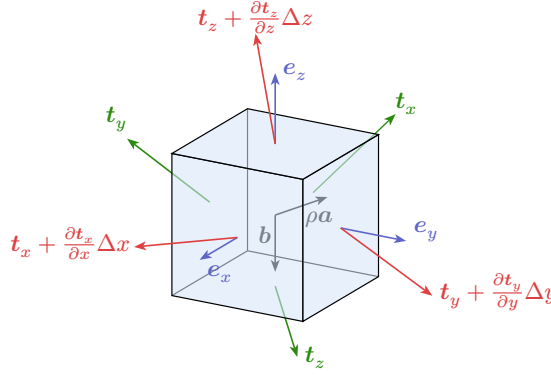


Figure 4.6: Traction forces in a continuum body.

Dividing by the element volume and taking the limit gives

$$\frac{\partial t_x}{\partial x} + \frac{\partial t_y}{\partial y} + \frac{\partial t_z}{\partial z} + \mathbf{b} = \rho \mathbf{a} \quad (4.56)$$

If the traction vectors are those acting on planes normal to the Cartesian axes, then

$$t_x = \boldsymbol{\sigma} \cdot \mathbf{e}_x, \quad t_y = \boldsymbol{\sigma} \cdot \mathbf{e}_y, \quad t_z = \boldsymbol{\sigma} \cdot \mathbf{e}_z \quad (4.57)$$

and the balance can be written in stress form as

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \rho \mathbf{a} \quad (4.58)$$

Eq. (4.58) is a set of three partial differential equations known as Cauchy's equations of motion. They apply to any continuum, whether solid or fluid.

In statics, $\mathbf{a} = \mathbf{0}$, so the equilibrium equation becomes

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0} \quad (4.59)$$

In Cartesian coordinates, the static equilibrium equations are

$$\begin{aligned}
 \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} + b_x &= 0 \\
 \frac{\partial \tau_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} + b_y &= 0 \\
 \frac{\partial \tau_{zx}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + b_z &= 0
 \end{aligned} \tag{4.60}$$

To complete the boundary-value problem, this equation must be accompanied by boundary conditions. Using Cauchy's formula for traction,

$$\mathbf{t} = \boldsymbol{\sigma} \cdot \hat{\mathbf{n}} \tag{4.61}$$

the static problem is written as

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0} \quad \text{in } \Omega \tag{4.62}$$

$$\mathbf{u} = \bar{\mathbf{u}} \quad \text{on } \Gamma_u \tag{4.63}$$

$$\boldsymbol{\sigma} \cdot \hat{\mathbf{n}} = \bar{\mathbf{t}} \quad \text{on } \Gamma_t \tag{4.64}$$

where $\hat{\mathbf{n}}$ is the outward unit normal vector.

If $\boldsymbol{\sigma}$ is considered as a vector instead of a tensor, the equilibrium equation can also be written in matrix form as

$$\bar{\nabla}^T \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0} \tag{4.65}$$

where

$$\bar{\nabla}^T = \begin{pmatrix} \frac{\partial}{\partial x} & 0 & 0 & 0 & \frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ 0 & \frac{\partial}{\partial y} & 0 & \frac{\partial}{\partial z} & 0 & \frac{\partial}{\partial x} \\ 0 & 0 & \frac{\partial}{\partial z} & \frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \end{pmatrix} \tag{4.66}$$

The strong form is well suited for stating the mechanics of the problem, but the finite element method is based on an integral, or weak, statement. To obtain it, consider a virtual displacement field $\delta \mathbf{u}$ that is kinematically admissible, that is,

$$\begin{aligned}
 \delta \mathbf{u} &= \mathbf{0} \quad \text{on } \Gamma_u \\
 \delta \mathbf{u} &\neq \mathbf{0} \quad \text{on } \Gamma_t
 \end{aligned} \tag{4.67}$$

Multiplying the strong form by $\delta \mathbf{u}$ and integrating over the body gives

$$\int_{\Omega} \delta \mathbf{u} \cdot (\nabla \cdot \boldsymbol{\sigma}) dV + \int_{\Omega} \delta \mathbf{u} \cdot \mathbf{b} dV = 0 \tag{4.68}$$

Now using the identity

$$\nabla \cdot (\delta \mathbf{u} \cdot \boldsymbol{\sigma}) = \delta \mathbf{u} \cdot (\nabla \cdot \boldsymbol{\sigma}) + \nabla \delta \mathbf{u} : \boldsymbol{\sigma} \tag{4.69}$$

and then applying the divergence theorem to transfer one spatial derivative from the stress field to the virtual displacement. This yields

$$\int_{\Omega} \nabla \delta \mathbf{u} : \boldsymbol{\sigma} dV = \int_{\Omega} \delta \mathbf{u} \cdot \mathbf{b} dV + \int_{\Gamma} \delta \mathbf{u} \cdot \mathbf{t} dS \tag{4.70}$$

Because the Cauchy stress tensor is symmetric, only the symmetric part of the virtual displacement gradient contributes to the internal virtual work. Defining the virtual strain tensor as

$$\delta \varepsilon = \frac{1}{2}(\nabla \delta \mathbf{u} + \nabla \delta \mathbf{u}^T) \quad (4.71)$$

we have

$$\nabla \delta \mathbf{u} : \boldsymbol{\sigma} = \delta \varepsilon : \boldsymbol{\sigma} \quad (4.72)$$

Also, since $\delta \mathbf{u} = 0$ on Γ_u , only the traction part of the boundary contributes to the surface integral. The weak form therefore becomes

$$\int_{\Omega} \delta \varepsilon : \boldsymbol{\sigma} dV = \int_{\Omega} \delta \mathbf{u} \cdot \mathbf{b} dV + \int_{\Gamma_t} \delta \mathbf{u} \cdot \bar{\mathbf{t}} dS \quad (4.73)$$

The left-hand side is the internal virtual work, while the terms on the right-hand side represent the external virtual work of body forces and boundary tractions.

At this point the role of the boundary conditions becomes transparent:

- Essential boundary conditions prescribe the primary variable $\mathbf{u}$ on Γ_u . They restrict the admissible displacement field and the virtual displacement field.
- Natural boundary conditions prescribe the traction vector on Γ_t . They appear explicitly in the right-hand side of the weak form.

This separation is one of the main reasons the weak form is so useful in finite elements. Essential conditions are enforced on the unknown field itself, whereas natural conditions enter as load terms. Concentrated forces are then applied directly at nodes, while distributed actions such as body forces and surface tractions are converted into equivalent nodal forces.

4.6 Finite element kinematics

Finite element kinematics links the nodal degrees of freedom of an element to the strain field inside it. The displacement field is approximated from the nodal values through the shape functions, and the strain field is then obtained by differentiating that interpolation. **Figure 4.7** illustrates this idea for an arbitrary element: once the relation between nodal displacements and strains is established, stresses follow from the constitutive law.

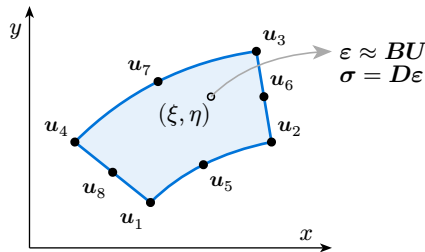


Figure 4.7: Strain at a point as a function of nodal displacements.

The starting point is the interpolation of the displacement field within an element:

$$\mathbf{u}(x, y, z) \approx \mathbf{N}(x, y, z)\mathbf{U}^{(e)} \quad (4.74)$$

where $\mathbf{U}^{(e)}$ collects the nodal displacement degrees of freedom of the element and $\mathbf{N}$ contains the shape functions arranged consistently with those degrees of freedom. At this stage, the continuum field is approximated by a finite set of nodal unknowns.

For a single displacement component, say u_x , the interpolation reads

$$u_{x(x,y,z)} = \sum_{i=1}^n N_{i(x,y,z)} u_{ix} \quad (4.75)$$

so that

$$\frac{\partial u_x}{\partial x} = \sum_{i=1}^n \frac{\partial N_i}{\partial x} u_{ix} \quad (4.76)$$

Hence, every strain component becomes a linear combination of the nodal displacements.

4.6.1 General strain-displacement relation

Using the small-strain relations introduced earlier, a two-dimensional continuum satisfies

$$\varepsilon_{xx} = \frac{\partial u_x}{\partial x}, \quad \varepsilon_{yy} = \frac{\partial u_y}{\partial y}, \quad \gamma_{xy} = \frac{\partial u_y}{\partial x} + \frac{\partial u_x}{\partial y} \quad (4.77)$$

Substituting the displacement interpolation into these expressions gives

$$\begin{aligned} \varepsilon_{xx} &= \frac{\partial N_1}{\partial x} u_{1x} + \frac{\partial N_2}{\partial x} u_{2x} + \dots + \frac{\partial N_n}{\partial x} u_{nx} \\ \varepsilon_{yy} &= \frac{\partial N_1}{\partial y} u_{1y} + \frac{\partial N_2}{\partial y} u_{2y} + \dots + \frac{\partial N_n}{\partial y} u_{ny} \\ \gamma_{xy} &= \frac{\partial N_1}{\partial y} u_{1x} + \frac{\partial N_1}{\partial x} u_{1y} + \frac{\partial N_2}{\partial y} u_{2x} + \frac{\partial N_2}{\partial x} u_{2y} + \dots + \frac{\partial N_n}{\partial y} u_{nx} + \frac{\partial N_n}{\partial x} u_{ny} \end{aligned} \quad (4.78)$$

Collecting these relations in matrix form gives

$$\begin{pmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{pmatrix} = \begin{pmatrix} \frac{\partial N_1}{\partial x} & 0 & \frac{\partial N_2}{\partial x} & 0 & \frac{\partial N_3}{\partial x} & 0 & \dots & \frac{\partial N_n}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \frac{\partial N_2}{\partial y} & 0 & \frac{\partial N_3}{\partial y} & \dots & 0 & \frac{\partial N_n}{\partial y} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial x} & \frac{\partial N_3}{\partial y} & \frac{\partial N_3}{\partial x} & \dots & \frac{\partial N_n}{\partial y} & \frac{\partial N_n}{\partial x} \end{pmatrix} \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ \vdots \\ u_{nx} \\ u_{ny} \end{pmatrix} \quad (4.79)$$

or, in compact form,

$$\varepsilon = \mathbf{B} \mathbf{U}^{(e)} \quad (4.80)$$

where $\mathbf{B}$ is the strain-displacement matrix. Since it contains spatial derivatives of the shape functions, it depends on the element geometry and, in general, on the evaluation point within the element.

As shown in Eq. (4.79), for 2D elements with two displacement components per node, each pair of columns of $\mathbf{B}$ is associated with one node. Thus,

$$\mathbf{B} = (\mathbf{B}_1 \quad \mathbf{B}_2 \quad \mathbf{B}_3 \quad \dots \quad \mathbf{B}_n) \quad (4.81)$$

For three-dimensional continua, the same construction extends to three displacement components per node and six strain components. Using engineering shear strains, we obtain

$$\begin{pmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{yz} \\ \gamma_{xz} \\ \gamma_{xy} \end{pmatrix} = \begin{pmatrix} \frac{\partial N_1}{\partial x} & 0 & 0 & \frac{\partial N_2}{\partial x} & 0 & 0 & \dots & \frac{\partial N_n}{\partial x} & 0 & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & 0 & \frac{\partial N_2}{\partial y} & 0 & \dots & 0 & \frac{\partial N_n}{\partial y} & 0 \\ 0 & 0 & \frac{\partial N_1}{\partial z} & 0 & 0 & \frac{\partial N_2}{\partial z} & \dots & 0 & 0 & \frac{\partial N_n}{\partial z} \\ 0 & \frac{\partial N_1}{\partial z} & \frac{\partial N_1}{\partial y} & 0 & \frac{\partial N_2}{\partial z} & \frac{\partial N_2}{\partial y} & \dots & 0 & \frac{\partial N_n}{\partial z} & \frac{\partial N_n}{\partial y} \\ \frac{\partial N_1}{\partial z} & 0 & \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial z} & 0 & \frac{\partial N_2}{\partial x} & \dots & \frac{\partial N_n}{\partial z} & 0 & \frac{\partial N_n}{\partial x} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & 0 & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial x} & 0 & \dots & \frac{\partial N_n}{\partial y} & \frac{\partial N_n}{\partial x} & 0 \end{pmatrix} \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{1z} \\ u_{2x} \\ u_{2y} \\ u_{2z} \\ \vdots \\ u_{nx} \\ u_{ny} \\ u_{nz} \end{pmatrix} \quad (4.82)$$

The derivatives appearing in $\mathbf{B}$ must be taken with respect to the physical Cartesian coordinates. However, in isoparametric elements the shape functions are usually defined in natural coordinates. For a given node i , the Cartesian derivatives of N_i are therefore obtained from the natural derivatives through the chain rule:

$$\begin{aligned} \frac{\partial N_i}{\partial x} &= \frac{\partial N_i}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial N_i}{\partial \eta} \frac{\partial \eta}{\partial x} + \frac{\partial N_i}{\partial \zeta} \frac{\partial \zeta}{\partial x} \\ \frac{\partial N_i}{\partial y} &= \frac{\partial N_i}{\partial \xi} \frac{\partial \xi}{\partial y} + \frac{\partial N_i}{\partial \eta} \frac{\partial \eta}{\partial y} + \frac{\partial N_i}{\partial \zeta} \frac{\partial \zeta}{\partial y} \\ \frac{\partial N_i}{\partial z} &= \frac{\partial N_i}{\partial \xi} \frac{\partial \xi}{\partial z} + \frac{\partial N_i}{\partial \eta} \frac{\partial \eta}{\partial z} + \frac{\partial N_i}{\partial \zeta} \frac{\partial \zeta}{\partial z} \end{aligned} \quad (4.83)$$

Using the inverse Jacobian matrix, these relations can be written more compactly as

$$\begin{pmatrix} \frac{\partial N_i}{\partial x} & \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial z} \end{pmatrix} = \begin{pmatrix} \frac{\partial N_i}{\partial \xi} & \frac{\partial N_i}{\partial \eta} & \frac{\partial N_i}{\partial \zeta} \end{pmatrix} \mathbf{J}^{-1} \quad (4.84)$$

Collecting the derivatives of all shape functions gives

$$\frac{\partial \mathbf{N}}{\partial \mathbf{x}} = \frac{\partial \mathbf{N}}{\partial \boldsymbol{\xi}} \mathbf{J}^{-1} \quad (4.85)$$

where

$$\frac{\partial \mathbf{N}}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial N_1}{\partial x} & \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial z} \\ \frac{\partial N_2}{\partial x} & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial z} \\ \vdots & \vdots & \vdots \\ \frac{\partial N_n}{\partial x} & \frac{\partial N_n}{\partial y} & \frac{\partial N_n}{\partial z} \end{pmatrix}, \quad \frac{\partial \mathbf{N}}{\partial \boldsymbol{\xi}} = \begin{pmatrix} \frac{\partial N_1}{\partial \xi} & \frac{\partial N_1}{\partial \eta} & \frac{\partial N_1}{\partial \zeta} \\ \frac{\partial N_2}{\partial \xi} & \frac{\partial N_2}{\partial \eta} & \frac{\partial N_2}{\partial \zeta} \\ \vdots & \vdots & \vdots \\ \frac{\partial N_n}{\partial \xi} & \frac{\partial N_n}{\partial \eta} & \frac{\partial N_n}{\partial \zeta} \end{pmatrix} \quad (4.86)$$

Note 4.1: In conventional isoparametric finite elements, the displacement interpolation is defined in natural coordinates, but the strains must be evaluated in the physical element coordinates. The Jacobian matrix provides the required transformation.

4.6.2 Matrix $\mathbf{B}$ for axisymmetric state

In axisymmetric analysis without torsion, the displacement field depends only on the radial and axial coordinates. Even so, the circumferential strain is not zero, because a radial displacement changes the circumference of the material ring. This is the main kinematic difference relative to a plane problem.

The strain components are

$$\varepsilon_{rr} = \frac{\partial u_r}{\partial r}, \quad \varepsilon_{zz} = \frac{\partial u_z}{\partial z}, \quad \varepsilon_{\theta\theta} = \frac{u_r}{r}, \quad \gamma_{rz} = \frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \quad (4.87)$$

The strain-displacement relation for an axisymmetric element is therefore

$$\begin{pmatrix} \varepsilon_{rr} \\ \varepsilon_{zz} \\ \varepsilon_{\theta\theta} \\ \gamma_{rz} \end{pmatrix} = \begin{pmatrix} \frac{\partial N_1}{\partial r} & 0 & \frac{\partial N_2}{\partial r} & 0 & \dots & \frac{\partial N_n}{\partial r} & 0 \\ 0 & \frac{\partial N_1}{\partial z} & 0 & \frac{\partial N_2}{\partial z} & \dots & 0 & \frac{\partial N_n}{\partial z} \\ \frac{N_1}{r} & 0 & \frac{N_2}{r} & 0 & \dots & \frac{N_n}{r} & 0 \\ \frac{\partial N_1}{\partial z} & \frac{\partial N_1}{\partial r} & \frac{\partial N_2}{\partial z} & \frac{\partial N_2}{\partial r} & \dots & \frac{\partial N_n}{\partial z} & \frac{\partial N_n}{\partial r} \end{pmatrix} \begin{pmatrix} u_{r1} \\ u_{z1} \\ u_{r2} \\ u_{z2} \\ \vdots \\ u_{rn} \\ u_{zn} \end{pmatrix} \quad (4.88)$$

Example 4.1: The strain in a 1D truss element is given by $\varepsilon = \frac{\partial u}{\partial x}$. Build the corresponding matrix $\mathbf{B}$ in terms of the nodal coordinates x_1 and x_2 .

Solution:

The displacement field in a 1D truss element is approximated as

$$u(\xi) = \sum_{i=1}^n N_i(\xi) u_i$$

Then, the axial strain is obtained by differentiation as

$$\varepsilon = \frac{\partial u}{\partial x} = \sum_{i=1}^n \frac{\partial N_i}{\partial x}(\xi) u_i$$

For two nodes:

$$\begin{aligned} \varepsilon &= \frac{\partial N_1}{\partial x} u_1 + \frac{\partial N_2}{\partial x} u_2 \\ &= \begin{pmatrix} \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial x} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \end{aligned}$$

where the $\mathbf{B}$ matrix is obtained as $\begin{pmatrix} \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial x} \end{pmatrix}$. The Cartesian derivatives can be computed using the chain rule:

$$\begin{aligned} \frac{\partial N_1}{\partial x} &= \frac{\partial N_1}{\partial \xi} \frac{\partial \xi}{\partial x} \\ \frac{\partial N_2}{\partial x} &= \frac{\partial N_2}{\partial \xi} \frac{\partial \xi}{\partial x} \end{aligned}$$

Note that the term $\frac{\partial \xi}{\partial x}$ represents the inverse of the Jacobian, and for a two-node element it is constant and equal to

$$J^{-1} = \frac{2}{x_2 - x_1}$$

For linear interpolation, the shape functions and corresponding derivatives are

$$\mathbf{N} = \begin{pmatrix} \frac{1}{2} - \frac{\xi}{2} \\ \frac{1}{2} + \frac{\xi}{2} \end{pmatrix}, \quad \frac{\partial \mathbf{N}}{\partial \xi} = \begin{pmatrix} -\frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

Then, the Cartesian derivatives are found as

$$\frac{\partial N_1}{\partial x} = -\frac{1}{x_2 - x_1}$$

$$\frac{\partial N_2}{\partial x} = \frac{1}{x_2 - x_1}$$

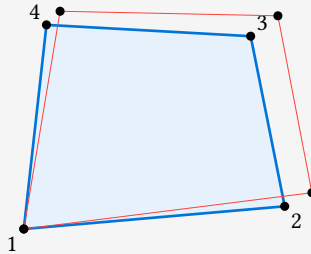
Finally, the $\mathbf{B}$ matrix for a two-node 1D truss element is

$$\mathbf{B} = \begin{pmatrix} -\frac{1}{x_2 - x_1} & \frac{1}{x_2 - x_1} \end{pmatrix}$$

Example 4.2: Consider the 4-node quadrilateral element with nodal coordinates and nodal displacements

$$\mathbf{C} = \begin{pmatrix} 0 & 0 \\ 2.3 & 0.2 \\ 2 & 1.7 \\ 0.2 & 1.8 \end{pmatrix} \text{ m} \quad \text{and} \quad \mathbf{U}^{(e)} = \begin{pmatrix} 0 \\ 0 \\ 0.04 \\ 0.02 \\ 0.04 \\ 0.03 \\ 0.02 \\ 0.02 \end{pmatrix} \text{ m}$$

shown in the figure. Using the small-strain approximation, compute the strain vectors at the natural coordinates $(\xi, \eta) = (0, 0)$ and $(\xi, \eta) = \left(-\sqrt{\frac{1}{3}}, \sqrt{\frac{1}{3}}\right)$.



Solution:

To compute the strain vector, we first evaluate the matrix $\mathbf{B}$ at the given natural coordinates. This requires the Cartesian derivatives of the shape functions, which are obtained from the natural derivatives through the Jacobian of the mapping.

For the given element, the natural derivatives of the shape functions are

$$\frac{\partial \mathbf{N}}{\partial \xi} = \begin{pmatrix} -\frac{1}{4}(1-\eta) & -\frac{1}{4}(1-\xi) \\ \frac{1}{4}(1-\eta) & -\frac{1}{4}(1+\xi) \\ \frac{1}{4}(1+\eta) & \frac{1}{4}(1+\xi) \\ -\frac{1}{4}(1+\eta) & \frac{1}{4}(1-\xi) \end{pmatrix} = \begin{pmatrix} -0.25 & -0.25 \\ 0.25 & -0.25 \\ 0.25 & 0.25 \\ -0.25 & 0.25 \end{pmatrix}$$

The Jacobian matrix is

$$\mathbf{J} = \mathbf{C}^T \frac{\partial \mathbf{N}}{\partial \xi} = \begin{pmatrix} 1.025 & -0.025 \\ 0.025 & 0.825 \end{pmatrix}$$

Then, the Cartesian derivatives of the shape functions are obtained as

$$\frac{\partial \mathbf{N}}{\partial \mathbf{x}} = \frac{\partial \mathbf{N}}{\partial \xi} \mathbf{J}^{-1} = \begin{pmatrix} -0.23634 & -0.31019 \\ 0.25111 & -0.29542 \\ 0.23634 & 0.31019 \\ -0.25111 & 0.29542 \end{pmatrix}$$

Filling the corresponding components of the $\mathbf{B}$ matrix gives

$$\mathbf{B} = \begin{pmatrix} -0.23634 & 0 & 0.25111 & 0 & 0.23634 & 0 & -0.25111 & 0 \\ 0 & -0.31019 & 0 & -0.29542 & 0 & 0.31019 & 0 & 0.29542 \\ -0.31019 & -0.23634 & -0.29542 & 0.25111 & 0.31019 & 0.23634 & 0.29542 & -0.25111 \end{pmatrix}$$

Finally, the strain vector at $(\xi, \eta) = (0, 0)$ is obtained as

$$\boldsymbol{\varepsilon} = \mathbf{B} \mathbf{U}^{(e)} = \begin{pmatrix} 0.014476 \\ 0.009306 \\ 0.013589 \end{pmatrix}$$

Repeating the same procedure at $(\xi, \eta) = \left(-\sqrt{\frac{1}{3}}, \sqrt{\frac{1}{3}}\right)$, the corresponding $\mathbf{B}$ matrix is

$$\mathbf{B} = \begin{pmatrix} -0.11949 & 0 & 0.10844 & 0 & 0.41576 & 0 & -0.40471 & 0 \\ 0 & -0.44766 & 0 & -0.12758 & 0 & 0.0991 & 0 & 0.47613 \\ -0.44766 & -0.11949 & -0.12758 & 0.10844 & 0.0991 & 0.41576 & 0.47613 & -0.40471 \end{pmatrix}$$

and the strain vector is

$$\boldsymbol{\varepsilon} = \mathbf{B} \mathbf{U}^{(e)} = \begin{pmatrix} 0.012874 \\ 0.009944 \\ 0.014931 \end{pmatrix}$$

Example 4.3: Write a code that computes the matrix $\mathbf{B}$ for a quadrilateral element at a given point in natural coordinates. Given a nodal-displacement vector, it should also compute the strain vector at the same point.

Solution:

The following code computes the matrix $\mathbf{B}$ and verifies it by evaluating the strain vector at $(\xi, \eta) = (0, 0)$ for a given set of element coordinates and nodal displacements.

```

using LinearAlgebra
Base.show(io::IO, M::Matrix) = show(io, "text/plain", M)

function quad4_derivs(xi, eta)
    return [ 0.25*(-1.0+eta)  0.25*(-1.0+xi)
            0.25*(+1.0-eta)  0.25*(-1.0-xi)
            0.25*(+1.0+eta)  0.25*(+1.0+xi)
            0.25*(-1.0-eta)  0.25*(+1.0-xi) ]
end

# Computes the B matrix for a quadrilateral element
function compute_B(C, xi, eta)
    # C: Matrix of nodal coordinates (nnodes x 2)
    # (ξ, η): Natural coordinates where B is evaluated

    nnodes = size(C, 1)
    ndofs = 2
    dN = quad4_derivs(xi, eta)
    J = C'*dN # Jacobian matrix
    dNdX = dN*inv(J) # Cartesian derivatives of shape functions
    B = zeros(3, ndofs*nnodes) # Initialize B matrix

    for i in 1:nnodes
        c = (i-1)*ndofs
        B[1, c+1] = dNdX[i,1]
        B[2, c+2] = dNdX[i,2]
        B[3, c+1] = dNdX[i,2]
        B[3, c+2] = dNdX[i,1]
    end
    return B
end

C = [ 0.0  0.0
      2.3  0.2
      2.0  1.7
      0.2  1.8 ] # Element coordinates
U = [ 0.0, 0.0, 0.04, 0.02, 0.04, 0.03, 0.02, 0.02 ] # Displacements vector

B = compute_B(C, 0.0, 0.0)
Strain = B*U

println("Matrix B at ξ=0 and η=0:")
println(B)
println("Strain vector at ξ=0 and η=0")
println(Strain)

```

Matrix B at ξ=0 and η=0:

3×8 Matrix{Float64}:

```

-0.236337  0.0      0.251108  0.0      0.236337  0.0      -0.251108  0.0
0.0        -0.310192  0.0      -0.295421  0.0      0.310192  0.0      0.295421
-0.310192  -0.236337  -0.295421  0.251108  0.310192  0.236337  0.295421  -0.251108

```

Strain vector at ξ=0 and η=0

```
[0.01447562776957164, 0.00930576070901034, 0.013589364844903988]
```

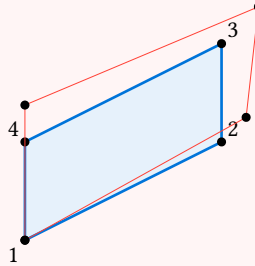
Exercises 4.2:

1. Explain the role of the strain-displacement matrix B in a finite element formulation.
2. Build the matrix B for a three-node bar element with nodal coordinates x_1 , x_2 , and x_3 .

- Find the matrix $\mathbf{B}$ for a three-node triangular element with nodal coordinates (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) . Why is matrix $\mathbf{B}$ constant along the element, and what does that imply?
- For the quadrilateral element shown below, and given its nodal coordinates and nodal-displacement vector

$$\mathbf{C} = \begin{pmatrix} 0 & 0 \\ 4 & 2 \\ 4 & 4 \\ 0 & 2 \end{pmatrix} \text{ m} \quad \text{and} \quad \mathbf{U}^{(e)} = \begin{pmatrix} 0 \\ 0 \\ 0.01 \\ 0.01 \\ 0.015 \\ 0.015 \\ 0 \\ 0.015 \end{pmatrix} \text{ m}$$

as shown in the figure. Approximate the strain vector at the natural coordinates $(\xi, \eta) = \left(-\sqrt{\frac{1}{3}}, \sqrt{\frac{1}{3}}\right)$.



$$\mathbf{A. B} = \begin{pmatrix} \frac{1}{8} & 0 & \frac{5}{24} & 0 & -\frac{1}{12} & 0 \\ 0 & -\frac{1}{8} & 0 & -\frac{1}{8} & 0 & \frac{1}{4} \\ -\frac{1}{8} & -\frac{1}{8} & -\frac{1}{8} & \frac{5}{24} & \frac{1}{4} & -\frac{1}{12} \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} 0.00322 \\ 0.00644 \\ -0.00216 \end{pmatrix}$$

- Find an expression for the matrix $\mathbf{B}$ of a three-node bar element in 2D as a function of the Jacobian and the local coordinate ξ . Consider $\varepsilon = \frac{\partial u}{\partial s}$, where s is the curvilinear coordinate along the bar path.

4.7 Element stiffness matrix

The element stiffness matrix measures how the internal nodal forces of an element change when its nodal displacements change. In the most general sense, it may be viewed as a tangent relation between infinitesimal displacement increments and the corresponding increments of the internal nodal-force vector:

$$\mathbf{K}^{(e)} d\mathbf{U}^{(e)} = d\mathbf{F}_{\text{int}}^{(e)} \quad (4.89)$$

Thus, each coefficient of $\mathbf{K}^{(e)}$ represents the sensitivity of one nodal force component to one nodal displacement component. For an element with n nodes and two degrees of freedom per node, this tangent matrix can be written symbolically as

$$\begin{pmatrix} \frac{\partial f_{1x}}{\partial u_{1x}} & \frac{\partial f_{1x}}{\partial u_{1y}} & \dots & \frac{\partial f_{1x}}{\partial u_{nx}} & \frac{\partial f_{1x}}{\partial u_{ny}} \\ \frac{\partial f_{1y}}{\partial u_{1x}} & \frac{\partial f_{1y}}{\partial u_{1y}} & \dots & \frac{\partial f_{1y}}{\partial u_{nx}} & \frac{\partial f_{1y}}{\partial u_{ny}} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{\partial f_{nx}}{\partial u_{1x}} & \frac{\partial f_{nx}}{\partial u_{1y}} & \dots & \frac{\partial f_{nx}}{\partial u_{nx}} & \frac{\partial f_{nx}}{\partial u_{ny}} \\ \frac{\partial f_{ny}}{\partial u_{1x}} & \frac{\partial f_{ny}}{\partial u_{1y}} & \dots & \frac{\partial f_{ny}}{\partial u_{nx}} & \frac{\partial f_{ny}}{\partial u_{ny}} \end{pmatrix} \begin{pmatrix} du_{1x} \\ du_{1y} \\ \vdots \\ du_{nx} \\ du_{ny} \end{pmatrix} = \begin{pmatrix} df_{1x} \\ df_{1y} \\ \vdots \\ df_{nx} \\ df_{ny} \end{pmatrix} \quad (4.90)$$

For a linear-elastic element, this relation is linear from the outset, so the internal nodal-force vector may be written as

$$\mathbf{F}_{\text{int}}^{(e)} = \mathbf{K}^{(e)} \mathbf{U}^{(e)} \quad (4.91)$$

In equilibrium, these internal nodal forces must balance the external nodal loads applied to the element. At element level, the stiffness matrix $\mathbf{K}^{(e)}$ is symmetric for standard linear elasticity, but it is singular because an isolated element still admits rigid-body motions. These motions produce no strain and therefore no strain energy, so they must belong to the null space of the element stiffness matrix.

The rank of $\mathbf{K}^{(e)}$ indicates how many independent displacement patterns are resisted by the element. After the rigid-body modes are excluded, every physically meaningful deformational mode should produce positive strain energy:

$$\mathbf{U}^T \mathbf{K}^{(e)} \mathbf{U} > 0 \quad (4.92)$$

for every nonzero non-rigid displacement vector $\mathbf{U}$ in the element displacement space. A unique global solution is obtained only after assembly and the imposition of sufficient essential boundary conditions, which remove the remaining global rigid-body motions and mechanisms.

4.7.1 Discrete equations from virtual work

To derive the element stiffness matrix, we first write the virtual-work statement in discrete form at the element level.

Starting from the weak form of equilibrium, the virtual work over an element domain $\Omega^{(e)}$ is

$$\int_{\Omega^{(e)}} \delta \boldsymbol{\varepsilon} \cdot \boldsymbol{\sigma} \, dV = \int_{\Omega^{(e)}} \delta \mathbf{u} \cdot \mathbf{b} \, dV + \int_{\Gamma_t^{(e)}} \delta \mathbf{u} \cdot \bar{\mathbf{t}} \, dS \quad (4.93)$$

where $\Gamma_t^{(e)}$ is the portion of the element boundary on which tractions are prescribed.

In the finite element approximation, the virtual displacement field inside the element is interpolated from the virtual nodal displacements:

$$\delta \mathbf{u} = \mathbf{N} \delta \mathbf{U}^{(e)} \quad (4.94)$$

Since strains are obtained by differentiating the displacement interpolation,

$$\delta \boldsymbol{\varepsilon} = \mathbf{B} \delta \mathbf{U}^{(e)} \quad (4.95)$$

Substituting these expressions into the virtual-work equation gives

$$(\delta \mathbf{U}^{(e)})^T \left(\int_{\Omega^{(e)}} \mathbf{B}^T \boldsymbol{\sigma} \, dV - \int_{\Omega^{(e)}} \mathbf{N}^T \mathbf{b} \, dV - \int_{\Gamma_t^{(e)}} \mathbf{N}^T \bar{\mathbf{t}} \, dS \right) = 0 \quad (4.96)$$

Because $\delta \mathbf{U}^{(e)}$ is arbitrary, the term inside parentheses must vanish:

$$\int_{\Omega^{(e)}} \mathbf{B}^T \boldsymbol{\sigma} \, dV = \int_{\Omega^{(e)}} \mathbf{N}^T \mathbf{b} \, dV + \int_{\Gamma_t^{(e)}} \mathbf{N}^T \bar{\mathbf{t}} \, dS \quad (4.97)$$

This is the discrete equilibrium equation for the element. The left-hand side is the internal nodal-force vector, and the right-hand side is the equivalent nodal-load vector due to body forces and prescribed tractions.

For small-strain linear elasticity, we have

$$\boldsymbol{\sigma} = \mathbf{D} \mathbf{B} \mathbf{U}^{(e)} \quad (4.98)$$

Hence,

$$\left(\int_{\Omega^{(e)}} \mathbf{B}^T \mathbf{D} \mathbf{B} \, dV \right) \mathbf{U}^{(e)} = \int_{\Omega^{(e)}} \mathbf{N}^T \mathbf{b} \, dV + \int_{\Gamma_t^{(e)}} \mathbf{N}^T \bar{\mathbf{t}} \, dS \quad (4.99)$$

which is usually written in condensed form as

$$\mathbf{K}^{(e)} \mathbf{U}^{(e)} = \mathbf{F}^{(e)} \quad (4.100)$$

with

$$\mathbf{K}^{(e)} = \int_{\Omega^{(e)}} \mathbf{B}^T \mathbf{D} \mathbf{B} \, dV \quad (4.101)$$

and

$$\mathbf{F}^{(e)} = \int_{\Omega^{(e)}} \mathbf{N}^T \mathbf{b} \, dV + \int_{\Gamma_t^{(e)}} \mathbf{N}^T \bar{\mathbf{t}} \, dS \quad (4.102)$$

Each element contributes its stiffness matrix $\mathbf{K}^{(e)}$ and load vector $\mathbf{F}^{(e)}$ to the global system. After assembly over all elements, we obtain

$$\mathbf{K} \mathbf{U} = \mathbf{F} \quad (4.103)$$

4.7.2 Element stiffness in common settings

Although the element stiffness matrix always has the form

$$\mathbf{K}^{(e)} = \int_{\Omega^{(e)}} \mathbf{B}^T \mathbf{D} \mathbf{B} \, dV \quad (4.104)$$

the volume differential dV depends on the dimensional setting and on the kinematic assumptions adopted for the element.

For a one-dimensional bar or truss element, the physical volume is the cross-sectional area times the element length, so

$$dV = A \, dx = A J \, d\xi \quad (4.105)$$

Hence,

$$\mathbf{K}_{1D} = A \int \mathbf{B}^T \mathbf{E} \mathbf{B} J \, d\xi \approx A \sum_{i=1}^{n_{ip}} \mathbf{B}_i^T \mathbf{E} \mathbf{B}_i J_i w_i \quad (4.106)$$

where E is the one-dimensional constitutive modulus and the subscript i indicates evaluation at the integration point.

For plane stress or plane strain, the element represents a slice of constant thickness h , so

$$dV = h dA = hJ d\xi d\eta \quad (4.107)$$

Therefore,

$$K_{2D} = h \int B^T DBJ d\xi d\eta \approx h \sum_{i=1}^{n_{ip}} B_i^T DB_i J_i w_i \quad (4.108)$$

In plane stress, h is the actual thickness of the body; in plane strain it is common to take a unit thickness, $h = 1$.

For axisymmetric elements, each differential area $dA = dr dz$ generates a ring of circumference $2\pi r$. Thus,

$$dV = 2\pi r dA = 2\pi rJ d\xi d\eta \quad (4.109)$$

and the stiffness matrix becomes

$$K_{\text{axisym}} = 2\pi \int B^T DBrJ d\xi d\eta \approx 2\pi \sum_{i=1}^{n_{ip}} B_i^T DB_i r_i J_i w_i \quad (4.110)$$

where r_i is the radial coordinate at the integration point.

In all these cases, $\mathbf{K}^{(e)}$ is symmetric for standard linear elasticity and positive semidefinite before essential boundary conditions are imposed.

4.7.3 Appropriate order of quadrature

The stiffness matrix is usually computed by numerical quadrature, because the integrand $B^T DBJ$ is rarely simple enough to integrate in closed form for a general element geometry. As previously introduced, Gauss quadrature can integrate exactly polynomials up to degree $2n - 1$ with n points.

In the numerical evaluation of the stiffness matrix, the two most common choices are **full** and **reduced** integration. The choice of quadrature rule has important implications for the accuracy, stability, and robustness of the element formulation.

The rule usually called **full integration** is the lowest-order rule that integrates the stiffness integrand exactly for the corresponding undistorted affine element with constant Jacobian determinant and constant material properties. **Figure 4.8** and **Figure 4.9** show the polynomial terms in the stiffness integrand for regular quadrilateral elements. Similarly, **Figure 4.10** shows the terms for a regular hexahedral element. Those terms are the ones that must be integrated exactly by the full-integration rule for the corresponding element type.

For distorted isoparametric elements, once the mapping becomes non-affine, J and the entries of B do not remain polynomial, so the stiffness integrand is no longer a polynomial and the standard full-integration rule is no longer exact. Thus, the term **full integration** should be understood as the conventional rule for the element, not as a guarantee of exact integration for every distorted element.

Using fewer integration points than the full-integration rule gives **reduced integration**. Reduced integration lowers the computational cost and can help relieve **locking** in some displacement-based elements, but it also changes which parts of the strain field are sampled by the stiffness integral. If an insufficient quadrature rule fails to sample the strain associated with one or more deformational displacement patterns, those patterns may receive zero stiffness. Thus, reduced integration may reduce

the **rank** of the element stiffness matrix and activate spurious zero-energy modes. The quadrature choice therefore affects not only accuracy, but also the stability and robustness of the element formulation.

Note 4.2: The rank of a matrix is the dimension of its row or column space, i.e., the number of linearly independent rows or columns. A full-rank square matrix has a rank equal to the number of its rows (or columns), while a rank-deficient matrix has a rank less than that number.

In finite element analysis, the rank of the (reduced) global stiffness matrix indicates whether the system is properly constrained.

- A full-rank stiffness matrix (after applying boundary conditions) ensures a unique solution for the nodal displacements.
- A rank-deficient matrix indicates the presence of rigid body modes or mechanisms, meaning that the structure is not sufficiently constrained.

Rank-deficient finite elements may also introduce spurious (non-physical) zero- or near-zero-energy modes.

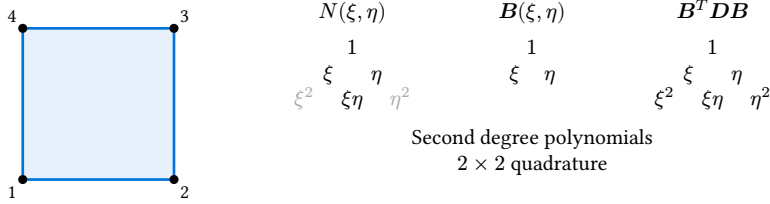


Figure 4.8: Polynomial terms in the integrand of the stiffness matrix for regular 4-node quadrilateral elements.

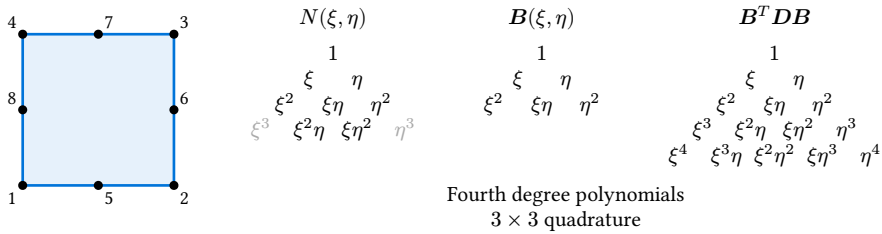


Figure 4.9: Polynomial terms in the integrand of the stiffness matrix for regular 8-node quadrilateral elements.

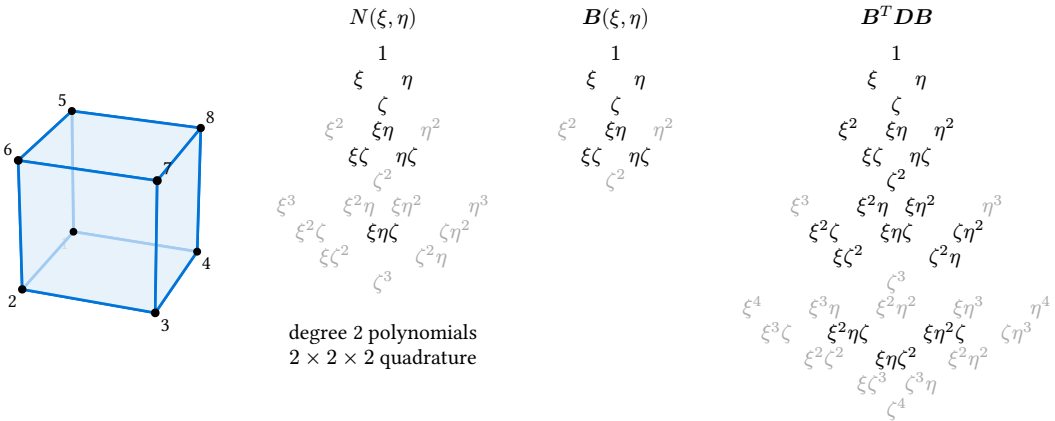


Figure 4.10: Polynomial terms in the integrand of the stiffness matrix for regular 8-node hexahedral elements.

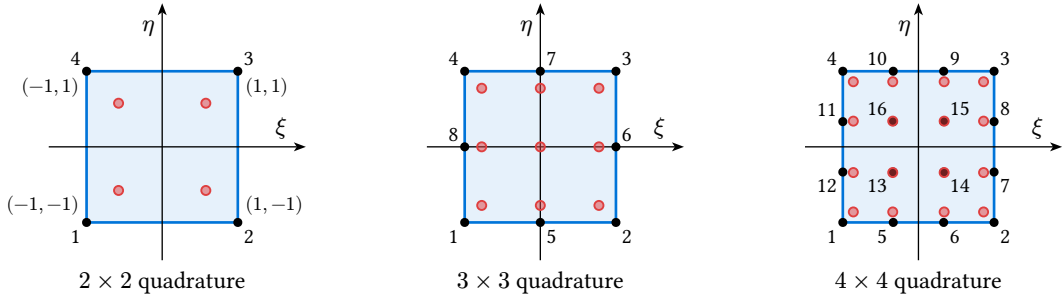


Figure 4.11: Full integration quadrature for quadrilateral elements.

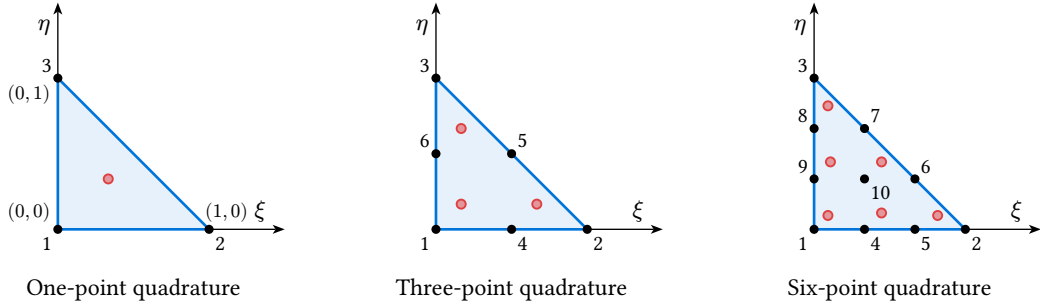


Figure 4.12: Full integration quadrature for triangular elements

4.8 Equivalent nodal loads

This section covers the conversion of distributed external actions such as body forces and surface tractions into **equivalent nodal loads** before assembly.

4.8.1 Body forces

Body forces act throughout the element volume. Self-weight is the most common example. Figure 4.13 shows the equivalent nodal forces corresponding to vertical body forces distributed over a quadrilateral element.

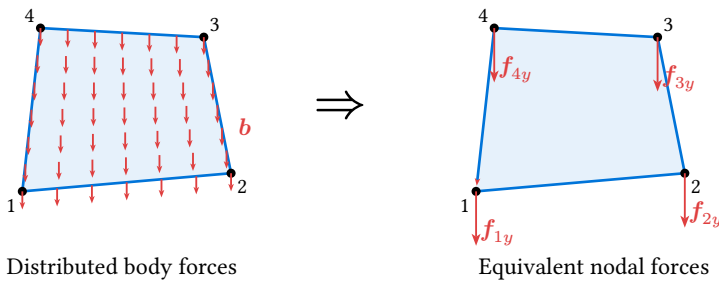


Figure 4.13: Equivalent nodal forces corresponding to vertical body forces.

To derive the equivalent nodal forces associated with body forces, we again use the principle of virtual work together with the interpolation relation between the displacement field inside the element and the nodal displacements:

$$\mathbf{u} = \mathbf{N}\mathbf{U} \quad (4.111)$$

where, in two dimensions, the matrix $\mathbf{N}$ is defined as

$$\mathbf{N} = \begin{pmatrix} N_1 & 0 & N_2 & 0 & \dots & N_n & 0 \\ 0 & N_1 & 0 & N_2 & \dots & 0 & N_n \end{pmatrix} \quad (4.112)$$

and in three dimensions as

$$\mathbf{N} = \begin{pmatrix} N_1 & 0 & 0 & N_2 & 0 & 0 & \dots & N_n & 0 & 0 \\ 0 & N_1 & 0 & 0 & N_2 & 0 & \dots & 0 & N_n & 0 \\ 0 & 0 & N_1 & 0 & 0 & N_2 & \dots & 0 & 0 & N_n \end{pmatrix} \quad (4.113)$$

By equating the virtual work of the body forces and that of the equivalent nodal forces,

$$\delta \mathbf{U} \cdot \mathbf{F}_b = \int_V \delta \mathbf{u} \cdot \mathbf{b} \, dV \quad (4.114)$$

where $\mathbf{b} = \begin{pmatrix} b_x \\ b_y \end{pmatrix}$ in two dimensions and $\mathbf{b} = \begin{pmatrix} b_x \\ b_y \\ b_z \end{pmatrix}$ in three dimensions is the body force vector.

Using the interpolation relation $\delta \mathbf{u} = \mathbf{N} \delta \mathbf{U}$,

$$\delta \mathbf{U}^T \mathbf{F}_b = \int_V \delta \mathbf{U}^T \mathbf{N}^T \mathbf{b} \, dV \quad (4.115)$$

and therefore

$$\mathbf{F}_b = \int_V \mathbf{N}^T \mathbf{b} \, dV \quad (4.116)$$

Using numerical integration,

$$\mathbf{F}_b \approx \sum_{i=1}^m \mathbf{N}_i^T \mathbf{b}_i J_i w_i \quad (4.117)$$

For two-dimensional elements of thickness h ,

$$\mathbf{F}_b \approx h \sum_{i=1}^m \mathbf{N}_i^T \mathbf{b}_i J_i w_i \quad (4.118)$$

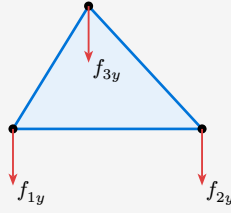
If the body force acts only in the vertical direction, the corresponding equivalent nodal-force vector may be written as

$$\mathbf{F}_v = \int_V \mathbf{N}^T b_v \, dV \approx \sum_{i=1}^m \mathbf{N}_i^T b_{v,i} J_i w_i \quad (4.119)$$

where b_v is the vertical component of the body force. Here $\mathbf{N}$ is the row matrix

$$\mathbf{N} = (N_1 \quad N_2 \quad N_3 \quad \dots \quad N_n) \quad (4.120)$$

Example 4.4: Compute analytically the equivalent nodal forces corresponding to a vertical body force b_v distributed over a three-node triangular element with thickness h and unit weight γ .



Solution:

The equivalent vertical nodal-force vector is

$$\mathbf{F}_v = h \int_{A'} \mathbf{N}^T b_v J d\eta d\xi$$

For a linear triangular element, the shape functions in parent coordinates (ξ, η) are

$$N_1 = 1 - \xi - \eta, \quad N_2 = \xi, \quad N_3 = \eta$$

Hence,

$$\mathbf{N} = (N_1 \quad N_2 \quad N_3) = (1 - \xi - \eta \quad \xi \quad \eta)$$

For three-node triangular elements, the Jacobian determinant is constant and equal to twice the element area, thus $J = 2A$. Also, the scalar body force is $b_v = -\gamma$. Then, the equivalent nodal-force vector is rewritten as

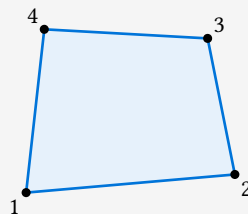
$$\mathbf{F}_v = -2Ah\gamma \int_0^1 \int_0^{1-\xi} \begin{pmatrix} 1 - \xi - \eta \\ \xi \\ \eta \end{pmatrix} d\eta d\xi$$

Evaluating the integral we get

$$\mathbf{F}_v = -2Ah\gamma \begin{pmatrix} \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \end{pmatrix} = -Ah\gamma \begin{pmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix}$$

Example 4.5: Consider a four-node quadrilateral element with coordinates

$$\mathbf{C} = \begin{pmatrix} 0 & 0 \\ 2.3 & 0.2 \\ 2 & 1.7 \\ 0.2 & 1.8 \end{pmatrix} \text{ m}$$



Compute the equivalent vertical body-force vector for thickness $h = 0.1$ and unit weight $\gamma = -25 \text{ kN/m}^3$.

Solution:

For integration point 1 with $(\xi, \eta) = (-0.577350269189626, -0.577350269189626)$:

$$\mathbf{N}_1 = (0.62201, 0.16667, 0.04466, 0.16667)$$

$$J_1 = 0.94945$$

$$\mathbf{F}_{v,1} = h\mathbf{N}_1^T \gamma J_1 w_i = \begin{pmatrix} -1.47642 \\ -0.39561 \\ -0.10601 \\ -0.39561 \end{pmatrix}$$

For integration point 2 with $(\xi, \eta) = (0.577350269189626, -0.577350269189626)$:

$$\mathbf{N}_2 = (0.16667, 0.62201, 0.16667, 0.04466)$$

$$J_2 = 0.86429$$

$$\mathbf{F}_{v,2} = h\mathbf{N}_2^T \gamma J_2 w_i = \begin{pmatrix} -0.36013 \\ -1.344 \\ -0.36013 \\ -0.0965 \end{pmatrix}$$

For integration point 3 with $(\xi, \eta) = (-0.577350269189626, 0.577350269189626)$:

$$\mathbf{N}_3 = (0.16667, 0.04466, 0.16667, 0.62201)$$

$$J_3 = 0.82821$$

$$\mathbf{F}_{v,3} = h\mathbf{N}_3^T \gamma J_3 w_i = \begin{pmatrix} -0.34509 \\ -0.09247 \\ -0.34509 \\ -1.28788 \end{pmatrix}$$

For integration point 4 with $(\xi, \eta) = (0.577350269189626, 0.577350269189626)$:

$$\mathbf{N}_4 = (0.04466, 0.16667, 0.62201, 0.16667)$$

$$J_4 = 0.74305$$

$$\mathbf{F}_{v,4} = h\mathbf{N}_4^T \gamma J_4 w_i = \begin{pmatrix} -0.08296 \\ -0.30961 \\ -1.15546 \\ -0.30961 \end{pmatrix}$$

Summing the four contributions gives

$$\mathbf{F}_v = \sum_{i=1}^4 \mathbf{F}_{v,i} = \begin{pmatrix} -2.2646 \\ -2.14169 \\ -1.96669 \\ -2.0896 \end{pmatrix} \text{ kN}$$

Example 4.6: Write a code that, given the geometry of a 4-node quadrilateral element, computes the equivalent vertical body-force vector due to a uniform unit weight.

Solution:

The following code evaluates the shape functions and the Jacobian at each integration point and then uses numerical integration to compute the equivalent vertical body-force vector. The code assumes the existence of functions `quad4_shape` and `quad4_derivs` that compute the shape functions and their derivatives for a 4-node quadrilateral element, respectively.

```
using LinearAlgebra

# Computes the equivalent nodal forces vector for a quadrilateral element
# due to vertical body forces
function body_forces(C, thickness, gamma, quadrature)
    npts = size(quadrature, 1)
    nnodes = size(C, 1)
    F = zeros(nnodes)

    for i in 1:npts
        xi = quadrature[i, 1]
        eta = quadrature[i, 2]
        w = quadrature[i, 3]
        N = quad4_shape(xi, eta)'
        dN = quad4_derivs(xi, eta)
        J = C'*dN
        F = F + N'*gamma*thickness*det(J)*w
    end

    return F
end

# Quadrature points and weights for a quadrilateral element
quadrature = [
    -0.577350269189626 -0.577350269189626 1.0
    0.577350269189626 -0.577350269189626 1.0
    -0.577350269189626 0.577350269189626 1.0
    0.577350269189626 0.577350269189626 1.0 ]

# Element coordinates
C = [ 0.0 0.0
      2.3 0.2
      2.0 1.7
      0.2 1.8 ]

thickness = 0.1
gamma = -25.0

F = body_forces(C, thickness, gamma, quadrature)

println("Vector with vertical body forces:")
printarray(F)
```

```
Vector with vertical body forces:
4-element Vector{Float64}:
-2.264583333333334
-2.1416666666666666
-1.9666666666666663
-2.0895833333333336
```

4.8.2 Surface tractions

Surface tractions act on element edges in two dimensions and on element faces in three dimensions.

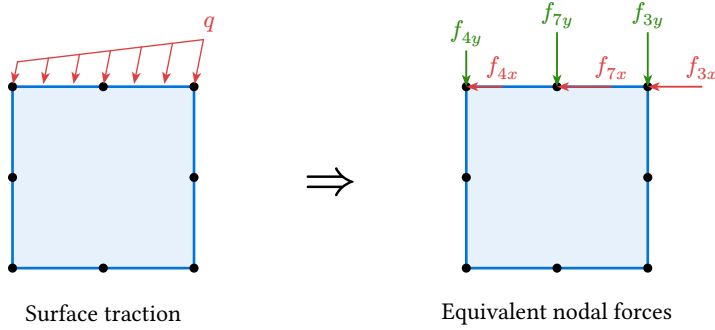


Figure 4.14: Equivalent nodal forces corresponding to surface tractions.

As with body forces, they are converted into equivalent nodal forces through virtual work:

$$\int_A \delta \mathbf{u} \cdot \mathbf{t} \, dA = \delta \mathbf{U} \cdot \mathbf{F}_t \quad (4.121)$$

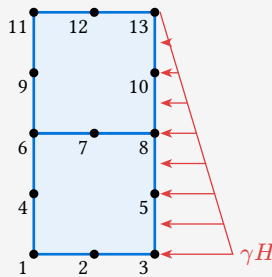
and therefore

$$\mathbf{F}_t = \int_A \mathbf{N}^T \mathbf{t} \, dA \quad (4.122)$$

Using numerical integration,

$$\mathbf{F}_t \approx \sum_{i=1}^m \mathbf{N}_i^T \mathbf{t}_i J_i w_i \quad (4.123)$$

Example 4.7: The mesh shown is part of a wall subjected to hydrostatic pressure on its right boundary. Find the equivalent nodal forces associated with this pressure.



Consider the total fluid height to be $H = 2$ m. Use a unit out-of-plane thickness $b = 1$ m, and take the unit weight of the fluid as $\gamma = 10$ kN/m³.

Solution:

The pressure depends on the y coordinate, thus

$$p = \gamma(H - y)$$

The corresponding surface traction acts in the negative x direction along the out-of-plane thickness, so the traction vector applied to the two-dimensional model is

$$\mathbf{t}(y) = \begin{pmatrix} t_x \\ t_y \end{pmatrix} = \begin{pmatrix} -\gamma(H - y) \\ 0 \end{pmatrix}$$

The equivalent nodal forces are

$$\mathbf{F}_t = \int_A \mathbf{N}^T \mathbf{t} \, dA$$

where the area differential can be expressed in parametric coordinates as $dA = Jb \, d\xi$.

For each loaded quadratic edge, use the local coordinate $\xi \in [-1, 1]$. Following the usual three-node line convention, the nodes are ordered as lower corner, upper corner, and midside node, with $\xi = -1$, $\xi = 1$, and $\xi = 0$, respectively. The one-dimensional quadratic shape functions are therefore

$$N_1 = \frac{1}{2}\xi(\xi - 1), \quad N_2 = \frac{1}{2}\xi(\xi + 1), \quad N_3 = 1 - \xi^2$$

The edge length is 1 m for each element, therefore $J = \frac{dy}{d\xi} = \frac{1}{2}$ m.

For the lower edge, whose global nodes are (3, 8, 5), the coordinate mapping is

$$y(\xi) = \frac{1 + \xi}{2}$$

and hence

$$t_x(\xi) = -\gamma(H - y(\xi)) = -15 + 5\xi \quad \text{kN/m}$$

Thus, in the nodal order (3, 8, 5),

$$\mathbf{f}_{3,8,5} = \int_{-1}^1 \begin{pmatrix} N_1 t_x \\ 0 \\ N_2 t_x \\ 0 \\ N_3 t_x \\ 0 \end{pmatrix} Jb \, d\xi = \begin{pmatrix} -\frac{10}{3} \\ 0 \\ -\frac{5}{3} \\ 0 \\ -10 \\ 0 \end{pmatrix} \text{ kN}$$

For the upper edge, whose global nodes are (8, 13, 10),

$$y(\xi) = 1 + \frac{1 + \xi}{2}$$

and therefore

$$t_{x(\xi)} = -\gamma(H - y(\xi))b = -5 + 5\xi \quad \text{kN/m}$$

Then, in the nodal order (8, 13, 10),

$$\mathbf{f}_{8,13,10} = \begin{pmatrix} -\frac{5}{3} \\ 0 \\ 0 \\ 0 \\ -\frac{10}{3} \\ 0 \end{pmatrix} \text{ kN}$$

Assembling the two edge contributions gives the loaded-boundary nodal forces

$$\mathbf{F}_\Gamma = \begin{pmatrix} f_{3x} \\ f_{3y} \\ f_{5x} \\ f_{5y} \\ f_{8x} \\ f_{8y} \\ f_{10x} \\ f_{10y} \\ f_{13x} \\ f_{13y} \end{pmatrix} = \begin{pmatrix} -\frac{10}{3} \\ 0 \\ -10 \\ 0 \\ -\frac{10}{3} \\ 0 \\ -\frac{10}{3} \\ 0 \\ 0 \\ 0 \end{pmatrix} \text{ kN}$$

The sum of the horizontal components is -20 kN, which matches the resultant of the triangular hydrostatic pressure distribution.

4.8.3 Normal surface traction

If a normal traction t_n is applied over a surface, then

$$\mathbf{t} = t_n \hat{\mathbf{n}} \quad (4.124)$$

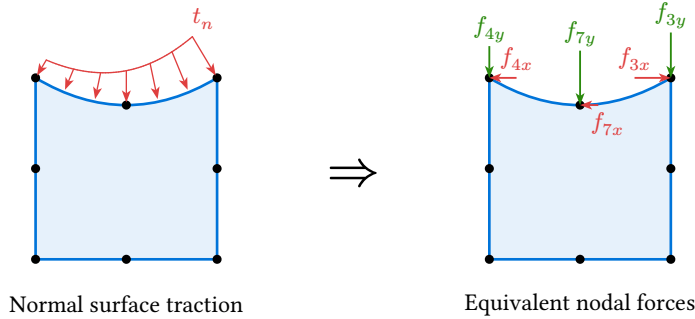


Figure 4.15: Equivalent nodal forces corresponding to surface tractions.

In two dimensions, the Jacobian of an edge gives a tangent vector, so a normal vector may be computed as

$$\mathbf{n} = \begin{pmatrix} J_2 \\ -J_1 \end{pmatrix} \quad (4.125)$$

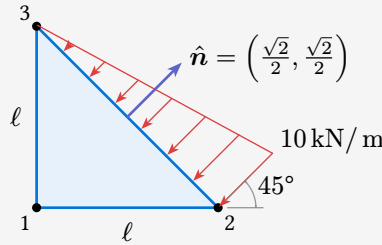
For a surface in three dimensions,

$$\mathbf{n} = \begin{pmatrix} J_{11} \\ J_{21} \\ J_{31} \end{pmatrix} \times \begin{pmatrix} J_{12} \\ J_{22} \\ J_{32} \end{pmatrix} \quad (4.126)$$

The equivalent nodal forces are then

$$\begin{aligned} \mathbf{F}_n &= \int_A \mathbf{N}^T t_n \hat{\mathbf{n}} dA \\ &\approx \sum_{i=1}^m \mathbf{N}_i^T t_{n,i} \hat{\mathbf{n}}_i J_i w_i \end{aligned} \quad (4.127)$$

Example 4.8: Find the equivalent nodal forces corresponding to a normal line load t_n applied along the edge connecting nodes 2 and 3 of the three-node triangular element shown in the figure. Assume the load varies linearly along the edge, from 10 kN/m at node 2 to zero at node 3. Also, consider the length $\ell = 1$ m.



Solution:

The equivalent nodal-force vector along the edge where the normal line load is applied is given by

$$\mathbf{F}_n = \int_{-1}^1 \mathbf{N}^T t_n \hat{\mathbf{n}} J d\xi$$

For the two-node edge where the normal line load is applied, take the local coordinate such that $\xi = -1$ at node 2 and $\xi = 1$ at node 3. Then,

$$N_1 = \frac{1}{2} - \frac{\xi}{2}, \quad N_2 = \frac{1}{2} + \frac{\xi}{2}$$

where N_1 and N_2 are the shape functions of a 2-node line element. Writing the equivalent nodal-force vector in the global order (2, 3), the shape functions matrix for the edge is

$$\mathbf{N} = \begin{pmatrix} N_1 & 0 & N_2 & 0 \\ 0 & N_1 & 0 & N_2 \end{pmatrix}$$

The Jacobian determinant is $J = \frac{\sqrt{2}\ell}{2}$. Since $\ell = 1$ m, then $J = \frac{\sqrt{2}}{2}$ m.

Since the load acts inward, its signed intensity varies linearly from -10 kN/m at node 2 to zero at node 3. Therefore,

$$t_n(\xi) = -10N_1(\xi) + 0N_2(\xi) = -10\left(\frac{1}{2} - \frac{\xi}{2}\right)$$

The normal vector remains constant along the edge, as shown in the figure. Then, the equivalent nodal-force vector can be computed as

$$\begin{aligned} \mathbf{F}_n &= \int_{-1}^1 \begin{pmatrix} N_1 & 0 \\ 0 & N_1 \\ N_2 & 0 \\ 0 & N_2 \end{pmatrix} (-10N_1) \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix} J d\xi \\ &= -5 \int_{-1}^1 \begin{pmatrix} N_1^2 \\ N_1^2 \\ N_1 N_2 \\ N_1 N_2 \end{pmatrix} d\xi \end{aligned}$$

This integral can be evaluated analytically or numerically. The result is

$$\mathbf{F}_n = 5 \begin{pmatrix} -\frac{2}{3} \\ -\frac{2}{3} \\ -\frac{1}{3} \\ -\frac{1}{3} \end{pmatrix} = \begin{pmatrix} -\frac{10}{3} \\ -\frac{10}{3} \\ -\frac{5}{3} \\ -\frac{5}{3} \end{pmatrix} \text{ kN}$$

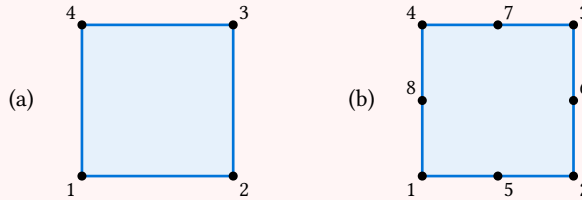
or, numerically,

$$\mathbf{F}_n \approx \begin{pmatrix} -3.333 \\ -3.333 \\ -1.667 \\ -1.667 \end{pmatrix} \text{ kN}$$

Therefore, these are the equivalent nodal forces at nodes 2 and 3.

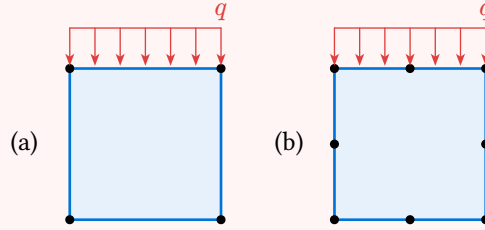
Exercises 4.3:

1. What is an equivalent nodal-load vector, and why is it needed in the finite element method?
2. Explain the difference between an externally applied nodal force and a consistent equivalent nodal load obtained from a distributed action.
3. How do body forces and surface tractions enter the weak form and the element load vector?
4. For the elements below, determine the nodal forces equivalent to body forces. Consider the element area A , thickness t , and unit weight γ .



A. $\mathbf{F}_{(a)} = \frac{\gamma A t}{4} (-1, -1, -1, -1)$, $\mathbf{F}_{(b)} = \frac{\gamma A t}{4} (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{4}{3}, -\frac{4}{3}, -\frac{4}{3}, -\frac{4}{3})$

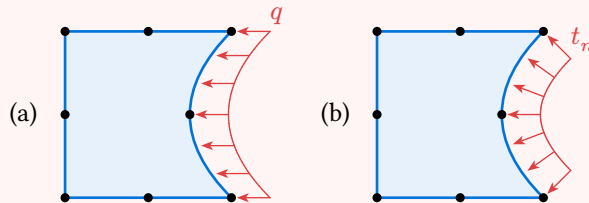
5. Determine the equivalent nodal forces corresponding to the distributed load per unit length q .



A. $\mathbf{F}_{(a)} = \frac{qL}{2}(-1, -1)$ kN, $\mathbf{F}_{(b)} = \frac{qL}{6}(-1, -1, -4)$ kN

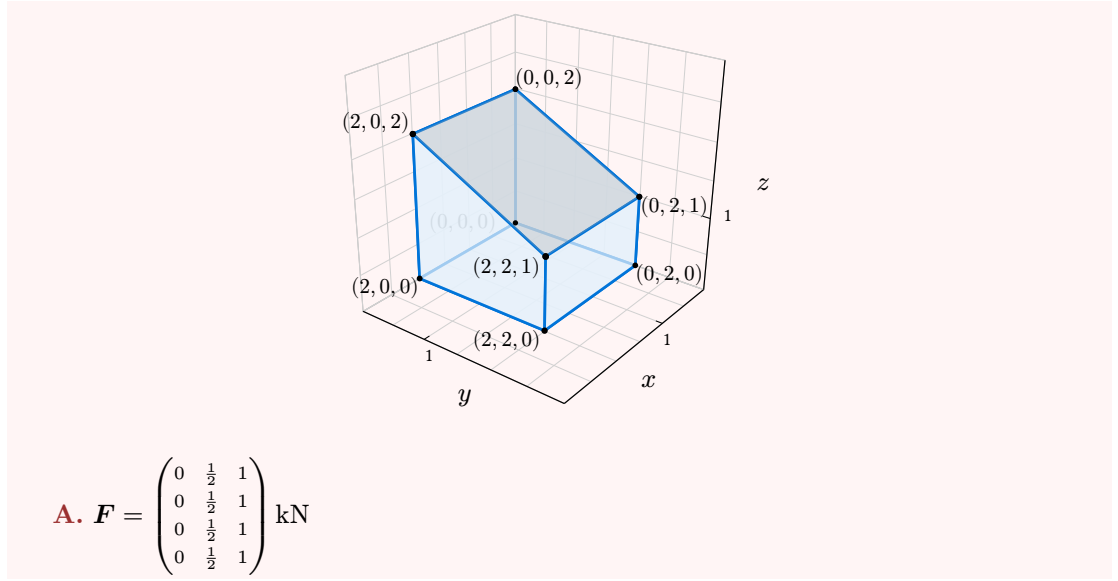
6. The nodal coordinates of a 3-node bar element are $(x_1, y_1) = (-1, -1)$, $(x_2, y_2) = (1, 1)$, and $(x_3, y_3) = (0, 1)$. If the unit weight is $\gamma = 70$ kN/m³ and the area is $A = 0.02$ m², calculate the equivalent nodal forces due to the element weight using numerical integration. **A.** $\mathbf{F}_y = (1.383, 0.413, 2.957)$ kN
7. Find the nodal forces corresponding to horizontal and normal tractions with $q = 1$ kN/m as shown below. Assume the element thickness is $h = 1$ m. The coordinates matrix is:

$$\mathbf{X} = \begin{pmatrix} 0 & 0 \\ 4 & 0 \\ 4 & 4 \\ 0 & 4 \\ 2 & 0 \\ 3 & 2 \\ 2 & 4 \\ 0 & 2 \end{pmatrix} \text{ m}$$



A. $\mathbf{F}_{(a)} = \dots, \mathbf{F}_{(b)} = \begin{pmatrix} -\frac{2}{3} & \frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} \\ -\frac{8}{3} & 0 \end{pmatrix}$ kN

8. For the element below, determine the corresponding nodal forces if a normal traction $t_n = 1$ kPa is applied on the shaded face.



4.9 Global assembly and system solution

Once the stiffness matrix and equivalent nodal load vector have been obtained for each element, the finite element model is turned into a single algebraic system for the whole structure. This global system is built by enforcing compatibility between adjacent elements through their shared nodal degrees of freedom and by accumulating all element contributions according to the mesh connectivity.

At this stage, the problem is no longer treated element by element, but at the structural level. The essential boundary conditions are imposed, the unknown nodal displacements are solved for, and the corresponding reactions and internal forces can then be recovered.

4.9.1 Assembly of the global system

The element equations are combined into a single system for the whole mesh:

$$\mathbf{KU} = \mathbf{F} \quad (4.128)$$

The vector $\mathbf{U}$ contains the nodal displacement degrees of freedom of the model, including those associated with prescribed displacements. The matrix $\mathbf{K}$ is the global stiffness matrix. In linear elasticity it is generally sparse and symmetric, and its size is equal to the total number of degrees of freedom in the mesh.

The global force vector $\mathbf{F}$ is assembled from body forces, surface tractions, and any concentrated nodal loads. The components of $\mathbf{F}$ associated with prescribed displacements correspond to reaction forces. These forces are not known a priori and are obtained as part of the solution process.

Figure 4.16 shows a small finite element model. Each node is labeled with its global node number, and the associated global degrees of freedom are shown in parentheses. In the figure, element 3 is highlighted for further description. The vector that contains the list of node numbers corresponding to an element is known as the **connectivity vector**. For element 3 we have

$$\mathbf{i}^{(3)} = (3, 4, 6) \quad (4.129)$$

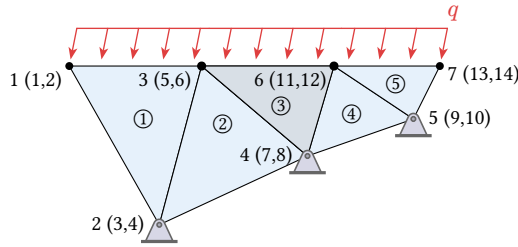


Figure 4.16: A small finite element mesh displaying the numbering of nodes, elements and degrees of freedom.

The vector that contains the list of global degrees of freedom corresponding to an element is known as the **location vector**. For element 3, the connectivity vector identifies the nodes 3, 4, and 6, which have global degrees of freedom (5,6), (7,8), and (11,12), respectively. Therefore, the corresponding location vector for its degrees of freedom is

$$\ell^{(3)} = (5, 6, 7, 8, 11, 12) \quad (4.130)$$

The vector $\ell^{(3)}$ specifies where the element contributions are placed in the global system: it gives the rows and columns used to assemble the element stiffness matrix into $\mathbf{K}$ and the entries used to assemble the element load vector into $\mathbf{F}$. Figure 4.17 illustrates the contribution of element 3 to the global stiffness matrix, and Figure 4.18 illustrates its contribution to the global load vector.

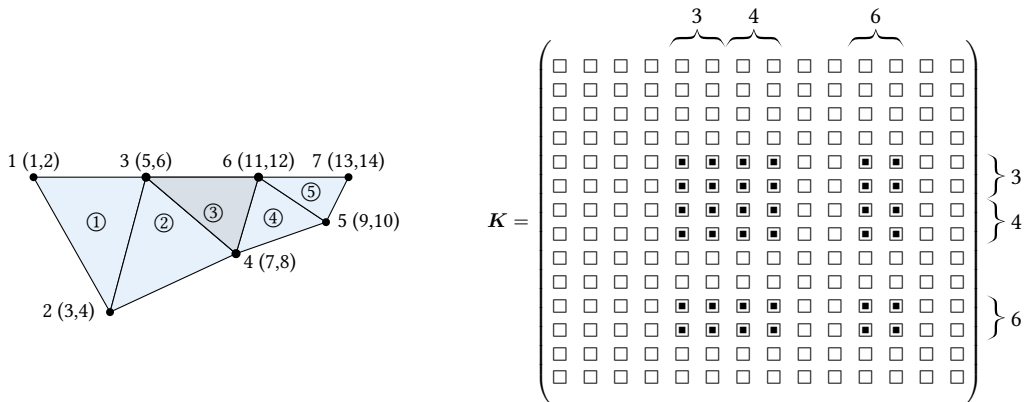


Figure 4.17: Contribution of element 3 to the global stiffness matrix $\mathbf{K}$.

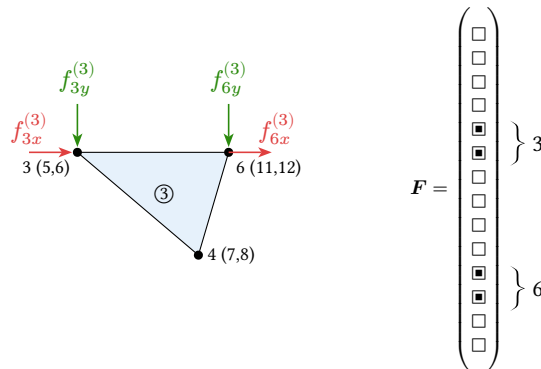


Figure 4.18: Contribution of nodal loads from element 3 to the global load vector $\mathbf{F}$.

The same location vector is used after solving to recover the element displacement vector from the global solution. For example,

$$\mathbf{U}^{(3)} = \mathbf{U}(\ell^{(3)}) \quad (4.131)$$

4.9.2 Solving procedure

The global system of equations

$$\mathbf{KU} = \mathbf{F} \quad (4.132)$$

can be reorganized as

$$\begin{pmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} \\ \mathbf{K}_{21} & \mathbf{K}_{22} \end{pmatrix} \begin{pmatrix} \mathbf{U}_1 \\ \mathbf{U}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{F}_1 \\ \mathbf{F}_2 \end{pmatrix} \quad (4.133)$$

where $\mathbf{U}_1$ contains the unknown displacements, $\mathbf{U}_2$ the prescribed displacements, $\mathbf{F}_1$ the prescribed forces, and $\mathbf{F}_2$ the reactions.

This leads to

$$\mathbf{K}_{11}\mathbf{U}_1 + \mathbf{K}_{12}\mathbf{U}_2 = \mathbf{F}_1 \quad (4.134)$$

$$\mathbf{K}_{21}\mathbf{U}_1 + \mathbf{K}_{22}\mathbf{U}_2 = \mathbf{F}_2 \quad (4.135)$$

The first equation is solved for $\mathbf{U}_1$:

$$\mathbf{K}_{11}\mathbf{U}_1 = \mathbf{F}_1 - \mathbf{K}_{12}\mathbf{U}_2 \quad (4.136)$$

and once $\mathbf{U}_1$ is known, the reactions $\mathbf{F}_2$ are obtained from Eq. (4.135). The reduced matrix $\mathbf{K}_{11}$ should be nonsingular if the structure is properly constrained; otherwise the model still contains rigid-body modes or mechanisms.

4.9.3 Internal nodal forces

After the system solution, the displacements at each element can be recovered as

$$\mathbf{U}^{(e)} = \mathbf{U}(\ell^{(e)}) \quad (4.137)$$

The strains and stresses are then recovered at the element integration points as

$$\begin{aligned} \boldsymbol{\varepsilon}_i &= \mathbf{B}_i \mathbf{U}^{(e)} \\ \boldsymbol{\sigma}_i &= \mathbf{D}_i \boldsymbol{\varepsilon}_i \end{aligned} \quad (4.138)$$

where the subscript i denotes an integration point. These stresses can then be converted into an equivalent nodal-force vector, called the **internal nodal-force vector**. This vector represents the nodal forces that are in equilibrium with the stress field inside the element. Figure 4.19 illustrates the stress state at the integration points of a quadrilateral element and the corresponding internal nodal forces.

Internal nodal forces are useful for checking equilibrium against the assembled external loads and for computing reactions. Unlike external equivalent nodal loads, they are evaluated after the system has been solved, because they depend on the stress field obtained from the displacement solution.

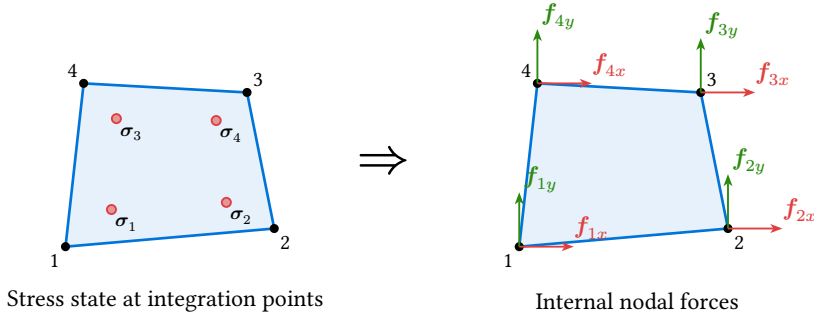


Figure 4.19: Stress state at the integration points of a quadrilateral element and the corresponding internal nodal forces.

To derive an expression for the internal nodal-force vector, consider a single element. From the principle of virtual work,

$$\delta U^{(e)} \cdot \mathbf{F}^{\text{int},(e)} = \int_V \delta \boldsymbol{\varepsilon} \cdot \boldsymbol{\sigma} \, dV \quad (4.139)$$

Since $\delta \boldsymbol{\varepsilon} = \mathbf{B} \delta \mathbf{U}^{(e)}$, this becomes

$$\delta U^{(e)} \cdot \mathbf{F}^{\text{int},(e)} = \delta U^{(e)} \cdot \int_V \mathbf{B}^T \boldsymbol{\sigma} \, dV \quad (4.140)$$

Because $\delta U^{(e)}$ is arbitrary, it follows that

$$\mathbf{F}^{\text{int},(e)} = \int_V \mathbf{B}^T \boldsymbol{\sigma} \, dV \quad (4.141)$$

Using numerical integration, the internal nodal-force vector is approximated by

$$\mathbf{F}^{\text{int},(e)} \approx \sum_{i=1}^{n_{\text{ip}}} \mathbf{B}_i^T \boldsymbol{\sigma}_i J_i w_i \quad (4.142)$$

or, for a two-dimensional element of thickness h ,

$$\mathbf{F}^{\text{int},(e)} \approx h \sum_{i=1}^{n_{\text{ip}}} \mathbf{B}_i^T \boldsymbol{\sigma}_i J_i w_i \quad (4.143)$$

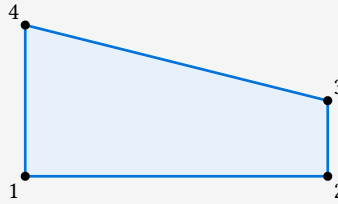
In these expressions, n_{ip} is the number of integration points, J_i is the determinant of the Jacobian at point i , and w_i is the corresponding quadrature weight.

The components of $\mathbf{F}^{\text{int},(e)}$ are arranged as

$$\mathbf{F}^{\text{int},(e)} = \begin{pmatrix} f_{1x} \\ f_{1y} \\ f_{2x} \\ f_{2y} \\ \vdots \\ f_{nx} \\ f_{ny} \end{pmatrix} \quad (4.144)$$

Example 4.9: The finite element shown in the figure has nodal coordinate matrix C and nodal displacement vector $U^{(e)}$ given by

$$C = \begin{pmatrix} 1 & 1 \\ 3 & 1 \\ 3 & 1.5 \\ 1 & 2 \end{pmatrix} \text{ m} \quad \text{and} \quad U^{(e)} = \begin{pmatrix} 0 \\ 0 \\ 0.04 \\ 0.02 \\ 0.04 \\ 0.03 \\ 0.02 \\ 0.02 \end{pmatrix} \text{ m}$$



Find the strain and stress components at the integration points. Then compute the internal nodal-force vector. Consider $E = 2 \times 10^3$ MPa and $\nu = 0.25$ and plane stress conditions.

Solution:

Given the elastic parameters and plane stress conditions, the constitutive matrix is computed as

$$\begin{aligned} D &= \frac{E}{1-\nu^2} \begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{pmatrix} \text{ MPa} \\ &= \begin{pmatrix} 2133.33333 & 533.33333 & 0 \\ 533.33333 & 2133.33333 & 0 \\ 0 & 0 & 800 \end{pmatrix} \text{ MPa} \end{aligned}$$

To compute the strain components at each integration point, we first evaluate the matrix B at those points in natural coordinates. Then, the strain vector is obtained as $\varepsilon = BU^{(e)}$ and the stress vector is obtained as $\sigma = D\varepsilon$.

For integration point 1 with $(\xi, \eta) = (-0.57735, -0.57735)$:

$$B = \begin{pmatrix} -0.44093 & 0 & 0.38185 & 0 & 0.11815 & 0 & -0.05907 & 0 \\ 0 & -0.88185 & 0 & -0.23629 & 0 & 0.23629 & 0 & 0.88185 \\ -0.88185 & -0.44093 & -0.23629 & 0.38185 & 0.23629 & 0.11815 & 0.88185 & -0.05907 \end{pmatrix}$$

$$\varepsilon = BU^{(e)} = (0.01882 \quad 0.02 \quad 0.02764)$$

$$\sigma = D\varepsilon = (50.816 \quad 52.704 \quad 22.112) \text{ MPa}$$

$$\det J = 0.44716875$$

For integration point 2 with $(\xi, \eta) = (0.57735, -0.57735)$:

$$\mathbf{B} = \begin{pmatrix} -0.41277 & 0 & 0.32554 & 0 & 0.17446 & 0 & -0.08723 & 0 \\ 0 & -0.34892 & 0 & -1.30217 & 0 & 1.30217 & 0 & 0.34892 \\ -0.34892 & -0.41277 & -1.30217 & 0.32554 & 1.30217 & 0.17446 & 0.34892 & -0.08723 \end{pmatrix}$$

$$\boldsymbol{\varepsilon} = \mathbf{B}\mathbf{U}^{(e)} = (0.01826 \quad 0.02 \quad 0.01698)$$

$$\boldsymbol{\sigma} = \mathbf{D}\boldsymbol{\varepsilon} = (49.62133 \quad 52.40533 \quad 13.584) \text{ MPa}$$

$$\det \mathbf{J} = 0.30283125$$

For integration point 3 with $(\xi, \eta) = (-0.57735, 0.57735)$:

$$\mathbf{B} = \begin{pmatrix} -0.27954 & 0 & 0.05907 & 0 & 0.44093 & 0 & -0.22046 & 0 \\ 0 & -0.88185 & 0 & -0.23629 & 0 & 0.23629 & 0 & 0.88185 \\ -0.88185 & -0.27954 & -0.23629 & 0.05907 & 0.23629 & 0.44093 & 0.88185 & -0.22046 \end{pmatrix}$$

$$\boldsymbol{\varepsilon} = \mathbf{B}\mathbf{U}^{(e)} = (0.01559 \quad 0.02 \quad 0.02764)$$

$$\boldsymbol{\sigma} = \mathbf{D}\boldsymbol{\varepsilon} = (43.92533 \quad 50.98133 \quad 22.112) \text{ MPa}$$

$$\det \mathbf{J} = 0.44716875$$

For integration point 4 with $(\xi, \eta) = (0.57735, 0.57735)$:

$$\mathbf{B} = \begin{pmatrix} -0.17446 & 0 & -0.15108 & 0 & 0.65108 & 0 & -0.32554 & 0 \\ 0 & -0.34892 & 0 & -1.30217 & 0 & 1.30217 & 0 & 0.34892 \\ -0.34892 & -0.17446 & -1.30217 & -0.15108 & 1.30217 & 0.65108 & 0.34892 & -0.32554 \end{pmatrix}$$

$$\boldsymbol{\varepsilon} = \mathbf{B}\mathbf{U}^{(e)} = (0.01349 \quad 0.02 \quad 0.01698)$$

$$\boldsymbol{\sigma} = \mathbf{D}\boldsymbol{\varepsilon} = (39.44533 \quad 49.86133 \quad 13.584) \text{ MPa}$$

$$\det \mathbf{J} = 0.30283125$$

Finally, the internal nodal-force vector is computed using numerical integration as

$$\mathbf{F}^{\text{int},(e)} = \sum_{i=1}^4 \mathbf{B}_i^T \boldsymbol{\sigma}_i J_i w_i$$

which yields

$$\mathbf{F}^{\text{int},(e)} = \begin{pmatrix} -44.11 \\ -61.23 \\ -2.46 \\ -46.21 \\ 37.13 \\ 60.21 \\ 9.44 \\ 47.23 \end{pmatrix} \text{ MN}$$

Example 4.10: Write a code that, given the geometry and nodal displacements of a quadrilateral element, computes the element internal force vector.

Solution:

The following code computes the matrix B , the strains ε , and the stresses σ at each integration point. It then uses numerical integration to compute the internal nodal-force vector F . The code assumes the existence of functions `quad4_derivs` and `compute_B` that compute the shape function derivatives and the B matrix, respectively.

```
using LinearAlgebra

# Computes the internal nodal-force vector for a quadrilateral element
function compute_F(C, U, quadrature)
    # C: Coordinate matrix of the element (nnodes x 2)
    # U: Displacements vector (nnodes*ndofs)

    npts = size(quadrature, 1)
    nnodes = size(C, 1)
    ndofs = 2
    F = zeros(nnodes*ndofs)
    for i in 1:npts
        xi = quadrature[i, 1]
        eta = quadrature[i, 2]
        w = quadrature[i, 3]
        B = compute_B(C, xi, eta)
        eps = B*U
        sig = D*eps

        dN = quad4_derivs(xi, eta)
        J = C'*dN
        F = F + B'*sig*det(J)*w
    end
    return F
end
```

```
# 2x2 quadrature points and weights for a quadrilateral element
quadrature = [ -0.577350269189626 -0.577350269189626 1.0
                0.577350269189626 -0.577350269189626 1.0
                -0.577350269189626 0.577350269189626 1.0
                0.577350269189626 0.577350269189626 1.0 ]

# Elastic parameters and constitutive matrix for plane stress
E = 2e3
nu = 0.25
D = E/(1-nu^2)*[ 1 nu 0
                 nu 1 0
                 0 0 (1-nu)/2 ]

C = [ 1 1; 3 1; 3 1.5; 1 2 ] # Element coordinates
U = [ 0.0, 0.0, 0.04, 0.02, 0.04, 0.03, 0.02, 0.02 ] # Displacements vector
F = compute_F(C, U, quadrature)

println("Internal nodal forces:")
println(F)
```

```
Internal nodal forces:
[-44.10256410256411, -61.23076923076924, -2.461538461538459, -46.205128205128204,
 37.12820512820513, 60.20512820512822, 9.43589743589744, 47.230769230769226]
```

4.9.4 Equilibrium residual and reactions

Once the displacement solution is known, the internal nodal-force vectors can be assembled in the same way as the element stiffness matrices and external load vectors. This gives the global internal force vector

$$\mathbf{F}^{\text{int}}(U) = \sum_e^{\text{asm}} \mathbf{F}_{\text{int}}^{(e)} \quad (4.145)$$

where each element contribution is computed from the stress field recovered at the integration points. The external force vector contains the prescribed nodal forces and the equivalent nodal loads obtained from body forces and surface tractions:

$$\mathbf{F}_{\text{ext}} = \mathbf{F}_{\text{nodal}} + \mathbf{F}_{\text{body}} + \mathbf{F}_{\text{traction}} \quad (4.146)$$

The difference between external and internal forces is the equilibrium residual:

$$\mathbf{R} = \mathbf{F}_{\text{ext}} - \mathbf{F}_{\text{int}} \quad (4.147)$$

For a converged static solution, the residual should vanish at all free degrees of freedom:

$$\mathbf{R}_f \approx \mathbf{0} \quad (4.148)$$

This statement is the discrete version of mechanical equilibrium. If the residual at a free degree of freedom is not small, the finite-element equations have not been satisfied there. This may indicate an unconverged nonlinear iteration, an inconsistent load vector, an assembly error, or an incorrectly imposed boundary condition.

At constrained degrees of freedom, the residual does not need to vanish because the support supplies whatever force is needed to enforce the prescribed displacement. With the sign convention adopted here, the reaction vector is

$$\mathbf{r}_c = \mathbf{F}_{\text{int},c} - \mathbf{F}_{\text{ext},c} = -\mathbf{R}_c \quad (4.149)$$

Thus reactions are not additional unknowns solved independently; they are obtained from the imbalance between internal and external nodal forces at restrained degrees of freedom after the displacement field has been found.

This equilibrium assessment should not be confused with error assessment. A small residual means that the discrete finite-element equations are balanced, but it does not necessarily mean that the finite-element solution is close to the exact continuum solution.

Note 4.3: The remaining difference between the exact continuum solution $\mathbf{u}$ and the finite-element approximation $\mathbf{u}_h$ is called **discretization error**:

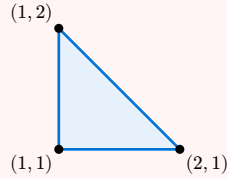
$$\mathbf{e}_h = \mathbf{u} - \mathbf{u}_h$$

It is caused by replacing the continuum problem with a finite mesh and finite-order interpolation, and depends on mesh size, interpolation order, element quality, numerical integration, and the regularity of the exact solution.

Exercises 4.4:

1. What is full and reduced integration in the computation of the stiffness matrix?
2. What are the connectivity and location vectors of an element, and how are they used in the assembly process?

3. What is the equilibrium residual, and why must it vanish only at free degrees of freedom?
4. How are support reactions recovered after the displacement field has been computed?
5. Using computer software, compute the stiffness matrix for the triangular element. Use $E = 20$ GPa and $\nu = 0.25$.



6. The nodal coordinates of a three-node bar element are

$$C = \begin{pmatrix} -1 & -1 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ m}$$

At the end of a finite-element analysis, the axial stresses at the three integration points are $\sigma_1 = 5$ kPa, $\sigma_2 = 10$ kPa, and $\sigma_3 = 15$ kPa. Compute the nodal internal forces vector. **A.** $\mathbf{F} = (-2.43, -7.54, 12.17, -2.68, -9.74, 10.21)$ kN

7. The nodal coordinates of a four-node quadrilateral element are

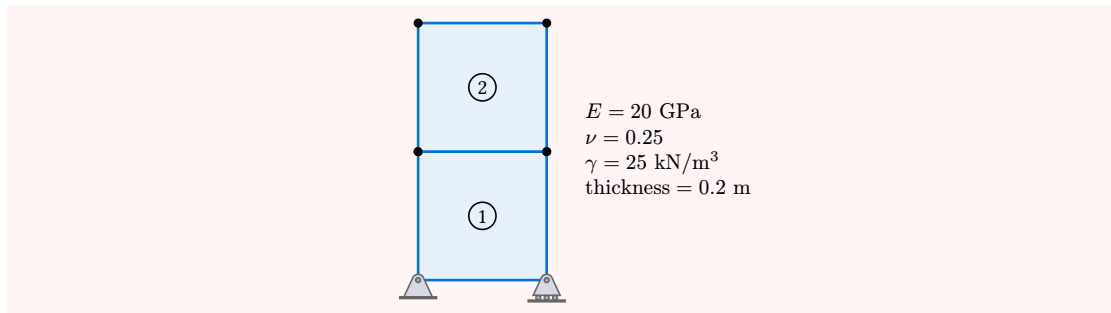
$$C = \begin{pmatrix} 0 & 0 \\ 2 & 0 \\ 2 & 2 \\ 0 & 2 \end{pmatrix} \text{ m}$$

and the nodal displacements are

$$\mathbf{U}^{(e)} = (0.0, 0.0, 0.001, 0.0, 0.002, -0.001, -0.001, -0.001)$$

Assuming full integration, compute the internal nodal-force vector. Use $E = 20$ GPa and $\nu = 0.1$. **A.** $\mathbf{F} = (-14.31, 8.081, 14.31, 8.081, 24.07, -8.081, -24.07, -8.081)$ MN

8. The two-element mesh was designed to simulate self-weight in a structure under plane stress conditions. Each element has dimensions 1×1 m. Considering full integration, compute:
 - a. The stiffness matrix for each element.
 - b. The equivalent nodal loads corresponding to the body force.
 - c. The global stiffness matrix and the global force vector.
 - d. The nodal displacements and reactions.
 - e. The strain and stress vectors at the integration points of element 1.
 - f. Check if the vertical stress at a lower integration point of element 1 is close to the analytical value γh .



4.10 Numerical issues

The finite element equations may be correct in form and still give poor numerical results if the element interpolation, numerical integration, mesh geometry, or boundary conditions introduce artificial constraints or instabilities. This section summarizes some issues that affect the displacement solution itself. Stress recovery, stress smoothing, and singular stress fields are post-processing topics and will be treated separately.

4.10.1 Locking

In some circumstances finite elements behave artificially stiffer than the continuum problem they are intended to approximate. This phenomenon is called **locking**.

Locking occurs when the finite element interpolation imposes parasitic strain constraints that are not required by the physical problem. As the mesh is refined, the solution may converge very slowly or approach the correct result only after an impractically fine mesh. The most common forms in solid mechanics are shear locking and volumetric locking.

4.10.1.1 Shear locking

Shear locking appears mainly in bending-dominated problems modeled with low-order displacement elements. In pure bending, the physical transverse shear strain should be small or zero, but the element interpolation may be unable to represent that state exactly. The element then develops spurious shear strain energy and becomes too stiff.

Typical symptoms are underestimated deflections and stresses that improve only slowly with mesh refinement. For example, a coarse mesh of bilinear quadrilateral elements may predict a cantilever beam that is much stiffer than the same mesh modeled with higher-order elements.

The benchmark used here is the cantilever beam of [Figure 4.20](#), clamped on the left edge and loaded by a uniform vertical traction on the right edge. The numerical response in [Figure 4.21](#) shows the effect clearly. The fully integrated 4-node element predicts the smallest deflection, whereas the reduced-integration 4-node and the fully integrated 8-node solutions are much closer to each other.

This is the typical signature of shear locking: the coarse bilinear element cannot reproduce the nearly zero transverse shear strain required by bending, so it introduces parasitic shear energy and becomes too stiff. The deformation patterns in [Figure 4.21](#) reflect this behavior, since the fully integrated 4-node mesh bends visibly less than the other two solutions under the same geometry, material, and loading.

Common remedies include using higher-order elements, reduced or selective reduced integration, assumed-strain formulations, or mixed formulations.

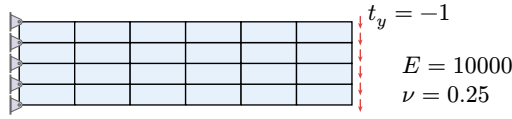


Figure 4.20: Benchmark used to illustrate shear locking: a 4×1 cantilever beam, clamped on the left edge and subjected to a uniform vertical traction on the right edge.

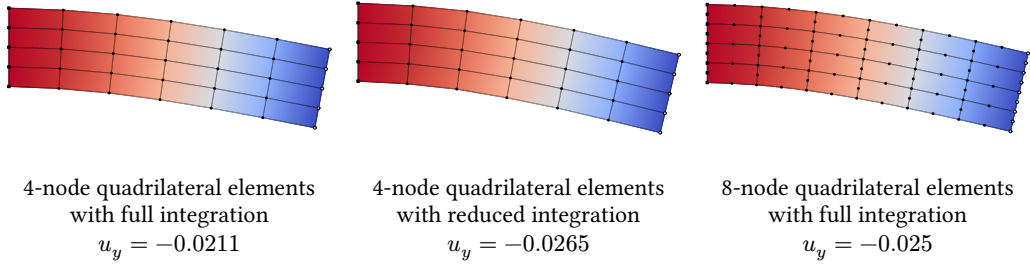


Figure 4.21: Vertical displacement field for the cantilever beam of Figure 4.20. The reported value is the minimum vertical displacement.

4.10.1.2 Volumetric locking

Volumetric locking appears when the material response is nearly incompressible, that is, when ν approaches 0.5 in isotropic linear elasticity. In this limit, the bulk modulus becomes very large and the deformation should occur with almost no volume change:

$$\varepsilon_v = \text{tr}(\varepsilon) \approx 0 \quad (4.150)$$

Standard low-order displacement elements often cannot represent this constraint without introducing excessive stiffness. The result is an artificially small displacement field and an overconstrained pressure or mean-stress response. This is why nearly incompressible elasticity, plasticity, and rubber-like materials often require special element technology.

The Cook panel benchmark in Figure 4.22 makes this effect visible because the skewed geometry and bending-dominated deformation are sensitive to artificial incompressibility constraints. The results in Figure 4.23 show a strong difference between the formulations. The fully integrated 4-node element predicts a much smaller displacement, while the reduced-integration 4-node and fully integrated 8-node solutions remain relatively close to each other.

The reason is that the bilinear displacement field cannot satisfy the near-incompressibility constraint well at all integration points in plane strain, so the element develops excessive volumetric stiffness. The displacement contours in Figure 4.23 show the same trend: the fully integrated 4-node mesh deforms much less than the reduced-integration and higher-order solutions.

Common remedies include mixed displacement-pressure elements, selective reduced integration of the volumetric part of the stiffness matrix, and enhanced-strain formulations.

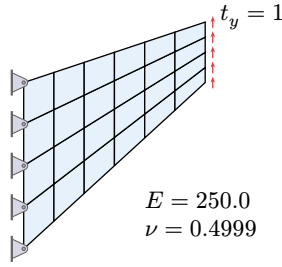


Figure 4.22: Benchmark used to illustrate volumetric locking: a Cook panel in plane strain, fixed on the left edge and subjected to a uniform vertical traction on the right edge.

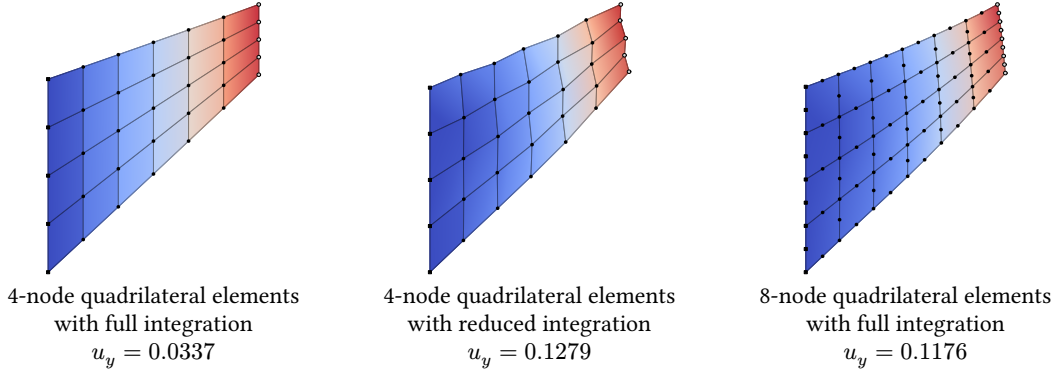


Figure 4.23: Vertical displacement field for the nearly incompressible Cook panel of Figure 4.22. The reported value is the top-right vertical displacement.

4.10.1.3 Reduced integration as a remedy

Reduced integration lowers the quadrature order used to evaluate the element stiffness matrix. It can relieve locking because some of the parasitic strain constraints are no longer enforced at all integration points. It also reduces computational cost.

However, reduced integration is a method with tradeoffs, not a numerical issue by itself. If too few integration points are used, the element stiffness matrix may lose rank and admit spurious zero-energy deformation modes. These modes are discussed next as hourglass modes.

4.10.2 Hourglass modes

An **hourglass mode** is a spurious deformation pattern that produces little or no strain at the integration points. It is most commonly associated with under-integrated low-order elements.

For example, a four-node bilinear quadrilateral in two dimensions has eight nodal degrees of freedom,

$$\mathbf{U}^{(e)} = \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \\ u_{4x} \\ u_{4y} \end{pmatrix} \quad (4.151)$$

Therefore its displacement space is eight-dimensional: any element displacement can be written as a linear combination of eight independent nodal patterns. These patterns are not additional assumptions; they are simply a convenient basis for the same interpolation space.

It is therefore natural to examine the element through a basis of independent displacement modes, as shown in [Figure 4.24](#). Different choices of basis are possible, but the modes in [Figure 4.24](#) fall into two groups. The first group consists of the rigid-body modes: two translations and one rotation. These should produce no strain energy. The second group consists of the deformational modes, which include regular strain-producing patterns and alternating distortions. All of these non-rigid modes should produce positive strain energy. The critical question is whether the quadrature rule detects the alternating deformational patterns or whether they accidentally produce zero strain at the sampled points.

The term zero-energy mode refers to nodal displacements $\mathbf{U}$ that do not correspond to rigid-body motion but nevertheless produce zero strain energy:

$$U_e = \frac{1}{2} \mathbf{U}^T \mathbf{K} \mathbf{U} = \frac{1}{2} \int \varepsilon^T \mathbf{D} \varepsilon dV = 0 \quad (4.152)$$

If this happens for a mode $\mathbf{U}$ different from a rigid-body mode, the mode is **spurious** and the corresponding element is unstable.

For a two-dimensional quadrilateral element, three zero-energy modes are expected: two rigid translations and one rigid rotation. The remaining five independent modes should receive positive stiffness. With sufficient integration, the stiffness matrix detects the strain energy of these non-rigid modes. With one-point reduced integration, however, only the strain at the element center is sampled. Some alternating displacement patterns can have zero strain at that point even though they deform the element away from the center. Those patterns then receive no stiffness from the element stiffness integral and become hourglass modes.

An element that displays such an additional mechanism is said to be **rank deficient**, meaning that the rank of $\mathbf{K}$ is less than the number of degrees of freedom minus the number of rigid-body modes. In a mesh, hourglass modes may appear as alternating nodal displacement patterns with little visible resistance from the element. This is why the independent-mode view is important: it separates the admissible zero-energy modes, which are rigid-body motions, from the inadmissible zero-energy modes introduced by the numerical integration.

Note 4.4: A **well posed** linear system has a unique solution for the unknowns. Accordingly, its coefficient matrix must have **full rank** (equal to the number of unknowns). If the coefficient matrix is rank deficient (singular), the system is not uniquely solvable: it has either no solution or infinitely many solutions. For an element stiffness matrix, **full rank** means the number of element degrees of freedom minus the admissible rigid-body modes.

The benchmark of [Figure 4.25](#) illustrates this behavior in a narrow rectangular panel fixed along its base and loaded by a vertical nodal force at the top-right corner. The two solutions in [Figure 4.26](#) show the qualitative difference between stable and unstable behavior. With full integration, the mesh deforms smoothly and the stress field varies in a regular way through the panel. With reduced integration, the mesh develops an alternating zig-zag pattern, even though the loading is simple. This oscillatory shape is characteristic of an hourglass mode: the deformation is visually pronounced, but the reduced quadrature does not assign enough strain energy to suppress it.

Hourglass modes can often be controlled by using full integration, using an element with hourglass stabilization, refining the mesh, or choosing an element formulation that is less sensitive to under-integration. Reduced integration should therefore be used intentionally, with an element formulation that is known to be stable for the problem being solved.

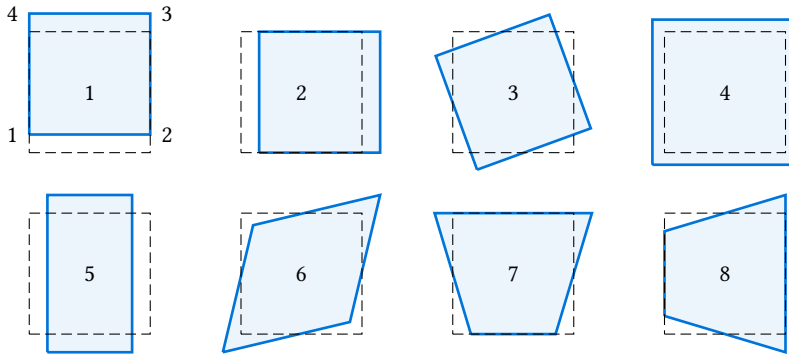


Figure 4.24: Independent displacement modes of a 4-node quadrilateral element.

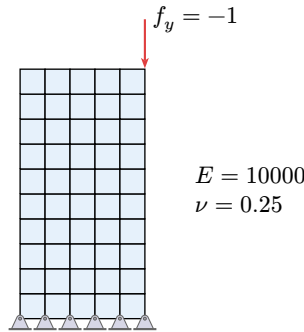


Figure 4.25: Benchmark used to illustrate hourglass modes: a rectangular panel fixed at the base and loaded by a vertical nodal force at the top-right corner.

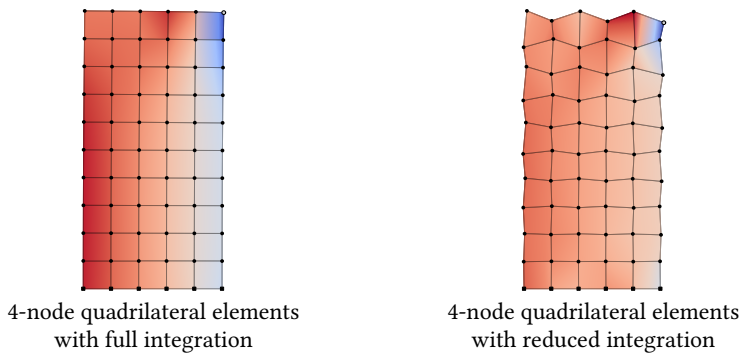


Figure 4.26: Vertical stress field of the panel in Figure 4.25. Full integration gives a smooth response, whereas reduced integration activates a spurious hourglass mode.

4.10.3 Mesh distortion

Isoparametric elements use a mapping from the parent domain to the physical element. If the physical element is highly skewed, stretched, warped, or nearly collapsed, this mapping becomes poor and the numerical accuracy deteriorates. The Jacobian matrix controls this mapping:

$$\mathbf{J} = \frac{\partial \mathbf{x}}{\partial \xi} \quad (4.153)$$

The determinant $\det \mathbf{J}$ should remain positive and should not become too small inside the element. If $\det \mathbf{J} \leq 0$ at an integration point, the element is inverted or degenerate and the stiffness contribution is invalid.

Distorted elements usually produce inaccurate strains, poor stiffness estimates, and slow convergence. Good meshes avoid extreme aspect ratios, very small angles, crossed elements, and strong warping in three-dimensional elements. Some distortion is unavoidable in practical meshes, but the numerical solution should be checked for sensitivity to mesh refinement and element quality.

4.10.4 Ill-conditioning and singular stiffness matrices

A linear finite element model leads to a system

$$\mathbf{K}\mathbf{U} = \mathbf{F} \quad (4.154)$$

whose solution depends on the numerical properties of $\mathbf{K}$. If $\mathbf{K}$ is ill-conditioned, small round-off errors or small changes in the data may produce noticeable changes in the computed displacement vector. A useful measure is the condition number

$$\kappa(\mathbf{K}) = \|\mathbf{K}\| \|\mathbf{K}^{-1}\| \quad (4.155)$$

Large condition numbers can arise from very distorted elements, large contrasts in material stiffness, badly scaled units, penalty constraints with excessively large penalty factors, or elements with very different sizes.

A singular stiffness matrix is a more severe problem: the system has no unique solution. This usually means that the model still contains an unconstrained rigid-body motion, a mechanism, disconnected parts, or a rank-deficient element formulation. For a properly supported linear elastic model, the reduced stiffness matrix after applying essential boundary conditions should be nonsingular.

Practical checks include inspecting supports, looking for disconnected nodes or element groups, checking element Jacobians, avoiding excessive penalty parameters, and verifying that the selected element and quadrature rule do not introduce mechanisms.

Exercises 4.5:

1. Explain the origin of shear locking and volumetric locking, and indicate in which types of problems each one is most likely to appear.
2. Why can reduced integration alleviate locking, and what numerical risk does it introduce?
3. What are hourglass modes, and why are they associated with underintegrated elements?
4. Why can element distortion degrade accuracy even when the interpolation order is unchanged?
5. What is meant by ill-conditioning of the stiffness matrix, and how can it affect the numerical solution?
6. Distinguish between an ill-conditioned stiffness matrix and a singular stiffness matrix.
7. Give four common modeling or formulation errors that can lead to a singular stiffness matrix.
8. Why may excessively large penalty parameters create numerical difficulties even though they enforce constraints more strongly?

4.11 Constraint enforcement

The global finite-element equations are written in terms of nodal degrees of freedom, but not all of these degrees of freedom are independent. Supports, prescribed displacements, symmetry conditions, rigid links, contact ties, and other modeling assumptions may restrict one displacement or relate several displacements to each other. Constraint enforcement is the process of modifying or augmenting the algebraic system so that these restrictions are satisfied while the remaining degrees of freedom still obey equilibrium.

In this section, the word **constraint** refers only to algebraic restrictions on essential degrees of freedom, such as u_x , u_y , u_z , or rotations when they are part of the element formulation. Typical constraints may be single-freedom, such as $u_x = 0$, or multifreedom, such as $u_x + 2u_y = 0$.

Note 4.5: Boundary conditions and constraints are related, but they are not identical. Essential boundary conditions prescribe displacement-like variables and therefore act as constraints on the primary unknowns. Natural boundary conditions prescribe force-like quantities, such as nodal forces or surface tractions, and enter the system as loads rather than constraints.

4.11.1 Multifreedom constraints

After assembly, constraints must be imposed on the global displacement vector. Prescribed displacements such as $u_{2x} = 0$ or $u_{3y} = 0.01$ are single-freedom constraints. In some cases, several displacement degrees of freedom are related, for example

$$u_{2x} = u_{4x} \quad \text{or} \quad u_{2x} + 2u_{2y} + u_{2z} = 0 \quad (4.156)$$

These are called multifreedom constraints (MFCs). They appear in inclined supports, rigid links, symmetry relations, tied interfaces, and other situations where a displacement component cannot be prescribed independently.

In matrix form, a set of linear constraints can be written as

$$\mathbf{A}\mathbf{U} = \mathbf{b} \quad (4.157)$$

where each row of $\mathbf{A}$ defines one constraint equation and $\mathbf{b}$ contains the prescribed values. For example, the constraint $u_4 - u_2 = 0$ can be written as

$$\begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} u_4 \\ u_2 \end{pmatrix} = 0 \quad (4.158)$$

The most common methods to enforce these constraints are:

- Master-slave elimination
- Penalty method
- Lagrange multipliers method

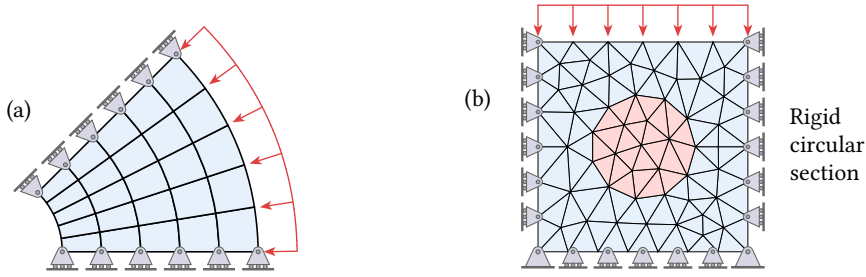


Figure 4.27: Examples of multi-degree-of-freedom constraints: (a) displacement constraints imposed by inclined supports along the boundary, and (b) a rigid inclusion embedded in a deformable domain.

4.11.2 Master-slave elimination

In master-slave elimination, some constrained degrees of freedom are removed from the system. The removed degrees of freedom are called **slave** degrees of freedom, and they are expressed in terms of retained **master** degrees of freedom. For example, if

$$u_s = a_1 u_1 + a_2 u_2 + b_s \quad (4.159)$$

then u_s does not need to be solved as an independent unknown.

More generally, the full displacement vector can be written in terms of a smaller vector of independent unknowns $\mathbf{q}$:

$$\mathbf{U} = \mathbf{T}\mathbf{q} + \mathbf{U}_0 \quad (4.160)$$

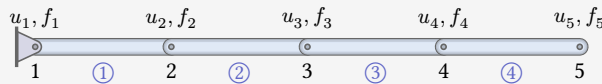
where $\mathbf{T}$ maps independent degrees of freedom to the full set of nodal degrees of freedom and $\mathbf{U}_0$ contains prescribed offsets. Substitution into the global equilibrium equation gives the reduced system

$$\mathbf{T}^T \mathbf{K} \mathbf{T} \mathbf{q} = \mathbf{T}^T (\mathbf{F} - \mathbf{K} \mathbf{U}_0) \quad (4.161)$$

This method enforces the constraints exactly and does not add unknowns. It also keeps the final system positive definite when the original constrained problem is well posed. However, the implementation can become more involved for many complex constraints, because the global numbering and transformation matrix must be handled consistently.

Example 4.11: For the same 1D truss system, subjected to the constraints

$$\begin{aligned} u_1 &= 0 \\ u_5 &= 0.01 \\ u_4 - u_2 &= 0 \end{aligned}$$



Find the modified system of equations that enforces these constraints using the master-slave elimination method.

Solution:

The constraint $u_4 - u_2 = 0$ gives

$$u_4 = u_2$$

Thus u_4 can be treated as a slave degree of freedom and u_2 as its master. The single-freedom constraints prescribe

$$u_1 = 0, \quad u_5 = 0.01$$

Therefore, the independent unknowns may be chosen as

$$\mathbf{q} = \begin{pmatrix} u_2 \\ u_3 \end{pmatrix}$$

The full displacement vector can be reconstructed from $\mathbf{q}$ as

$$\mathbf{U} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u_2 \\ u_3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0.01 \end{pmatrix}$$

or, in compact form,

$$\mathbf{U} = \mathbf{T}\mathbf{q} + \mathbf{U}_0$$

Before imposing the constraints, the global equilibrium equations are

$$\begin{pmatrix} k_{11} & k_{12} & 0 & 0 & 0 \\ k_{21} & k_{22} & k_{23} & 0 & 0 \\ 0 & k_{32} & k_{33} & k_{34} & 0 \\ 0 & 0 & k_{43} & k_{44} & k_{45} \\ 0 & 0 & 0 & k_{54} & k_{55} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \end{pmatrix}$$

Substituting $\mathbf{U} = \mathbf{T}\mathbf{q} + \mathbf{U}_0$ and premultiplying by $\mathbf{T}^T$ gives

$$\mathbf{T}^T \mathbf{K} \mathbf{T} \mathbf{q} = \mathbf{T}^T (\mathbf{F} - \mathbf{K} \mathbf{U}_0)$$

For this problem, the reduced system is

$$\begin{pmatrix} k_{22} + k_{44} & k_{23} + k_{43} \\ k_{32} + k_{34} & k_{33} \end{pmatrix} \begin{pmatrix} u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} f_2 + f_4 - 0.01k_{45} \\ f_3 \end{pmatrix}$$

After solving this reduced system, the eliminated and prescribed degrees of freedom are recovered as

$$u_1 = 0, \quad u_4 = u_2, \quad u_5 = 0.01$$

This method enforces all three constraints exactly and solves only for the independent displacement degrees of freedom.

4.11.3 Penalty method

The penalty method enforces constraints approximately by adding a large stiffness against constraint violation. For the constraint equation

$$\mathbf{A}\mathbf{U} = \mathbf{b} \tag{4.162}$$

the penalized potential includes the additional term

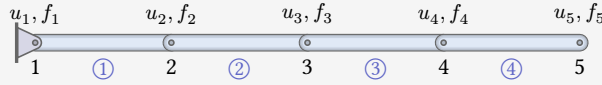
$$\frac{1}{2}\alpha(\mathbf{A}\mathbf{U} - \mathbf{b})^T(\mathbf{A}\mathbf{U} - \mathbf{b}) \quad (4.163)$$

where α is a large positive penalty parameter. The resulting linear system is

$$(\mathbf{K} + \alpha\mathbf{A}^T\mathbf{A})\mathbf{U} = \mathbf{F} + \alpha\mathbf{A}^T\mathbf{b} \quad (4.164)$$

Example 4.12: For the same 1D truss system, subjected to the constraints

$$\begin{aligned} u_1 &= 0 \\ u_5 &= 0.01 \\ u_4 - u_2 &= 0 \end{aligned}$$



Find the modified system of equations that enforces these constraints using the penalty method.

Solution:

The first constraint is a multifreedom constraint and will be addressed using the penalty method. The last two constraints are single-freedom constraints and will be imposed by separating the prescribed and unknown displacements.

The constraint $u_4 - u_2 = 0$ can be written as

$$\begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} u_4 \\ u_2 \end{pmatrix} = 0$$

Premultiplying by $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and introducing the penalty parameter α gives

$$\alpha \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} u_4 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This equation represents the stiffness matrix of a fictitious link element. The term α is the penalty weight. If $\alpha \rightarrow \infty$, the link behaves as rigid and the solution approaches $u_2 = u_4$.

For the truss before adding the fictitious link, the global system may be written as

$$\begin{pmatrix} k_{11} & k_{12} & 0 & 0 & 0 \\ k_{21} & k_{22} & k_{23} & 0 & 0 \\ 0 & k_{32} & k_{33} & k_{34} & 0 \\ 0 & 0 & k_{43} & k_{44} & k_{45} \\ 0 & 0 & 0 & k_{54} & k_{55} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \end{pmatrix}$$

The penalty contribution is assembled in the rows and columns associated with u_2 and u_4 . Since the global constraint can be written as

$$\mathbf{A}\mathbf{U} = 0, \quad \mathbf{A} = \begin{pmatrix} 0 & -1 & 0 & 1 & 0 \end{pmatrix}$$

the added stiffness is $\alpha\mathbf{A}^T\mathbf{A}$. Therefore, after adding the fictitious link element, the system becomes

$$\begin{pmatrix} k_{11} & k_{12} & 0 & 0 & 0 \\ k_{21} & k_{22} + \alpha & k_{23} & -\alpha & 0 \\ 0 & k_{32} & k_{33} & k_{34} & 0 \\ 0 & -\alpha & k_{43} & k_{44} + \alpha & k_{45} \\ 0 & 0 & 0 & k_{54} & k_{55} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \end{pmatrix}$$

The single-freedom constraints prescribe $u_1 = 0$ and $u_5 = 0.01$. Therefore, the remaining unknowns are u_2, u_3 , and u_4 . Moving the known displacement terms to the right-hand side gives

$$\begin{pmatrix} k_{22} + \alpha & k_{23} & -\alpha \\ k_{32} & k_{33} & k_{34} \\ -\alpha & k_{43} & k_{44} + \alpha \end{pmatrix} \begin{pmatrix} u_2 \\ u_3 \\ u_4 \end{pmatrix} = \begin{pmatrix} f_2 \\ f_3 \\ f_4 - 0.01k_{45} \end{pmatrix}$$

This is the modified system to be solved for the free degrees of freedom. After the solution is obtained, the constrained degrees of freedom are recovered as $u_1 = 0$ and $u_5 = 0.01$.

In theory, the larger the value of α , the more closely the multifreedom constraint is enforced. In practice, α must remain finite because excessively large values make the global stiffness matrix ill-conditioned.

4.11.4 Lagrange multipliers

The Lagrange multiplier method enforces the constraints exactly by adding one additional unknown for each constraint equation.

Note 4.6: In mathematical optimization, the Lagrange multiplier method is used to find the extrema of a function subject to constraints. For example, consider

$$\begin{aligned} &\text{maximize} && f(x, y) \\ &\text{subject to} && g(x, y) = 0 \end{aligned}$$

If f and g have continuous first derivatives, the Lagrangian may be defined as

$$\mathcal{L}(x, y, \lambda) = f(x, y) - \lambda g(x, y)$$

where λ is a new variable called a Lagrange multiplier. The stationarity condition

$$\nabla \mathcal{L} = \mathbf{0}$$

gives the extrema of f subject to the constraint $g = 0$. Equivalently,

$$\begin{cases} \nabla f(x, y) = \lambda \nabla g(x, y) \\ g(x, y) = 0 \end{cases}$$

The same idea extends to functions with n variables and m constraints by introducing one multiplier for each constraint:

$$\nabla \mathcal{L}(x_1, x_2, \dots, x_n, \lambda_1, \lambda_2, \dots, \lambda_m) = \mathbf{0}$$

From a variational viewpoint, the constrained equilibrium problem is written using the Lagrangian

$$\mathcal{L}(\mathbf{U}, \mathbf{\Lambda}) = \Pi + \mathbf{\Lambda}^T (\mathbf{A}\mathbf{U} - \mathbf{b}) \quad (4.165)$$

where Π is the total potential energy and $\mathbf{\Lambda}$ is the vector of Lagrange multipliers.

Stationarity with respect to U and Λ gives the augmented system

$$\begin{pmatrix} K & A^T \\ A & 0 \end{pmatrix} \begin{pmatrix} U \\ \Lambda \end{pmatrix} = \begin{pmatrix} F \\ b \end{pmatrix} \quad (4.166)$$

The second row enforces $AU = b$ exactly. The multipliers are related to the forces required to impose the constraints. This method does not require a penalty parameter and is exact at the discrete level, but it introduces additional unknowns. For m independent constraint equations, m additional multiplier unknowns are added to the system.

The augmented matrix contains a zero block on the multiplier equations and is generally indefinite. Therefore, solvers that rely on positive definiteness, such as Cholesky-type solvers for symmetric positive definite matrices, are not appropriate. One must use a solver that can handle indefinite systems.

The constraint equations must also be linearly independent. A constraint should not be specified twice, and no constraint equation should be obtainable as a linear combination of the others. Otherwise, the augmented matrix may become singular.

Example 4.13: Consider again the 1D truss system used in the penalty-method example, with constraints

$$u_4 - u_2 = 0, \quad u_1 = 0, \quad u_5 = 0.01$$

Find the modified system of equations that enforces these constraints using the Lagrange multiplier method.

Solution:

Define

$$U = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{pmatrix}$$

The three constraints can then be written as

$$AU = b$$

with

$$A = \begin{pmatrix} 0 & -1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \\ 0.01 \end{pmatrix}$$

The unconstrained global equilibrium equations are

$$\begin{pmatrix} k_{11} & k_{12} & 0 & 0 & 0 \\ k_{21} & k_{22} & k_{23} & 0 & 0 \\ 0 & k_{32} & k_{33} & k_{34} & 0 \\ 0 & 0 & k_{43} & k_{44} & k_{45} \\ 0 & 0 & 0 & k_{54} & k_{55} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \end{pmatrix}$$

Introducing the vector of multipliers

$$\Lambda = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix}$$

gives the augmented system

$$\begin{pmatrix} \mathbf{K} & \mathbf{A}^T \\ \mathbf{A} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{U} \\ \Lambda \end{pmatrix} = \begin{pmatrix} \mathbf{F} \\ \mathbf{b} \end{pmatrix}$$

or, written explicitly,

$$\begin{pmatrix} k_{11} & k_{12} & 0 & 0 & 0 & 0 & 1 & 0 \\ k_{21} & k_{22} & k_{23} & 0 & 0 & -1 & 0 & 0 \\ 0 & k_{32} & k_{33} & k_{34} & 0 & 0 & 0 & 0 \\ 0 & 0 & k_{43} & k_{44} & k_{45} & 1 & 0 & 0 \\ 0 & 0 & 0 & k_{54} & k_{55} & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \\ 0 \\ 0 \\ 0.01 \end{pmatrix}$$

This is the system to be solved for the unknown displacements and multipliers. This system is larger than the original system and is no longer positive definite due to the zero entries on the diagonal of the coefficient matrix.

Note that each Lagrange multiplier can be interpreted as the force required to enforce the corresponding constraint. For instance, the first multiplier can be interpreted from the first row of $\mathbf{A}$. Since

$$\mathbf{A}_1 = (0 \quad -1 \quad 0 \quad 1 \quad 0)$$

the term $\mathbf{A}_1^T \lambda_1$ adds $-\lambda_1$ to the equilibrium equation of node 2 and λ_1 to the equilibrium equation of node 4. Equivalently, moving those terms to the right-hand side gives

$$\mathbf{K}\mathbf{U} = \mathbf{F} - \mathbf{A}_1^T \lambda_1 = \begin{pmatrix} f_1 \\ f_2 + \lambda_1 \\ f_3 \\ f_4 - \lambda_1 \\ f_5 \end{pmatrix}$$

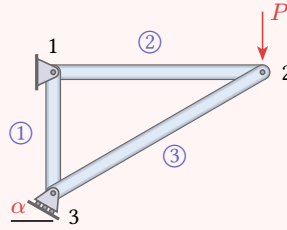
Thus λ_1 can be imagined as the magnitude of the force pair required to keep $u_4 - u_2 = 0$. Similarly, λ_2 and λ_3 are associated with the forces required to enforce $u_1 = 0$ and $u_5 = 0.01$.

The forces f_1 , f_2 , f_4 , and f_5 may include externally applied nodal forces at constrained degrees of freedom. They should not be confused with the reaction forces, which are obtained after solving from the equilibrium imbalance at the constrained degrees of freedom.

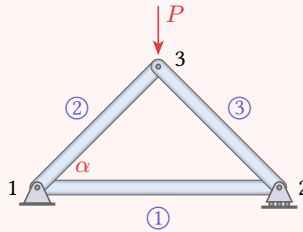
Exercises 4.6:

1. List five use-cases of multifreedom constraints in finite-element modeling.
2. Compare master-slave elimination, the penalty method, and the Lagrange multiplier method for enforcing constraints. State one advantage and one drawback of each.

3. Find all nodal displacements in the truss using the penalty method and the Lagrange multiplier method. Consider $EA = 600 \text{ kN}$, $P = 10 \text{ kN}$, and $\alpha = 30^\circ$. The lengths of elements 1, 2, and 3 are 3 m, 4 m, and 5 m, respectively. Compare the results.



4. In the symmetric truss below, element 3 is rigid. For the other elements, $EA = 600 \text{ kN}$. Find all nodal displacements using the Lagrange multiplier method. Consider the length of element 1 equal to 2 m, $\alpha = 45^\circ$, and $P = 10 \text{ kN}$.



5 SPECIAL ELEMENTS

This section introduces finite elements used for structures with special geometry or special mechanical behavior. We first present surface elements such as plates, membranes, and shells. Then we discuss curved beam elements and interface elements.

Plate, membrane and shell elements are a family of surface elements used to model thin structures. They are commonly used in the analysis of buildings, bridges, aircraft, and other structures where one dimension (thickness) is much smaller than the other two dimensions (length and width). Thus, the thickness is not explicitly modeled, but its effect is included in the element's stiffness properties. They are used to model structures that resist loads through bending, in-plane forces, or a combination of both. Figure 5.1 show the typical load types that can be applied to these elements, including in-plane forces, normal forces, shear forces, and moments. One advantage of these elements is that they can capture complex stress distributions and deformations in thin structures without the need for a full 3D solid model, which can be computationally expensive. Also, the mesh generation for surface elements is often simpler than for solid elements, especially for complex geometries.

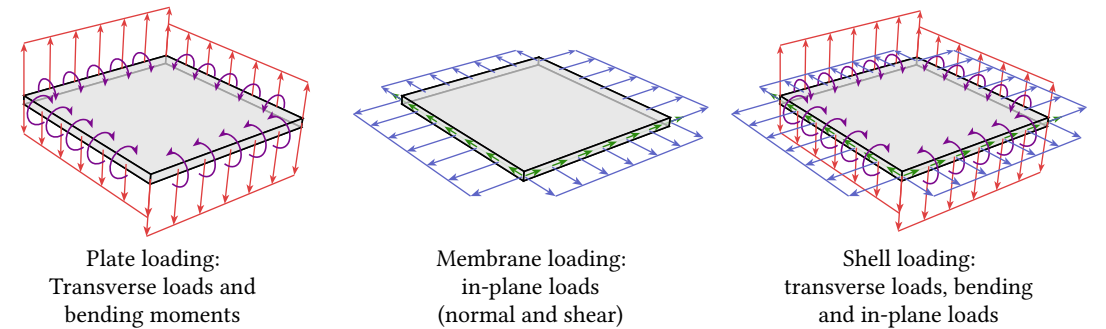


Figure 5.1: Typical loading modes for plate, membrane, and shell structural idealizations.

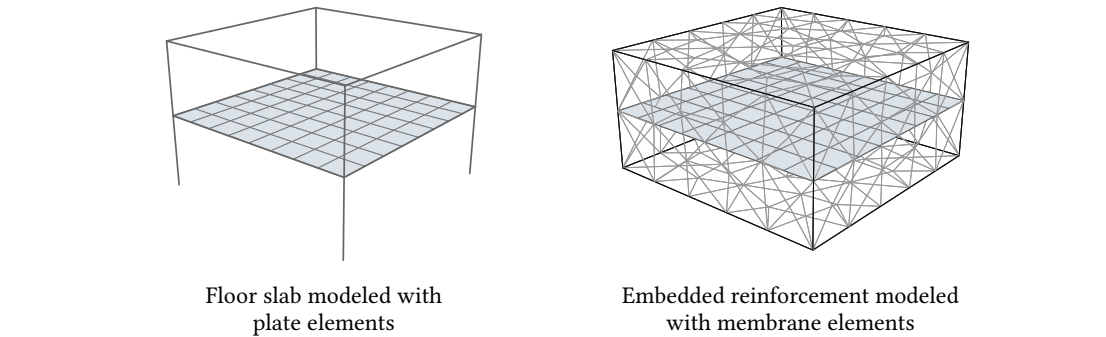


Figure 5.2: Examples of structural idealizations using plate and membrane elements.

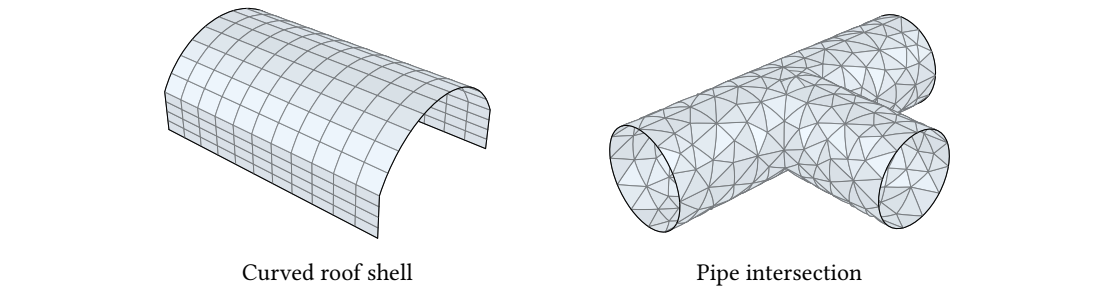


Figure 5.3: Examples of curved surface structures modeled with shell elements.

Other useful element is the curved beam element, which is used to model curved structures that primarily resist loads through bending. They are commonly used in the analysis of arches, curved beams, and other structures with significant curvature. Figure 5.4 shows a curved beam-grid idealization of a roof structure.

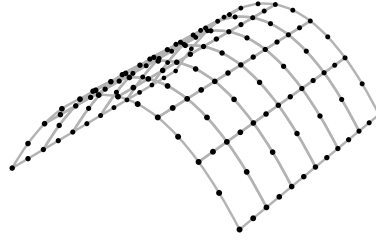


Figure 5.4: Curved beam-grid idealization of a roof structure.

Among special elements there is also the interface element, which is used to model the interaction between two different materials or structural components. They are commonly used in the analysis of composite structures, where different materials are bonded together, in the simulation of joints and cracking.

5.1 Plate element

Plate elements are used to model thin, flat structures that carry loads normal to their mid-surface mainly through bending. They are appropriate when the thickness is small compared with the other two dimensions, as in slabs, bridge decks, and floor panels. The geometry is described on the plate mid-surface, so standard 2D interpolation families, such as triangles and quadrilaterals, can be used. For example, Figure 5.5 shows a four-node quadrilateral plate element.

In the basic plate idealization, in-plane membrane deformation is not included. Each node therefore usually has three degrees of freedom: one transverse displacement and two rotations. For a node i , the nodal displacement and force-resultant vectors are

$$\begin{aligned} \mathbf{U}_i &= (u_{iz}, \theta_{ix}, \theta_{iy}) \\ \mathbf{F}_i &= (f_{iz}, M_{ix}, M_{iy}) \end{aligned} \quad (5.1)$$

where f_{iz} is the transverse force and M_{ix} and M_{iy} are bending moments. Figure 5.6 shows these nodal resultants.

The formulation below uses the components

$$\begin{aligned} \boldsymbol{\varepsilon} &= (\varepsilon_x, \varepsilon_y, \gamma_{yz}, \gamma_{xz}, \gamma_{xy}) \\ \boldsymbol{\sigma} &= (\sigma_x, \sigma_y, \tau_{yz}, \tau_{xz}, \tau_{xy}) \end{aligned} \quad (5.2)$$

The in-plane terms arise from bending, while the transverse shear terms appear in the Reissner-Mindlin theory.

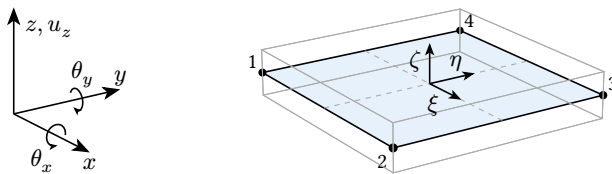


Figure 5.5: A four-node plate element and degrees of freedom.

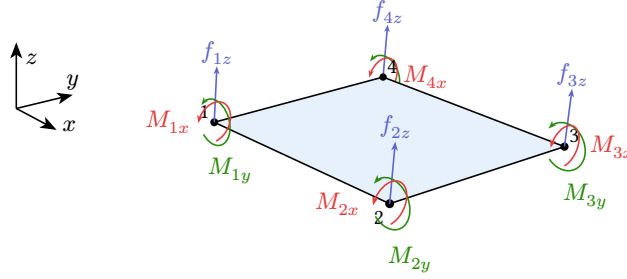


Figure 5.6: A four-node plate element with nodal force and moment resultants.

Two classical kinematic theories are commonly used for plate elements: Kirchhoff-Love theory and Reissner-Mindlin theory.

Kirchhoff-Love theory: This theory is intended for thin plates. It assumes that lines initially normal to the mid-surface remain straight and perpendicular to the mid-surface after deformation. As a result, transverse shear deformation is neglected. Its main assumptions are:

- Normals to the mid-surface remain straight after deformation.
- Normals to the mid-surface remain perpendicular to the mid-surface after deformation.
- Transverse shear deformation is neglected.

For a thin, isotropic, homogeneous elastic plate, the governing equation is

$$D \left(\frac{\partial^4 w}{\partial x^4} + 2 \frac{\partial^4 w}{\partial x^2 \partial y^2} + \frac{\partial^4 w}{\partial y^4} \right) = q(x, y) \quad \text{or} \quad (5.3)$$

$$D \nabla^4 w = q(x, y)$$

where $D = \frac{E h^3}{12(1-\nu^2)}$, E is Young's modulus, h is the plate thickness, ν is Poisson's ratio, and $q(x, y)$ is the applied transverse load per unit area. The operator $\nabla^4 = \frac{\partial^4}{\partial x^4} + 2 \frac{\partial^4}{\partial x^2 \partial y^2} + \frac{\partial^4}{\partial y^4}$ is called the biharmonic operator.

Reissner-Mindlin theory: This theory is suitable for moderately thick plates because it includes transverse shear deformation. Its main assumptions are:

- Normals to the mid-surface remain straight after deformation.
- Normals to the mid-surface do not need to remain perpendicular to the mid-surface.
- Transverse shear strains are included.

The element formulation developed below follows the Reissner-Mindlin kinematics.

5.1.1 Isoparametric horizontal plate element

For a flat horizontal plate, the mid-surface lies in the x - y plane, so only the x and y coordinates are needed at each node. Thus a standard 2D isoparametric mapping can be used. For such an element, the Jacobian matrix is

$$\mathbf{J} = \begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \end{pmatrix} \quad (5.4)$$

which can be computed in the usual way as

$$\mathbf{J} = \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \xi} \quad (5.5)$$

where the coordinate matrix and the derivatives of the shape functions are

$$\mathbf{X} = \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ \vdots & \vdots \\ x_n & y_n \end{pmatrix} \quad \frac{\partial \mathbf{N}}{\partial \xi} = \begin{pmatrix} \frac{\partial N_1}{\partial \xi} & \frac{\partial N_1}{\partial \eta} \\ \frac{\partial N_2}{\partial \xi} & \frac{\partial N_2}{\partial \eta} \\ \vdots & \vdots \\ \frac{\partial N_n}{\partial \xi} & \frac{\partial N_n}{\partial \eta} \end{pmatrix} \quad (5.6)$$

No z coordinates are needed in the geometry, even though the element still represents out-of-plane deformation through u_z , θ_x , and θ_y .

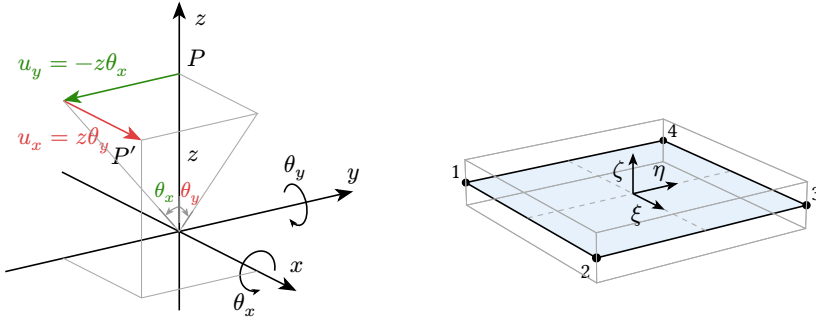


Figure 5.7: Displacements of a point P relative to the mid-surface of a plate.

Consider a point located at a distance z from the mid-surface, with

$$z = \frac{h}{2} \zeta, \quad \zeta \in [-1, 1] \quad (5.7)$$

where h is the plate thickness. For an isoparametric plate element, the displacement at that point is interpolated from the nodal degrees of freedom as

$$\begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix} = \sum_{i=1}^n N_i \begin{pmatrix} z\theta_{iy} \\ -z\theta_{ix} \\ u_{iz} \end{pmatrix} \quad (5.8)$$

The first two components describe the in-plane displacement caused by bending, while the third component is the transverse displacement.

From this displacement field, the bending-related in-plane strain components are

$$\begin{aligned} \varepsilon_x &= \frac{\partial u_x}{\partial x} = z \sum \frac{\partial N_i}{\partial x} \theta_{iy} \\ \varepsilon_y &= \frac{\partial u_y}{\partial y} = -z \sum \frac{\partial N_i}{\partial y} \theta_{ix} \\ \gamma_{xy} &= \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} = z \sum \frac{\partial N_i}{\partial y} \theta_{iy} - z \sum \frac{\partial N_i}{\partial x} \theta_{ix} \end{aligned} \quad (5.9)$$

In the Reissner-Mindlin theory, the transverse shear components are also included:

$$\begin{aligned}\gamma_{yz} &= \frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y} = -\sum N_i \theta_{ix} + \sum \frac{\partial N_i}{\partial y} u_{iz} \\ \gamma_{xz} &= \frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} = \sum N_i \theta_{iy} + \sum \frac{\partial N_i}{\partial x} u_{iz}\end{aligned}\quad (5.10)$$

Only five components appear because membrane strains are not included in this plate idealization.

The strain contribution associated with an individual node i can be written as

$$\begin{pmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{yz} \\ \gamma_{xz} \\ \gamma_{xy} \end{pmatrix}_i = \underbrace{\begin{pmatrix} 0 & 0 & z \frac{\partial N_i}{\partial x} \\ 0 & -z \frac{\partial N_i}{\partial y} & 0 \\ \frac{\partial N_i}{\partial y} & -N_i & 0 \\ \frac{\partial N_i}{\partial x} & 0 & N_i \\ 0 & -z \frac{\partial N_i}{\partial x} & z \frac{\partial N_i}{\partial y} \end{pmatrix}}_{B_i} \begin{pmatrix} u_{iz} \\ \theta_{ix} \\ \theta_{iy} \end{pmatrix} \quad (5.11)$$

where B_i is the strain-deformation matrix associated with node i .

The full strain-deformation matrix is assembled as

$$B = [B_1 \ B_2 \ B_3 \ \dots \ B_n] \quad (5.12)$$

Then the strain vector can be obtained in the conventional finite element form as

$$\varepsilon = BU \quad (5.13)$$

The cartesian derivatives in B_i are obtained using the chain rule

$$\frac{\partial N}{\partial x} = \frac{\partial N}{\partial \xi} J^{-1} \quad (5.14)$$

where N is a vector containing the shape functions at a given point.

Assuming a constant thickness h , the stiffness matrix is written as

$$\begin{aligned}K &= \int_A \int_{-\frac{h}{2}}^{\frac{h}{2}} B^T D B \, dz \, dA \\ &= \int_{A'} \int_{-1}^1 B^T D B \frac{h}{2} |J| \, d\zeta \, d\eta\end{aligned}\quad (5.15)$$

Because B depends on the thickness coordinate, the integration through the thickness must still be performed even when h is constant. Here D is the reduced 5×5 constitutive matrix, with a plane-stress block for bending and a transverse-shear block,

$$D = \begin{pmatrix} \frac{E}{1-\nu^2} & \frac{E\nu}{1-\nu^2} & 0 & 0 & 0 \\ \frac{E\nu}{1-\nu^2} & \frac{E}{1-\nu^2} & 0 & 0 & 0 \\ 0 & 0 & \alpha_s \frac{E}{2(1+\nu)} & 0 & 0 \\ 0 & 0 & 0 & \alpha_s \frac{E}{2(1+\nu)} & 0 \\ 0 & 0 & 0 & 0 & \frac{E}{2(1+\nu)} \end{pmatrix} \quad (5.16)$$

The constant $\alpha_s = \frac{5}{6}$ is a shear-correction factor used to improve the prediction of transverse shear stresses.

Note 5.1: The constant α_s arises because the shear strain is assumed to be constant across the thickness, which is only an approximation for real plates.

Using numerical integration, the stiffness matrix is approximated by

$$K \approx \sum_{j=1}^{n_{ip}} B_j^T D B_j \frac{h}{2} |J_j| w_j \quad (5.17)$$

where the integration points include both the in-plane coordinates and the thickness coordinate ζ .

5.2 Membrane element

Membrane elements are used to model thin structures that carry only in-plane forces. They are appropriate when the thickness is much smaller than the other dimensions and bending stiffness is not relevant. Typical applications include reinforcement sheets, geotextiles, and tensile roofs.

Even though the mechanical behavior is two-dimensional, membrane elements are usually placed in three-dimensional space. Each node therefore usually has three translational degrees of freedom. For a node i , the nodal displacement and force vectors in global coordinates are

$$\begin{aligned} U_i &= (u_{ix}, u_{iy}, u_{iz}) \\ F_i &= (f_{ix}, f_{iy}, f_{iz}) \end{aligned} \quad (5.18)$$

To describe the in-plane deformation, a local coordinate system is defined on the membrane surface. The axes x' and y' are tangent to the surface, and the axis z' is normal to it. The membrane strains and stresses are then written in this local system as

$$\begin{aligned} \epsilon &= (\epsilon_{x'}, \epsilon_{y'}, \gamma_{x'y'}) \\ \sigma &= (\sigma_{x'}, \sigma_{y'}, \tau_{x'y'}) \end{aligned} \quad (5.19)$$

Figure 5.8 shows a quadratic membrane element together with the global, local, and natural coordinate systems.

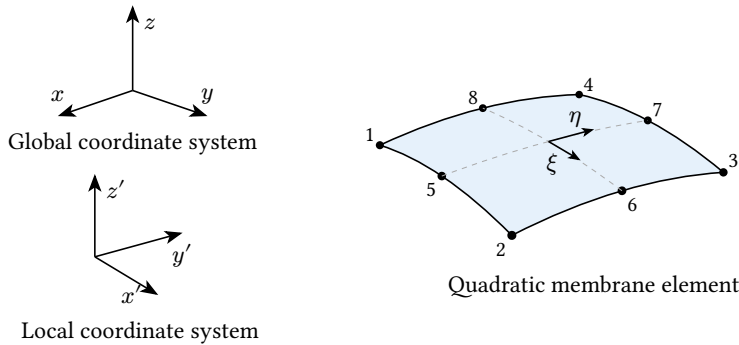


Figure 5.8: A quadratic membrane element displaying three different coordinate systems.

5.2.1 Isoparametric membrane element

Because a membrane element may be curved in space, the mapping from the natural coordinates to the surface is written in the general form

$$\mathbf{J} = \begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \\ \frac{\partial z}{\partial \xi} & \frac{\partial z}{\partial \eta} \end{pmatrix} = \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \xi} \quad (5.20)$$

A local coordinate system is then introduced on the surface. The corresponding Jacobian matrix is

$$\mathbf{J}' = \begin{pmatrix} \frac{\partial x'}{\partial \xi} & \frac{\partial x'}{\partial \eta} \\ \frac{\partial y'}{\partial \xi} & \frac{\partial y'}{\partial \eta} \end{pmatrix} = \mathbf{R} \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \xi} \quad (5.21)$$

where $\mathbf{R}$ is a rotation matrix.

For the membrane formulation, only the two tangent directions are needed, so

$$\mathbf{R} = \begin{pmatrix} l_x & l_y & l_z \\ m_x & m_y & m_z \end{pmatrix} \quad (5.22)$$

The rows of $\mathbf{R}$ are the components of two unit tangent vectors $\mathbf{l}$ and $\mathbf{m}$ defined on the membrane surface. These vectors are obtained from the Jacobian matrix. If $\mathbf{a}$ and $\mathbf{b}$ are the columns of $\mathbf{J}$, the surface normal is

$$\mathbf{n} = \mathbf{a} \times \mathbf{b} \quad (5.23)$$

and the two tangent directions can be built from the unit vectors along $\mathbf{a}$ and $\mathbf{n} \times \mathbf{a}$.

The displacement field is obtained by interpolating the nodal displacements. In the local coordinate system,

$$\begin{pmatrix} u_{x'} \\ u_{y'} \\ u_{z'} \end{pmatrix} = \sum_{i=1}^n N_i \begin{pmatrix} u_{ix'} \\ u_{iy'} \\ u_{iz'} \end{pmatrix} \quad (5.24)$$

Only the in-plane components $u_{x'}$ and $u_{y'}$ contribute to the membrane strains.

The strain components are

$$\begin{aligned} \varepsilon_{x'} &= \frac{\partial u_{x'}}{\partial x'} = \sum \frac{\partial N_i}{\partial x'} u_{ix'} \\ \varepsilon_{y'} &= \frac{\partial u_{y'}}{\partial y'} = \sum \frac{\partial N_i}{\partial y'} u_{iy'} \\ \gamma_{x'y'} &= \frac{\partial u_{x'}}{\partial y'} + \frac{\partial u_{y'}}{\partial x'} = \sum \frac{\partial N_i}{\partial y'} u_{ix'} + \sum \frac{\partial N_i}{\partial x'} u_{iy'} \end{aligned} \quad (5.25)$$

The strain contribution associated with node i can be written as

$$\begin{aligned}
 \begin{pmatrix} \varepsilon_{x'} \\ \varepsilon_{y'} \\ \gamma_{x'y'} \end{pmatrix}_i &= \begin{pmatrix} \frac{\partial N_i}{\partial x'} & 0 \\ 0 & \frac{\partial N_i}{\partial y'} \\ \frac{\partial N_i}{\partial y'} & \frac{\partial N_i}{\partial x'} \end{pmatrix} \begin{pmatrix} u_{ix'} \\ u_{iy'} \end{pmatrix} \\
 &= \underbrace{\begin{pmatrix} \frac{\partial N_i}{\partial x'} & 0 \\ 0 & \frac{\partial N_i}{\partial y'} \\ \frac{\partial N_i}{\partial y'} & \frac{\partial N_i}{\partial x'} \end{pmatrix}}_{B'_i} \underbrace{\begin{pmatrix} l_x & l_y & l_z \\ m_x & m_y & m_z \end{pmatrix}}_{R_i} \underbrace{\begin{pmatrix} u_{ix} \\ u_{iy} \\ u_{iz} \end{pmatrix}}_{U_i}
 \end{aligned} \tag{5.26}$$

The strain-deformation matrix associated with node i is

$$B_i = B'_i R_i \tag{5.27}$$

The full strain-deformation matrix is assembled as

$$B = [B_1 \ B_2 \ B_3 \ \dots \ B_n] \tag{5.28}$$

and the strain vector is written in the usual finite element form as

$$\varepsilon = BU \tag{5.29}$$

The components of B'_i depend on derivatives with respect to the local coordinates. Applying the chain rule,

$$\frac{\partial N}{\partial x'} = \frac{\partial N}{\partial \xi} J'^{-1} \tag{5.30}$$

The stiffness matrix is then written as

$$K = \int_A B^T D B h \, dA \tag{5.31}$$

where h is the membrane thickness and D is the 3×3 constitutive matrix for plane stress.

$$D = \begin{pmatrix} \frac{E}{1-\nu^2} & \frac{E\nu}{1-\nu^2} & 0 \\ \frac{E\nu}{1-\nu^2} & \frac{E}{1-\nu^2} & 0 \\ 0 & 0 & \frac{E}{2(1+\nu)} \end{pmatrix} \tag{5.32}$$

Using numerical integration, the stiffness matrix is approximated by

$$K \approx \sum_{j=1}^{n_{ip}} B_j^T D B_j h |J'_j| w_j \tag{5.33}$$

Since only membrane deformation is considered, the stiffness matrix is proportional to the thickness h and only one layer of integration points is needed through the thickness.

Note 5.2: Membrane elements have no bending stiffness and no out-of-plane stiffness. Therefore, if a mesh made only of membrane elements is not properly constrained out of plane, the system may become hypostatic.

5.3 Shell element

Shell elements are used to model thin curved or flat structures that carry both in-plane forces and bending effects. They combine the main features of membrane and plate elements in a single surface formulation. Because they replace a three-dimensional body by its mid-surface plus a thickness parameter, they are usually more efficient than full 3D solid elements. Typical applications include pressure vessels, silos, aircraft fuselages, pipes, and other thin-walled structures.

In practice, the membrane part describes stretching and in-plane shear, while the plate part describes bending and transverse shear. Depending on the formulation, the bending behavior may follow either Kirchhoff-Love theory or Reissner-Mindlin theory. The formulation developed here follows the Reissner-Mindlin approach.

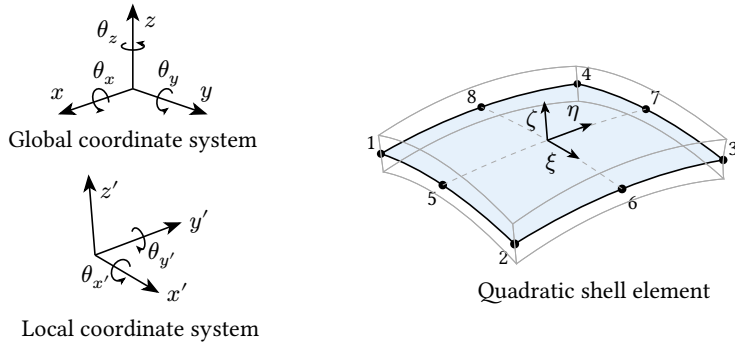


Figure 5.9: A quadratic shell element displaying three different coordinate systems and the degrees of freedom for a node in the global and local coordinate systems.

Many practical shell elements use six nodal degrees of freedom: three translations and three rotations. For a node i , the nodal displacement and force-resultant vectors are

$$\begin{aligned} \mathbf{U}_i &= (u_{ix}, u_{iy}, u_{iz}, \theta_{ix}, \theta_{iy}, \theta_{iz}) \\ \mathbf{F}_i &= (f_{ix}, f_{iy}, f_{iz}, M_{ix}, M_{iy}, M_{iz}) \end{aligned} \quad (5.34)$$

The rotation about the shell normal, often called the drilling rotation, is commonly included as an additional nodal degree of freedom even though it does not appear directly in the basic strain field written below.

As in the membrane formulation, a local coordinate system is defined on the shell surface. The local strain and stress components at a point are

$$\begin{aligned} \boldsymbol{\varepsilon} &= (\varepsilon_{x'}, \varepsilon_{y'}, \gamma_{y'z'}, \gamma_{x'z'}, \gamma_{x'y'}) \\ \boldsymbol{\sigma} &= (\sigma_{x'}, \sigma_{y'}, \tau_{y'z'}, \tau_{x'z'}, \tau_{x'y'}) \end{aligned} \quad (5.35)$$

5.3.1 Isoparametric shell element

Isoparametric shell elements are often called degenerated shell elements because they can be viewed as a reduced form of a 3D solid element, with the thickness introduced as a parameter.

The mid-surface mapping is first defined from the natural coordinates to the global coordinates by

$$\mathbf{J} = \frac{\partial \mathbf{x}}{\partial \boldsymbol{\xi}} = \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \boldsymbol{\xi}} \quad (5.36)$$

where

$$\mathbf{X} = \begin{pmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \\ \vdots & \vdots & \vdots \\ x_n & y_n & z_n \end{pmatrix} \quad \frac{\partial \mathbf{N}}{\partial \boldsymbol{\xi}} = \begin{pmatrix} \frac{\partial N_1}{\partial \xi} & \frac{\partial N_1}{\partial \eta} \\ \frac{\partial N_2}{\partial \xi} & \frac{\partial N_2}{\partial \eta} \\ \vdots & \vdots \\ \frac{\partial N_n}{\partial \xi} & \frac{\partial N_n}{\partial \eta} \end{pmatrix} \quad (5.37)$$

At each point of the mid-surface, a local orthonormal basis $\mathbf{l}$, $\mathbf{m}$, and $\mathbf{n}$ is introduced. The corresponding rotation matrix is

$$\mathbf{R} = \begin{pmatrix} l_x & l_y & l_z \\ m_x & m_y & m_z \\ n_x & n_y & n_z \end{pmatrix} \quad (5.38)$$

The vectors $\mathbf{l}$ and $\mathbf{m}$ are tangent to the shell surface, and $\mathbf{n}$ is normal to it. They are obtained from the Jacobian matrix $\mathbf{J}$ in the same way as for the membrane element.

The local Jacobian matrix, including the thickness coordinate, is then written as

$$\mathbf{J}'(\xi, \eta, \zeta) = \begin{pmatrix} \frac{\partial x'}{\partial \xi} & \frac{\partial x'}{\partial \eta} & \frac{\partial x'}{\partial \zeta} \\ \frac{\partial y'}{\partial \xi} & \frac{\partial y'}{\partial \eta} & \frac{\partial y'}{\partial \zeta} \\ \frac{\partial z'}{\partial \xi} & \frac{\partial z'}{\partial \eta} & \frac{\partial z'}{\partial \zeta} \end{pmatrix} = \begin{pmatrix} \mathbf{R} \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \boldsymbol{\xi}} & \begin{pmatrix} 0 \\ 0 \\ \frac{h}{2} \end{pmatrix} \end{pmatrix} \quad (5.39)$$

Consider a point located at a distance z from the mid-surface, with

$$z = \frac{h}{2} \zeta, \quad \zeta \in [-1, 1] \quad (5.40)$$

For a Reissner-Mindlin shell, the displacement field in local coordinates is

$$\begin{pmatrix} u_{x'} \\ u_{y'} \\ u_{z'} \end{pmatrix} = \sum_{i=1}^n N_i \begin{pmatrix} u_{ix'} \\ u_{iy'} \\ u_{iz'} \end{pmatrix} + \sum_{i=1}^n N_i z \begin{pmatrix} \theta_{iy'} \\ -\theta_{ix'} \\ 0 \end{pmatrix} \quad (5.41)$$

The first term describes the displacement of the mid-surface, and the second term accounts for the displacement away from the mid-surface caused by the nodal rotations.

From this displacement field, the in-plane strain components are

$$\begin{aligned} \varepsilon_{x'} &= \frac{\partial u_{x'}}{\partial x'} = \sum \frac{\partial N_i}{\partial x'} u_{ix'} + z \sum \frac{\partial N_i}{\partial x'} \theta_{iy'} \\ \varepsilon_{y'} &= \frac{\partial u_{y'}}{\partial y'} = \sum \frac{\partial N_i}{\partial y'} u_{iy'} - z \sum \frac{\partial N_i}{\partial y'} \theta_{ix'} \\ \gamma_{x'y'} &= \frac{\partial u_{x'}}{\partial y'} + \frac{\partial u_{y'}}{\partial x'} = \sum \frac{\partial N_i}{\partial y'} u_{ix'} + \sum \frac{\partial N_i}{\partial x'} u_{iy'} + z \sum \frac{\partial N_i}{\partial y'} \theta_{iy'} - z \sum \frac{\partial N_i}{\partial x'} \theta_{ix'} \end{aligned} \quad (5.42)$$

The transverse shear components are

$$\begin{aligned}\gamma_{y'z'} &= \frac{\partial u_{y'}}{\partial z'} + \frac{\partial u_{z'}}{\partial y'} = -\sum N_i \theta_{ix'} + \sum \frac{\partial N_i}{\partial y'} u_{iz'} \\ \gamma_{x'z'} &= \frac{\partial u_{x'}}{\partial z'} + \frac{\partial u_{z'}}{\partial x'} = \sum N_i \theta_{iy'} + \sum \frac{\partial N_i}{\partial x'} u_{iz'}\end{aligned}\quad (5.43)$$

Collecting these terms in matrix form gives

$$\varepsilon = BU \quad (5.44)$$

where B is the strain-deformation matrix of the shell element.

The strain contribution associated with node i is

$$\begin{pmatrix} \varepsilon_{x'} \\ \varepsilon_{y'} \\ \gamma_{y'z'} \\ \gamma_{x'z'} \\ \gamma_{x'y'} \end{pmatrix}_i = \underbrace{\begin{pmatrix} \frac{\partial N_i}{\partial x'} & 0 & 0 & 0 & z \frac{\partial N_i}{\partial x'} \\ 0 & \frac{\partial N_i}{\partial y'} & 0 & -z \frac{\partial N_i}{\partial y'} & 0 \\ 0 & 0 & \frac{\partial N_i}{\partial y'} & -N_i & 0 \\ 0 & 0 & \frac{\partial N_i}{\partial x'} & 0 & N_i \\ \frac{\partial N_i}{\partial y'} & \frac{\partial N_i}{\partial x'} & 0 & -z \frac{\partial N_i}{\partial x'} & z \frac{\partial N_i}{\partial y'} \end{pmatrix}}_{B'_i} \underbrace{\begin{pmatrix} l_x & l_y & l_z & 0 & 0 & 0 \\ m_x & m_y & m_z & 0 & 0 & 0 \\ n_x & n_y & n_z & 0 & 0 & 0 \\ 0 & 0 & 0 & l_x & l_y & l_z \\ 0 & 0 & 0 & m_x & m_y & m_z \end{pmatrix}}_{\bar{R}_i} \begin{pmatrix} u_{ix} \\ u_{iy} \\ u_{iz} \\ \theta_{ix} \\ \theta_{iy} \\ \theta_{iz} \end{pmatrix} \quad (5.45)$$

In this expression, B'_i is the local strain-deformation matrix and $\bar{R}_i$ transforms translations and rotations from the global coordinate system to the local coordinate system. Thus,

$$B_i = B'_i \bar{R}_i \quad (5.46)$$

The full strain-deformation matrix is assembled as

$$B = [B_1 \ B_2 \ B_3 \ \dots \ B_n] \quad (5.47)$$

The derivatives with respect to the local coordinates are obtained from

$$\frac{\partial N}{\partial x'} = \frac{\partial N}{\partial \xi} J'^{-1} \quad (5.48)$$

The stiffness matrix is written as

$$K = \int_V B^T D B dV \quad (5.49)$$

where D is the reduced constitutive matrix for plane stress with transverse shear, as in the plate formulation.

Using numerical integration,

$$K \approx \sum_{j=1}^{n_{ip}} B_j^T D B_j w_j \det(J'_j) \quad (5.50)$$

Note 5.3: In many shell formulations, the rotation about the normal direction has little or no physical stiffness. This rotation is called the **drilling rotation**. If it is not treated properly, the global system may become singular or nearly singular, especially when adjacent shell elements have similar

orientations. Common remedies include stabilization terms, additional constraints, or shell elements with drilling control.

5.4 Curved beam elements

Curved beam elements are used to model members whose axis is curved in two-dimensional or three-dimensional space. They capture axial deformation, bending, torsion, and shear effects associated with both the geometry and the applied loads. Typical applications include arches, rings, and other curved structural members.

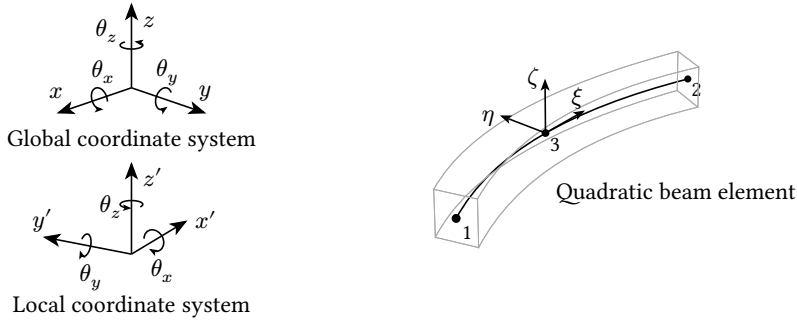


Figure 5.10: A curved beam element with global, local, and natural coordinate systems.

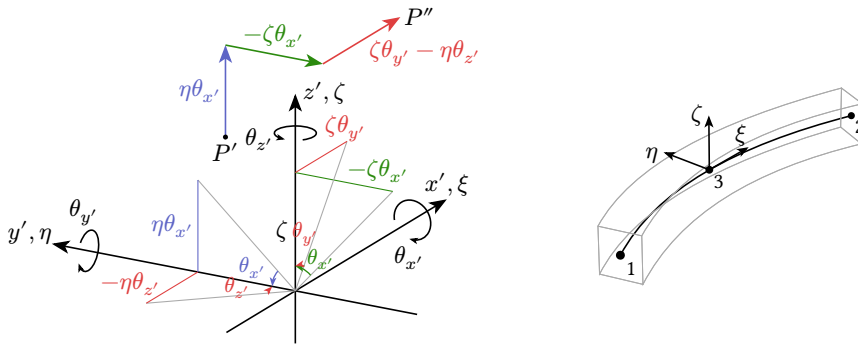


Figure 5.11: Displacements of a point in the cross-section of a curved beam.

In the local coordinate system of the beam, the strain and stress components are

$$\begin{aligned}\epsilon &= (\epsilon_{x'}, \gamma_{x'z'}, \gamma_{x'y'}) \\ \sigma &= (\sigma_{x'}, \tau_{x'z'}, \tau_{x'y'})\end{aligned}\tag{5.51}$$

As in shell elements, each node of a beam element usually has six degrees of freedom: three translations and three rotations. For a node i , the nodal displacement and force-resultant vectors are

$$\begin{aligned}U_i &= (u_{ix}, u_{iy}, u_{iz}, \theta_{ix}, \theta_{iy}, \theta_{iz}) \\ F_i &= (f_{ix}, f_{iy}, f_{iz}, m_{ix}, m_{iy}, m_{iz})\end{aligned}\tag{5.52}$$

5.4.1 Isoparametric beam element

In a degenerated beam formulation, the element is described by its axis and by the dimensions of its cross-section. The local coordinate system x' , y' , and z' is chosen so that x' follows the beam axis, while y' and z' describe the cross-section.

For an isoparametric beam element, the Jacobian associated with the beam axis is the vector

$$\mathbf{J} = \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \xi} \quad (5.53)$$

The local geometry is then approximated by

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} x'(\xi) \\ r_{y'}(\xi)\eta \\ r_{z'}(\xi)\zeta \end{pmatrix} \quad (5.54)$$

where $r_{y'}(\xi)$ and $r_{z'}(\xi)$ are the half-dimensions of the cross-section in the y' and z' directions. For a prismatic beam, these quantities are constant; for a tapered beam, they may vary with ξ . In the simplified formulation presented here, the beam is assumed to be prismatic, so $r_{y'}$ and $r_{z'}$ are constant.

With this simplified mapping, the local Jacobian matrix is

$$\mathbf{J}' = \begin{pmatrix} \frac{\partial x'}{\partial \xi} & 0 & 0 \\ 0 & r_{y'} & 0 \\ 0 & 0 & r_{z'} \end{pmatrix} \quad (5.55)$$

Since x' is interpolated along the beam axis,

$$J = \frac{\partial x'}{\partial \xi} = \left\| \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \xi} \right\| \quad (5.56)$$

and therefore

$$\mathbf{J}' = \begin{pmatrix} J & 0 & 0 \\ 0 & r_{y'} & 0 \\ 0 & 0 & r_{z'} \end{pmatrix} \quad (5.57)$$

The displacement field is obtained by combining the interpolation of the nodal displacements with the displacement of points away from the beam axis caused by the section rotations:

$$\begin{pmatrix} u_{x'} \\ u_{y'} \\ u_{z'} \end{pmatrix} = \sum_{i=1}^n N_i \begin{pmatrix} u_{ix'} \\ u_{iy'} \\ u_{iz'} \end{pmatrix} + \sum_{i=1}^n N_i \begin{pmatrix} -r_{y'}\eta\theta_{iz'} + r_{z'}\zeta\theta_{iy'} \\ -r_{z'}\zeta\theta_{ix'} \\ r_{y'}\eta\theta_{ix'} \end{pmatrix} \quad (5.58)$$

The axial strain is

$$\varepsilon_{x'} = \frac{\partial u_{x'}}{\partial x'} = \sum \frac{\partial N_i}{\partial x'} u_{ix'} - \sum \frac{\partial N_i}{\partial x'} r_{y'}\eta\theta_{iz'} + \sum \frac{\partial N_i}{\partial x'} r_{z'}\zeta\theta_{iy'} \quad (5.59)$$

The shear strain components are

$$\begin{aligned} \gamma_{x'z'} &= \frac{\partial u_{x'}}{\partial z'} + \frac{\partial u_{z'}}{\partial x'} = \sum N_i \theta_{iy'} + \sum \frac{\partial N_i}{\partial x'} u_{iz'} + \sum \frac{\partial N_i}{\partial x'} r_{y'}\eta\theta_{ix'} \\ \gamma_{x'y'} &= \frac{\partial u_{x'}}{\partial y'} + \frac{\partial u_{y'}}{\partial x'} = -\sum N_i \theta_{iz'} + \sum \frac{\partial N_i}{\partial x'} u_{iy'} - \sum \frac{\partial N_i}{\partial x'} r_{z'}\zeta\theta_{ix'} \end{aligned} \quad (5.60)$$

These equations can be written in matrix form. In the local system,

$$\begin{pmatrix} \varepsilon_{x'} \\ \gamma_{x'z'} \\ \gamma_{x'y'} \end{pmatrix} = \mathbf{B}' \begin{pmatrix} u_{1x'} \\ u_{1y'} \\ u_{1z'} \\ \theta_{1x'} \\ \theta_{1y'} \\ \theta_{1z'} \\ \vdots \end{pmatrix} \quad (5.61)$$

The matrix $\mathbf{B}'$ is assembled from the nodal submatrices $\mathbf{B}'_i$, where

$$\mathbf{B}'_i = \begin{pmatrix} \frac{\partial N_i}{\partial x'} & 0 & 0 & 0 & \frac{\partial N_i}{\partial x'} r_{z'} \zeta & -\frac{\partial N_i}{\partial x'} r_{y'} \eta \\ 0 & 0 & \frac{\partial N_i}{\partial x'} & \frac{\partial N_i}{\partial x'} r_{y'} \eta & N_i & 0 \\ 0 & \frac{\partial N_i}{\partial x'} & 0 & -\frac{\partial N_i}{\partial x'} r_{z'} \zeta & 0 & -N_i \end{pmatrix} \quad (5.62)$$

The terms $\frac{\partial N_i}{\partial x'}$ are obtained from

$$\frac{\partial N}{\partial x'} = \frac{\partial N}{\partial \xi} \frac{1}{J} \quad (5.63)$$

where $J = \|\mathbf{J}\|$.

To express the strains in terms of the nodal displacements written in the global system, the local matrix is multiplied by a transformation matrix:

$$\mathbf{B}_i = \mathbf{B}'_i \bar{\mathbf{R}}_i \quad (5.64)$$

with

$$\bar{\mathbf{R}}_i = \begin{pmatrix} l_x & l_y & l_z & 0 & 0 & 0 \\ m_x & m_y & m_z & 0 & 0 & 0 \\ n_x & n_y & n_z & 0 & 0 & 0 \\ 0 & 0 & 0 & l_x & l_y & l_z \\ 0 & 0 & 0 & m_x & m_y & m_z \\ 0 & 0 & 0 & n_x & n_y & n_z \end{pmatrix} \quad (5.65)$$

The full strain-deformation matrix is then

$$\mathbf{B} = [\mathbf{B}_1 \mathbf{B}_2 \mathbf{B}_3 \dots \mathbf{B}_n] \quad (5.66)$$

The stiffness matrix of the beam element is written as

$$\mathbf{K}_{\text{beam}} = \int_V \mathbf{B}^T \mathbf{D} \mathbf{B} dV \quad (5.67)$$

where $\mathbf{B}$ is the deformation matrix and $\mathbf{D}$ is the constitutive matrix. For an elastic material,

$$\mathbf{D} = \begin{pmatrix} E & 0 & 0 \\ 0 & \alpha_s \frac{E}{2(1+\nu)} & 0 \\ 0 & 0 & \alpha_s \frac{E}{2(1+\nu)} \end{pmatrix} \quad (5.68)$$

Using numerical integration,

$$K_{\text{beam}} = \sum_{i=1}^{n_{\text{ip}}} B_i^T D B_i w_i J_i' \quad (5.69)$$

where $J_i' = \left\| \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \xi} \right\| r_{y'} r_{z'}$. For this type of element, at least two integration points are usually required in the directions normal to the beam axis.

5.5 Interface elements

Interface elements, also called zero-thickness elements, are used to model the mechanical behavior of surfaces that connect two bodies or two parts of the same body. They are useful for joints, bonded interfaces, contact zones, and potential crack paths.

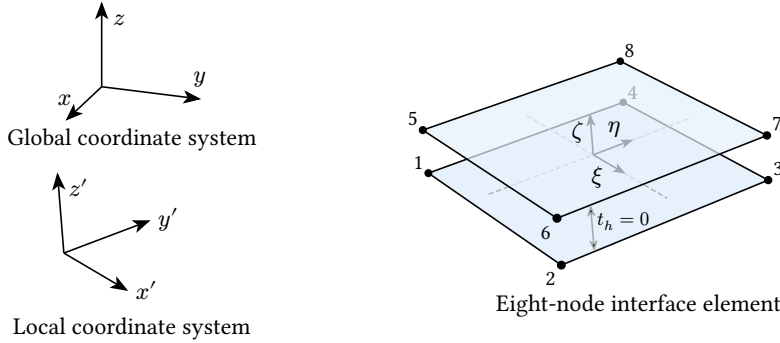


Figure 5.12: An interface element with global, local, and natural coordinate systems.

These elements can represent opening and sliding between the two faces of the interface. They are therefore suitable for modeling mode-I opening, mode-II shear, friction, cohesion, and debonding. Specialized constitutive laws, such as cohesive zone models, usually relate interface tractions to relative displacements.

The response is described in a local coordinate system x' , y' , and z' , where z' is normal to the interface and x' and y' lie on the interface surface. At a point of the interface, the traction vector is

$$\mathbf{t} = \begin{pmatrix} \tau_{x'} \\ \tau_{y'} \\ \sigma_{z'} \end{pmatrix} \quad (5.70)$$

For a node i , the nodal displacement and force vectors are

$$\begin{aligned} \mathbf{U}_i &= (u_{ix}, u_{iy}, u_{iz}) \\ \mathbf{F}_i &= (f_{ix}, f_{iy}, f_{iz}) \end{aligned} \quad (5.71)$$

5.5.1 Isoparametric interface element

Because an interface element may be curved in space, the surface mapping is written in the general form

$$\mathbf{J} = \begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \\ \frac{\partial z}{\partial \xi} & \frac{\partial z}{\partial \eta} \end{pmatrix} = \mathbf{X}^T \frac{\partial \mathbf{N}}{\partial \xi} \quad (5.72)$$

Because the two faces may separate or slide relative to each other, the response is described by relative displacements rather than by continuum strains.

Half of the nodes belong to the bottom face and the other half belong to the top face. The displacement field on the bottom face is

$$\begin{pmatrix} u_{x'} \\ u_{y'} \\ u_{z'} \end{pmatrix}_{\text{bottom}} = \sum_{i=1}^{n/2} N_i \begin{pmatrix} u_{ix'} \\ u_{iy'} \\ u_{iz'} \end{pmatrix} \quad (5.73)$$

and the displacement field on the top face is

$$\begin{pmatrix} u_{x'} \\ u_{y'} \\ u_{z'} \end{pmatrix}_{\text{top}} = \sum_{i=n/2+1}^n N_i \begin{pmatrix} u_{ix'} \\ u_{iy'} \\ u_{iz'} \end{pmatrix} \quad (5.74)$$

The relative displacement field, defined as top minus bottom, is

$$\begin{pmatrix} w_{x'} \\ w_{y'} \\ w_{z'} \end{pmatrix} = \sum_{i=n/2+1}^n N_i \begin{pmatrix} u_{ix'} \\ u_{iy'} \\ u_{iz'} \end{pmatrix} - \sum_{i=1}^{n/2} N_i \begin{pmatrix} u_{ix'} \\ u_{iy'} \\ u_{iz'} \end{pmatrix} \quad (5.75)$$

This expression can be written as

$$\mathbf{w} = \mathbf{N}\mathbf{U}' \quad (5.76)$$

where $\mathbf{U}'$ is the vector of nodal displacements in the local system and $\mathbf{N}$ is

$$\mathbf{N} = \begin{pmatrix} -N_1 & 0 & 0 & \cdots & -N_{\frac{n}{2}} & 0 & 0 & N_1 & 0 & 0 & \cdots & N_{\frac{n}{2}} & 0 & 0 \\ 0 & -N_1 & 0 & \cdots & 0 & -N_{\frac{n}{2}} & 0 & 0 & N_1 & 0 & \cdots & 0 & N_{\frac{n}{2}} & 0 \\ 0 & 0 & -N_1 & \cdots & 0 & 0 & -N_{\frac{n}{2}} & 0 & 0 & N_1 & \cdots & 0 & 0 & N_{\frac{n}{2}} \end{pmatrix} \quad (5.77)$$

Using a rotation matrix, the relative displacement vector can be written in terms of the nodal displacements in the global coordinate system as

$$\mathbf{w}' = \mathbf{R}\mathbf{N}\mathbf{U} \quad (5.78)$$

where $\mathbf{R}$ is given by

$$\mathbf{R} = \begin{pmatrix} l_x & l_y & l_z \\ m_x & m_y & m_z \\ n_x & n_y & n_z \end{pmatrix} \quad (5.79)$$

Again, the components of the rotation matrix are obtained from the Jacobian matrix.

Therefore,

$$\mathbf{B} = \mathbf{R}\mathbf{N} \quad (5.80)$$

and the kinematic relation becomes

$$\mathbf{w}' = \mathbf{B}\mathbf{U} \quad (5.81)$$

The matrix $\mathbf{B}$ transforms nodal displacements into relative displacements along the interface.

The interface tractions are related to the relative displacements through the constitutive matrix $\mathbf{D}$:

$$D = \begin{pmatrix} k_s & 0 & 0 \\ 0 & k_s & 0 \\ 0 & 0 & k_n \end{pmatrix} \quad (5.82)$$

Here k_s is the shear stiffness and k_n is the normal stiffness, both with units of traction per unit displacement.

The stiffness matrix of the interface element is written as

$$K = \int_A B^T D B dA \quad (5.83)$$

Using numerical integration, this matrix is approximated by

$$K = \sum_{i=1}^{n_{ip}} B_i^T D_i B_i w_i J_i \quad (5.84)$$

Since the element has zero thickness, the integration is performed over the interface area.

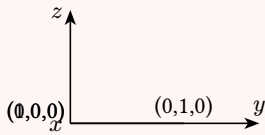
Since J is not a square matrix, the surface Jacobian is obtained from the norm of the cross product of its two columns:

$$J = \|a \times b\| \quad (5.85)$$

where a and b are the columns of the Jacobian matrix J .

Exercises 5.1:

1. List four applications for each of the following element types: plate, membrane, shell, and interface.
2. Compute the B matrix of the element below at the parametric coordinate $(\xi, \eta, \zeta) = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{\sqrt{3}}\right)$:
(a) as a membrane element; (b) as a shell element. Consider the thickness equal to $t = 0.1$.



3. Simplify the 3D formulation for isoparametric beam elements in order to derive the formulation for an isoparametric 2D beam element.
4. Derive the formulation for a 2D interface element.

6 Pre- and post-processing

Finite element analysis depends not only on the mathematical formulation and the solver, but also on the quality of the steps that precede and follow the numerical solution. Before the system of equations can be solved, the analyst must define an appropriate geometric model, select material properties, generate a mesh, and prescribe loads and boundary conditions. After the solution is obtained, the results at integration points must be recovered at nodes, visualized, and interpreted critically so that the model can be verified and, if necessary, improved. This chapter introduces the main tasks involved in pre-processing and post-processing and highlights their role in producing reliable finite element simulations. The main stages of this workflow are summarized in [Figure 6.1](#).

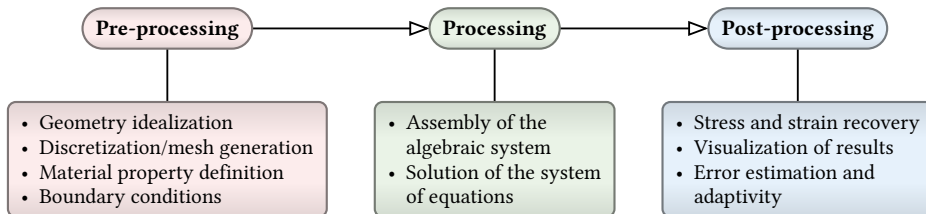


Figure 6.1: Main stages of a finite element analysis, from model definition and discretization to system solution and post-processing of results.

6.1 Analysis geometry modeling and idealization

In finite element analysis, one often starts directly from a reduced analysis geometry rather than from the full physical geometry. Depending on the problem, the model may be represented by lines, surfaces, or simplified solids chosen to capture the dominant mechanical behavior with an appropriate computational cost. Slender members are commonly modeled as lines and analyzed with beam or truss elements, thin-walled components are often represented by surfaces and analyzed with plate or shell elements, and fully three-dimensional continua may still be simplified before meshing. The objective is to choose the lowest-dimensional or simplest geometric representation that still captures the relevant deformation modes and stress resultants while preserving the features needed to define dimensions, interfaces, supports, and loading regions.

The analysis geometry may be created directly in the pre-processor or derived from a more detailed CAD model and then prepared for finite element analysis. In either case, the model is typically organized in terms of edges, surfaces, and volumes together with their topological connectivity. An example of a three-dimensional geometry already prepared for analysis is shown in [Figure 6.2](#).

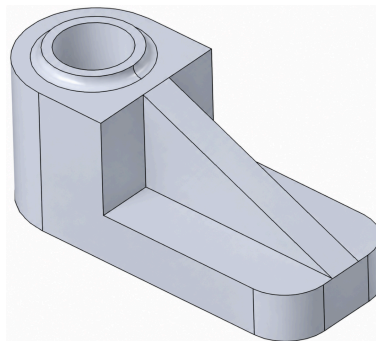


Figure 6.2: Geometry model of a bearing support prepared for finite element analysis.

When the finite element model is derived from a detailed CAD model, common preparation tasks include defeaturing and topology cleanup. Small geometric details that have little influence on the structural response are often removed when they would force unnecessary mesh refinement, and the

topology is cleaned to obtain a consistent model for meshing. In many analysis-oriented workflows, however, the geometry is created directly in a simplified form suitable for discretization, which reduces or eliminates the need for later defeaturing. Once the analysis geometry has been defined, **partitioning** and **imprinting** are often required to create clean subregions and interfaces where loads, supports, material changes, contact conditions, or local mesh refinement must be applied.

6.2 Discretization and mesh generation

Discretization converts the analysis geometry into a finite set of elements connected at nodes, replacing the continuous domain with a discrete model suitable for numerical solution. Starting from the same geometry, different meshes may be generated depending on the selected element family, interpolation order, and element size distribution. The conversion of a prepared geometric model into a finite element mesh is illustrated in **Figure 6.3**.

The meshing strategy must balance accuracy, robustness, and computational cost. A coarse mesh may be sufficient in regions where the solution varies smoothly, whereas local refinement is usually required near stress concentrations, sharp gradients, contact zones, crack tips, material interfaces, and boundary layers. Gradual transitions in element size are generally preferred, since abrupt changes can reduce accuracy and deteriorate element quality.

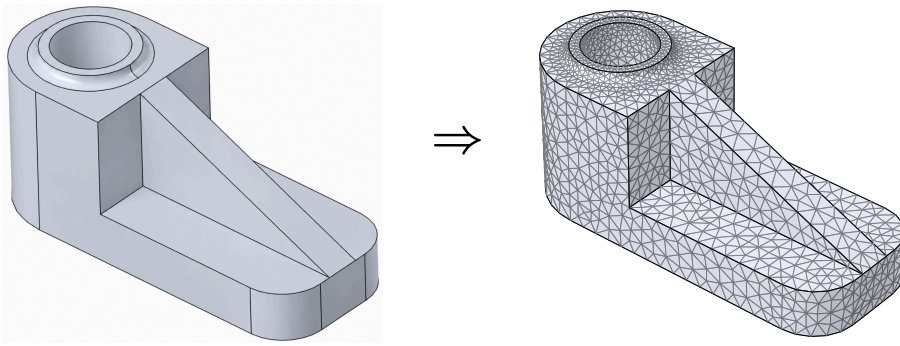


Figure 6.3: Conversion of the CAD geometry into a finite element mesh.

6.2.1 Mesh generation techniques

Mesh generation may follow either **structured** or **unstructured** strategies. Structured meshes follow a regular connectivity pattern and are common in simple domains, especially when quadrilateral or hexahedral elements are desired. They usually provide better control of alignment and element quality, but they are less flexible for complex geometries. Unstructured meshes are more adaptable to arbitrary shapes and are therefore the standard choice for complicated engineering models. Examples of these approaches are shown in **Figure 6.4**.

Automatic mesh generators often rely on **advancing-front**, **Delaunay**, or **octree-based** algorithms. The first two are especially common in unstructured surface and volume meshing, whereas octree-based methods are particularly attractive for complex three-dimensional domains, especially when combined with surface projection and local refinement rules.

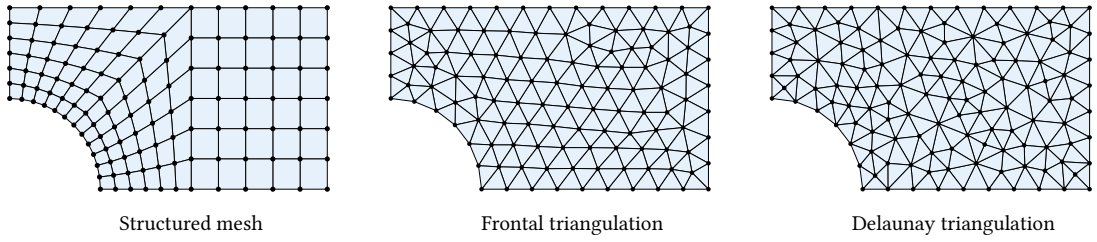


Figure 6.4: Mesh discretization techniques.

Among the unstructured techniques illustrated in Figure 6.4, frontal triangulation, also called advancing-front triangulation, starts from the discretized boundary and progressively fills the interior by attaching elements to the current front. This usually provides excellent control of boundary conformity, element size grading, and local directional refinement. Delaunay triangulation, in contrast, is based on the **empty-circumcircle criterion**; in two dimensions it tends to maximize the minimum angle, reducing the occurrence of very acute triangles and leading to robust automatic meshing procedures.

Several important variants are built on the same Delaunay idea. **Constrained Delaunay** triangulation forces prescribed boundary segments, holes, or internal interfaces to appear in the final mesh, which is essential when geometric boundaries or material interfaces must be represented exactly. In three dimensions, the corresponding procedure is **Delaunay tetrahedralization**, which is widely used for automatic tetrahedral mesh generation, although additional post-processing may still be needed to remove sliver elements and improve mesh quality.

6.2.2 Element quality metrics

Element quality metrics are used to assess whether the mesh is suitable for accurate and stable finite element computations. Poor-quality elements may lead to inaccurate stresses, slow convergence, or even invalid mappings between the parent and physical elements. Common indicators include aspect ratio, skewness, angular distortion, warping, and Jacobian-based distortion measures.

The aspect ratio measures how elongated an element is and is often written as

$$q_a = \frac{l_{\min}}{l_{\max}} \quad (6.1)$$

where $l_{\min}$ and $l_{\max}$ are the minimum and maximum characteristic lengths. For triangular elements, a common area-based shape metric is

$$q = \frac{4\sqrt{3}A}{\sum l_i^2} \quad (6.2)$$

which is equal to 1 for an equilateral triangle and decreases as the element becomes distorted. More generally, perimeter-based measures may be defined from the perimeter of an ideal reference polygon with the same area:

$$q_2(e) = \left(\frac{p_r}{p} \right)^\beta \quad (6.3)$$

where p is the perimeter of the element, p_r is the perimeter of the regular polygon with the same area, and $\beta > 1$ is a tuning exponent.

Among these indicators, the Jacobian determinant is especially important because it must remain positive throughout the element. At the sampling or integration points, a common normalized measure is

$$q_J = \frac{\min_k J_k}{\max_k J_k} \quad (6.4)$$

with $J_k > 0$ for all points k . For shell and quadrilateral-type elements, warping measures the loss of planarity of the element faces and may be defined as

$$q_w = |\mathbf{n}_1 \cdot \mathbf{n}_2| = \cos \omega \quad (6.5)$$

where $\mathbf{n}_1$ and $\mathbf{n}_2$ are normals to two triangles formed from the same quadrilateral face.

For three-dimensional elements, surface-based measures compare the area s of the element boundary with the area s_r of an ideal reference polyhedron with the same volume:

$$q_3(e) = \left(\frac{s_r}{s} \right)^\beta \quad (6.6)$$

In practice, these quantities are interpreted together with the element type and the expected deformation pattern: elongated elements may be acceptable when aligned with the solution field, whereas strongly distorted or inverted elements are unacceptable.

6.2.3 Mesh smoothing

After an initial mesh has been generated, additional operations may be applied to improve element quality without changing the physical model. Mesh smoothing relocates nodes in order to reduce distortion, improve shape regularity, and produce a more uniform quality field.

One of the simplest and most widely used techniques is **Laplacian smoothing**, in which an interior node is moved toward an average position defined by the neighboring nodes in the patch of elements that share that node. This procedure is inexpensive and often effective for mild distortions, but if used without care it may degrade boundary fidelity or even create invalid elements.

More advanced approaches include **optimization-based** and **physics-based** smoothing. In optimization-based smoothing, the nodal positions are updated so as to improve a selected quality functional, usually related to aspect ratio, Jacobian distortion, or angular regularity. In physics-based smoothing, the mesh is treated as a fictitious elastic or diffusive continuum, and the nodal displacements are obtained from an auxiliary deformation analysis. This tends to produce smoother global redistributions while better respecting geometric constraints. An example of physics-based smoothing applied to a coarse hexahedral mesh is shown in **Figure 6.5**, where the quality field becomes more uniform and the number of low-quality elements is reduced.

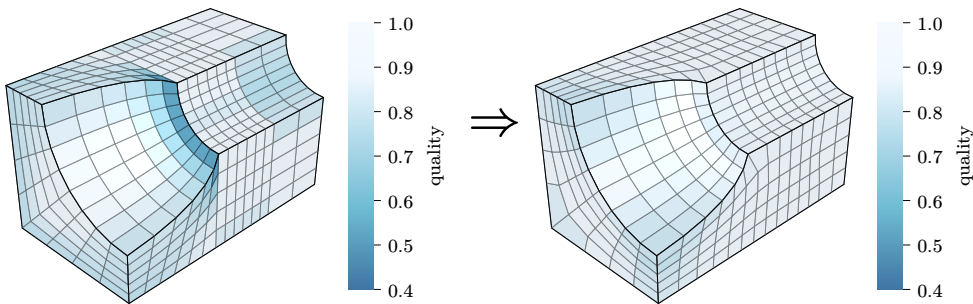


Figure 6.5: Comparison of the element quality field before and after mesh smoothing.

6.2.4 Node and Element Numbering

Node and element numbering affect storage requirements, solver efficiency, and implementation consistency. The local numbering within each element must preserve a consistent orientation so that

the Jacobian remains positive and boundary conditions, loads, and post-processing quantities are assembled correctly.

At the global level, a proper nodal numbering reduces the **bandwidth** and profile of the sparse stiffness matrix, which lowers memory consumption and can accelerate the solution of the algebraic system, especially with direct solvers. The Cuthill-McKee algorithm, and especially its reverse variant, is widely used for this purpose because it reorders the connectivity graph so that coupled degrees of freedom are placed closer together in the matrix.

The resulting numbering does not necessarily follow an obvious geometric progression such as left-to-right or bottom-to-top. Instead, the reordered sequence reflects the adjacency structure of the mesh graph and is chosen to improve matrix sparsity properties. This effect is illustrated in **Figure 6.6**, where neighboring nodes tend to receive nearby labels even though the visible numbering pattern may appear irregular from a purely geometric point of view.

Element numbering is also important from a computational point of view. When neighboring elements are stored in contiguous or nearby regions of memory, element-level operations such as assembly, stress recovery, and internal-variable updates tend to access data more efficiently, which improves cache usage and reduces memory-access cost.

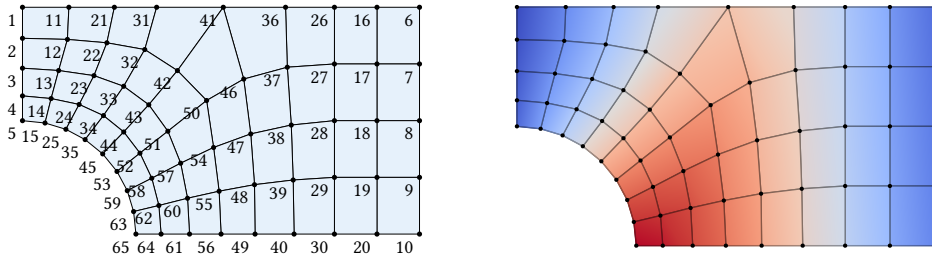


Figure 6.6: Effect of reverse Cuthill-McKee reordering on the node numbering of a finite element mesh. The corresponding nodal order is shown as a color field.

6.3 Material property definition

Material property definition specifies the constitutive law that relates stresses, strains, and, when relevant, temperature, time, damage, or other internal variables. The chosen model must be compatible with the expected response of the structure and with the available experimental data.

In many analyses, a linear elastic model is sufficient and is defined by constants such as Young's modulus E , Poisson's ratio ν , and, in some cases, the shear modulus G . When permanent deformation is important, elastoplastic models are used to represent yielding, hardening, and possible unloading and reloading behavior. For rubber-like materials and large-strain applications, hyperelastic models are more appropriate because they describe strongly nonlinear stress-strain relations.

The directional dependence of the material must also be defined. Isotropic materials have the same properties in every direction, orthotropic materials have distinct properties along three preferred orthogonal directions, and anisotropic materials have general direction-dependent behavior. This choice is essential in composites, wood, layered media, rolled metals, soils, rocks, and other materials whose internal structure governs the mechanical response.

6.4 Boundary conditions

Boundary conditions define how the structure interacts with its surroundings and are essential for obtaining a well-posed finite element model. They restrict rigid-body motions, represent supports and connections, and prescribe the loads or kinematic constraints that drive the response.

Whenever the geometry, loading, and material distribution are symmetric, symmetry conditions may be used to reduce the size of the model. On a symmetry plane, the displacement normal to the plane is set to zero, whereas the tangential components remain free unless additional physical constraints are present. This reduces computational cost without changing the solution, provided that the symmetry assumptions are valid. **Figure 6.7** illustrates this idea for a uniformly loaded beam: the full model can be replaced by half of the domain, and the cut boundary is assigned roller-type constraints that suppress only the displacement normal to the plane of symmetry.

In geotechnical analyses, the computational domain often represents only a finite portion of a much larger soil mass, so artificial boundaries must be placed sufficiently far from the region of interest. A common choice is to impose roller supports on the vertical lateral boundaries, preventing displacement normal to the boundary while allowing vertical movement. At the bottom boundary, vertical displacement is typically restrained, and horizontal displacement may also be restrained when a fully fixed base is justified. These conventional boundary conditions aim to reproduce the confinement of the surrounding ground while minimizing spurious boundary effects near the zone of interest. The recommended arrangement is shown in **Figure 6.8** for a slope problem, where the lateral boundaries are placed several characteristic heights away from the region of interest and only the normal displacement is restrained along those artificial sides.

The same reasoning applies to laboratory specimens and other three-dimensional solids when geometric and loading symmetries are present. In such cases, only a fraction of the specimen needs to be modeled, provided that the corresponding symmetry planes are assigned the appropriate displacement constraints. This reduction is illustrated in **Figure 6.9**, where a cylindrical sample is replaced by a reduced block-like model with $u_x = 0$, $u_y = 0$, and $u_z = 0$ imposed on the symmetry planes and base, while the external loading is preserved.

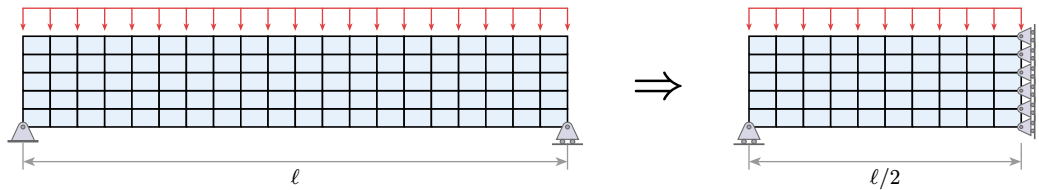


Figure 6.7: Equivalent boundary conditions for symmetry.

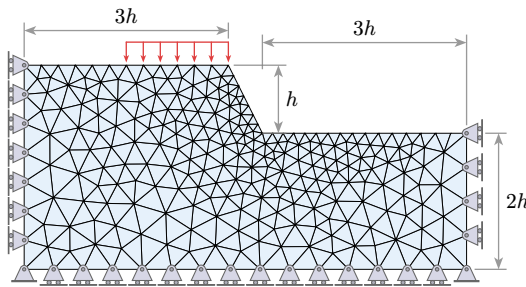


Figure 6.8: Recommended domain extent and boundary conditions for slope analysis..

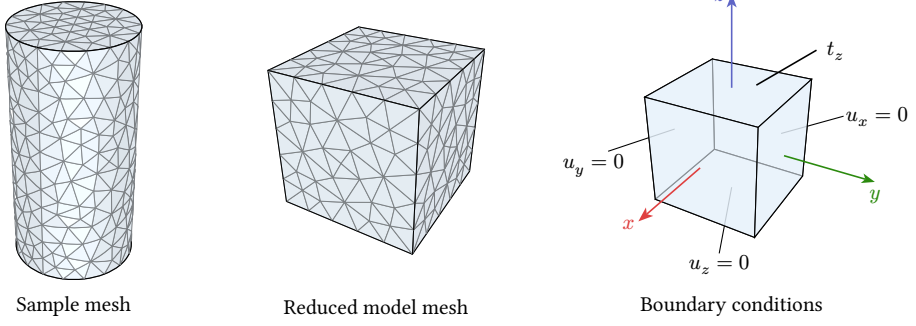


Figure 6.9: Cylindrical sample mesh, reduced model obtained by exploiting symmetry, and boundary conditions imposed on the reduced model.

6.5 Recovery of nodal values

Once a finite element solution has been obtained, it is often necessary to evaluate derived quantities, such as stresses, at the nodes for visualization, error estimation, or further analysis. Many of these quantities are first computed at sampling points inside the elements, most commonly at integration points, and are therefore not directly available at the nodes. In deformation analysis, this procedure is often referred to as **stress recovery**. In that case, the stresses are computed at the integration points from the displacement field as

$$\sigma = DBU \quad (6.7)$$

where B and D are evaluated at the corresponding integration point.

More generally, the same procedure applies to any quantity evaluated at sampling points within the elements, such as strains, fluxes, or gradients of the primary variables. The goal is to reconstruct a nodal field from these sampled values.

6.5.1 Local recovery:

In the local recovery approach, the nodal values are recovered element by element using only information from the element itself. Consider, for example, a four-node quadrilateral element with four integration points. If the stress field (from a stress component, e.g. σ_x) is approximated with the same shape functions used over the element, then the stresses sampled at the integration points satisfy

$$\begin{aligned} \tilde{\sigma}^{(1)} &= \sum N_i^{(1)} \sigma_i \\ \tilde{\sigma}^{(2)} &= \sum N_i^{(2)} \sigma_i \\ \tilde{\sigma}^{(3)} &= \sum N_i^{(3)} \sigma_i \\ \tilde{\sigma}^{(4)} &= \sum N_i^{(4)} \sigma_i \end{aligned} \quad (6.8)$$

where $\tilde{\sigma}^{(j)}$ are the stress values at the integration points, σ_i are the nodal stress values, and $N_i^{(j)}$ is the shape function of node i evaluated at integration point j .

This system can be written in matrix form as

$$\begin{pmatrix} \tilde{\sigma}^{(1)} \\ \tilde{\sigma}^{(2)} \\ \tilde{\sigma}^{(3)} \\ \tilde{\sigma}^{(4)} \end{pmatrix} = \begin{pmatrix} N_1^{(1)} & N_2^{(1)} & N_3^{(1)} & N_4^{(1)} \\ N_1^{(2)} & N_2^{(2)} & N_3^{(2)} & N_4^{(2)} \\ N_1^{(3)} & N_2^{(3)} & N_3^{(3)} & N_4^{(3)} \\ N_1^{(4)} & N_2^{(4)} & N_3^{(4)} & N_4^{(4)} \end{pmatrix} \begin{pmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \end{pmatrix} \quad (6.9)$$

and in condensed form as

$$\tilde{\sigma} = N\sigma \quad (6.10)$$

Since N is square and invertible in this example, the nodal values follow directly from

$$\sigma = N^{-1}\tilde{\sigma} \quad (6.11)$$

This procedure applies when the number of sampling points matches the number of nodal unknowns. More generally, the system may be overdetermined or underdetermined.

If the number of sampling points exceeds the number of nodal unknowns, the system is overdetermined and a least-squares solution can be used:

$$\sigma = N^+\tilde{\sigma} \quad (6.12)$$

where N^+ is the pseudo-inverse of N , given by

$$N^+ = (N^T N)^{-1} N^T \quad (6.13)$$

If the number of sampling points is smaller than the number of nodal unknowns, the system is underdetermined and the nodal values cannot be obtained uniquely from the interpolation equations alone. In that case, it is preferable to fit the sampled values with a lower-dimensional polynomial basis. For a two-dimensional element, typical choices are

$$\begin{aligned} \sigma(\xi, \eta) &= a_0 + a_1\xi + a_2\eta \\ \sigma(\xi, \eta) &= a_0 + a_1\xi + a_2\eta + a_3\xi\eta \\ \sigma(\xi, \eta) &= a_0 + a_1\xi + a_2\eta + a_3\xi^2 + a_4\xi\eta + a_5\eta^2 \end{aligned} \quad (6.14)$$

where the coefficients are determined from the sampled data.

To obtain a well-posed fit, the number of coefficients should not exceed the number of sampling points. For example, a bilinear basis has four coefficients and therefore requires at least four sampling points. The regression problem for a bilinear basis can then be written as

$$\begin{pmatrix} 1 & \xi^{(1)} & \eta^{(1)} & \xi^{(1)}\eta^{(1)} \\ 1 & \xi^{(2)} & \eta^{(2)} & \xi^{(2)}\eta^{(2)} \\ 1 & \xi^{(3)} & \eta^{(3)} & \xi^{(3)}\eta^{(3)} \\ 1 & \xi^{(4)} & \eta^{(4)} & \xi^{(4)}\eta^{(4)} \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} \tilde{\sigma}^{(1)} \\ \tilde{\sigma}^{(2)} \\ \tilde{\sigma}^{(3)} \\ \tilde{\sigma}^{(4)} \\ \vdots \end{pmatrix} \quad (6.15)$$

Once the coefficients have been determined, the fitted polynomial can be evaluated at any point in the element, including the nodes:

$$\sigma_i = \sigma(\xi_i, \eta_i) \quad (6.16)$$

6.5.2 Superconvergent patch recovery (SPR):

The Superconvergent Patch Recovery (SPR) method is a more accurate recovery technique, particularly for stresses. It is based on the observation that certain points inside the elements, called superconvergent points, often yield more accurate stress values than the nodes themselves. For many low-order elements, these points coincide with the integration points, although this is not always the case. In practice, the integration points are commonly used as the sampling points for the recovery procedure.

In SPR, a patch of surrounding elements is defined for each interior node. The sampled stress values from all points in the patch are collected, and a polynomial is fitted to them in a least-squares sense.

Unlike purely local recovery, the fit is performed over the entire patch, usually in terms of the physical coordinates of the sampling points:

$$\begin{aligned}\sigma(x, y) &= a_0 + a_1x + a_2y \\ \sigma(x, y) &= a_0 + a_1x + a_2y + a_3xy \\ \sigma(x, y) &= a_0 + a_1x + a_2y + a_3x^2 + a_4xy + a_5y^2\end{aligned}\tag{6.17}$$

A second-degree polynomial basis is often used in SPR because it captures stress variations more accurately than a linear basis, especially in regions with strong gradients. Higher-degree polynomials may also be used, but they can introduce overfitting and spurious oscillations.

Figure 6.10 illustrates typical patches associated with the recovery of nodal values at an interior node, a boundary-edge node, and a corner node. In practice, boundary-edge and corner nodes often do not require their own patches, since their values can be recovered from neighboring interior patches that contain them. Likewise, midside-node values may be obtained from more than one patch and then combined by averaging.

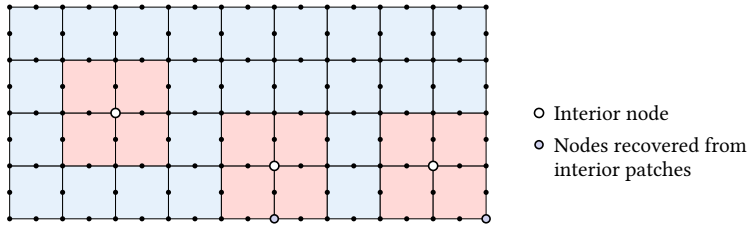


Figure 6.10: Different SPR patches associated with an interior node, a boundary-edge node, and a corner node.

After the polynomial coefficients have been determined, the nodal value is obtained by evaluating the fitted polynomial at the nodal coordinates.

$$\sigma_i = \sigma(\xi_i, \eta_i)\tag{6.18}$$

6.6 Convergence of the numerical solution

Convergence assessment is used to determine whether the algebraic solution procedure has stabilized. This issue is distinct from discretization error: a nonlinear solution may satisfy the convergence criteria of the iterative solver and still be inaccurate if the mesh or interpolation order is inadequate.

6.6.1 Residual-based convergence criteria

In nonlinear analysis, convergence is often monitored through the residual force vector,

$$\mathbf{R} = \mathbf{F}_{\text{ext}} - \mathbf{F}_{\text{int}}\tag{6.19}$$

which measures the imbalance between external and internal nodal forces. A common stopping criterion requires

$$\|\mathbf{R}\| \leq tol\tag{6.20}$$

or, in relative form,

$$\frac{\|\mathbf{R}\|_{\infty}}{\|\mathbf{F}_{\text{int}}\|_{\infty}} \leq tol\tag{6.21}$$

where tol is a prescribed tolerance. Residual-based criteria are especially meaningful because they test whether equilibrium has effectively been achieved. There are other convergence checks more appropriate for nonlinear analyses.

6.6.2 Additional convergence checks

In practice, residual norms are often complemented by other indicators, such as the norm of the displacement correction or the incremental work done during an iteration. Using more than one criterion is useful because a small residual alone does not always guarantee that the iterative process is numerically stable or that all variables have evolved consistently.

6.7 Error estimation and discretization accuracy

Error estimation is concerned with the accuracy of the finite element approximation itself. Its purpose is to assess the effect of discretization, identify regions where the error is large, and guide mesh or interpolation refinement.

6.7.1 Nature of discretization error

Even when the algebraic solver has converged, the finite element solution still differs from the exact continuum solution because of the discretization. This error depends mainly on the mesh density, the interpolation order of the elements, and the regularity of the exact solution. In post-processing, it is therefore important to distinguish solver convergence from approximation accuracy.

6.7.2 A posteriori error estimation

A posteriori error estimation uses the computed solution to estimate the discretization error after the analysis has been performed. These estimates may be based on residuals, on comparisons between solutions obtained with different approximation orders, or on recovered quantities such as stresses obtained by recovery procedures like SPR. Besides providing a global measure of accuracy, they can also indicate which elements or regions should be refined.

6.7.3 Mesh convergence studies

A practical way to assess discretization accuracy is to repeat the analysis with successively refined meshes or higher-order elements and monitor quantities of interest, such as displacements, stresses, reaction forces, or strain energy. If these quantities approach stable values, the solution is said to exhibit mesh convergence. This type of study is particularly important in regions with steep gradients, stress concentrations, or possible singularities.

6.7.4 p and h-refinement

In **h**-refinement, the element size h is reduced by subdividing the mesh. This is the most common refinement strategy and is especially effective near localized stress gradients and singular regions.

In **p**-refinement, the mesh topology is kept fixed while the polynomial order of the interpolation functions is increased. This is attractive when the solution is smooth, because accuracy can improve substantially without introducing many new elements.

In practice, many analyses benefit from a combination of both strategies. This combined **hp**-refinement uses small elements where the solution varies rapidly and higher-order interpolation where the field is smooth. Refinement is ideally guided by error estimation rather than by trial and error alone. The distinction between these approaches is illustrated in [Figure 6.11](#): **h**-refinement increases the number of elements by subdividing the mesh, whereas **p**-refinement keeps the same geometric partition while enriching the interpolation within each element.

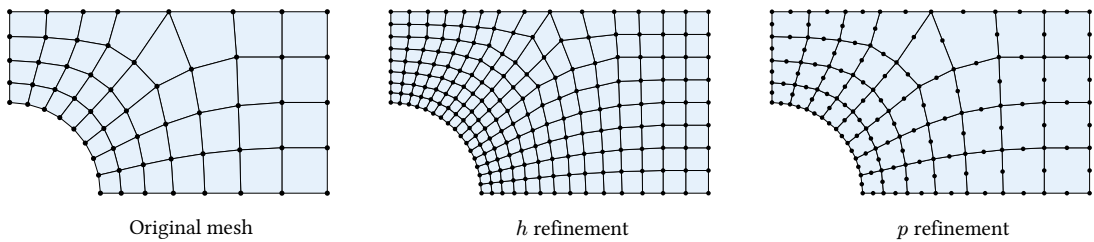


Figure 6.11: Types of mesh refinement.

For discretization error, the simplest procedure is a **mesh convergence** study. The analysis is repeated with successively finer meshes, and a representative quantity of interest, such as a displacement, reaction, strain energy, or peak stress away from singular points, is plotted against the number of degrees of freedom or a characteristic mesh size. If the quantity approaches a stable value, the mesh is considered sufficiently refined for that response. This process is particularly important because local peak stresses near corners, point loads, or crack tips may not converge in the classical sense.

Adaptive remeshing automates this process by refining only the regions that need improvement. In **h**-adaptive methods, the mesh is locally refined by reducing the element size. In **p**-adaptive methods, the polynomial order is increased while keeping the mesh topology unchanged. Combined **hp**-adaptive procedures use both strategies and are often the most efficient when the solution contains both smooth regions and localized steep gradients.

Adaptivity is typically guided by an error indicator or an a posteriori error estimator, often based on stress recovery, residuals, or jumps of fluxes or stresses across element boundaries. The purpose is to distribute the discretization error more uniformly and achieve the desired accuracy with the lowest possible computational cost.

6.8 Visualization techniques

Visualization techniques are used to interpret the numerical results and communicate the physical response of the model. Post-processing should always be read together with the mesh, the boundary conditions, and the expected mechanics of the problem, since visually smooth plots may still hide discretization errors or singular behavior.

Scalar field plots display spatial distributions of scalar quantities over the domain, such as displacement magnitude, temperature, pressure, equivalent plastic strain, or Von Mises stress. They may be represented as contour lines, filled contours, or continuous color maps. They are among the most common visualization tools because they reveal spatial trends, high-gradient regions, and local concentrations. When nodal averaging or smoothing is used, the analyst should remember that the displayed field may look smoother than the raw elementwise solution.

Vector plots represent directional quantities such as displacement, velocity, heat flux, seepage velocity, or principal stress directions. They help identify flow paths, dominant deformation modes, and load-transfer mechanisms. Because dense vector fields can become visually cluttered, they are often combined with sampling, glyph scaling, or streamlines.

Deformed-shape plots show the displaced configuration of the structure. In most structural analyses, the deformation is scaled by a visualization factor because the true displacements are often too small to be seen clearly. These plots are useful for checking boundary conditions, identifying rigid-body motions, and understanding the global deformation pattern, but the scale factor must always be indicated to avoid misinterpretation.

These three complementary views are compared in Figure 6.12. The scalar plots highlight the spatial distribution of displacement and stress, the deformed-shape plot emphasizes the overall kinematics of

the member, and the vector plot makes the local direction and relative magnitude of the displacement field easier to interpret.

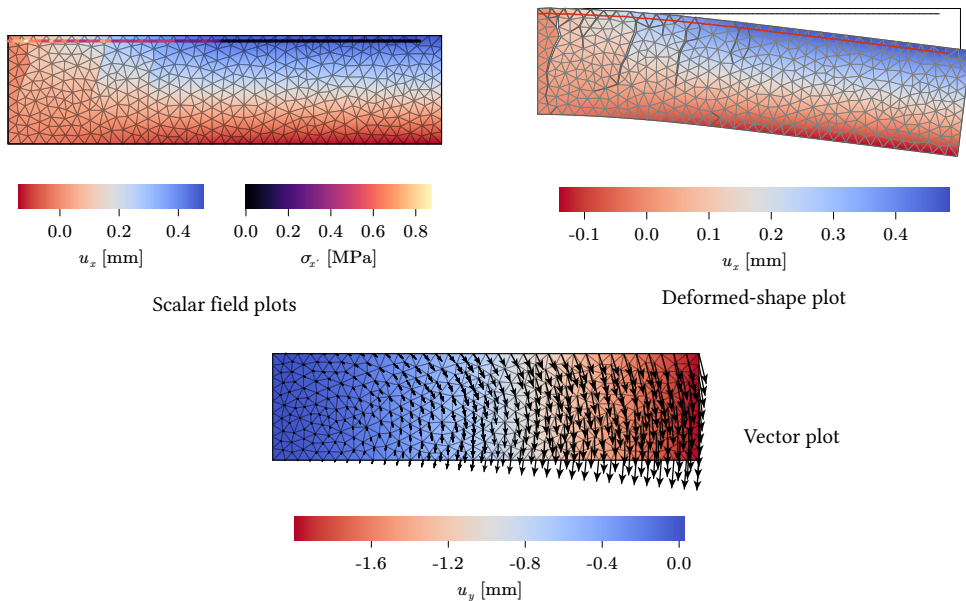


Figure 6.12: Visualization of finite element results for a reinforced cantilever beam combining scalar fields, deformed shape, and vector plots.

Other common techniques include animations over time or load steps, iso-surfaces, section cuts and clipping planes. These are especially useful in three-dimensional analyses, where internal stress paths and localized phenomena cannot be understood from the outer surface alone.

Exercises 6.1:

1. During geometry idealization in finite element analysis, what criteria should guide the choice between line, surface, and solid models?
2. Compare structured and unstructured meshes. Discuss the main advantages and limitations of each approach and indicate situations in which one is usually preferred over the other.
3. Describe the role of element quality metrics in pre-processing. Why can poor-quality elements lead to inaccurate results or numerical difficulties even when the solver converges?
4. Explain the difference between local element numbering and global node numbering. Why must local numbering preserve orientation, and why can global renumbering improve computational efficiency?
5. Under what assumptions can a full model be replaced by a reduced symmetric model without changing the solution?
6. Explain the objective of stress recovery or, more generally, nodal-value recovery. Why are many derived quantities not directly available at the nodes after the finite element solution is obtained?
7. Compare local recovery and superconvergent patch recovery (SPR). What additional information does SPR use, and why is it often more accurate than purely local recovery?

8. Distinguish clearly between convergence of the algebraic solver and accuracy of the finite element approximation. Why is it possible for a nonlinear analysis to satisfy the convergence criteria and still produce an inaccurate solution?
9. Define **h**-, **p**-, and **hp**-refinement. For each strategy, describe the type of solution behavior for which it is most effective.
10. What is the purpose of an a posteriori error estimator? How can recovered stresses, residuals, or flux jumps be used to guide adaptive remeshing?
11. Discuss the interpretation of post-processing plots. Why should scalar contours, vector plots, and deformed-shape plots always be examined together with the mesh and boundary conditions?
12. The nodal coordinates of a three-node bar element are $(x_1, y_1) = (-1, -1)$, $(x_2, y_2) = (1, 1)$, and $(x_3, y_3) = (0, 1)$. At the end of a finite-element analysis, the axial stresses at the three integration points are $\sigma_1 = 5$ kPa, $\sigma_2 = 10$ kPa, and $\sigma_3 = 15$ kPa. Recover the nodal stress values. **A.** $\sigma = (3.545, 16.455, 10)$ kPa

7 NONLINEAR ANALYSIS AND CONSTITUTIVE MODELS

Nonlinear analyses account for structural behaviors where the relationship between loads and responses is not proportional, as seen in cases involving large deformations, material inelasticity, or changing boundary conditions. These analyses require iterative numerical techniques to trace the equilibrium path as the system evolves. To accurately capture such responses, constitutive models are used to describe how materials behave under different loading conditions, including plastic deformation, damage, and other nonlinear effects. Nonlinear analysis and constitutive modeling are fundamental components of modern computational mechanics.

7.1 Nonlinear analysis

Nonlinear analysis using the finite element method (FEM) is essential for accurately simulating structural behaviors that extend beyond the assumptions of linearity. While linear models assume a direct proportionality between loads and displacements, many real-world problems involve **nonlinear relationships between loads and displacements**. These arise in situations such as material yielding (plasticity), structural instability (buckling), time-dependent deformation (creep), and large elastic deformations (hyperelasticity).

Nonlinearities in FEM can be broadly classified into three types: geometric, material, and boundary condition nonlinearities. **Material nonlinearity** accounts for deviations from linear elastic behavior, including plastic deformation, viscoelastic effects, and nonlinear elastic responses. **Figure 7.1** shows typical stress-strain curves, illustrating how materials can exhibit different behaviors under stress. **Geometric nonlinearity** appears when deformations are large enough that the structure's configuration significantly changes, affecting internal force distribution. **Figure 7.2** illustrates a cantilever column experiencing geometric nonlinear behavior, where the deflection leads to a change in the load path and stiffness. **Boundary condition nonlinearity** occurs when contact, friction, or changing constraints alter the equilibrium path, introducing additional complexities to the solution process.

To solve nonlinear finite element problems, advanced numerical methods are required. Iterative algorithms, such as the Newton-Raphson method, are commonly used to resolve the nonlinear equilibrium equations. In cases involving snap-through or limit points, arc-length or path-following techniques help maintain convergence. For time-dependent phenomena, appropriate time integration schemes ensure stability and accuracy. Together, these methods form a robust computational framework for analyzing a wide range of nonlinear applications.

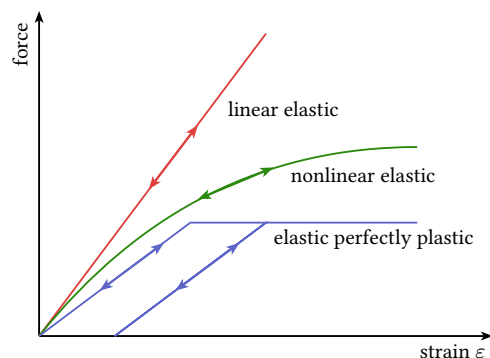


Figure 7.1: Stress-strain curves for different types of materials.

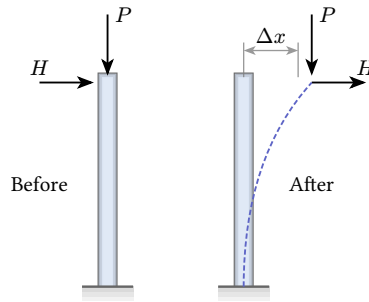


Figure 7.2: Geometric nonlinear behavior in a cantilever column

7.1.1 Equilibrium path

The equilibrium path describes the relationship between an applied load and the resulting displacement at a representative point in a structure during a loading process. It is typically visualized as a curve, where each point corresponds to a distinct equilibrium configuration of the system—meaning that internal and external forces are balanced at that state.

The starting point of this analysis is the reference state, defined as the initial, unstressed, and undeformed configuration from which loads and displacements are measured; see Figure 7.3a. The equilibrium path that evolves continuously from this reference state under increasing load is known as the fundamental path. This path reflects the primary sequence of equilibrium states the structure experiences during the initial loading phase.

Critical points along the equilibrium path mark significant changes in the structural response, such as the onset of buckling, maximum load capacity, or the initiation of failure, as shown in Figure 7.3b. In addition to the fundamental path, structures may exhibit secondary paths, which are alternative equilibrium branches that intersect or bifurcate from the fundamental path. These paths often emerge in problems involving instability or post-critical behavior and are key to understanding complex phenomena like snap-through, bifurcation, and load redistribution.

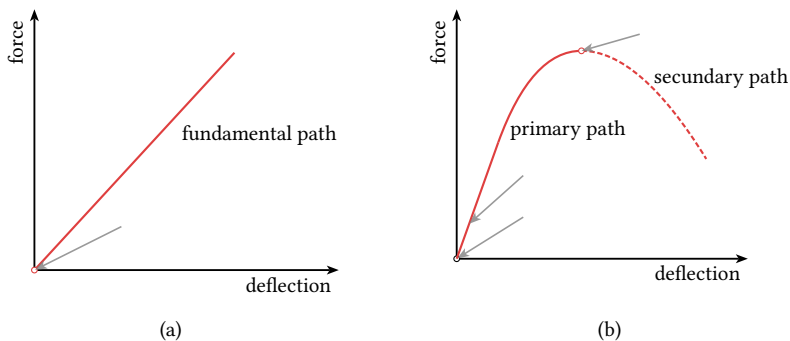


Figure 7.3: Equilibrium paths: (a) linear path; (b) nonlinear path.

Critical points are classified as:

Limit point: A local maximum or minimum in the load-displacement curve.

Bifurcation point: A point where the equilibrium path splits, leading to multiple possible equilibrium configurations under the same load level.

Turning point: A configuration where the tangent to the equilibrium path is vertical, indicating a reversal in the direction of load-displacement progression.

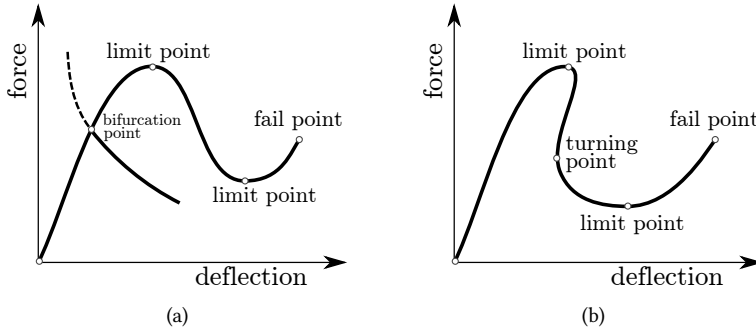


Figure 7.4: Critical points and instability paths: (a) snap through; (b) snap back.

Failure point: A terminal point on the path where the structure can no longer sustain equilibrium due to material or structural failure.

Depending on the presence and arrangement of these critical points, certain distinct nonlinear behaviors may occur:

Snap through: Characterized by the presence of at least two limit points, this phenomenon involves a sudden transition between two stable configurations.

Snap back: Occurs when a turning point is present, resulting in a segment of the path where displacement decreases as load increases

7.1.2 Incremental analysis

To address nonlinear problems, incremental analysis is commonly employed. This method applies external loads and prescribed displacements in small steps, enabling a step-by-step solution process that captures the system's evolving nonlinear behavior. By breaking the total loading into increments, the structural response can be predicted more accurately throughout the analysis.

At each increment, the nonlinear equilibrium equations are solved iteratively, typically using methods like Newton-Raphson. The displacements and internal forces are updated, and the process continues until the full load is applied or a predefined convergence criterion is satisfied. This approach is essential for managing changes in stiffness, boundary conditions, or material states that arise during nonlinear response.

In nonlinear finite element analysis, both the incremental displacement vector $\Delta \mathbf{U}_{\text{ext}}$ and the incremental load vector $\Delta \mathbf{F}_{\text{ext}}$ are computed at each step. The stiffness matrix $\mathbf{K}$ must also be updated to reflect the current configuration or material state. The resulting system of equations to solve is:

$$\mathbf{K} \Delta \mathbf{U}_{\text{ex}} = \Delta \mathbf{F}_{\text{ex}} \quad (7.1)$$

This process repeats incrementally until the desired loading or displacement conditions are reached.

7.1.3 Simple incremental (Forward-Euler) method

The simple incremental method, often referred to as the Forward-Euler method, is a first-order convergence technique. In this approach, external loads are applied incrementally, and the response is computed step by step. The tangent stiffness matrix is updated at each increment, but no corrections are made within the step. While the method is straightforward and easy to implement, it lacks accuracy and may introduce significant errors, especially in highly nonlinear problems.

Alg. 7.1 shows the algorithm for the Forward-Euler method, where *nincs* is the number of increments in the load application and *tol* is the allowed tolerance for each increment.

Alg. 7.1: Forward-Euler algorithm.

```

input:  $F_{\text{ext}}, U_{\text{ext}}, n_{\text{incs}}, tol$ 
 $T = 0$  //pseudo time
 $\Delta T = \frac{1}{n_{\text{incs}}}$  //pseudo time per increment
while  $T < 1$ 
    Compute  $\Delta F_{\text{ext}}$  and  $\Delta U$  //boundary conditions increments
    for each element
        | Mount stiffness  $K^{(e)}$ 
    end
    Assemble global stiffness  $K$ 
    Solve system  $K \Delta U = \Delta F_{\text{ext}}$ 
    for each element
        | Compute element displacements  $\Delta U^{(e)}$ 
        | Compute strain incrementes  $\Delta \epsilon$  at integration points
        | Compute stresses incrementes  $\Delta \sigma$ 
        | Compute element internal forces  $\Delta F_{\text{int}}^{(e)}$ 
    end
    Assemble the global internal forces increment  $\Delta F_{\text{int}}$ 
    Calculate residuals  $R = \Delta F_{\text{ext}} - \Delta F_{\text{int}}$ 
    if  $\|R\| > tol$  //convergence check
        | error
    end
     $T = T + \Delta T$ 
end

```

7.1.4 Newton-Raphson method

The Newton-Raphson method is a widely used iterative technique with second-order convergence, offering greater accuracy and robustness in nonlinear analysis. At each load increment, the equilibrium equations are solved iteratively, and the residual forces are re-evaluated and re-applied to update the solution. The tangent stiffness matrix is updated at each iteration to reflect the current configuration and material state, as shown in Figure 7.5.

In a nonlinear analysis, the residual vector between internal and external forces is defined as:

$$R(U) = \underbrace{F_{\text{ext}} - F_{\text{int}}(U)}_{\Delta F} \quad (7.2)$$

Using a Taylor expansion of the residual around iteration k , we have:

$$R^{k+1} \approx R^k + \frac{\partial R^k}{\partial U^k} \underbrace{(U^{k+1} - U^k)}_{\Delta U} = 0 \quad (7.3)$$

Noting that $\frac{\partial R}{\partial U} = -\frac{\partial F_{\text{int}}}{\partial U}$, this leads to:

$$R^k - \frac{\partial F_{\text{int}}^k}{\partial U^k} \Delta U = 0 \quad (7.4)$$

Since $\frac{\partial F_{\text{int}}^k}{\partial U^k} = K^k$, we obtain the Newton-Raphson iteration system:

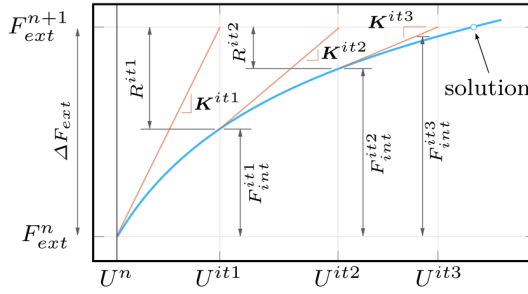


Figure 7.5: Newton-Raphson iterations

$$\mathbf{K}^k \Delta \mathbf{U} = \mathbf{R}^k \quad (7.5)$$

Once $\Delta \mathbf{U}$ is computed, the solution is updated as:

$$\mathbf{U}^{k+1} = \mathbf{U}^k + \Delta \mathbf{U} \quad (7.6)$$

And the internal force vector is recalculated by assembling the contributions from all elements:

$$\mathbf{F}_{\text{int}}^{k+1} = \sum_{\text{asm}} \mathbf{F}_{\text{int}}^{(e)} \quad (7.7)$$

Alg. 7.2 shows the algorithm for the Newton-Raphson method, where *nincs* is the number of increments in the load application, *maxits* is the maximum number of iterations allowed in each increment and *tol* is the convergence tolerance. The algorithm iterates until either convergence is achieved or the maximum number of iterations is reached.

7.1.5 Modified Newton-Raphson method

The Modified Newton-Raphson method is a variant of the standard Newton-Raphson approach in which the tangent stiffness matrix is computed only once per load increment and kept constant during all subsequent iterations, as shown in Figure 7.6. This reduces computational cost by avoiding repeated matrix assembly and factorization, making it efficient for large-scale problems. While residual forces are updated at each iteration, the fixed stiffness can lead to slower convergence, especially in problems with strong nonlinearity where the system response changes significantly within an increment. As a result, more iterations may be needed, and convergence may be less reliable compared to the full Newton-Raphson method. Nonetheless, it offers a practical compromise between efficiency and accuracy in moderately nonlinear analyses.

7.1.6 Convergence criteria

n iterative methods such as Newton-Raphson or Modified Newton-Raphson, different criteria can be chosen to verify the convergence of each load increment. These criteria help ensure that the solution has reached a sufficiently accurate equilibrium state.

Force residuals: This criterion checks whether the maximum component of the difference between external and internal forces is within a predefined tolerance *tol*:

$$\|\mathbf{R}\| = \|\mathbf{F}_{\text{ext}} - \mathbf{F}_{\text{int}}\| \leq \text{tol} \quad (7.8)$$

or, in a relative form:

$$\frac{\|\mathbf{F}_{\text{ext}} - \mathbf{F}_{\text{int}}\|_{\infty}}{\|\mathbf{F}_{\text{int}}\|_{\infty}} \leq \text{tol}\% \quad (7.9)$$

Alg. 7.2: Newton-Raphson algorithm.

```

input:  $F_{\text{ext}}, U_{\text{ext}}, \text{maxits}, \text{nincs}$ 
 $T = 0$  //pseudo time
 $\Delta T = \frac{1}{\text{nincs}}$  //pseudo time per increment
while  $T < 1$ 
    Compute  $\Delta F_{\text{ext}}$  and  $\Delta U$  //boundary conditions increments
     $R = \Delta F_{\text{ext}}$  //initial residual
    for  $k$  from 1 to  $\text{maxits}$ 
        for each element
            Mount stiffness  $K^{(e)}$ 
        end
        Assemble global stiffness  $K$ 
        Solve system  $K \Delta U = R$ 
        for each element
            Compute element displacements  $\Delta U^{(e)}$ 
            Compute strain incrementes  $\Delta \varepsilon$  at integration points
            Compute stresses incrementes  $\Delta \sigma$ 
            Compute element internal forces  $\Delta F_{\text{int}}^{(e)}$ 
        end
        Assemble the global internal forces increment  $\Delta F_{\text{int}}$ 
        Calculate residuals  $R = \Delta F_{\text{ext}} - \Delta F_{\text{int}}$ 
        if  $\|R\| \leq \text{tol}$  //convergence check
            break
        end
         $T = T + \Delta T$ 
    end
end

```

Relative change in displacements: This criterion verifies whether the relative change in nodal displacements between successive iterations is within a given tolerance $\text{atol}\%$:

$$\frac{\|U^{k+1} - U^k\|}{\|U^{k+1}\|} \leq \text{atol}\% \quad (7.10)$$

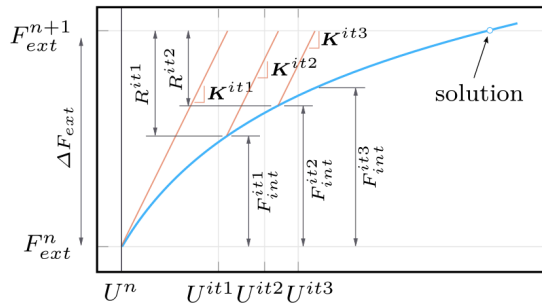


Figure 7.6: Modified Newton-Raphson iterations

Relative change in displacement work: This criterion provides a measure of energy balance. It evaluates the work done by the displacement increment relative to the total internal work:

$$\frac{\Delta \mathbf{F}_{\text{ext}} \cdot \Delta \mathbf{U}}{\mathbf{F}_{\text{int}} \cdot \mathbf{U}} \leq \text{wtol}\% \quad (7.11)$$

Energy change: This criterion evaluates convergence based on the total energy associated with the system. It is more robust and less sensitive to localized errors or oscillations, offering more stable convergence than criteria based on forces or displacements.

$$\sum (U_i - W_i) \leq \text{wtol} \quad (7.12)$$

where

$$U_i = \sum \int_V \frac{1}{2} \boldsymbol{\sigma} \cdot \boldsymbol{\varepsilon} \, dV \quad (7.13)$$

is the strain energy stored in the volume of each element, and W_i is the work potential from each external force and is calculated by:

$$W_i = \mathbf{F}_{\text{ext}} \cdot \mathbf{U} \quad (7.14)$$

7.2 Constitutive modeling

A constitutive model is a **mathematical formulation** that defines the relationship between applied loads and the resulting deformation of a material. These models incorporate key mechanical properties — such as stiffness, strength, ductility, and viscoelasticity— to accurately represent material behavior under different loading conditions.

In numerical simulations, constitutive models are essential for predicting the mechanical response of structures. In the finite element method, they are used to compute the constitutive matrix $\mathbf{D}$, which contributes to the stiffness matrix $\mathbf{K}^{(e)}$ of each element. Additionally, they are responsible for calculating the stress increments $\Delta \boldsymbol{\sigma}$ from strain increments $\Delta \boldsymbol{\varepsilon}$ at each integration point. These stresses are then used to assemble the internal force vector increment $\Delta \mathbf{F}_{\text{int}}^{(e)}$ of the element. **Figure 7.7** illustrates these tasks.

Several formulations exist for developing physically nonlinear constitutive models, including elasticity, continuum damage mechanics, viscoelasticity, hyperelasticity, and plasticity theory.

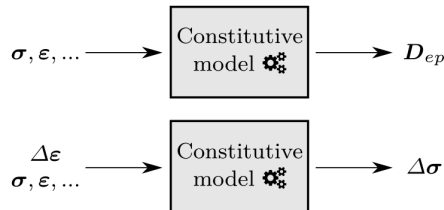


Figure 7.7: Representation of the tasks of a constitutive model in a finite element analysis

7.2.1 Principal stresses and directions

Principal stresses are the normal stresses acting on planes where the shear stresses vanish. These directions are particularly important because they provide a coordinate system in which the stress tensor is diagonal.

Mathematically, the principal stresses and principal directions are the eigenvalues and eigenvectors of σ :

$$\sigma \mathbf{v}_i = \sigma_i \mathbf{v}_i \quad (7.15)$$

with principal stresses ordered as $\sigma_1 \geq \sigma_2 \geq \sigma_3$.

The characteristic equation is

$$\det(\sigma - \lambda \mathbf{I}) = 0 \quad (7.16)$$

whose roots, σ_1 , σ_2 and σ_3 , are the principal stresses. This corresponds to finding the eigenvalues of the stress tensor. The corresponding eigenvectors $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$, are the principal directions. Figure 7.8 shows the principal stresses and directions.

In two-dimensional stress states such as plane stress or plane strain, the in-plane principal stresses may be written directly as

$$\sigma_{1;2} = \frac{\sigma_{11} + \sigma_{22}}{2} \pm \sqrt{\left(\frac{\sigma_{11} - \sigma_{22}}{2}\right)^2 + \sigma_{12}^2} \quad (7.17)$$

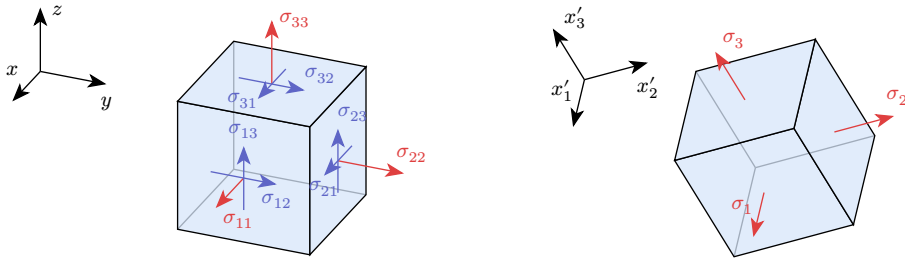


Figure 7.8: Stress components in the original and principal strain direction systems

7.2.2 Stress invariants

Quantities that remain unchanged under a rotation of axes are called stress invariants. The principal invariants of the stress tensor are

$$\begin{aligned} I_1 &= \sigma_1 + \sigma_2 + \sigma_3 \\ I_2 &= \sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1 \\ I_3 &= \sigma_1\sigma_2\sigma_3 \end{aligned} \quad (7.18)$$

Equivalently, I_1 is the trace of the stress tensor:

$$I_1 = \sigma_{xx} + \sigma_{yy} + \sigma_{zz} \quad (7.19)$$

The mean stress, also called hydrostatic stress, is

$$p = \frac{1}{3}I_1 \quad (7.20)$$

These invariants are useful because they characterize the stress state independently of the chosen coordinate system and appear naturally in constitutive models and failure criteria.

Note 7.1: Principal stresses and stress invariants are especially useful because they characterize the stress state independently of the chosen coordinate system and connect directly to many strength criteria.

7.2.3 Spherical and deviatoric decomposition

The stress tensor can be decomposed into a volumetric part and a distortional part. This separation is useful because many constitutive and failure models depend differently on mean stress and shear distortion.

Given a stress tensor σ , it can be decomposed as

$$\sigma = \frac{I_1}{3} \mathbf{I} + s \quad (7.21)$$

where $\frac{I_1}{3} \mathbf{I} = p \mathbf{I}$ is the spherical or hydrostatic part, and s is the deviatoric part.

In components,

$$\begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{pmatrix} = \begin{pmatrix} p & 0 & 0 \\ 0 & p & 0 \\ 0 & 0 & p \end{pmatrix} + \begin{pmatrix} \sigma_{11} - p & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} - p & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} - p \end{pmatrix} \quad (7.22)$$

The deviatoric stress tensor is therefore

$$s = \sigma - \frac{I_1}{3} \mathbf{I} \quad (7.23)$$

The spherical part controls purely volumetric effects, while the deviatoric part governs distortional effects.

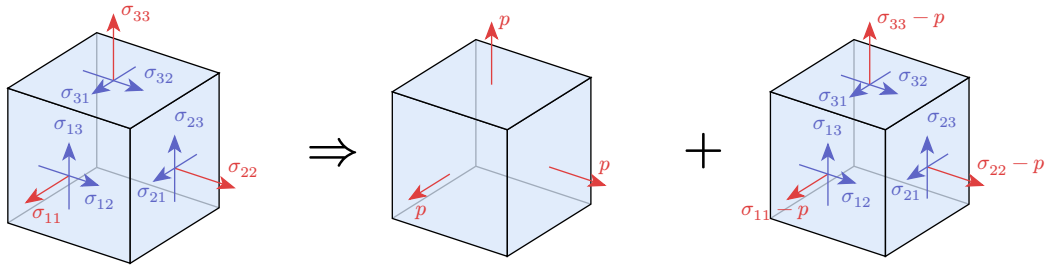


Figure 7.9: Stress tensor decomposition into the spherical and deviatoric parts.

7.2.4 Spherical and deviatoric decomposition of the stress tensor

The spherical and deviatoric decomposition of the stress tensor is a method used to separate the stress tensor into two distinct components: the spherical (or hydrostatic) part and the deviatoric part. This decomposition enables the analysis of volume changes, captured by the spherical component, and shape changes (distortion), captured by the deviatoric component. This approach is widely used in constitutive modeling and material failure analysis as many material yield and failure criteria are based on the deviatoric stress. Meanwhile, the spherical part is important for studying volumetric effects, such as compression or expansion.

Given a stress tensor σ , it can be decomposed as

$$\sigma = \frac{I_1}{3} \mathbf{I} + s \quad (7.24)$$

where $\frac{I_1}{3}\mathbf{I}$ is the spherical (hydrostatic) part, representing the isotropic (volumetric) stress, and $\mathbf{s}$ is the deviatoric part, representing the distortional or shear stress that causes shape changes without altering the volume. The explicit representation of this decomposition is given as

$$\begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{pmatrix} = \begin{pmatrix} p & 0 & 0 \\ 0 & p & 0 \\ 0 & 0 & p \end{pmatrix} + \begin{pmatrix} \sigma_{11} - p & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} - p & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} - p \end{pmatrix} \quad (7.25)$$

where $p = \text{trace}(\boldsymbol{\sigma})/3$ is the mean stress (hydrostatic pressure).

Based on Eq. (7.21), the deviatoric stress tensor $\mathbf{s}$ is defined as

$$\mathbf{s} = \boldsymbol{\sigma} - \frac{I_1}{3}\mathbf{I} \quad (7.26)$$

7.2.5 Fundamental invariants of the deviatoric tensor

The invariants of the deviatoric tensor $\mathbf{s}$ are useful for characterizing the stress state in terms of the distortional (shear) component of the stress tensor. These invariants are commonly denoted J_1 , J_2 and J_3 . In particular, the second and third invariants are widely used in material modeling.

The first deviatoric invariant J_1 is defined as

$$J_1 = \text{trace}(\mathbf{s}) \quad (7.27)$$

By definition, the trace of the deviatoric stress tensor is always zero, $J_1 = 0$

The second invariant J_2 of the deviatoric stress tensor is defined as

$$J_2 = \frac{1}{2} \text{trace}(\mathbf{s}^2) \quad (7.28)$$

In terms of the principal stresses, the second deviatoric invariant can be expressed as

$$J_2 = \frac{1}{6} \left((\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \right) \quad (7.29)$$

This expression shows that J_2 is independent of hydrostatic stress and depends only on differences between the principal stresses, which reflect pure shear effects. This invariant is widely used in the definition of failure criteria because it is directly related to the material strength. For instance, the von Mises yield criterion states that yielding occurs when the von Mises stress $\sigma_{\text{vm}} = \sqrt{3J_2}$ reaches a critical value f_y , i.e.

$$f_y = \sqrt{3J_2} \quad (7.30)$$

where f_y is the uniaxial yield stress of the material. The **von Mises stress** is widely used to identify the onset of yielding in ductile materials under multiaxial loading conditions.

The third deviatoric invariant J_3 of the deviatoric tensor $\mathbf{s}$ is defined as

$$J_3 = \det(\mathbf{s}) \quad (7.31)$$

In terms of principal stresses, J_3 captures the asymmetry of the deviatoric stress tensor, reflecting the influence of the relative ordering of the principal stresses (shear skewness). While J_3 is less commonly used than J_2 , it plays an important role in advanced failure criteria and in describing complex shear behavior.

7.2.6 The Haigh–Westergaard stress space

The Haigh–Westergaard stress space is a three-dimensional geometric representation used to describe the state of stress at a point. This space is particularly useful for visualizing stress invariants and for understanding material behavior under multiaxial loading, such as yielding and failure in isotropic materials. It employs a coordinate system based on stress invariants, making it independent of the body's coordinate system.

Figure 7.10a illustrates the space of principal stresses, where the three axes represent the principal stresses σ_1 , σ_2 , and σ_3 . The hydrostatic stress component lies along the line $\sigma_1 = \sigma_2 = \sigma_3$, known as the hydrostatic axis. Meanwhile, Figure 7.10b shows the Π -plane, also called the deviatoric plane, defined by the condition $\sigma_1 + \sigma_2 + \sigma_3 = 0$.

To decouple the hydrostatic (spherical) and deviatoric (distortional) components of the stress tensor, the coordinates of a point in this space are expressed using three quantities derived from stress invariants:

- the spherical radius ξ , as measure of the hydrostatic stress
- the deviatoric radius ρ , as measure of the magnitude of the deviatoric stress, and
- the Lode angle θ as a measure of the asymmetry of the deviatoric stress state.

These invariants are defined as:

$$\begin{aligned}\xi &= \sqrt{3} p \\ \rho &= \sqrt{2J_2} = \|s\| \\ \theta &= \frac{1}{3} \cos^{-1} \left(\frac{3\sqrt{3}}{2} \frac{J_3}{\sqrt{J_2^3}} \right)\end{aligned}\tag{7.32}$$

7.2.7 Failure criteria

A failure criterion defines the conditions under which a material loses its **load-bearing capacity**, meaning it can no longer sustain applied loads due to fracture, excessive deformation, or other forms of failure. Failure can be classified as brittle (fracture) or ductile (excessive plastic deformation leading to collapse).

A failure criterion is mathematically given by function $f(\sigma)$ expressed in terms of the stress tensor and material parameters. Thus the material behavior can be classified based on the sign of f as

- $f(\sigma) < 0$: No failure;
- $f(\sigma) = 0$: The material is at the failure limit;
- $f(\sigma) > 0$: Material cannot stand this level of stress.

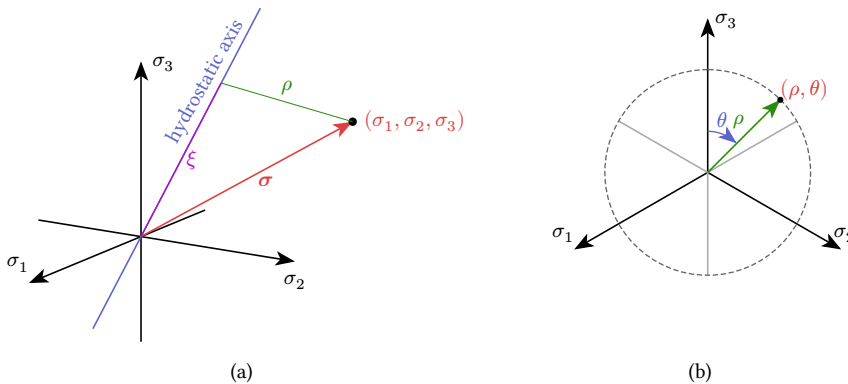


Figure 7.10: Haigh–Westergaard space: (a) principal stresses; (b) projection onto the Π plane.

Failure is identified when the stress state reaches the failure surface defined by

$$f(\boldsymbol{\sigma}) = 0 \quad (7.33)$$

7.2.8 Yield criteria

A yield criterion defines the condition under which a material begins to undergo **plastic deformation**, marking the onset of yielding. Yielding occurs when the material experiences permanent deformation under applied stress, without requiring any further increase in load. In a one-dimensional stress state, for example, the elastic domain is bounded by the yield stress σ_y . Yielding begins when the stress reaches this threshold, i.e., $\sigma = \sigma_y$. Beyond this point, the material response becomes plastic, and the stress–strain relationship becomes nonlinear.

Depending on the material properties, the elastic domain can expand (hardening), contract (softening), or remain unchanged (perfectly plastic). Consequently, the yield stress σ_y evolves according to the size of the elastic domain, as illustrated in **Figure 7.11**.

In a general stress state, the yield criterion is described by a yield surface, a geometric boundary in stress space denoted by $f(\boldsymbol{\sigma}, h)$, where h is an internal variable that governs the current size of the elastic domain (**Figure 7.12**). This surface distinguishes elastic behavior (inside the surface) from plastic behavior (on the surface). As plastic deformation progresses and the material hardens, the internal variable h evolves, causing the yield surface to expand. This concept is illustrated in **Figure 7.19**, which shows the evolution of the yield surface in the ξ – ρ space.

Examples of failure criteria defined in the space of principal stresses are the Tresca, von Mises, Mohr-Coulomb and Drucker-Prager surfaces. **Figures 7.13** and **7.14** show the Mohr-Coulomb and Drucker-Prager failure surfaces, respectively, in the Haigh-Westergaard space.

The yield function of the Mohr-Coulomb criterion is given by

$$f(\boldsymbol{\sigma}) = (\sigma_1 - \sigma_3) + (\sigma_1 + \sigma_3) \sin \varphi - 2c \cos \varphi \quad (7.34)$$

where c is the cohesion and φ is the internal friction angle of the material. In turn, the Drucker-Prager criterion is given by

$$f(\boldsymbol{\sigma}) = \sqrt{J_2} + mI_1 - \kappa \quad (7.35)$$

where m and κ are material parameters related to cohesion and friction.

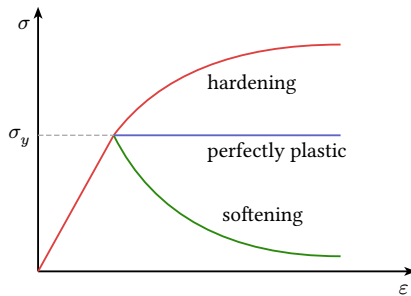


Figure 7.11: Typical uniaxial stress-strain responses illustrating elastic loading followed by hardening, perfectly plastic behavior, or softening after yielding.

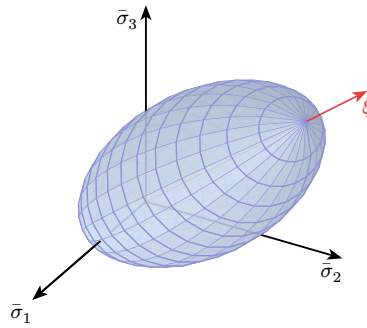


Figure 7.12: Idealized closed yield surface represented in Haigh-Westergaard stress space.

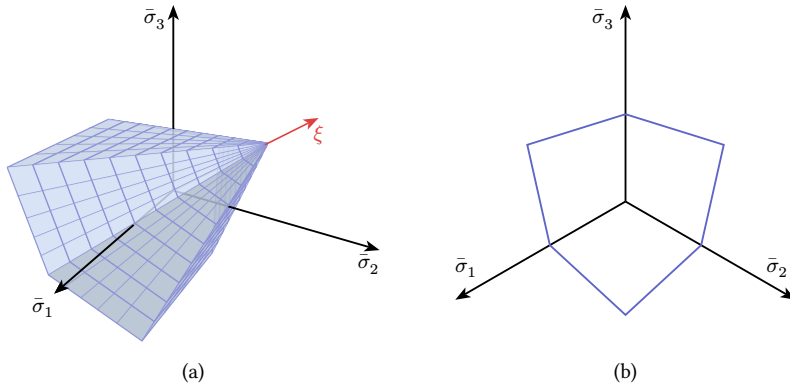


Figure 7.13: Mohr-Coulomb criterion: (a) failure surface; (b) cross-section view.

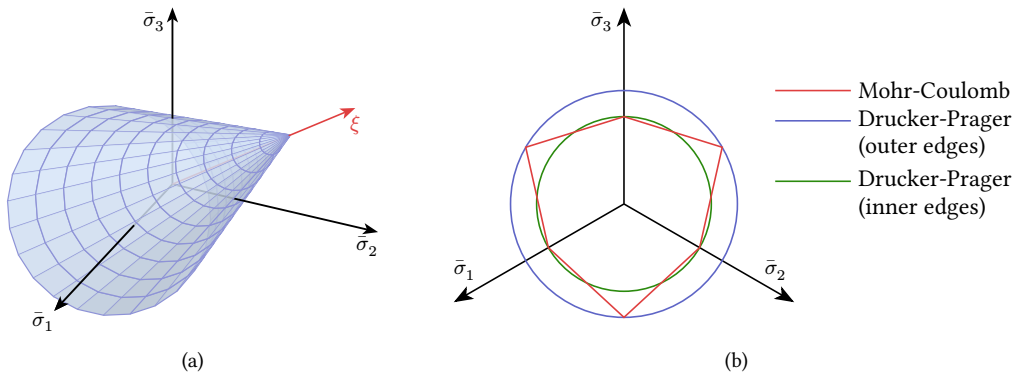


Figure 7.14: The Drucker-Prager criterion: (a) yield surface; (b) outer and inner edges fit relative to the Mohr-Coulomb criterion.

7.3 Continuum damage mechanics

Continuum Damage Mechanics (CDM) is a theoretical framework used to describe and analyze the **progressive degradation and failure** of materials under various loading conditions. It is a branch of solid mechanics that focuses on modeling the **evolution of damage** in continuous and homogeneous materials.

Damage is associated with microstructural defects, such as cracks, voids, or dislocations, which lead to observable macroscopic effects (Figure 7.15). The quantification of damage is performed using internal damage variables that reflect the material's deteriorating state.

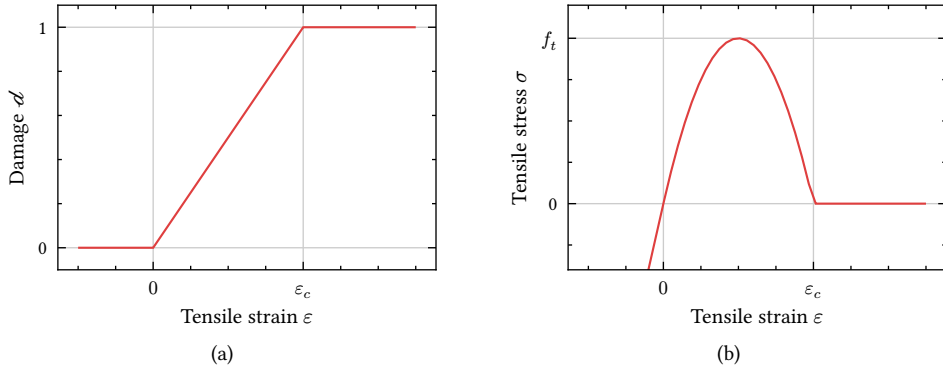


Figure 7.16: Simple 1D tensile model: (a) damage evolution law; (b) strain-stress curve.

The progression of damage under different loading conditions is governed by damage evolution laws. These laws are typically formulated based on experimental data or derived from physical mechanisms that describe the growth and interaction of micro-defects. As the material is loaded, the damage variables increase, representing the accumulation of irreversible changes in the material's microstructure.

Constitutive models in damage mechanics establish relationships between stress, strain, and damage variables, incorporating the degradation of mechanical properties. Most models use a single scalar damage variable $\mathcal{d}$, which ranges from 0 (undamaged) to 1 (fully damaged), and is hypothetically defined as:

$$\mathcal{d} = \frac{A_d}{A_0} \quad (7.36)$$

where A_0 is the original, undamaged cross-sectional area of a material sample, and A_d is the effective load-carrying area considering the presence of damage-induced defects.

7.3.1 Simple 1D tensile model with linear damage

A 1D damage model can be used to simulate a rod with cross-sectional area A subjected to axial tensile loads.

This model accounts for the progressive degradation of the material's stiffness by reducing the Young's modulus through a damage variable. The secant modulus is defined as:

$$E_s = E_0(1 - \mathcal{d}) \quad (7.37)$$

where E_0 is the initial (undamaged) Young's modulus, and $\mathcal{d}$ is the damage variable.

The model assumes linear damage evolution, meaning that damage increases linearly with tensile strain. The damage law is given by:

$$\mathcal{d}(\varepsilon) = \begin{cases} 0 & \text{if } \varepsilon < 0 \\ c\varepsilon & \text{if } 0 \leq \varepsilon < \varepsilon_c \\ 1 & \text{otherwise} \end{cases} \quad (7.38)$$

Here, c is a material parameter that controls the damage rate, and ε is the applied strain. According to Figure 7.16a, the damage variable reaches 1 when the strain reaches the critical value ε_c , so we define $c = \frac{1}{\varepsilon_c}$.

The stress-strain relationship is expressed using the secant modulus:

Figure 7.15: Damage evolution in a sample

$$\sigma = E_s \varepsilon = E_0(1 - \mathcal{d})\varepsilon \quad (7.39)$$

Substituting the damage law, the piecewise expression for stress becomes:

$$\sigma(\varepsilon) = \begin{cases} E_0 \varepsilon & \text{if } \varepsilon < 0 \\ E_0 \left(\varepsilon - \frac{\varepsilon^2}{\varepsilon_c} \right) & \text{if } 0 \leq \varepsilon < \varepsilon_c \\ 0 & \text{otherwise} \end{cases} \quad (7.40)$$

as illustrated in [Figure 7.16b](#).

The tangent modulus at any point along the stress–strain curve is the derivative of stress with respect to strain:

$$E = \frac{d\sigma}{d\varepsilon} = \begin{cases} E_0 & \text{if } \varepsilon < 0 \\ E_0 \left(1 - 2\frac{\varepsilon}{\varepsilon_c} \right) & \text{if } 0 \leq \varepsilon < \varepsilon_c \\ 0 & \text{otherwise} \end{cases} \quad (7.41)$$

Given the current state of stress σ^i and strain ε^i , and a strain increment $\Delta\varepsilon$, the updated state is computed as:

$$\begin{aligned} \varepsilon^{i+1} &= \varepsilon^i + \Delta\varepsilon \\ \sigma^{i+1} &= \sigma(\varepsilon^{i+1}) \end{aligned} \quad (7.42)$$

Finally, the stress increment is given by

$$\Delta\sigma = \sigma^{i+1} - \sigma^i \quad (7.43)$$

7.3.2 1D compressive damage model

This model follows the same stiffness degradation formulation as the tensile case:

$$E_s = E_0(1 - \mathcal{d}) \quad (7.44)$$

Under compression, the model assumes an initial linear elastic response followed by nonlinear softening. The damage evolution law is defined as:

$$\mathcal{d}(\varepsilon) = \begin{cases} 1 & \text{if } \varepsilon < \varepsilon_{c2} \\ 1 - \left(\frac{\varepsilon - \varepsilon_{c2}}{\varepsilon_{c1} - \varepsilon_{c2}} \right)^\alpha & \text{if } \varepsilon_{c2} \leq \varepsilon < \varepsilon_{c1} \\ 0 & \text{otherwise} \end{cases} \quad (7.45)$$

Here, $\varepsilon_{\{c1\}}$ and $\varepsilon_{\{c2\}}$ are characteristic compressive strain values ($\varepsilon_{c2} < \varepsilon_{c1} < 0$), and $\alpha > 1$ is a material parameter that controls the shape of the softening curve. This is illustrated in [Figure 7.17a](#).

By substituting E_s and $\mathcal{d}$ into the stress definition $\sigma = E_s \varepsilon$, the stress–strain relationship becomes:

$$\sigma = \begin{cases} 0 & \text{if } \varepsilon < \varepsilon_{c2} \\ E_0 \varepsilon \left(\frac{\varepsilon - \varepsilon_{c2}}{\varepsilon_{c1} - \varepsilon_{c2}} \right)^\alpha & \text{if } \varepsilon_{c2} \leq \varepsilon \leq \varepsilon_{c1} \\ E_0 \varepsilon & \text{otherwise} \end{cases} \quad (7.46)$$

The tangent Young's modulus is then calculated as:

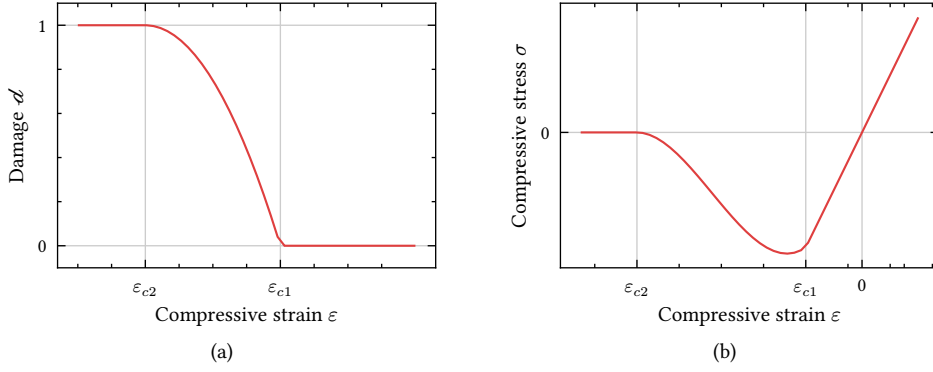


Figure 7.17: 1D compressive damage model: (a) damage evolution law; (b) strain-stress curve.

$$E = \frac{d\sigma}{d\varepsilon} = \begin{cases} 0 & \text{if } \varepsilon < \varepsilon_{c2} \\ E_0 \frac{(\varepsilon_{c2} - (\alpha+1)\varepsilon) \left(\frac{\varepsilon - \varepsilon_{c2}}{\varepsilon_{c1} - \varepsilon_{c2}} \right)^\alpha}{\varepsilon_{c2} - \varepsilon} & \text{if } \varepsilon_{c2} \leq \varepsilon \leq \varepsilon_{c1} \\ E_0 & \text{otherwise} \end{cases} \quad (7.47)$$

The resulting stress-strain behavior, including the nonlinear softening region, is shown in Figure 7.17b.

Finally, given current state σ^i, ε^i and the increment $\Delta\varepsilon$, the updated state and the stress increment $\Delta\sigma$ are computed as in the previous model.

7.3.3 Simple 3D damage model

This is a simple hypothetic damage model based on the volumetric strain. The stiffness degradation law is given by:

$$D_s = D_0(1 - d) \quad (7.48)$$

where D_0 is the elastic constitutive matrix of the undamaged material, and D_s is the secant constitutive matrix that incorporates damage effects.

The material is allowed to accumulate damage under both tensile and compressive loading. In this simplified model, damage evolves based on characteristic values of the volumetric strain $\varepsilon_v = \text{trace}(\varepsilon)$. The damage evolution law, shown in Figure 7.18, is given by:

$$d = \begin{cases} 1 & \text{if } \varepsilon_v < \varepsilon_{c2} \\ 1 - \left(\frac{\varepsilon_v - \varepsilon_{c2}}{\varepsilon_{c1} - \varepsilon_{c2}} \right)^\alpha & \text{if } \varepsilon_{c2} \leq \varepsilon_v \leq \varepsilon_{c1} \\ 0 & \text{if } \varepsilon_{c1} \leq \varepsilon_v \leq \varepsilon_{t1} \\ 1 - \left(\frac{\varepsilon_{t2} - \varepsilon_v}{\varepsilon_{t2} - \varepsilon_{t1}} \right)^\beta & \text{if } \varepsilon_{t1} \leq \varepsilon_v \leq \varepsilon_{t2} \\ 1 & \text{otherwise} \end{cases} \quad (7.49)$$

where α and β are material parameters that control the shape of the softening curves, and $\varepsilon_{c1}, \varepsilon_{c2}, \varepsilon_{t1}$, and ε_{t2} are characteristic strain values for compression and tension.

Next, the tangent stiffness matrix is computed as $D = \frac{\partial \sigma}{\partial \varepsilon}$. Given the current state of stress σ^i and strain ε^i , and a strain increment $\Delta\varepsilon$, the updated state is computed as:

$$\begin{aligned} \varepsilon^{i+1} &= \varepsilon^i + \Delta\varepsilon \\ \sigma^{i+1} &= D_s \varepsilon^{i+1} \end{aligned} \quad (7.50)$$

Finally, the corresponding stress increment is given by

$$\Delta \sigma = \sigma^{i+1} - \sigma^i \quad (7.51)$$

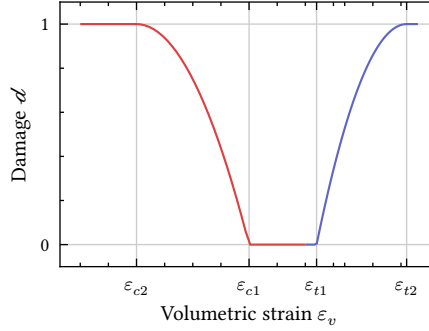


Figure 7.18: Damage evolution law for the simple 3D damage model.

Exercises 7.1:

1. A simple 1D damage model that simulates perfect plastic behavior under both tension and compression uses the following damage evolution law:

$$d = \begin{cases} 0 & \text{if } \varepsilon < |\varepsilon_c| \\ 1 - \frac{f_y}{E_0 |\varepsilon|} & \text{otherwise} \end{cases}$$

where E_0 is the initial Young's modulus, ε_c is the strain limit for elastic behavior, and f_y is the tensile strength. Find the expressions for the strain-stress response $\sigma(\varepsilon)$ and the corresponding tangent modulus $E = \frac{d\sigma}{d\varepsilon}$. Finally, plot the evolution of the damage variable d and the stress function $\sigma(\varepsilon)$.

2. During a finite element increment in a 1D truss under tension, the increment of nodal displacements is given by $\Delta U = (0.002, 0.005)$ m. The truss is modeled using a single integration point and the state at the beginning of the increment is $\sigma = 10$ kPa and $\varepsilon = 0$. The material follows a 1D tensile damage model with the following damage evolution law:

$$d = \begin{cases} 0 & \text{if } \varepsilon < 0 \\ \frac{\varepsilon}{\varepsilon_c} & \text{if } 0 \leq \varepsilon < \varepsilon_c \\ 1 & \text{otherwise} \end{cases}$$

Given the material and geometric properties $E = 200$ GPa, $A = 0.01$ m², $L = 1$ m and $\varepsilon_c = 0.004$, compute the internal force vector of the element at the end of the increment.

3. A sample under isotropic compression is being modeled by a simple 3D-damage model. At a given integration point, the current state is $\sigma = (-30, -30, -30, 0, 0, 0)$ MPa and $\varepsilon = (0.001, 0.001, 0.001, 0, 0, 0)$. If a strain increment $\Delta \varepsilon = (-0.003, -0.003, -0.003, 0, 0, 0)$ is applied, determine the final state at the end of the increment. The material follows the damage evolution law:

$$d = \begin{cases} 1 & \text{if } \varepsilon_v < \varepsilon_{c2} \\ 1 - \left(\frac{\varepsilon_v - \varepsilon_{c2}}{\varepsilon_{c1} - \varepsilon_{c2}} \right)^2 & \text{if } \varepsilon_{c2} \leq \varepsilon_v \leq \varepsilon_{c1} \\ 0 & \text{otherwise} \end{cases}$$

where ε_v is the volumetric strain, and the critical volumetric strain values are $\varepsilon_{c_1} = -0.002$ and $\varepsilon_{c_2} = -0.015$. Assume the initial material properties: Young's modulus $E_0 = 25$ GPa and Poisson ratio $\nu_0 = 0.2$.

7.4 Plasticity theory

Plasticity refers to the permanent deformation or flow of a material under the action of external forces, even after those forces are removed. It is a fundamental concept in solid mechanics and plays a significant role in understanding material behavior under load.

In plasticity-based constitutive modeling, two primary tasks must be addressed:

Determination of the elastoplastic constitutive matrix D_{ep} : This matrix defines the relationship between stress and strain rates as

$$\dot{\sigma} = D_{ep} : \dot{\varepsilon} \quad (7.52)$$

and includes the elastic and plastic components of the material response.

Development of the stress update procedure: It represents the development of update rules for stress evolution given a strain increment $\Delta\varepsilon$ and a set of state variables $\mathbf{h}$. This is expressed as:

$$\sigma^{i+1} = \sigma(\varepsilon^i + \Delta\varepsilon, \mathbf{h}) \quad (7.53)$$

where n represents the current state and $i + 1$ the updated state.

7.4.1 Fundamental hypotheses of plasticity

The theory of plasticity for small strain is based on several hypotheses as follows:

Small strains assumption: This hypothesis assumes that the deformations experienced by the material are sufficiently small so that, during a load increment, the material retains a linear response. This allows plasticity theory to use the Cauchy stress tensor and assume small-strain kinematics. In finite strain plasticity, this assumption is relaxed, and more advanced measures such as the logarithmic strain tensor are used.

Additive decomposition of strains: The total strain ε is assumed to be the sum of elastic and plastic components:

$$\varepsilon = \varepsilon_e + \varepsilon_p \quad (7.54)$$

The elastic strain is recoverable while the plastic strain is permanent. This assumption is valid in small-strain plasticity but is replaced by multiplicative decomposition in finite strain plasticity.

Existence of a yield criterion: The yield criterion defines the onset of plastic deformation through a yield function: $f(\sigma, \mathbf{h})$, where $\mathbf{h}$ is an internal variable that tracks the evolution of plastic deformation.

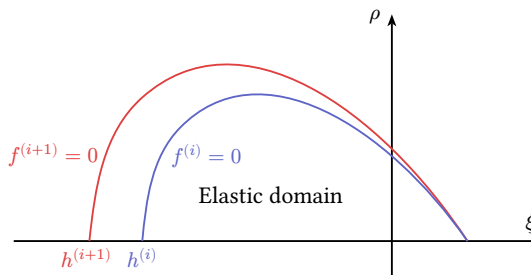


Figure 7.19: Yield surface evolution in the $\xi - \rho$ space.

mation. If $f(\sigma) < 0$, the material is in the elastic regime. If $f(\sigma) = 0$, the material is at the onset or in the plastic regime.

Flow rule: The flow rule governs the direction of plastic strain increments. It defines the rate of plastic strain as:

$$\dot{\epsilon}_p = \dot{\lambda} \frac{\partial g}{\partial \sigma} \quad (7.55)$$

where $g(\sigma)$ is a **plastic potential function** and $\dot{\lambda}$ is a positive scalar called **plastic multiplier**. If $g \equiv f$, the flow rule is associated, while if $g \neq f$ the flow rule is non-associated.

Yield function law: The evolution of the yield function is governed by a **hardening/softening function** that defines the values of the internal variable:

$$h = h(\bar{\epsilon}_p) \quad (7.56)$$

$\bar{\epsilon}_p$ is an equivalent plastic strain measure. Typically, if h increases, the material exhibits hardening (increased strength). Conversely, if h decreases, the material undergoes softening (loss of strength).

As plastic deformation occurs, the yield surface may expand, contract, or translate, depending on the hardening or softening behavior:

- **Isotropic hardening/softening:** The yield surface expands/contracts uniformly in all directions. Used in monotonic plastic deformation of metals.
- **Kinematic hardening/softening:** The yield surface moves in the stress space without changing size. Frequently used in cyclic plasticity.
- **Combined hardening/softening:** The yield surface expands/contracts and moves in the direction of the plastic strain.
- **Anisotropic hardening/softening:** The yield surface expands/contracts at different rates in different directions. Common in brittle materials.

Loading and unloading conditions (Kuhn-Tucker): The Kuhn–Tucker conditions describe the interaction between plasticity and loading/unloading conditions. The conditions are:

- **Yield condition $f \leq 0$:** The stress state must be on or inside the yield surface. There is not a possible stress state for which $f(\sigma)$ is greater than zero.
- **Non-negativity of the plastic multiplier $\dot{\lambda} \geq 0$:** Since the plastic strain cannot decrease, the plastic multiplier must be non-negative.
- **Complementary condition $\dot{\lambda} f = 0$:** The product of the plastic multiplier and the yield function must be zero. This condition ensures that the material is either in the elastic or plastic regime. Thus, if $f = 0$, then $\dot{\lambda} > 0$; otherwise, if $f < 0$, then $\dot{\lambda} = 0$.

Plastic work and dissipation: Plastic deformation is irreversible, meaning energy is dissipated as heat or internal damage. The rate of plastic work is given by

$$\dot{W}_p = \dot{\sigma} : \dot{\epsilon}_p \geq 0 \quad (7.57)$$

ensuring that plastic deformation does not reduce energy, which guarantees stable plastic behavior. This is also called as Drucker's stability postulate.

7.4.2 Consistency condition

The Consistency condition, also called the Prager consistency condition, ensures that when the stress state reaches the yield surface, it remains there during plastic flow. It can be expressed as

$$\dot{f}(\sigma, h) = 0 \quad (7.58)$$

Expanding using the chain rule

$$\frac{\partial f}{\partial \boldsymbol{\sigma}} : \dot{\boldsymbol{\sigma}} + \frac{\partial f}{\partial h} \dot{h} = 0 \quad (7.59)$$

This condition is used to determine the plastic multiplier $\dot{\lambda}$, ensuring that stress increments do not violate the yield surface.

7.4.3 General derivation of the continuous elastoplastic tensor D_{ep}

The stress rate can be expressed as

$$\dot{\boldsymbol{\sigma}} = \mathbf{D}_e : \dot{\boldsymbol{\varepsilon}}_e \quad (7.60)$$

where $\mathbf{D}_e$ is the elastic constitutive tensor. Using the additive decomposition of strains, $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_e + \boldsymbol{\varepsilon}_p$, we rewrite

$$\dot{\boldsymbol{\sigma}} = \mathbf{D}_e : (\dot{\boldsymbol{\varepsilon}} - \dot{\boldsymbol{\varepsilon}}_p) \quad (7.61)$$

Substituting the flow rule, $\boldsymbol{\varepsilon}_p = \dot{\lambda} \frac{\partial g}{\partial \boldsymbol{\sigma}}$, we obtain

$$\dot{\boldsymbol{\sigma}} = \mathbf{D}_e : \left(\dot{\boldsymbol{\varepsilon}} - \dot{\lambda} \frac{\partial g}{\partial \boldsymbol{\sigma}} \right) \quad (7.62)$$

To establish a stress-strain relationship, the plastic multiplier $\dot{\lambda}$ must be determined. This is done using the consistency condition, which ensures that the stress state remains on the yield surface during plastic flow.

The equivalent plastic strain is a scalar measure of the accumulated plastic deformation. It can be expressed as

$$\bar{\varepsilon}_p = \int_0^t \|\dot{\boldsymbol{\varepsilon}}_p\| dt \quad (7.63)$$

Then the rate of the equivalent plastic strain rate is

$$\dot{\bar{\varepsilon}}_p = \|\dot{\boldsymbol{\varepsilon}}_p\| \quad (7.64)$$

Substituting the flow rule it is rewritten as

$$\dot{\bar{\varepsilon}}_p = \dot{\lambda} \left\| \frac{\partial g}{\partial \boldsymbol{\sigma}} \right\| \quad (7.65)$$

This expression is used to define the evolution of the internal state variable h , which governs hardening or softening. Thus,

$$\begin{aligned} \dot{h} &= \frac{\partial h}{\partial \bar{\varepsilon}_p} \dot{\bar{\varepsilon}}_p \\ &= \frac{\partial h}{\partial \bar{\varepsilon}_p} \dot{\lambda} \left\| \frac{\partial g}{\partial \boldsymbol{\sigma}} \right\| \end{aligned} \quad (7.66)$$

The consistency condition states that during plastic flow, the yield function remains zero. Substituting Eq. (7.62) and Eq. (7.66) into the consistency condition we have

$$\begin{aligned} \frac{\partial f}{\partial \boldsymbol{\sigma}} : \dot{\boldsymbol{\sigma}} + \frac{\partial f}{\partial h} \dot{h} &= 0 \\ \frac{\partial f}{\partial \boldsymbol{\sigma}} : \mathbf{D} : \left(\dot{\boldsymbol{\varepsilon}} - \dot{\lambda} \frac{\partial g}{\partial \boldsymbol{\sigma}} \right) + \frac{\partial f}{\partial h} \frac{\partial h}{\partial \varepsilon_p} \dot{\lambda} \left\| \frac{\partial g}{\partial \boldsymbol{\sigma}} \right\| &= 0 \end{aligned} \quad (7.67)$$

Then solving for the plastic multiplier we get

$$\dot{\lambda} = \frac{\frac{\partial f}{\partial \boldsymbol{\sigma}} : \mathbf{D}_e : \dot{\boldsymbol{\varepsilon}}}{\frac{\partial f}{\partial \boldsymbol{\sigma}} : \mathbf{D}_e : \frac{\partial g}{\partial \boldsymbol{\sigma}} - \frac{\partial f}{\partial h} \frac{\partial h}{\partial \varepsilon_p} \left\| \frac{\partial g}{\partial \boldsymbol{\sigma}} \right\|}} \quad (7.68)$$

Substituting back into the stress rate equation, Eq. (7.62), we get

$$\begin{aligned} \dot{\boldsymbol{\sigma}} &= \mathbf{D}_e : \dot{\boldsymbol{\varepsilon}} - \dot{\lambda} \mathbf{D}_e : \frac{\partial g}{\partial \boldsymbol{\sigma}} \\ &= \mathbf{D}_e : \dot{\boldsymbol{\varepsilon}} - \frac{\frac{\partial f}{\partial \boldsymbol{\sigma}} : \mathbf{D}_e : \dot{\boldsymbol{\varepsilon}}}{\frac{\partial f}{\partial \boldsymbol{\sigma}} : \mathbf{D}_e : \frac{\partial g}{\partial \boldsymbol{\sigma}} - \frac{\partial f}{\partial h} \frac{\partial h}{\partial \varepsilon_p} \left\| \frac{\partial g}{\partial \boldsymbol{\sigma}} \right\|}} \mathbf{D}_e : \frac{\partial g}{\partial \boldsymbol{\sigma}} \\ &= \mathbf{D}_e : \dot{\boldsymbol{\varepsilon}} - \frac{\mathbf{D}_e : \frac{\partial g}{\partial \boldsymbol{\sigma}} \left(\frac{\partial f}{\partial \boldsymbol{\sigma}} : \mathbf{D}_e : \dot{\boldsymbol{\varepsilon}} \right)}{\frac{\partial f}{\partial \boldsymbol{\sigma}} : \mathbf{D}_e : \frac{\partial g}{\partial \boldsymbol{\sigma}} - \frac{\partial f}{\partial h} \frac{\partial h}{\partial \varepsilon_p} \left\| \frac{\partial g}{\partial \boldsymbol{\sigma}} \right\|}} \end{aligned} \quad (7.69)$$

This allows us to express the elastoplastic constitutive tensor as

$$\mathbf{D}_{ep} = \mathbf{D}_e - \frac{\mathbf{D}_e : \frac{\partial g}{\partial \boldsymbol{\sigma}} \otimes \frac{\partial f}{\partial \boldsymbol{\sigma}} : \mathbf{D}_e}{\frac{\partial f}{\partial \boldsymbol{\sigma}} : \mathbf{D}_e : \frac{\partial g}{\partial \boldsymbol{\sigma}} - \frac{\partial f}{\partial h} \frac{\partial h}{\partial \varepsilon_p} \left\| \frac{\partial g}{\partial \boldsymbol{\sigma}} \right\|}} \quad (7.70)$$

7.4.4 Stress update

The stress update procedure aims to determine the stress state at the end of a load increment ($i + 1$) given the stress state at the beginning of the increment (i) and the strain increment $\Delta \varepsilon$. Figure 7.20 illustrates the stress update procedure for both associated and non-associated flow rules. This procedure is expressed in terms of increments rather than rates. Given a strain increment $\Delta \varepsilon$, we must first determine whether the material remains in the elastic regime or transitions into the plastic regime.

To check for yielding, we define a trial stress assuming purely elastic behavior:

$$\boldsymbol{\sigma}^{\text{trial}} = \boldsymbol{\sigma}^i + \mathbf{D}_e : \Delta \boldsymbol{\varepsilon} \quad (7.71)$$

If the trial stress satisfies the yield condition $f(\boldsymbol{\sigma}^{\text{trial}}) < 0$, the material remains elastic, and the final stress state is

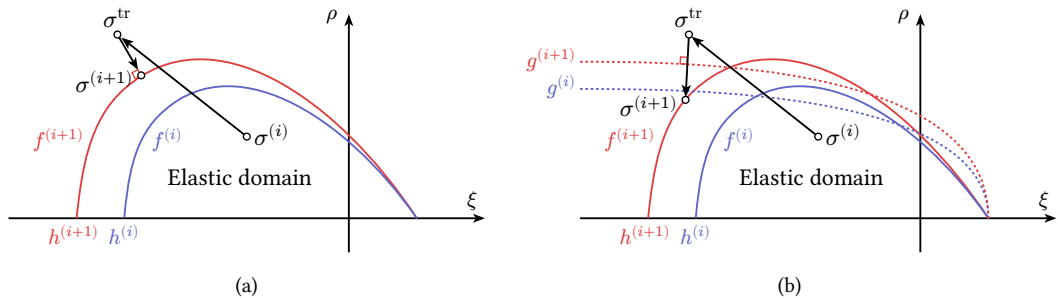


Figure 7.20: Stress update procedure: (a) associated flow rule; (b) non-associated flow rule.

$$\boldsymbol{\sigma}^{i+1} = \boldsymbol{\sigma}^i + \mathbf{D}_e : \Delta \boldsymbol{\varepsilon} \quad (7.72)$$

On the other hand, if $f(\boldsymbol{\sigma}^{\text{trial}}) \geq 0$, the material yields, and plasticity must be accounted for in the stress update. In this case, the stress state at the end of the increment can be expressed as

$$\begin{aligned} \boldsymbol{\sigma}^{i+1} &= \boldsymbol{\sigma}^i + \mathbf{D}_e : (\Delta \boldsymbol{\varepsilon} - \Delta \boldsymbol{\varepsilon}_p) \\ &= \boldsymbol{\sigma}^i + \mathbf{D}_e : \Delta \boldsymbol{\varepsilon} - \mathbf{D}_e : \Delta \boldsymbol{\varepsilon}_p \end{aligned} \quad (7.73)$$

which can also be rewritten using the plastic strain increment

$$\boldsymbol{\sigma}^{i+1} = \boldsymbol{\sigma}^{\text{trial}} - \Delta \lambda \mathbf{D}_e : \left. \frac{\partial g}{\partial \boldsymbol{\sigma}} \right|^{i+1} \quad (7.74)$$

where $\Delta \lambda$ is the plastic multiplier, which must be determined.

The internal state variables must also be updated at each increment

$$h^{i+1} = h(\boldsymbol{\sigma}^{i+1}, \bar{\varepsilon}_p^{i+1}) \quad (7.75)$$

The equivalent plastic strain is updated as:

$$\begin{aligned} \bar{\varepsilon}_p^{i+1} &= \bar{\varepsilon}_p^i + \|\Delta \boldsymbol{\varepsilon}_p\| \\ &= \bar{\varepsilon}_p^i + \Delta \lambda \left\| \frac{\partial g}{\partial \boldsymbol{\sigma}} \right\| \end{aligned} \quad (7.76)$$

Note that if the flow rule is associated, then $\frac{\partial g}{\partial \boldsymbol{\sigma}} = \frac{\partial f}{\partial \boldsymbol{\sigma}}$.

To determine the plastic multiplier $\Delta \lambda$, we substitute Eq. (7.74) and Eq. (7.76) into the yield function at the end of the increment

$$f(\boldsymbol{\sigma}^{i+1}, \bar{\varepsilon}_p^{i+1}) = f(\Delta \lambda) = 0 \quad (7.77)$$

This typically results in a nonlinear equation in terms of $\Delta \lambda$, which must be solved either analytically (for simple cases) or numerically (for complex plasticity models). Once, $\Delta \lambda$ is found, $\boldsymbol{\sigma}^{i+1}$ and $\bar{\varepsilon}_p^{i+1}$ can be computed.

7.4.5 Von Mises plasticity model

The von Mises plasticity model is one of the most widely used yield criteria in structural and mechanical engineering, particularly for ductile materials such as metals. It provides a robust and mathematically convenient way to determine the onset of plastic deformation in materials subjected to complex stress states.

This model is not suitable for brittle or pressure-dependent materials such as concrete and soils where the strength depends on the hydrostatic stress component.

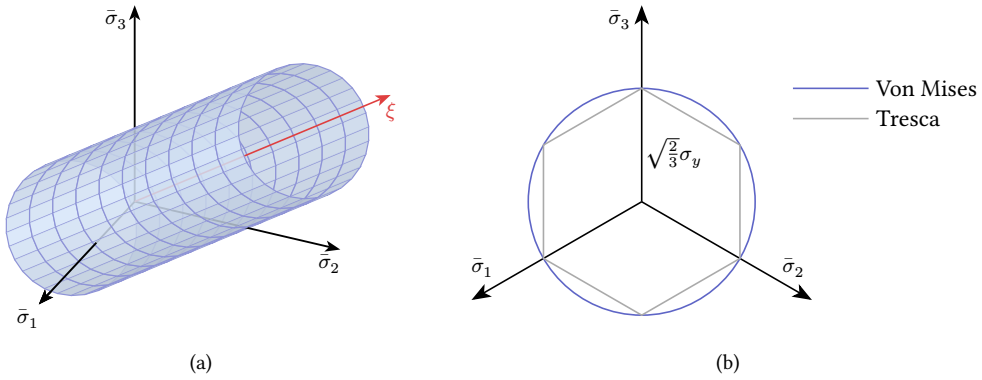


Figure 7.21: The von Mises yield criterion: (a) yield surface; (b) cross-section comparison with the Tresca criterion.

Yield function

In the context of plasticity modeling, the von Mises yield function with linear hardening is defined as

$$f(\boldsymbol{\sigma}, \varepsilon_p) = \sqrt{3J_2} - \underbrace{(f_y + H\bar{\varepsilon}_p)}_{h(\bar{\varepsilon}_p)} \quad (7.78)$$

where H is the hardening modulus. **Figure 7.21** illustrates the von Mises yield surface in the principal stress space and its cross-section compared to the Tresca yield surface.

Yield function derivatives

The derivative with respect to the stress tensor is

$$\begin{aligned} \frac{\partial f}{\partial \boldsymbol{\sigma}} &= \frac{\partial f}{\partial J_2} \frac{\partial J_2}{\partial \boldsymbol{\sigma}} \\ &= \frac{3}{2\sqrt{J_2}} \mathbf{s} \\ &= \sqrt{\frac{3}{2}} \frac{\mathbf{s}}{\|\mathbf{s}\|} \end{aligned} \quad (7.79)$$

The derivative with respect to the plastic strain is

$$\frac{\partial f}{\partial \varepsilon_p} = -H \quad (7.80)$$

Plastic strain evolution

Considering an associated flow rule, $g \equiv f$, the plastic strain rate is

$$\begin{aligned} \dot{\varepsilon}_p &= \dot{\lambda} \frac{\partial f}{\partial \boldsymbol{\sigma}} \\ &= \dot{\lambda} \sqrt{\frac{3}{2}} \hat{\mathbf{s}} \end{aligned} \quad (7.81)$$

and the equivalent plastic strain is

$$\begin{aligned}\dot{\bar{\epsilon}}_p &= \|\dot{\epsilon}_p\| \\ &= \sqrt{\frac{3}{2}} \dot{\lambda}\end{aligned}\tag{7.82}$$

Consistency condition

Using the consistency condition

$$\frac{\partial f}{\partial \boldsymbol{\sigma}} : \dot{\boldsymbol{\sigma}} + \frac{\partial f}{\partial \epsilon_p} \dot{\bar{\epsilon}}_p = 0\tag{7.83}$$

leads to

$$\frac{\partial f}{\partial \boldsymbol{\sigma}} : \left(\mathbf{D}_e : \dot{\boldsymbol{\epsilon}} - \mathbf{D}_e \dot{\lambda} : \frac{\partial g}{\partial \boldsymbol{\sigma}} \right) - H \sqrt{\frac{3}{2}} \dot{\lambda} = 0\tag{7.84}$$

Solving for $\dot{\lambda}$ we get the plastic multiplier as

$$\dot{\lambda} = \frac{\frac{\partial f}{\partial \boldsymbol{\sigma}} : \mathbf{D}_e : \dot{\boldsymbol{\epsilon}}}{\frac{\partial f}{\partial \boldsymbol{\sigma}} : \mathbf{D}_e : \frac{\partial f}{\partial \boldsymbol{\sigma}} + \sqrt{\frac{3}{2}} H}\tag{7.85}$$

Elastoplastic constitutive tensor

Substituting $\dot{\lambda}$ into the stress rate equation, we obtain

$$\mathbf{D}_{ep} = \mathbf{D}_e - \frac{\mathbf{D}_e : \frac{\partial f}{\partial \boldsymbol{\sigma}} \otimes \frac{\partial f}{\partial \boldsymbol{\sigma}} : \mathbf{D}_e}{\frac{\partial f}{\partial \boldsymbol{\sigma}} : \mathbf{D}_e : \frac{\partial f}{\partial \boldsymbol{\sigma}} + \sqrt{\frac{3}{2}} H}\tag{7.86}$$

Stress update

Given a strain increment $\Delta \boldsymbol{\epsilon}$, the stress at the end of the increment is given by

$$\begin{aligned}\boldsymbol{\sigma}^{i+1} &= \boldsymbol{\sigma}^i + \mathbf{D}_e : \Delta \boldsymbol{\epsilon} - \mathbf{D}_e : \Delta \boldsymbol{\epsilon}_p \\ &= \boldsymbol{\sigma}^{\text{tr}} - \Delta \lambda \mathbf{D}_e : \frac{\partial g}{\partial \boldsymbol{\sigma}} \Big|^{i+1} \\ &= \boldsymbol{\sigma}^{\text{tr}} - \Delta \lambda \mathbf{D}_e : \sqrt{\frac{3}{2}} \hat{\mathbf{s}}\end{aligned}\tag{7.87}$$

Expressing the elastic constitutive tensor in terms of the bulk modulus K and the shear modulus G , we have

$$\begin{aligned}
 \boldsymbol{\sigma}^{i+1} &= \boldsymbol{\sigma}^{\text{tr}} - \Delta\lambda(K\mathbf{I} \otimes \mathbf{I} + 2G\mathbf{P}_{\text{sd}}) : \sqrt{\frac{3}{2}}\hat{\mathbf{s}} \\
 &= \boldsymbol{\sigma}^{\text{tr}} - \Delta\lambda \left(K\mathbf{I} \text{trace} \left(\sqrt{\frac{3}{2}}\hat{\mathbf{s}} \right) + 2G\sqrt{\frac{3}{2}}\hat{\mathbf{s}} \right) \\
 &= \boldsymbol{\sigma}^{\text{tr}} - 2G\Delta\lambda\sqrt{\frac{3}{2}}\hat{\mathbf{s}} \\
 &= \boldsymbol{\sigma}^{\text{tr}} - \sqrt{6}G\Delta\lambda \frac{\mathbf{s}^{i+1}}{\|\mathbf{s}^{i+1}\|} \\
 &= \boldsymbol{\sigma}^{\text{tr}} - \sqrt{6}G\Delta\lambda\hat{\mathbf{s}}^{i+1}
 \end{aligned} \tag{7.88}$$

In turn, the deviatoric stress at the end of the increment can be obtained with the aid of the deviatoric projector tensor as

$$\mathbf{P}_{\text{sd}} : \boldsymbol{\sigma}^{i+1} = \mathbf{P}_{\text{sd}} : \boldsymbol{\sigma}^{\text{tr}} - \mathbf{P}_{\text{sd}} : (\sqrt{6}G\Delta\lambda\hat{\mathbf{s}}^{i+1}) \tag{7.89}$$

Then,

$$\mathbf{s}^{i+1} = \mathbf{s}^{\text{tr}} - \sqrt{6}G\Delta\lambda\hat{\mathbf{s}}^{i+1} \tag{7.90}$$

Considering that $\mathbf{s}^{i+1}$ has the same direction as $\mathbf{s}^{\text{tr}}$, we have

$$\begin{aligned}
 \mathbf{s}^{i+1} &= \mathbf{s}^{\text{tr}} - \sqrt{6}G\Delta\lambda\hat{\mathbf{s}}^{\text{tr}} \\
 &= \left(1 - \frac{\sqrt{6}G\Delta\lambda}{\|\mathbf{s}^{\text{tr}}\|} \right) \mathbf{s}^{\text{tr}} \\
 &= \left(1 - \frac{\sqrt{6}G\Delta\lambda}{\sqrt{2J_2^{\text{tr}}}} \right) \mathbf{s}^{\text{tr}} \\
 &= \left(1 - \frac{\sqrt{3}G\Delta\lambda}{\sqrt{J_2^{\text{tr}}}} \right) \mathbf{s}^{\text{tr}}
 \end{aligned} \tag{7.91}$$

Then, $\sqrt{J_2}$ can be expressed as

$$\begin{aligned}
 \sqrt{J_2^{i+1}} &= \left(1 - \frac{\sqrt{3}G\Delta\lambda}{\sqrt{J_2^{\text{tr}}}} \right) \sqrt{J_2^{\text{tr}}} \\
 &= \sqrt{J_2^{\text{tr}}} - \sqrt{3}G\Delta\lambda \geq 0
 \end{aligned} \tag{7.92}$$

Plastic strain update

For the plastic strain update, we have

$$\begin{aligned}
 \bar{\varepsilon}_p^{i+1} &= \bar{\varepsilon}_p^i + \Delta\bar{\varepsilon}_p \\
 &= \bar{\varepsilon}_p^i + \|\Delta\boldsymbol{\varepsilon}_p\| \\
 &= \bar{\varepsilon}_p^i + \sqrt{\frac{3}{2}}\Delta\lambda
 \end{aligned} \tag{7.93}$$

Plastic multiplier

The yield function at increment $i + 1$ is written as

$$f(\sigma^{i+1}, \bar{\varepsilon}_p^{i+1}) = \sqrt{3J_2^{i+1}} - (f_y + H\bar{\varepsilon}_p^{i+1}) \quad (7.94)$$

Substituting the expressions for $\sqrt{J_2^{i+1}}$ and $\bar{\varepsilon}_p^{i+1}$, we have

$$\begin{aligned} f(\sigma^{i+1}, \bar{\varepsilon}_p^{i+1}) &= \sqrt{3} \left(\sqrt{J_2^{\text{tr}}} - \sqrt{3} G \Delta\lambda \right) - f_y - H \left(\bar{\varepsilon}_p^i + \sqrt{\frac{3}{2}} \Delta\lambda \right) \\ &= \underbrace{\sqrt{3J_2^{\text{tr}}} - f_y - H\bar{\varepsilon}_p^i}_{f^{\text{tr}}(\sigma^{\text{tr}}, \bar{\varepsilon}_p)} - 3G\Delta\lambda - \sqrt{\frac{3}{2}} \Delta\lambda H \\ &= f^{\text{tr}} - 3G\Delta\lambda - \sqrt{\frac{3}{2}} \Delta\lambda H \end{aligned} \quad (7.95)$$

Solving for $\Delta\lambda$ we get

$$\Delta\lambda = \frac{f^{\text{tr}}}{3G + \sqrt{\frac{3}{2}}H} \quad (7.96)$$

Summary

Finally, at the end of a load increment, the stress and internal variables are updated according to

$f^{\text{tr}} \leq 0$	$f^{\text{tr}} > 0$
$\Delta\lambda = 0$	$\Delta\lambda = \frac{f^{\text{tr}}}{3G + \sqrt{\frac{3}{2}}H}$
$\sigma^{i+1} = \sigma^{\text{tr}}$	$\sigma^{i+1} = \sigma^{\text{tr}} - \sqrt{6} G \Delta\lambda \hat{s}^{\text{tr}}$
$\bar{\varepsilon}_p^{i+1} = \bar{\varepsilon}_p^n$	$\Delta\varepsilon_p = \sqrt{\frac{3}{2}} \hat{s}^{\text{tr}} \Delta\lambda$
$\Delta\bar{\varepsilon}_p = 0$	$\bar{\varepsilon}_p^{i+1} = \bar{\varepsilon}_p^n + \sqrt{\frac{3}{2}} \Delta\lambda$

7.4.6 A 1D plasticity model for tension and compression

A 1D plasticity model is useful for the simulation of metals subjected to uniaxial loading where the material present the same strength in tension and compression.

Yield function

In this model, the yield function is given by

$$f(\sigma, \bar{\varepsilon}_p) = |\sigma| - \underbrace{(\sigma_y + H\bar{\varepsilon}_p)}_{h(\bar{\varepsilon}_p)} \quad (7.97)$$

where σ is the uniaxial stress, σ_y is the yield stress (valid for compression and tension), H is the hardening modulus and $\bar{\varepsilon}_p$ is the equivalent plastic strain given by

$$\bar{\varepsilon}_p = \int_0^t |\dot{\varepsilon}_p| dt \approx \sum |\Delta\varepsilon_p| \quad (7.98)$$

The yield function is written in terms of $|\sigma|$ to consider the same strength in tension and compression.

Figure 7.22 shows the stress–strain curve for this model.

Yield function derivatives

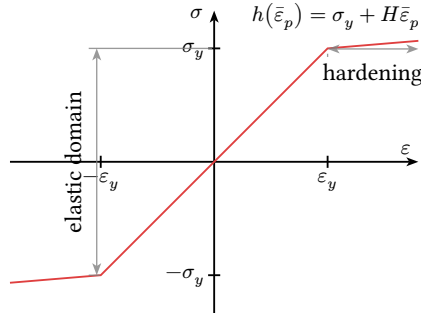


Figure 7.22: Stress-strain curve for the 1D plasticity model.

The derivative with respect to the uniaxial stress is

$$\frac{\partial f}{\partial \sigma} = \text{sgn}(\sigma) \quad (7.99)$$

In turn, the derivative with respect to the plastic strain is

$$\frac{\partial f}{\partial \bar{\varepsilon}_p} = -H \quad (7.100)$$

Flow rule

Considering an associated flow rule, we have

$$\dot{\varepsilon}_p = \dot{\lambda} \frac{\partial f}{\partial \sigma} \quad (7.101)$$

Since $\frac{\partial f}{\partial \sigma} = \text{sgn}(\sigma)$ and, by definition, $\dot{\lambda} > 0$, we have

$$\dot{\lambda} = |\dot{\varepsilon}_p| = \dot{\bar{\varepsilon}}_p \quad (7.102)$$

Consistency condition

For this case, the consistency condition is given by

$$\frac{\partial f}{\partial \sigma} \dot{\sigma} + \frac{\partial f}{\partial \bar{\varepsilon}_p} \dot{\bar{\varepsilon}}_p = 0 \quad (7.103)$$

Using the additive decomposition of strains, the stress rate is given by

$$\begin{aligned} \dot{\sigma} &= E(\dot{\varepsilon} - \dot{\varepsilon}_p) \\ &= E(\dot{\varepsilon} - \dot{\lambda} \text{sgn}(\sigma)) \end{aligned} \quad (7.104)$$

Substituting into the consistency condition we have

$$\text{sgn}(\sigma) E(\dot{\varepsilon} - \dot{\lambda} \text{sgn}(\sigma)) - H \dot{\lambda} = 0 \quad (7.105)$$

Solving for $\dot{\lambda}$ we obtain the plastic multiplier as

$$\dot{\lambda} = \frac{E}{E + H} \text{sgn}(\sigma) \dot{\varepsilon} \quad (7.106)$$

Elastoplastic constitutive scalar

Substituting $\dot{\lambda}$ into the stress rate equation we get

$$\begin{aligned}\dot{\sigma} &= \left(E - \frac{E^2}{E + H} \right) \dot{\varepsilon} \\ &= \frac{EH}{E + H} \dot{\varepsilon}\end{aligned}\tag{7.107}$$

Thus, the elastoplastic constitutive scalar is given by

$$D_{ep} = \frac{EH}{E + H}\tag{7.108}$$

Stress update

Given a strain increment $\Delta\varepsilon$, the stress at the end of the increment is given by

$$\begin{aligned}\sigma^{i+1} &= \sigma^i + E\Delta\varepsilon - E\Delta\varepsilon_p \\ &= \sigma^{\text{tr}} - \Delta\lambda E \operatorname{sgn}(\sigma^{i+1})\end{aligned}\tag{7.109}$$

Assuming $\operatorname{sgn}(\sigma^{\text{tr}}) = \operatorname{sgn}(\sigma^{i+1})$,

$$\sigma^{i+1} = \sigma^{\text{tr}} - \Delta\lambda E \operatorname{sgn}(\sigma^{\text{tr}})\tag{7.110}$$

Also, multiplying both sides by $\operatorname{sgn}(\sigma^{\text{tr}})$ we have

$$|\sigma^{i+1}| = |\sigma^{\text{tr}}| - \Delta\lambda E\tag{7.111}$$

The equivalent plastic strain is updated as

$$\begin{aligned}\bar{\varepsilon}_p^{i+1} &= \bar{\varepsilon}_p^i + |\Delta\varepsilon_p| \\ &= \bar{\varepsilon}_p^i + \Delta\lambda\end{aligned}\tag{7.112}$$

Plastic multiplier

The yield function at the end of the increment is

$$f(\sigma^{i+1}, \bar{\varepsilon}_p) = |\sigma^{i+1}| - (\sigma_y + H\bar{\varepsilon}_p^{i+1})\tag{7.113}$$

Substituting the expression for $|\sigma^{i+1}|$ and $\bar{\varepsilon}_p^{i+1}$ we have

$$\begin{aligned}f(\sigma^{i+1}, \bar{\varepsilon}_p) &= |\sigma^{\text{tr}}| - \Delta\lambda E - \sigma_y - H(\bar{\varepsilon}_p^i + \Delta\lambda) \\ &= \underbrace{|\sigma^{\text{tr}}| - \sigma_y - H\bar{\varepsilon}_p^i}_{f(\sigma^{\text{tr}}, \bar{\varepsilon}_p^i)} - \Delta\lambda E - \Delta\lambda H\end{aligned}\tag{7.114}$$

Solving for $\Delta\lambda$ we get

$$\Delta\lambda = \frac{f^{\text{tr}}}{E + H}\tag{7.115}$$

Summary

Finally, at the end of a load increment, the stress and internal variables are updated according to

$$\begin{array}{cc} f^{\text{tr}} \leq 0 & f^{\text{tr}} > 0 \\ \hline \Delta\lambda = 0 & \Delta\lambda = \frac{f^{\text{tr}}}{E+H} \\ \Delta\bar{\varepsilon}_p = 0 & \Delta\varepsilon_p = \Delta\lambda \operatorname{sgn}(\sigma^{\text{tr}}) \\ \bar{\varepsilon}_p^{i+1} = \bar{\varepsilon}_p^n & \bar{\varepsilon}^{i+1} = \bar{\varepsilon}_p^n + \Delta\lambda \\ \sigma^{i+1} = \sigma^{\text{tr}} & \sigma^{i+1} = \sigma^{\text{tr}} - E\Delta\varepsilon_p \end{array}$$

7.4.7 A 2D contact plasticity model

This model aims to simulate the interaction between two surfaces. The model considers the normal and shear stresses (σ_n and τ) at the interface as well as the normal and shear relative displacements (w_n and w_s).

In the elastic regime, the stress components are related to the relative displacements by the elastic constitutive matrix as $\boldsymbol{\sigma} = \mathbf{D}_e \mathbf{w}$, which can be written in full form as

$$\begin{pmatrix} \sigma_n \\ \tau \end{pmatrix} = \begin{pmatrix} k_n & 0 \\ 0 & k_t \end{pmatrix} \begin{pmatrix} w_n \\ w_s \end{pmatrix} \quad (7.116)$$

where k_n and k_t are, respectively, the normal and tangential stiffness parameters of the interface.

For the elastoplastic regime, the rate of relative displacements are decomposed into elastic and plastic components as

$$\dot{w} = \dot{w}_e + \dot{w}_p \quad (7.117)$$

thus, the stress rate can be written as

$$\begin{aligned}\dot{\sigma} &= D_e : (\dot{w} - \dot{w}_p) \\ &= D_e : \left(\dot{w} - \dot{\lambda} \frac{\partial g}{\partial \sigma} \right)\end{aligned}\tag{7.118}$$

where g is the plastic potential function.

Yield function

In this model, the yield function is given by

$$f(\boldsymbol{\sigma}, \sigma_{\max}) = |\tau| + (\sigma_n - \sigma_{\max}(\bar{w}_p)) \mu \quad (7.119)$$

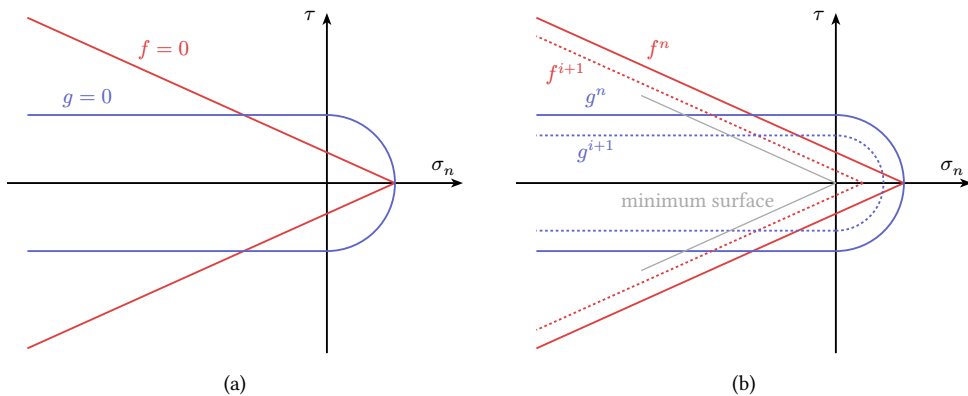


Figure 7.23: 2D contact plasticity model: (a) yield, f , and plastic potential, g , functions; (b) evolution of f and g .

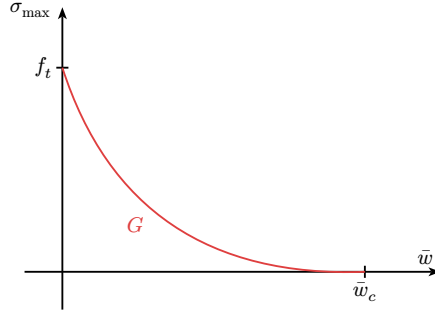


Figure 7.24: Contact strength $\sigma_{\max}$ as a function of the equivalent displacement $\bar{w}$.

where $\sigma = \langle \sigma_n, \tau \rangle$, μ is the friction coefficient of the interface and $\sigma_{\max}$ is a state variable that depends on the equivalent plastic displacement $\bar{w}_p$.

Yield function derivatives

The derivative of the yield function with respect to the stress vector is

$$\frac{\partial f}{\partial \sigma} = \begin{pmatrix} \mu \\ \text{sgn}(\tau) \end{pmatrix} \quad (7.120)$$

In turn, the derivative with respect to the equivalent plastic displacement is

$$\begin{aligned} \frac{\partial f}{\partial \sigma_{\max}} &= \frac{\partial f}{\partial \sigma_{\max}} \frac{\partial \sigma_{\max}}{\partial \bar{w}_p} \\ &= -\mu \frac{\partial \sigma_{\max}}{\partial \bar{w}_p} \end{aligned} \quad (7.121)$$

Plastic potential function and derivatives

The model uses a non associated flow rule with a piece-wise plastic potential function defined as

$$g(\sigma, \sigma_{\max}) = \begin{cases} \tau^2 - \sigma_{\max}^2 = 0 & \text{for } \sigma_n \leq 0 \\ \sigma_n^2 + \tau^2 - \sigma_{\max}^2 = 0 & \text{for } \sigma_n > 0 \end{cases} \quad (7.122)$$

The derivative with respect to the stress vector is

$$\frac{\partial g}{\partial \sigma} = \begin{cases} \begin{pmatrix} 2\sigma_n \mu^2 \\ 2\tau \end{pmatrix} & \text{for } \sigma_n \leq 0 \\ \begin{pmatrix} 0 \\ 2\tau \end{pmatrix} & \text{for } \sigma_n > 0 \end{cases} \quad (7.123)$$

Flow rule

The rate of plastic relative displacements is given by

$$\dot{w}_p = \dot{\lambda} \frac{\partial g}{\partial \sigma} \quad (7.124)$$

Then, an equivalent plastic displacement is defined as

$$\dot{\bar{w}}_p = \dot{\lambda} \left\| \frac{\partial g}{\partial \sigma} \right\| \quad (7.125)$$

Consistency condition

The consistency condition is written as

$$\frac{\partial f}{\partial \boldsymbol{\sigma}} \cdot \dot{\boldsymbol{\sigma}} + \frac{\partial f}{\partial \sigma_{\max}} \frac{\partial \sigma_{\max}}{\partial \bar{w}_p} \dot{\bar{w}}_p = 0 \quad (7.126)$$

Substituting the expressions for the rate of stress, the equivalent plastic displacement and the corresponding derivatives we have

Later, using the additive decomposition of relative displacements and expanding the second term, we have

$$\frac{\partial f}{\partial \boldsymbol{\sigma}} \cdot \left(D_e \dot{\mathbf{w}} - \dot{\lambda} D_e \frac{\partial g}{\partial \boldsymbol{\sigma}} \right) + \frac{\partial f}{\partial \sigma_{\max}} \frac{\partial \sigma_{\max}}{\partial \bar{w}_p} \dot{\lambda} \left\| \frac{\partial g}{\partial \boldsymbol{\sigma}} \right\| = 0 \quad (7.127)$$

Solving for $\dot{\lambda}$ we get the plastic multiplier as

$$\dot{\lambda} = \frac{\frac{\partial f}{\partial \boldsymbol{\sigma}} \cdot D_e \dot{\mathbf{w}}}{\frac{\partial f}{\partial \boldsymbol{\sigma}} \cdot D_e \frac{\partial g}{\partial \boldsymbol{\sigma}} - \frac{\partial f}{\partial \sigma_{\max}} \frac{\partial \sigma_{\max}}{\partial \bar{w}_p} \left\| \frac{\partial g}{\partial \boldsymbol{\sigma}} \right\|} \quad (7.128)$$

Elastoplastic constitutive matrix

Substituting the expression for $\dot{\lambda}$ into the stress rate equation we get the elastoplastic constitutive matrix as

$$D_{ep} = D_e - \frac{D_e \frac{\partial g}{\partial \boldsymbol{\sigma}} \frac{\partial f}{\partial \boldsymbol{\sigma}}^T D_e}{\frac{\partial f}{\partial \boldsymbol{\sigma}}^T D_e \frac{\partial g}{\partial \boldsymbol{\sigma}} - \frac{\partial f}{\partial \sigma_{\max}} \frac{\partial \sigma_{\max}}{\partial \bar{w}_p} \left\| \frac{\partial g}{\partial \boldsymbol{\sigma}} \right\|} \quad (7.129)$$

Stress update

Given an increment of relative displacements $\Delta \mathbf{w}$, the stress at the end of the increment is given by

$$\begin{aligned} \boldsymbol{\sigma}^{i+1} &= \boldsymbol{\sigma}^i + D_e \Delta \mathbf{w} - \Delta \lambda D_e \frac{\partial g}{\partial \boldsymbol{\sigma}} \Big|^{i+1} \\ &= \boldsymbol{\sigma}^{\text{tr}} - \Delta \lambda D_e \frac{\partial g}{\partial \boldsymbol{\sigma}} \Big|^{i+1} \end{aligned} \quad (7.130)$$

The internal variables are updated as

$$\begin{aligned} \sigma_{\max}^{i+1} &= \sigma_{\max}^i + \frac{\partial \sigma_{\max}}{\partial \bar{w}_p} \Delta \bar{w}_p^{i+1} \\ \bar{w}_p^{i+1} &= \bar{w}_p^i + \Delta \lambda \left\| \frac{\partial g}{\partial \boldsymbol{\sigma}} \Big|^{i+1} \right\| \end{aligned} \quad (7.131)$$

Plastic multiplier

To find the plastic multiplier we consider the yield function at the end of the increment

$$f(\boldsymbol{\sigma}^{i+1}, \sigma_{\max}^{i+1}) = |\tau^{i+1}| + (\sigma_n^{i+1} - \sigma_{\max}^{i+1})\mu = 0 \quad (7.132)$$

where

$$\begin{aligned}
 \sigma_n^{i+1} &= \sigma_n^{\text{tr}} - \Delta\lambda k_n \left. \frac{\partial g}{\partial \sigma_n} \right|^{i+1} \\
 \tau^{i+1} &= \tau^{\text{tr}} - \Delta\lambda k_s \left. \frac{\partial g}{\partial \tau} \right|^{i+1} \\
 \sigma_{\text{max}}^{i+1} &= \sigma_{\text{max}}^i + \frac{\partial \sigma_{\text{max}}}{\partial \bar{w}_p} \Delta\lambda \left\| \left. \frac{\partial g}{\partial \sigma} \right|^{i+1} \right\|
 \end{aligned} \tag{7.133}$$

$\Delta\lambda$ is found by solving $f(\sigma^{i+1}, \sigma_{\text{max}}^{i+1}) = 0$ which can be performed numerically.

After $\Delta\lambda$ is found, the stress vector can be written as

$$\begin{pmatrix} \sigma^{i+1} \\ \tau^{i+1} \end{pmatrix} = \begin{pmatrix} \sigma^{\text{tr}} \\ \tau^{\text{tr}} \end{pmatrix} - \Delta\lambda \begin{pmatrix} k_n & 0 \\ 0 & k_s \end{pmatrix} \begin{pmatrix} \left. \frac{\partial g}{\partial \sigma_n} \right|^{i+1} \\ \left. \frac{\partial g}{\partial \tau} \right|^{i+1} \end{pmatrix} \tag{7.134}$$

Note that the derivatives of the plastic potential function also depend on the stress components σ_n^{i+1} and τ^{i+1} . Since g is a piece-wise function, two cases are obtained for the stresses at the end of the increment. Solving for the stress components and considering $\text{sgn}(\sigma_n^{i+1}) = \text{sgn}(\sigma_n^{\text{tr}})$ we have

$$\sigma_n^{i+1} = \begin{cases} \left(\frac{\sigma_n}{\frac{1+2\Delta\lambda k_n \mu^2}{1+2\Delta\lambda k_s}} \right) & \text{for } \sigma_n^{\text{tr}} > 0 \\ \left(\frac{\sigma_n}{\frac{\tau}{1+2\Delta\lambda k_s}} \right) & \text{for } \sigma_n^{\text{tr}} \leq 0 \end{cases} \tag{7.135}$$

Finally, at the end of a load increment, the internal state is updated according to

$$\begin{array}{cc}
 f^{\text{tr}} \leq 0 & f^{\text{tr}} > 0 \\
 \hline
 \Delta\lambda = 0 & \Delta\lambda \text{ is found by solving } f^{i+1} = 0 \\
 \sigma^{i+1} = \sigma^{\text{tr}} & \sigma^{i+1} = \begin{cases} \left(\frac{\sigma_n}{\frac{1+2\Delta\lambda k_n \mu^2}{1+2\Delta\lambda k_s}} \right) & \text{for } \sigma_n^{\text{tr}} > 0 \\ \left(\frac{\sigma_n}{\frac{\tau}{1+2\Delta\lambda k_s}} \right) & \text{for } \sigma_n^{\text{tr}} \leq 0 \end{cases} \\
 \Delta w_p = 0 & \Delta w_p = \Delta\lambda \left. \frac{\partial g}{\partial \sigma} \right|^{i+1} \\
 \bar{w}_p^{i+1} = \bar{w}_p^i & \bar{w}_p^{i+1} = \bar{w}_p^i + \Delta\lambda \left\| \left. \frac{\partial g}{\partial \sigma} \right|^{i+1} \right\|
 \end{array}$$

Exercises 7.2:

1. The von Mises model is being used to simulate the mechanical response of a metal. The material properties are $E = 200$ GPa, $\nu = 0.3$, $\sigma_y = 500$ MPa and $H = 100$ MPa. The current state is defined by the stress σ and the equivalent plastic strain $\bar{\epsilon}_p = 0$. Given the strain increment $\Delta\epsilon$, determine whether the material remains in the elastic regime or transitions into the plastic regime. In either case, compute the stress tensor and the equivalent plastic strain at the end of the increment. Consider

$$\sigma = \begin{pmatrix} 195 & 0 & 0 \\ 0 & 195 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad \Delta\epsilon = \begin{pmatrix} 0.0001 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

2. A 1D plasticity model is being developed to simulate softening behavior under tension. The material behaves as linear-elastic up to the strain $\epsilon = \epsilon_c$, where the tension reaches the tensile strength f_y . Afterwards, the material presents the following softening rule:

$$\sigma_y = f_y \left(\frac{\varepsilon_c}{\varepsilon_c + H\varepsilon_p} \right)$$

where H is a material parameter. Considering E_0 as the initial Young's modulus,

- draw a sketch of the stress-strain curve for the material;
- find the expression for the elastoplastic constitutive scalar D_{ep} ;
- find the equations to perform the stress-update step.

8 Variational methods

Methods of weighted residuals and variational methods are two approaches used to develop finite element formulations.

A weighted residual method is an approximation technique in which the solution of a **differential equation** is represented by a linear combination of basis functions. This approximation is then substituted into the differential equation, producing an error, or residual. Finally, the residual is driven to zero in an average sense over the solution domain by means of weighted integrals.

Weak formulations are obtained by integrating the weighted residual so that the order of the derivatives of the dependent variable is reduced.

In the variational approach, the physical problem is restated through a **variational principle**, such as the principle of minimum potential energy. This approach is used to derive finite element equations whenever a **variational statement is available**. Its main disadvantage is that many physical problems do not admit a known variational statement.

The method of virtual work can also be considered a variational approach. It uses an equilibrium statement based on virtual displacements to derive finite element equations.

8.1 Forms of boundary value problems

Boundary value problems can be written in several forms, such as the strong form and the weak form.

8.1.1 Strong form

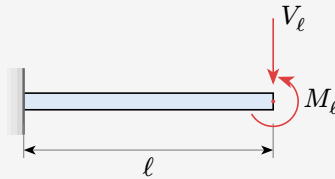
This form is essentially the BVP written directly as a differential equation together with its boundary conditions. It requires equilibrium, kinematics, and the material law to be satisfied at every point in the domain. For example, for a differential equation $\mathcal{D}(u(x)) = p(x)$, the strong form is given by:

$$\{\mathcal{D}(u(x)) = p(x) \quad \text{for } x \in \Omega, u(a) = u_a, u(b) = u_b\} \quad (8.1)$$

where $\mathcal{D}$ is a linear differential operator, $p(x)$ is a known function, and $u(x)$ is the unknown function to be determined. The boundary conditions are given by $u(a) = u_a$ and $u(b) = u_b$.

Most strong forms in physical problems do not have an analytical solution. Therefore, the solution is usually approximated using numerical methods.

Example 8.1: Considering the BVP of a beam clamped at the left end and subjected to a shear force and a bending moment at the right end, write the corresponding strong form.



Solution:

Assuming constant E and I , the Euler-Bernoulli theory can be expressed in terms of the fourth-order differential equation for the deflection $v(x)$ of a beam with length ℓ and subjected to a distributed load $q(x)$:

$$EI \frac{d^4 v}{dx^4} - q(x) = 0 \quad \text{for } x \in (0, \ell)$$

where E is the Young's modulus, I is the section moment of inertia.

By integration of the differential equation, for $q(x) = 0$, we can express the bending moment $M(x)$, shear force $V(x)$, and rotation $\theta(x)$ in terms of the deflection $v(x)$ as follows:

$$\begin{aligned} V(x) &= EI \frac{d^3 v}{dx^3} \\ M(x) &= EI \frac{d^2 v}{dx^2} \\ \theta(x) &= \frac{dv}{dx} \end{aligned}$$

Finally, the strong form of the BVP is obtained by combining the differential equation with the corresponding boundary conditions:

$$\begin{cases} EI \frac{d^4 v}{dx^4} = 0 & \text{for } x \in (0, \ell) \\ v(0) = 0 \\ \frac{dv}{dx}(0) = 0 \\ EI \frac{d^2 v}{dx^2}(\ell) = M_\ell \\ EI \frac{d^3 v}{dx^3}(\ell) = V_\ell \end{cases}$$

8.1.2 Weighted residual form

This form is usually obtained by writing a weighted integral of the differential equation over the domain. A weighted integral of the differential equation is given by:

$$\int_{\Omega} w(x) (\mathcal{D}(u(x)) - p(x)) dx = 0 \quad (8.2)$$

where $w(x)$ is an arbitrary weight function.

An approximate solution $u(x)$ can be defined as a linear combination of trial functions $N_i(x)$:

$$u(x) = \sum_{i=1}^n c_i N_i(x) \quad (8.3)$$

where c_i are constants to be determined.

Trial functions usually take the form of polynomials. Their main requirement is that they be **admissible functions**; that is, the functions and their derivatives appearing in the integrand must be continuous enough for the formulation, and the resulting approximation must be able to satisfy the boundary conditions.

The resulting solution is approximate because it satisfies the differential equation only in an average integral sense.

Example 8.2: For the Euler-Bernoulli beam example, find the weighted residual form of the BVP.

Solution:

Assuming constant E and I , the differential equation of the BVP is

$$EI \frac{d^4 v}{dx^4} - q(x) = 0 \quad \text{for } x \in (0, \ell)$$

Assuming an arbitrary weight function $w(x)$, the weighted residual form of the BVP is given by:

$$\int_0^\ell \left(EI \frac{d^4 v}{dx^4} - q(x) \right) w(x) dx = 0$$

with an approximate solution of the form:

$$v(x) = \sum_{i=1}^n c_i N_i(x)$$

where $N_{i(x)}$ are trial functions and c_i are constants to be determined.

8.1.3 Weak Form

The weak form is typically derived by applying **integration by parts** to the weighted residual form of a boundary value problem (BVP), followed by the application of natural (Neumann-type) boundary conditions. This process reduces the order of the highest derivatives acting on the unknown (dependent) function, making it suitable for broader classes of trial functions.

In essence, the weak form replaces the requirement to satisfy a differential equation pointwise (as in the strong form) with the requirement that an associated integral equation holds for a set of test (weight) functions. This allows the formulation to accommodate less regular solutions.

In statics, the weak form can also be derived from the principle of virtual work, and in dynamics, from d'Alembert's principle.

The weak form is mathematically equivalent to the strong form under sufficient regularity conditions. A function that satisfies the strong form also satisfies the weak form, and vice versa. However, in numerical methods such as the finite element method (FEM), approximate solutions are obtained by selecting specific trial (basis) and test (weight) functions. These approximations generally satisfy the weak form, and therefore the original BVP, only in an average or weighted integral sense rather than exactly.

Example 8.3: For the Euler-Bernoulli beam example, find the weak form of the BVP.

Solution:

Starting from the weighted residual form of the BVP, we write:

$$EI \int_0^\ell \frac{d^4 v}{dx^4} w(x) dx = \int_0^\ell q(x) w(x) dx$$

The next step is to integrate the first integral by parts. Using the integration by parts formula:

$$\int_0^\ell \bar{u} d\bar{v} = [\bar{u}\bar{v}]_0^\ell - \int_0^\ell \bar{v} d\bar{u}$$

Taking $\bar{u} = w(x)$ and $d\bar{v} = \frac{d^4 v}{dx^4} dx$, the integral becomes

$$\int_0^\ell \frac{d^4 v}{dx^4} w(x) dx = \left[w(x) \frac{d^3 v}{dx^3} \right]_0^\ell - \int_0^\ell \frac{d^3 v}{dx^3} \frac{dw}{dx} dx$$

Applying integration by parts once again to the last integral, we obtain:

$$\int_0^\ell \frac{d^3 v}{dx^3} \frac{dw}{dx} dx = \left[\frac{dw}{dx} \frac{d^2 v}{dx^2} \right]_0^\ell - \int_0^\ell \frac{d^2 v}{dx^2} \frac{d^2 w}{dx^2} dx$$

Substituting this result into the weighted residual form, we have:

$$\int_0^\ell EI \frac{d^2 w}{dx^2} \frac{d^2 v}{dx^2} dx - \int_0^\ell wq dx + \left[wEI \frac{d^3 v}{dx^3} \right]_0^\ell - \left[\frac{dw}{dx} EI \frac{d^2 v}{dx^2} \right]_0^\ell = 0$$

or, more compactly,

$$\int_0^\ell EI \frac{d^2 w}{dx^2} \frac{d^2 v}{dx^2} dx - \int_0^\ell wq dx + [wV]_0^\ell - \left[\frac{dw}{dx} M \right]_0^\ell = 0$$

To obtain the weak form for this cantilever beam problem, the test functions must satisfy the essential boundary conditions:

$$w(0) = 0, \quad \frac{dw}{dx}(0) = 0$$

Using these admissibility conditions together with the natural boundary conditions $M(\ell) = M_\ell$ and $V(\ell) = V_\ell$, the weak form becomes:

$$\int_0^\ell EI \frac{d^2 w}{dx^2} \frac{d^2 v}{dx^2} dx = \int_0^\ell wq dx - w(\ell)V_\ell + \frac{dw}{dx}(\ell)M_\ell$$

This is the weak form of the BVP.

8.1.4 Variational Form

In the variational form, the problem is formulated using a variational principle, such as the principle of minimum potential energy in statics or Hamilton's principle in dynamics.

This formulation applies only to **conservative systems**, and it requires the governing equations to be satisfied in an average or integral sense rather than pointwise.

Note 8.1: A system is said to be conservative when its internal and external forces can be derived from potential energy functions. In such systems, no mechanical energy is dissipated, the work of the forces is path-independent, and the stored energy can be recovered upon unloading.

Like the weak form, the variational form is mathematically equivalent to the strong form under appropriate smoothness assumptions. However, unlike the weak form, a variational formulation is not always available: most physical problems do not admit a known variational principle. When a variational form does exist, the weak, strong, and weighted residual formulations can typically be derived from it.

Methods that solve boundary value problems using a variational formulation are often called energy methods. For example, in the case of a Bernoulli–Euler beam, the variational form arises from the total potential energy, which includes the **internal elastic energy** and the **potential of the external load**:

$$\Pi(v) = \int_0^\ell \left(\frac{1}{2} EI \left(\frac{d^2 v}{dx^2} \right)^2 - qv \right) dx \quad (8.4)$$

where Π is the total potential energy functional.

Equilibrium is achieved when the **first variation** of the functional vanishes:

$$\delta \Pi = 0 \quad (8.5)$$

This condition expresses a stationary condition on the total potential energy functional. For stable equilibrium in a conservative system, this stationary value is a minimum.

Note 8.2: The first variation is the infinitesimal change of the functional $\Pi(v)$ caused by an infinitesimal admissible change in the unknown function $v(x)$.

Exercises 8.1:

1. Find the weak formulation of the BVP:

$$\begin{aligned} \frac{d^2 u}{dx^2} &= f(x) \quad \text{for } x \in (0, \ell) \\ u(0) &= u_0 \\ \left(x \frac{du}{dx} \right) \Big|_{x=\ell} &= Q \end{aligned}$$

2. A bar of length ℓ , fixed at its left end, is subjected to an axial force f at the right end and a distributed load $q(x)$ along its length. This BVP is governed by the following equation, where E is the Young's modulus and A is the cross-sectional area of the bar. Find the corresponding weak formulation.

$$\begin{aligned} EA \frac{d^2 u}{dx^2} + q(x) &= 0 \quad \text{for } x \in (0, \ell) \\ u(0) &= 0 \\ EA \frac{du}{dx}(\ell) &= f \end{aligned}$$

3. Write the variational form for the problem of displacements of a bar due to axial load.

8.2 Weighted Residuals

Weighted residual methods approximate the solution of a differential equation by a linear combination of basis functions. The resulting error is then driven to zero in an average sense over the solution domain by means of weighted integrals.

8.2.1 Weighted Integrals

A weight function is often used to compute integrals or averages of other functions. In this way, it defines a weighted integral or a weighted average.

For instance, a weighted average for a function $f(x)$ in the domain (a, b) is computed as:

$$\bar{f}|_a^b = \frac{\int_a^b f(x)w(x) dx}{\int_a^b w(x) dx} \quad (8.6)$$

where $w(x)$ is a weight function. Using this concept, a measure of the error of the approximate solution of a BVP can be given by the integral of the error function $R(x)$ over the domain.

$$E = \int_a^b w(x)R(x) dx \quad (8.7)$$

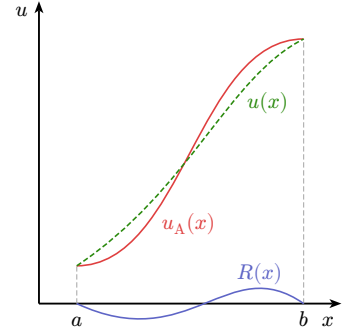


Figure 8.1: Illustration of the true and approximate solutions of a BVP, and the residual function.

8.2.2 Approximate solution of a BVP

Weighted integrals can be used to approximate the solution of a boundary value problem (BVP) by forcing weighted averages of the residual function to vanish. For instance, given the differential equation:

$$\mathcal{D}(u(x)) = p(x) \quad (8.8)$$

where $\mathcal{D}$ is a linear differential operator, we can approximate the solution $u(x)$ by a linear combination of basis functions $N_i(x)$, also called trial functions, as follows:

$$u(x) \approx \sum_{i=1}^n c_i N_i(x) + N_0(x) \quad (8.9)$$

where c_i are constants to be determined. The $N_i(x)$ functions are chosen such that the specified boundary conditions of the problem can be satisfied by $u(x)$.

When $u(x)$ is substituted into the differential equation, the equality is generally not satisfied, which leads to an error, or residual:

$$R(x) = \mathcal{D}(u(x)) - p(x) \neq 0 \quad (8.10)$$

The central idea of the Weighted Residual Method (WRM) is to reduce the residual to zero over the domain in an average sense as:

$$\int_{\Omega} w_i(x)R(x) dx = 0 \quad i = 1, 2, 3, \dots, n \quad (8.11)$$

where $w_i(x)$ are weight functions that can be chosen arbitrarily. The number of weight functions should be equal to the number of unknown constants c_i .

Example 8.4: Using weighted integrals, find the approximate solution of the following boundary value problem (BVP):

$$\begin{aligned} \frac{d^2 u}{dx^2} + u &= 0 \quad \text{for } x \in (0, 1) \\ u(0) &= 0 \\ u(1) &= 1 \end{aligned}$$

For reference, consider the analytical solution:

$$u_A(x) = \frac{\sin(x)}{\sin(1)}$$

Solution:

To compute an approximate solution $u(x)$ over the domain $(0, 1)$, we use a linear combination of basis functions taken from a linearly independent set $N_i(x)$:

$$u(x) = N_0(x) + c_1 N_1(x)$$

Let us take $N_0(x) = x$ and $N_1(x) = x(x - 1)$. Then the approximate solution becomes:

$$u(x) = x + c_1 x(x - 1)$$

The corresponding residual function is:

$$\begin{aligned} R(x) &= \frac{d^2 u}{dx^2} + u(x) \\ &= x + c_1(2 - x + x^2) \end{aligned}$$

To determine c_1 , we impose the weighted residual condition over the interval $(0, 1)$. For this, we select an arbitrary weight function, e.g., $w(x) = x$. Then, the weighted residual condition is expressed as:

$$\begin{aligned} \int_0^1 w(x) R(x) dx &= 0 \\ \int_0^1 x R(x) dx &= 0 \\ \int_0^1 x(x + c_1(2 - x + x^2)) dx &= 0 \end{aligned}$$

Evaluating the integral yields the equation

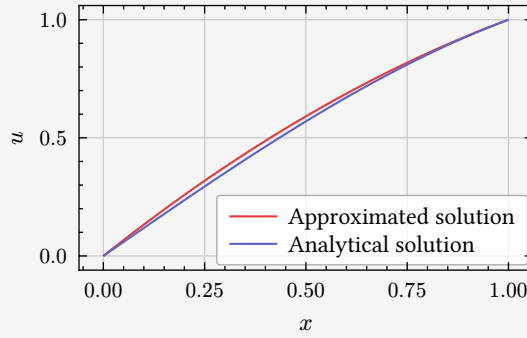
$$\frac{1}{3} + c_1 \left(1 - \frac{1}{3} + \frac{1}{4} \right) = 0$$

which leads to $c_1 = -\frac{4}{11}$.

Finally, the approximate solution and the residual functions are given by:

$$\begin{aligned} u(x) &= \frac{15}{11}x - \frac{4}{11}x^2 \\ R(x) &= -\frac{8}{11} + \frac{15}{11}x - \frac{4}{11}x^2 \end{aligned}$$

The figure below shows the approximate solution in comparison with the analytical one. Note that the approximate solution satisfies the boundary conditions $u(0) = 0$ and $u(1) = 1$.



Example 8.5: Improve the approximate solution of the previous example using an additional trial function.

Solution:

Let's now include an additional trial function, $N_2(x) = x^2(x - 1)$, which is a third-degree polynomial. With this, the approximate solution $u(x)$ becomes:

$$\begin{aligned} u(x) &= N_0(x) + c_1 N_1(x) + c_2 N_2(x) \\ u(x) &= x + c_1 x(x - 1) + c_2 x^2(x - 1) \end{aligned}$$

The corresponding residual function is:

$$\begin{aligned} R(x) &= \frac{d^2 u}{dx^2} + u(x) \\ &= c_1 x^2 - c_1 x + 2c_1 + c_2 x^3 - c_2 x^2 + 6c_2 x - 2c_2 + x \end{aligned}$$

Since there are two unknown constants, c_1 and c_2 , we require two additional equations. These are obtained using weighted integrals of the residual. For instance, we can choose the following weight functions:

$$w_1(x) = 1 \quad \text{and} \quad w_2(x) = x$$

Thus, the extra equations are:

$$\begin{aligned} \int_0^1 1(c_1 x^2 - c_1 x + 2c_1 + c_2 x^3 - c_2 x^2 + 6c_2 x - 2c_2 + x) dx &= 0 \\ \int_0^1 x(c_1 x^2 - c_1 x + 2c_1 + c_2 x^3 - c_2 x^2 + 6c_2 x - 2c_2 + x) dx &= 0 \end{aligned}$$

Finally, after evaluating the two weighted integrals, we obtain the following system of equations:

$$\begin{aligned} \frac{11}{6}c_1 + \frac{11}{12}c_2 + \frac{1}{2} &= 0 \\ \frac{11}{12}c_1 + \frac{19}{20}c_2 + \frac{1}{3} &= 0 \end{aligned}$$

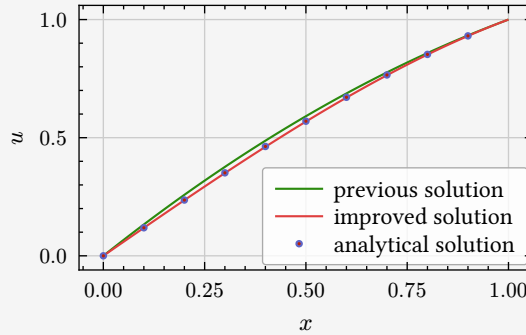
that leads to the solutions

$$c_1 = \frac{-122}{649} \quad \text{and} \quad c_2 = \frac{-10}{59}$$

Finally, the approximate solution and the residual functions are given by:

$$u(x) = -\frac{10}{59}x^3 - \frac{12}{649}x^2 + \frac{771}{649}x$$

$$R(x) = -\frac{10}{59}x^3 - \frac{12}{649}x^2 + \frac{111}{649}x - \frac{24}{649}$$



Exercises 8.2:

- Find an approximate solution for the BVP using weighted integrals using two basis function and the weight function $w(x) = 1$. Then, improve the solution using an additional basis function and the weight functions $w_1(x) = 1$ and $w_2(x) = x$.

$$\frac{d^2u}{dx^2} + u = 1 \quad \text{for } x \in (0, 1)$$

$$u(0) = 1$$

$$u(1) = 0$$

- Find an approximate solution for the BVP using two basis functions. Choose the corresponding weight function.

$$\frac{d^2u}{dx^2} + u = 0 \quad \text{for } x \in (0, 1)$$

$$u(0) = 1$$

$$\left[x \frac{du}{dx} \right]_{x=1} = 0$$

8.3 Weighted Residuals Methods

There is a well-known family of weighted residual methods (WRM) used to approximate the solution of a boundary value problem (BVP). The main idea is to represent the solution as a linear combination of basis (trial) functions and then reduce the residual in an average sense using weighted integrals.

The choice of the weight functions defines the specific method of weighted residuals. Among the most common methods are:

- Collocation method
- Least squares method
- Galerkin method
- Petrov-Galerkin method
- Sub-domain method

- Method of moments

We will focus on the first three methods.

8.3.1 Collocation Method

In the collocation method, the residual function is forced to be zero at specific points x_i within the domain. Thus, we obtain a set of equations by evaluating the residual function at these points:

$$R(x_i) = 0, \quad i = 1, 2, \dots, n \quad (8.12)$$

To fit this method into the framework of weighted residuals, we need to find corresponding weight functions $w_i(x)$ such that the equations above are satisfied. For this purpose, the weighting functions are chosen as Dirac delta functions centered at the points x_i . Thus:

$$w_i(x) = \bar{\delta}(x - x_i) \quad (8.13)$$

Note 8.3: The Dirac's δ function is defined as:

$$\bar{\delta}(t) = \begin{cases} 0 & \text{if } t \neq 0 \\ \infty & \text{if } t = 0 \end{cases} \quad \text{and} \quad \int_{-\infty}^{+\infty} \delta(t) dt = 1$$

Also, the **sampling property** of the Dirac delta function states that:

$$\int_{-\infty}^{+\infty} \bar{\delta}(t - x_i) f(t) dt = f(x_i)$$

Using Dirac's delta functions as weight functions, we have:

$$\int_{\Omega} \bar{\delta}(x - x_i) R(x) dx = R(x_i) = 0 \quad (8.14)$$

This reduces to forcing the residual to zero at specific points x_i in the domain

$$R(x_i) = 0 \quad (8.15)$$

8.3.2 Least Squares Method

In this method, the solution of a BVP is approximated by minimizing the following integral:

$$S(c_i) = \int_{\Omega} [R(x)]^2 dx \quad (8.16)$$

where c_i are the unknown parameters in the residual function $R(x)$.

To minimize this scalar functional, we take the derivative of S with respect to each parameter c_i and set it to zero:

$$\begin{aligned}
 \frac{\partial S}{\partial c_i} &= 0 \\
 \frac{\partial}{\partial c_i} \int_{\Omega} [R(x)]^2 dx &= 0 \\
 2 \int_{\Omega} R(x) \frac{\partial R(x)}{\partial c_i} dx &= 0 \\
 \int_{\Omega} R(x) \frac{\partial R(x)}{\partial c_i} dx &= 0
 \end{aligned} \tag{8.17}$$

By comparing this result with the general form of a weighted residual, we identify the weighting functions used in this method as:

$$w_i(x) = \frac{\partial R(x)}{\partial c_i} \tag{8.18}$$

8.3.3 Galerkin Method

The Galerkin method is one of the most widely used techniques for obtaining approximate solutions to BVPs. Here, the weighting functions are chosen as the derivatives of the approximate solution $u(x)$ with respect to the unknown parameters c_i :

$$\begin{aligned}
 w_i(x) &= \frac{\partial u(x)}{\partial c_i} \\
 &= \frac{\partial}{\partial c_i} \left(\sum_{i=1}^n c_i N_i(x) + N_0(x) \right)
 \end{aligned} \tag{8.19}$$

Thus, in the Galerkin method, the weighting functions are given by the trial functions themselves:

$$w_i(x) = N_i(x) \tag{8.20}$$

Example 8.6:

Given the following boundary value problem (BVP), find an approximate solution using the collocation, least squares, and Galerkin methods:

$$\begin{aligned}
 \frac{d^2 u}{dx^2} - \frac{du}{dx} &= 0, \quad x \in (0, 1) \\
 u(0) &= 0 \\
 u(1) &= 1
 \end{aligned}$$

For reference, consider the analytical solution:

$$u_A(x) = \frac{e^x - 1}{e - 1}$$

Solution:

Note that the boundary conditions are not homogeneous. In this case, we can establish an approximate solution of the form:

$$u(x) = \sum_{i=1}^n c_i N_i(x) + N_0(x)$$

Let's consider two basis functions for this example, thus the trial function assumes the following form:

$$u(x) = c_1 N_1(x) + c_2 N_2(x) + N_0(x)$$

$N_1(x)$ and $N_2(x)$ should satisfy the homogeneous boundary conditions and $N_0(x)$ is chosen to satisfy the nonhomogeneous condition. In this case we have:

$$N_1(x) = x(x-1)$$

$$N_2(x) = x^2(x-1)$$

$$N_0(x) = x$$

Therefore the trial solution is given by

$$u(x) = c_1 x(x-1) + c_2 x^2(x-1) + x$$

To find the residual function, we substitute $u(x)$ into the differential equation, thus

$$R(x) = -3c_2 x^2 + 8c_2 x - 2c_1 x + 3c_1 - 2c_2 - 1$$

Now we need to define the weighting functions to be used. In this case, since we have two unknown constants, we will need two weighting functions $w_1(x)$ and $w_2(x)$. They will be selected according to the weighted residual method.

Collocation Method

For this method, we choose the points along the domain: $x = \frac{1}{3}$ and $\frac{2}{3}$. Substituting the first point into the residual expression we have:

$$R\left(\frac{1}{3}\right) = 0$$

$$\frac{7}{3}c_1 + \frac{1}{3}c_2 - 1 = 0$$

Likewise, for the second point we have

$$R\left(\frac{2}{3}\right) = 0$$

$$\frac{5}{3}c_1 + 2c_2 - 1 = 0$$

Those are the two equations that compose the system to find c_1 and c_2 . Solving this system we obtain $c_1 = \frac{15}{37}$ and $c_2 = \frac{6}{37}$.

Finally, the approximate solution can be written as

$$u(x) = \frac{6}{37}x^3 + \frac{9}{37}x^2 + \frac{22}{37}x$$

and the residual function as

$$R(x) = \frac{1}{37}(-4 + 18x - 18x^2)$$

Note that the residual is in fact zero at the collocation points.

Least Squares Method

The weighting functions are found by deriving the residual function. For the first weighting function we have

$$\begin{aligned}w_1(x) &= \frac{\partial R}{\partial c_1} \\&= \frac{\partial}{\partial c_1}(-3c_2x^2 + 8c_2x - 2c_1x + 3c_1 - 2c_2 - 1) \\&= 3 - 2x\end{aligned}$$

and the second one

$$\begin{aligned}w_2(x) &= \frac{\partial R}{\partial c_2} \\&= \frac{\partial}{\partial c_2}(-3c_2x^2 + 8c_2x - 2c_1x + 3c_1 - 2c_2 - 1) \\&= -3x^2 + 8x - 2\end{aligned}$$

Later, evaluating the corresponding weighted integrals we have

$$\begin{aligned}\int_0^1 w_1(x)R(x) dx &= 0 \\ \int_0^1 (3 - 2x)(-3c_2x^2 + 8c_2x - 2c_1x + 3c_1 - 2c_2 - 1) dx &= 0 \\ \frac{13}{3}c_1 + \frac{7}{6}c_2 - 2 &= 0\end{aligned}$$

and

$$\begin{aligned}\int_0^1 w_2(x)R(x) dx &= 0 \\ \int_0^1 (-3x^2 + 8x - 2)(-3c_2x^2 + 8c_2x - 2c_1x + 3c_1 - 2c_2 - 1) dx &= 0 \\ \frac{7}{6}c_1 + \frac{47}{15}c_2 - 1 &= 0\end{aligned}$$

Solving the system of equations we have $c_1 = \frac{306}{733}$ and $c_2 = \frac{120}{733}$. Finally, the approximate solution using the least squares method is:

$$u(x) = \frac{120}{733}x^3 + \frac{186}{733}x^2 + \frac{427}{733}x$$

Galerkin Method

In the Galerkin method the weight functions are the same as the basis functions, so

$$\begin{aligned}w_1(x) &= N_1(x) = x(x - 1) \\ w_2(x) &= N_2(x) = x^2(x - 1)\end{aligned}$$

Performing the weighted integrals for each case we have

$$\int_0^1 w_1(x)R(x) dx = 0$$

$$\int_0^1 x(x-1)(-3c_2x^2 + 8c_2x - 2c_1x + 3c_1 - 2c_2 - 1) dx = 0$$

$$-\frac{1}{3}c_1 - \frac{11}{60}c_2 + \frac{1}{6} = 0$$

and

$$\int_0^1 w_2(x)R(x) dx = 0$$

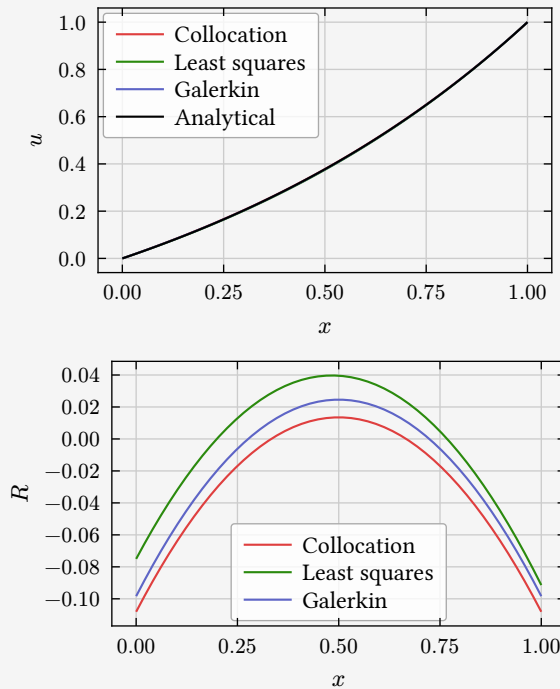
$$\int_0^1 x^2(x-1)(-3c_2x^2 + 8c_2x - 2c_1x + 3c_1 - 2c_2 - 1) dx = 0$$

$$-\frac{3}{20}c_1 - \frac{2}{15}c_2 + \frac{1}{12} = 0$$

Solving these equations, the constants in the trial function are found as $c_1 = \frac{25}{61}$ and $c_2 = \frac{10}{61}$. Then, the approximate solution using the Galerkin method is given by

$$u(x) = \frac{10}{61}x^3 + \frac{15}{61}x^2 + \frac{36}{61}x$$

The next figures compare the results obtained by the three weighted residual methods. The first figure compares the approximate solutions, while the second shows the residual functions obtained by each method.



Exercises 8.3:

1. Solve the boundary value problem stated below using the three studied methods of weighted residuals and one basis function.

$$\begin{aligned}\frac{d^2y}{dx^2} + y &= 2x, \quad x \in (0, 1) \\ y(0) &= 0 \\ y(1) &= 0\end{aligned}$$

Make a plot with the three approximate functions obtained and a second plot with the three corresponding residual functions.

2. Solve the problem above using the Galerkin method with two basis functions. You may need symbolic manipulation software in order to speed up your calculations.
3. Solve the boundary value problem using the Galerkin method and two basis functions.

$$\begin{aligned}\frac{d^2u}{dx^2} - \frac{du}{dx} &= 2x, \quad x \in (0, 1) \\ u(0) &= 1 \\ u(1) &= 0\end{aligned}$$

4. Compare the results with the analytical solution. Explain why the Galerkin method was able to match the analytical result.

8.3.4 Application of interpolation functions and the weak formulation

Instead of using general basis functions, we can use interpolation polynomials to approximate the solution of a BVP. Thus, considering the BVP

$$\begin{aligned}\mathcal{D}(u(x)) + f(x) &= 0 \quad x \in (x_1, x_2) \\ u(x_1) &= u_1 \\ u(x_2) &= u_2\end{aligned} \tag{8.21}$$

the approximate solution can be expressed as

$$u(x) = u_1 N_1(x) + u_2 N_2(x) \tag{8.22}$$

where $N_1(x)$ and $N_2(x)$ can be, for example, Lagrange interpolation polynomials.

For example, if we consider linear interpolation in the interval (x_1, x_2) , we have:

$$\begin{aligned}N_1(x) &= \frac{x_2 - x}{x_2 - x_1} \\ N_2(x) &= \frac{x - x_1}{x_2 - x_1}\end{aligned} \tag{8.23}$$

Note that these interpolation functions N_i satisfy the **Kronecker** condition:

$$\begin{aligned}N_1(x_1) &= 1 & N_1(x_2) &= 0 \\ N_2(x_1) &= 0 & N_2(x_2) &= 1\end{aligned} \tag{8.24}$$

Note that, depending on the degree of the interpolation polynomials, the approximate solution may not satisfy the differential equation in its strong form, since it might lack the differentiability required to

evaluate higher-order derivatives present in the equation. To address this, the **weak formulation** of the BVP is employed to reduce the order of the derivatives through integration by parts, thereby allowing the use of lower-degree interpolation polynomials that require less smoothness.

Example 8.7: Using linear interpolation functions and the Galerkin method combined with the weak formulation, derive the system of equations for a two-node element associated with the differential equation:

$$\frac{d^2 u}{dx^2} + f(x) = 0 \quad x \in (x_1, x_2)$$

Let u_1 and u_2 denote the nodal values of the interpolated field at x_1 and x_2 , respectively.

Solution:

First, let's consider the approximate solution for the element:

$$u(x) = u_1 N_1(x) + u_2 N_2(x)$$

The residual function for the element is given by:

$$R(x) = \frac{d^2}{dx^2}(u_1 N_1(x) + u_2 N_2(x)) + f(x)$$

Applying the Galerkin method using the weight functions $w_1(x) = N_1(x)$ and $w_2(x) = N_2(x)$, we have:

$$\int_{x_1}^{x_2} N_i(x) R(x) dx = \int_{x_1}^{x_2} N_i(x) \frac{d^2 u}{dx^2} dx + \int_{x_1}^{x_2} N_i(x) f(x) dx = 0 \quad \text{for } i = 1, 2$$

Applying integration by parts, we obtain the weak form of the element equation:

$$\left[N_i \frac{du}{dx} \right]_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{dN_i}{dx} \frac{du}{dx} dx + \int_{x_1}^{x_2} N_i f(x) dx = 0 \quad \text{for } i = 1, 2$$

Evaluating the boundary term using the interpolation functions at the endpoints, the resulting equations, for $i = 1, 2$, are:

$$\begin{aligned} \int_{x_1}^{x_2} \frac{dN_1}{dx} \frac{du}{dx} dx &= \int_{x_1}^{x_2} N_1 f(x) dx - \left[\frac{du}{dx} \right]_{x_1} \\ \int_{x_1}^{x_2} \frac{dN_2}{dx} \frac{du}{dx} dx &= \int_{x_1}^{x_2} N_2 f(x) dx + \left[\frac{du}{dx} \right]_{x_2} \end{aligned}$$

or, substituting the approximate solution $u(x)$:

$$\begin{aligned} \int_{x_1}^{x_2} \frac{dN_1}{dx} \left(u_1 \frac{dN_1}{dx} + u_2 \frac{dN_2}{dx} \right) dx &= \int_{x_1}^{x_2} N_1 f(x) dx - \left[\frac{du}{dx} \right]_{x_1} \\ \int_{x_1}^{x_2} \frac{dN_2}{dx} \left(u_1 \frac{dN_1}{dx} + u_2 \frac{dN_2}{dx} \right) dx &= \int_{x_1}^{x_2} N_2 f(x) dx + \left[\frac{du}{dx} \right]_{x_2} \end{aligned}$$

Introducing $\mathbf{U} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $\mathbf{N} = \begin{pmatrix} N_1 \\ N_2 \end{pmatrix}$, and joining both integral equations using matrix-vector notation, we can write:

$$\int_{x_1}^{x_2} \frac{dN^T}{dx} \frac{dN}{dx} dx U = \int_{x_1}^{x_2} N^T f(x) dx + \begin{pmatrix} -\left[\frac{du}{dx}\right]_{x_1} \\ \left[\frac{du}{dx}\right]_{x_2} \end{pmatrix}$$

Expanding the products in the first integral, we have:

$$\int_{x_1}^{x_2} \begin{pmatrix} \frac{dN_1}{dx} \frac{dN_1}{dx} & \frac{dN_1}{dx} \frac{dN_2}{dx} \\ \frac{dN_2}{dx} \frac{dN_1}{dx} & \frac{dN_2}{dx} \frac{dN_2}{dx} \end{pmatrix} dx \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \int_{x_1}^{x_2} \begin{pmatrix} N_1 \\ N_2 \end{pmatrix} f(x) dx + \begin{pmatrix} -\left[\frac{du}{dx}\right]_{x_1} \\ \left[\frac{du}{dx}\right]_{x_2} \end{pmatrix}$$

Finally, computing the derivatives of the linear interpolation functions, the system of equations to solve for u_1 and u_2 is given by:

$$\frac{1}{x_2 - x_1} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \int_{x_1}^{x_2} \begin{pmatrix} N_1 \\ N_2 \end{pmatrix} f(x) dx + \begin{pmatrix} -\left[\frac{du}{dx}\right]_{x_1} \\ \left[\frac{du}{dx}\right]_{x_2} \end{pmatrix}$$

8.3.5 Domain subdivision

In boundary value problems where the solution is complex or the domain is large, it is often beneficial to divide the domain into smaller subdomains. In this approach, the subdomains are referred to as **finite elements**, and the points that define the boundaries of these subdomains are called **nodes**.

For example, if we have a boundary value problem (BVP) defined on the interval $[a, b]$, we can divide it into n subdomains, each defined by nodes $x_1, x_2, \dots, x_{n+1}$. Thus, the solution is approximated by a piecewise function as:

$$u(x) = \begin{cases} u^{(1)}(x) = u_1 N_1^{(1)}(x) + u_2 N_2^{(1)}(x) & \text{for } x \in [x_1, x_2] \\ u^{(2)}(x) = u_2 N_1^{(2)}(x) + u_3 N_2^{(2)}(x) & \text{for } x \in [x_2, x_3] \\ \dots & \\ u^{(n)}(x) = u_n N_1^{(n)}(x) + u_{n+1} N_2^{(n)}(x) & \text{for } x \in [x_n, x_{n+1}] \end{cases} \quad (8.25)$$

where the superscript (e) indicates the element number. Note that each element has its own set of basis functions $N_i^{(e)}(x)$, which are defined locally.

For each element we can mount a system of equations similar to the one presented in the previous section, but now considering the local numbering of nodes and interpolation functions. Then, a global system of equations is obtained by joining the local systems of each element using the continuity conditions at the nodes. The solution of this global system provides the constants u_i for each element, which can be used to construct the approximate solution for the entire domain.

Example 8.8:

Solve the BVP below using two finite elements.

$$\begin{aligned} x \frac{d^2 u}{dx^2} + \frac{du}{dx} - x &= 0 & x \in [1, 4] \\ u(1) &= 0 \\ u(4) &= 1 \end{aligned}$$

For convenience, note that the above differential equation can also be written as:

$$\frac{d}{dx} \left(x \frac{du}{dx} \right) - x = 0$$

For reference, consider the analytical solution:

$$u(x) = \frac{x^2}{4} - \frac{11 \ln(x)}{8 \ln(2)} - \frac{1}{4}$$

Solution

Dividing the domain $[1, 4]$ into two, we have the subdomains $[1, 2.5]$ and $[2.5, 4]$, corresponding to the first and second elements, respectively.

Assuming linear basis functions and local numbering, the basis functions for each element are given by:

$$N_1(x) = \frac{x_2 - x}{x_2 - x_1}$$

$$N_2(x) = \frac{x - x_1}{x_2 - x_1}$$

Also, the approximate solution for each element is written as:

$$u(x)^{(e)} = u_1 N_1(x) + u_2 N_2(x)$$

Thus, the approximate solution for the entire domain is given by:

$$u(x) = \begin{cases} u^{(1)}(x) = u_1 N_1^{(1)}(x) + u_2 N_2^{(1)}(x) & \text{for } x \in [1, 2.5] \\ u^{(2)}(x) = u_2 N_1^{(2)}(x) + u_3 N_2^{(2)}(x) & \text{for } x \in [2.5, 4] \end{cases}$$

The residual function for each element is given by:

$$R^{(e)}(x) = x \frac{d^2 u}{dx^2} + \frac{du}{dx} - x$$

Applying the Galerkin method and integrating by parts we have:

$$\int_{x_1}^{x_2} N_i \left(\frac{d}{dx} \left(x \frac{du}{dx} \right) - x \right) dx = 0 \quad i = 1, 2$$

$$\left[N_i x \frac{du}{dx} \right]_{x_1}^{x_2} - \int_{x_1}^{x_2} x \frac{dN_i}{dx} \frac{du}{dx} dx - \int_{x_1}^{x_2} x N_i dx = 0$$

$$\int_{x_1}^{x_2} x \frac{dN_i}{dx} \frac{du}{dx} dx = \left[N_i x \frac{du}{dx} \right]_{x_1}^{x_2} - \int_{x_1}^{x_2} x N_i dx$$

Substituting with the approximate solution for each node $i = 1, 2$ we have

$$\int_{x_1}^{x_2} x \frac{dN_1}{dx} \left(u_1 \frac{dN_1}{dx} + u_2 \frac{dN_2}{dx} \right) dx = \left[x N_1 \frac{du^{(e)}}{dx} \right]_{x_1}^{x_2} - \int_{x_1}^{x_2} x N_1 dx$$

$$\int_{x_1}^{x_2} x \frac{dN_2}{dx} \left(u_1 \frac{dN_1}{dx} + u_2 \frac{dN_2}{dx} \right) dx = \left[x N_2 \frac{du^{(e)}}{dx} \right]_{x_1}^{x_2} - \int_{x_1}^{x_2} x N_2 dx$$

Then we join both equations into matrix form as

$$\int_{x_1}^{x_2} x \begin{pmatrix} \frac{dN_1}{dx} \\ \frac{dN_2}{dx} \end{pmatrix} \begin{pmatrix} \frac{dN_1}{dx} \\ \frac{dN_2}{dx} \end{pmatrix}^T dx \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \left[x \begin{pmatrix} N_1 \\ N_2 \end{pmatrix} \frac{du^{(e)}}{dx} \right]_{x_1}^{x_2} - \int_{x_1}^{x_2} x \begin{pmatrix} N_1 \\ N_2 \end{pmatrix} dx$$

or

$$\int_{x_1}^{x_2} x \frac{dN^T}{dx} \frac{dN}{dx} dx \mathbf{u}^{(e)} = \left[x N^T \frac{du^{(e)}}{dx} \right]_{x_1}^{x_2} - \int_{x_1}^{x_2} x N^T dx$$

Applying this equation for the first element, where $x \in [1, 2.5]$, we obtain the local system of equations

$$\begin{pmatrix} \frac{7}{6} & -\frac{7}{6} \\ -\frac{7}{6} & \frac{7}{6} \end{pmatrix} \begin{pmatrix} u_1^{(1)} \\ u_2^{(1)} \end{pmatrix} = \begin{pmatrix} -\frac{du}{dx} \Big|_{x=1} - \frac{9}{8} \\ 2.5 \frac{du}{dx} \Big|_{x=2.5} - \frac{3}{2} \end{pmatrix}$$

Likewise, for the second element, where $x \in [2.5, 4]$, we have

$$\begin{pmatrix} \frac{13}{6} & -\frac{13}{6} \\ -\frac{13}{6} & \frac{13}{6} \end{pmatrix} \begin{pmatrix} u_1^{(2)} \\ u_2^{(2)} \end{pmatrix} = \begin{pmatrix} -2.5 \frac{du}{dx} \Big|_{x=2.5} - \frac{9}{4} \\ 4 \frac{du}{dx} \Big|_{x=4} - \frac{21}{8} \end{pmatrix}$$

Using the superposition principle, we arrive at the following system of equations:

$$\begin{pmatrix} \frac{7}{6} & -\frac{7}{6} & 0 \\ -\frac{7}{6} & \frac{10}{3} & -\frac{13}{6} \\ 0 & -\frac{13}{6} & \frac{13}{6} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} -\frac{du}{dx} \Big|_{x=1} - \frac{9}{8} \\ -\frac{15}{4} \\ 4 \frac{du}{dx} \Big|_{x=4} - \frac{21}{8} \end{pmatrix}$$

Since both boundary conditions are essential, we impose them directly on the nodal values:

$$u_1 = 0 \quad \text{and} \quad u_3 = 1$$

Therefore, the only unknown nodal value is obtained from the second equation:

$$\begin{aligned} -\frac{7}{6}u_1 + \frac{10}{3}u_2 - \frac{13}{6}u_3 &= -\frac{15}{4} \\ \frac{10}{3}u_2 - \frac{13}{6} &= -\frac{15}{4} \\ u_2 &= -\frac{19}{40} = -0.475 \end{aligned}$$

If desired, the boundary fluxes can be recovered from the first and third equations as reaction terms:

$$\frac{du}{dx} \Big|_{x=1} = -\frac{403}{240} \quad \text{and} \quad \frac{du}{dx} \Big|_{x=4} = \frac{1397}{960}$$

The finite element solution for the BVP is then given by the piece-wise function:

$$u(x) = \begin{cases} u^{(1)}(x) = u_1 N_1^{(1)}(x) + u_2 N_2^{(1)}(x) & \text{for } x \in [1, 2.5] \\ u^{(2)}(x) = u_2 N_1^{(2)}(x) + u_3 N_2^{(2)}(x) & \text{for } x \in [2.5, 4] \end{cases}$$

or

$$u(x) = \begin{cases} u^{(1)}(x) = -\frac{19}{60}(x-1) & \text{for } x \in [1, 2.5] \\ u^{(2)}(x) = -\frac{19}{40}\left(\frac{4-x}{1.5}\right) + \left(\frac{x-2.5}{1.5}\right) & \text{for } x \in [2.5, 4] \end{cases}$$

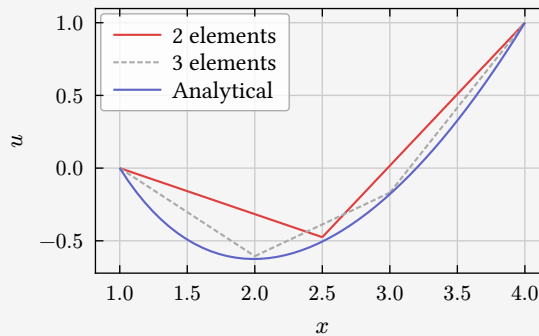
If three elements were used instead of two, the solution would be given by:

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{43}{71} \\ -\frac{12}{71} \\ 1 \end{pmatrix}$$

with nodes at $x = 1, 2, 3$, and 4 , so that

$$u(x) = \begin{cases} u^{(1)}(x) = -\frac{43}{71}(x-1) & \text{for } x \in [1, 2] \\ u^{(2)}(x) = -\frac{43}{71}(3-x) - \frac{12}{71}(x-2) & \text{for } x \in [2, 3] \\ u^{(3)}(x) = -\frac{12}{71}(4-x) + (x-3) & \text{for } x \in [3, 4] \end{cases}$$

The figure below shows a comparison between the analytical solution and the approximate solutions obtained by the Galerkin method for each element. Note that the numerical solution is a piece-wise linear function and only provides a fair approximation of the analytical solution. The use of more elements or higher order basis functions would lead to a better approximation.



Exercises 8.4:

1. Solve the following BVP using the Galerkin method and two elements.

$$\begin{aligned} \frac{d^2 u}{dx^2} - x &= 0 \\ u(0) &= 0 \\ u(3) &= 1 \end{aligned}$$

2. Solve the problem above using three elements and compare the results with the two elements solution and the analytical solution.

3. Solve the following BVP using the Galerkin method and two elements.

$$\begin{aligned}\frac{d}{dx}\left(x\frac{du}{dx}\right) &= 6x \\ u(1) &= 1 \\ \frac{du}{dx}(2) &= 2\end{aligned}$$

8.3.6 Application of the Galerkin method to a 1D bar element

The Galerkin method can be used to derive the element equations for the one-dimensional elasticity problem. The resulting element is the so-called 1D bar element.

Figure 8.3 shows the degrees of freedom and the external forces acting on a 1D bar element. The axial displacements at both ends are represented by u_1 and u_2 , while the corresponding axial forces are represented by F_1 and F_2 .

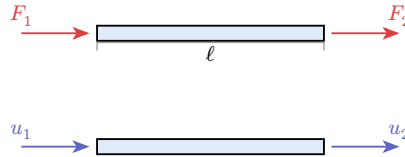


Figure 8.3: External forces and degrees of freedom of a 1D bar element.

For a prismatic bar element with length ℓ , constant cross section A , and constant Young's modulus E , the differential equation for the static equilibrium in terms of the axial stress is given by

$$A\frac{d\sigma_x}{dx} = 0, \quad x \in [0, \ell] \quad (8.26)$$

where σ_x is the axial stress. Considering the constitutive relation $\sigma_x = E\varepsilon$, and the axial strain definition $\varepsilon_x = \frac{du}{dx}$, the differential equation can be rewritten in terms of displacement as:

$$EA\frac{d^2u}{dx^2} = 0, \quad x \in [0, \ell] \quad (8.27)$$

Using an approximate displacement field with two basis functions and two unknown parameters u_1 and u_2 , we write:

$$u(x) = u_1N_1(x) + u_2N_2(x) \quad (8.28)$$

Assuming the endpoints are at $x = 0$ and $x = \ell$, we define the interpolation functions as

$$\begin{aligned}N_1(x) &= 1 - \frac{x}{\ell} \\ N_2(x) &= \frac{x}{\ell}\end{aligned} \quad (8.29)$$

Note that, in this case, x is a local coordinate for the element, so N_1 and N_2 are defined in the interval $[0, \ell]$.

Because the interpolation functions are linear, the displacement field for a set of interconnected bar elements is represented by linear segments. Since only displacement continuity is guaranteed, and not continuity of its derivatives, this element is said to have C^0 continuity.

Applying the Galerkin method, we obtain the weighted integral over the bar element:

$$\int_0^\ell N_i(x) \left(EA \frac{d^2 u}{dx^2} \right) dx = 0 \quad i = 1, 2 \quad (8.30)$$

After integrating by parts, we obtain the weak form of the bar element:

$$EA \int_0^\ell \frac{dN_i}{dx} \frac{du}{dx} dx = \left[N_i EA \frac{du}{dx} \right]_0^\ell \quad \text{for } i = 1, 2 \quad (8.31)$$

where $EA \frac{du}{dx}$ represents the axial force at a specific point x .

Substituting the approximate solution, we obtain the following two equations:

$$\begin{aligned} EA \int_0^\ell \frac{dN_1}{dx} \left(u_1 \frac{dN_1}{dx} + u_2 \frac{dN_2}{dx} \right) dx &= - \left[EA \frac{du}{dx} \right]_{x=0} \\ EA \int_0^\ell \frac{dN_2}{dx} \left(u_1 \frac{dN_1}{dx} + u_2 \frac{dN_2}{dx} \right) dx &= \left[EA \frac{du}{dx} \right]_{x=\ell} \end{aligned} \quad (8.32)$$

The terms on the right-hand side are associated with the natural boundary conditions, that is, the external forces.

Writing the result in matrix form, and using $E \frac{du}{dx} = \sigma_x$ and $\sigma_x A = F$, we obtain the following system of equations:

$$EA \int_0^\ell \begin{pmatrix} \frac{dN_1}{dx} \frac{dN_1}{dx} & \frac{dN_1}{dx} \frac{dN_2}{dx} \\ \frac{dN_2}{dx} \frac{dN_1}{dx} & \frac{dN_2}{dx} \frac{dN_2}{dx} \end{pmatrix} dx \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} \quad (8.33)$$

Since the interpolation functions are linear, all derivatives are constant. Therefore, the system becomes

$$\frac{EA}{\ell} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} \quad (8.34)$$

This system of equations is the same as the one derived using the direct stiffness method.

8.3.7 Application of the Galerkin method to a beam element

A beam element is intended to estimate deflections due to bending. In this case, axial strain is neglected.

Considering a beam element with length ℓ , the essential boundary conditions are given by displacements and rotations at the endpoints. The distributed load $q(x)$ acts over the element domain, while the natural boundary conditions are given by endpoint moments and shear forces.

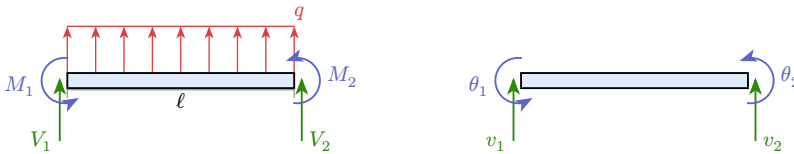


Figure 8.4: External forces and degrees of freedom in a beam element.

The governing equation for the beam-flexure problem is the general fourth-order Euler-Bernoulli differential equation. For variable flexural rigidity $EI = EI(x)$, it reads:

$$\frac{d^2}{dx^2} \left(EI \frac{d^2 v}{dx^2} \right) = q(x) \quad (8.35)$$

To provide C^1 continuity, the approximate solution is defined with four interpolation functions. These account for both displacements and rotations at the beam endpoints, resulting in a Hermite interpolation:

$$v(x) = v_1 N_1(x) + \theta_1 N_2(x) + v_2 N_3(x) + \theta_2 N_4(x) \quad (8.36)$$

Note 8.4: Hermite interpolation constructs a polynomial that not only passes through a given set of points but also matches the function's derivatives at those points. This is particularly useful when both function values and derivative information are known.

Suppose we have n distinct points $x_1, x_2, \dots, x_n$, and for each point x_i we know the function value y_i and its first derivatives $y'_i, y''_i, \dots, y_i^{(m)}$. Hermite interpolation then seeks a polynomial $f(x)$ such that:

$$\begin{aligned} f(x_1) &= y_1, & f'(x_1) &= y'_1, & f''(x_1) &= y''_1, & \dots, & f^{(m)}(x_1) &= y_1^{(m)} \\ f(x_2) &= y_2, & f'(x_2) &= y'_2, & f''(x_2) &= y''_2, & \dots, & f^{(m)}(x_2) &= y_2^{(m)} \\ &\vdots \\ f(x_n) &= y_n, & f'(x_n) &= y'_n, & f''(x_n) &= y''_n, & \dots, & f^{(m)}(x_n) &= y_n^{(m)} \end{aligned}$$

This results in $n(m+1)$ interpolation conditions, so the interpolating polynomial has degree $n(m+1) - 1$. It can be written as:

$$f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_{(m+1)n-1} x^{(m+1)n-1}$$

or more compactly as:

$$f(x) = \sum_{i=1}^{(m+1)n} a_{i-1} x^{i-1}$$

For instance, for the basis functions in a beam element, we have $n = 2$ (at 0 and ℓ) and $m = 1$ (first derivative).

Assuming $x \in [0, \ell]$, for any deformation state in a beam element, the interpolation should satisfy

$$v(0) = v_1, \quad v(\ell) = v_2, \quad v'(0) = \theta_1, \quad v'(\ell) = \theta_2 \quad (8.37)$$

Thus, the interpolation functions should satisfy

$$\begin{aligned} N_1(0) &= 1, & N_1(\ell) &= 0, & \frac{dN_1}{dx}(0) &= 0, & \frac{dN_1}{dx}(\ell) &= 0 \\ N_2(0) &= 0, & N_2(\ell) &= 0, & \frac{dN_2}{dx}(0) &= 1, & \frac{dN_2}{dx}(\ell) &= 0 \\ N_3(0) &= 0, & N_3(\ell) &= 1, & \frac{dN_3}{dx}(0) &= 0, & \frac{dN_3}{dx}(\ell) &= 0 \\ N_4(0) &= 0, & N_4(\ell) &= 0, & \frac{dN_4}{dx}(0) &= 0, & \frac{dN_4}{dx}(\ell) &= 1 \end{aligned} \quad (8.38)$$

Applying Hermite interpolation to each of the four cases above, we obtain

$$\begin{aligned}
 N_1(x) &= 1 - \frac{3x^2}{\ell^2} + \frac{2x^3}{\ell^3} \\
 N_2(x) &= x - \frac{2x^2}{\ell} + \frac{x^3}{\ell^2} \\
 N_3(x) &= \frac{3x^2}{\ell^2} - \frac{2x^3}{\ell^3} \\
 N_4(x) &= \frac{x^3}{\ell^2} - \frac{x^2}{\ell}
 \end{aligned} \tag{8.39}$$

These functions are commonly referred to as cubic Hermite polynomials for beam elements.

For a prismatic beam element, where EI is constant, applying the Galerkin method to the beam differential equation gives

$$\int_0^\ell N_i(x) \left[EI \frac{d^4 v}{dx^4} - q(x) \right] dx = 0 \quad i = 1, \dots, 4 \tag{8.40}$$

After integrating by parts twice, we obtain the weak form of the beam element:

$$EI \int_0^\ell \frac{d^2 N_i}{dx^2} \frac{d^2 v}{dx^2} dx = \int_0^\ell N_i q(x) dx - \left[EIN_i \frac{d^3 v}{dx^3} \right]_0^\ell + \left[EI \frac{dN_i}{dx} \frac{d^2 v}{dx^2} \right]_0^\ell \quad i = 1, \dots, 4 \tag{8.41}$$

Using the convention in [Figure 8.4](#), the nodal actions are related to the internal shear force and bending moment by $V_1 = V(0)$, $M_1 = -M(0)$, $V_2 = -V(\ell)$, and $M_2 = M(\ell)$.

After substituting the approximate solution $v(x)$ and carrying out the integrations, we obtain the finite element formulation of the beam. Assuming a constant load $q(x) = q$, and using matrix notation, this formulation is:

$$\frac{EI}{\ell^3} \begin{pmatrix} 12 & 6\ell & -12 & 6\ell \\ 6\ell & 4\ell^2 & -6\ell & 2\ell^2 \\ -12 & -6\ell & 12 & -6\ell \\ 6\ell & 2\ell^2 & -6\ell & 4\ell^2 \end{pmatrix} \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} V_1 \\ M_1 \\ V_2 \\ M_2 \end{pmatrix} + \frac{q\ell}{12} \begin{pmatrix} 6 \\ \ell \\ 6 \\ -\ell \end{pmatrix} \tag{8.42}$$

where the definition of shear force, $V = EI \frac{d^3 v}{dx^3}$, and bending moment, $M = EI \frac{d^2 v}{dx^2}$, were considered. Note that shear forces and moments are not continuous across elements. Also, along each element, shear forces present a constant distribution while moments present a linear distribution.

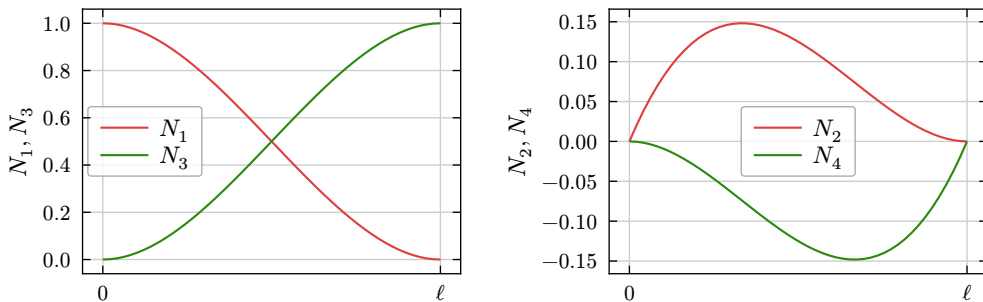


Figure 8.5: Cubic Hermite interpolation functions

Exercises 8.5:

1. Consider a uniaxial member subjected to axial loads where the cross-sectional area varies according to $A = A_0(1 + \frac{x}{\ell})$ with $x \in [0, \ell]$. Given the governing equation $\frac{d}{dx}(EA(x)\frac{du}{dx}) = 0$, determine the Galerkin finite element equations.
2. A 2 m length rod fixed at the left end is being pulled at the right end by a force $F_A = 2$ kN. At the middle of the rod another force $F_B = 1$ kN is also applied and is pulling to the right. Plot the resulting displacement field for the whole rod. Consider $EA = 2$ kN for the left half of the rod and $EA = 1$ kN for the other half.
3. A cantilever beam is embedded at the left end and subject to a vertical load $q(x) = -1$ kN/m. Plot the resulting deflection field for the beam. Consider $EI = 1$ kN m² and $\ell = 1$ m.
4. For the last exercise, find the functions for the shear force and moment along the beam based on the approximate solution for the deflection. Plot both functions together with the corresponding analytical expressions.

8.4 Calculus of variations

Calculus of variations is a branch of mathematical analysis concerned with finding functions that maximize or minimize **functionals**. A functional is an operator that maps a function, such as $y(x)$, to a real number. In this sense, it can be viewed as a **function of functions**. More precisely, a functional maps functions from a linear vector space to the real numbers.

Functionals are typically expressed as integrals involving a given function $F(x, y, y')$, where the goal is to determine the function y that makes the integral stationary (i.e., a minimum or maximum). This function y is known as the **extremal** or **stationary function**, and plays a central role in the study of steady-state and optimal control problems.

Consider the one-dimensional functional:

$$I(y) = \int_a^b F(x, y, y') dx, \quad \text{with } y = y(x) \quad \text{and } y' = \frac{dy}{dx} \quad (8.43)$$

Here, each admissible function $y(x)$ defines a real value of $I(y)$.

Functionals can also be defined over higher-dimensional domains. For example, in two dimensions:

$$I(u, v) = \int_{\Omega} F\left(x, y, u, v, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}\right) d\Omega \quad (8.44)$$

with

$$u = u(x, y); \quad v = v(x, y) \quad (8.45)$$

Some examples of functionals include:

$$\begin{aligned}
 A(y) &= \int_a^b y(x) \, dx \\
 B(y) &= \int_a^b \sqrt{1 + (y')^2} \, dx \\
 I(u) &= \int_a^b \left(p(x) \frac{du}{dx} + q(x)u \right) dx \\
 I(u, v) &= \int_{\Omega} \left(p(x, y) \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} + u + v \right) d\Omega
 \end{aligned} \tag{8.46}$$

8.4.1 Admissible functions

In the calculus of variations, admissible functions are those functions that are allowed as candidates in the search for an extremum of a given functional. To be admissible, a function must satisfy two key requirements:

Smoothness: The function must be sufficiently smooth so that all terms in the functional are well-defined. For example, if the functional includes terms involving y' , then $y(x)$ must be differentiable, and $y'(x)$ must be integrable over the domain.

Boundary conditions: The function must satisfy any imposed boundary conditions.

8.4.2 The Gâteaux variation

The Gâteaux variation describes how a functional responds to small perturbations in its input function. It plays a role analogous to that of the directional derivative in finite-dimensional calculus.

Note 8.5: A directional derivative measures the rate of change of a vectorial function in the direction of a vector. Thus, the directional derivative of a function $f(\mathbf{x})$ in the direction of a vector $\mathbf{v}$ at a point $\mathbf{x}$ is defined as:

$$D_{\mathbf{v}}f = \lim_{\varepsilon \rightarrow 0} \frac{f(\mathbf{x} + \varepsilon \mathbf{v}) - f(\mathbf{x})}{\varepsilon}$$

Let $I(y)$ be a functional, where $y = y(x)$. The Gâteaux variation of I at y in the direction of another function $\eta(x)$ is defined as:

$$\delta I(y) = \lim_{\varepsilon \rightarrow 0} \frac{I(y + \varepsilon \eta) - I(y)}{\varepsilon} \tag{8.47}$$

where $\eta(x)$ is called the variation function (or test function) and $\varepsilon \eta(x)$ is a small perturbation of $y(x)$.

If the functional is differentiable in the direction η , the Gâteaux variation, also known as the first variation, can be expressed using a standard derivative evaluated at $\varepsilon = 0$:

$$\delta I(y) = \left[\frac{d}{d\varepsilon} I(y + \varepsilon \eta) \right]_{\varepsilon=0} \tag{8.48}$$

In this form, the Gâteaux variation represents the linearized change in the functional due to an infinitesimal variation in the function y . If $\delta I(y) = 0$ for all admissible η , then y is called a stationary function or extremal, potentially corresponding to a minimum or maximum of the functional.

Using the Gâteaux variation definition, we have:

$$\begin{aligned}
 F(x, y, y') = x &\Rightarrow \delta F = 0 \\
 F(x, y, y') = y &\Rightarrow \delta F = \eta \\
 F(x, y, y') = y' &\Rightarrow \delta F = \eta'
 \end{aligned}
 \tag{8.49}$$

8.4.3 Properties of the First Variation

The first variation operator δ behaves similarly to a derivative and satisfies several important properties.

It commutes with both differentiation and integration:

$$\begin{aligned}
 \delta \frac{dF}{dx} &= \frac{d}{dx}(\delta F) \\
 \delta \int_a^b F dx &= \int_a^b \delta F dx
 \end{aligned}
 \tag{8.50}$$

It also follows rules analogous to standard calculus operations. For two functionals F_1 and F_2 :

$$\begin{aligned}
 \delta(F_1 \pm F_2) &= \delta F_1 \pm \delta F_2 \\
 \delta(F_1 F_2) &= \delta F_1 F_2 + F_1 \delta F_2 \\
 \delta\left(\frac{F_1}{F_2}\right) &= \frac{\delta F_1 F_2 - F_1 \delta F_2}{F_2^2} \\
 \delta(F_1)^n &= n(F_1)^{n-1} \delta F_1
 \end{aligned}
 \tag{8.51}$$

Finally, applying the multivariable chain rule yields:

$$\begin{aligned}
 \delta F(x, y, y') &= \frac{\partial F}{\partial y} \delta y + \frac{\partial F}{\partial y'} \delta y' \\
 \delta F(x, y, y') &= \frac{\partial F}{\partial y} \eta + \frac{\partial F}{\partial y'} \eta'
 \end{aligned}
 \tag{8.52}$$

Example 8.9: Find the first variation of the functional

$$I(y) = \int_a^b yy' dx$$

Solution:

Applying the definition of first variation, we have:

$$\begin{aligned}
 \delta I(y) &= \left[\frac{d}{d\varepsilon} I(y + \varepsilon\eta) \right]_{\varepsilon=0} \\
 &= \left[\frac{d}{d\varepsilon} \int_a^b (y + \varepsilon\eta)(y' + \varepsilon\eta') dx \right]_{\varepsilon=0}
 \end{aligned}$$

Then, commuting the derivative with the integral, we have:

$$= \int_a^b \left[\frac{d}{d\varepsilon} (yy' + y\varepsilon\eta' + y'\varepsilon\eta + \varepsilon^2\eta\eta') \right]_{\varepsilon=0} dx$$

$$\begin{aligned}
 &= \int_a^b [y\eta' + y'\eta + 2\varepsilon\eta\eta']_{\varepsilon=0} dx \\
 &= \int_a^b (y\eta' + y'\eta) dx
 \end{aligned}$$

Thus, the variation of the functional $I(y)$ in the direction of $\eta(x)$ is given by $\delta I(y) = \int_a^b (y\eta' + y'\eta) dx$.

8.4.4 Fundamental Lemma of the Calculus of Variations

This fundamental lemma states that if a continuous function $F(x)$ defined on an open interval (a, b) satisfies

$$\int_a^b F(x)\eta(x) dx = 0 \quad (8.53)$$

for every continuous function $\eta(x)$ on (a, b) , then it must be that

$$F(x) = 0 \quad \text{in } (a, b) \quad (8.54)$$

A simple verification follows by choosing $\eta(x) = F(x)$, since $\eta(x)$ is arbitrary. Substituting, we get:

$$\int_a^b (F(x))^2 dx = 0 \quad (8.55)$$

Because the integrand is non-negative, the integral equals zero only if $F(x) = 0$ for all $x \in (a, b)$.

This lemma is essential in the calculus of variations and is frequently used in boundary value problems to convert weak formulations into strong formulations.

Example 8.10: Find the function $y(x)$ which is the shortest curve that connects two points (x_1, y_1) and (x_2, y_2) in Euclidean space.

Solution:

There are infinitely many curves that connect two points. However, we are interested in the one with the shortest length (a minimization problem).

The problem can be stated as a functional (the arc length of a curve) with two boundary conditions that ensure the curve passes through the points (x_1, y_1) and (x_2, y_2) :

$$\begin{aligned}
 I(y) &= \int_{x_1}^{x_2} \sqrt{1 + [y'(x)]^2} dx \quad \text{for } x \in [x_1, x_2] \\
 y(x_1) &= y_1 \\
 y(x_2) &= y_2
 \end{aligned}$$

In order to find $y(x)$ that minimizes $I(y)$, we need to solve:

$$\delta I = 0$$

Using the definition of the first variation, we have:

$$\begin{aligned}
 \delta I &= \left[\frac{d}{d\varepsilon} \int_{x_1}^{x_2} \sqrt{1 + [(y + \varepsilon\eta)']^2} dx \right]_{\varepsilon=0} \\
 &= \left[\int_{x_1}^{x_2} \frac{(y' + \varepsilon\eta')\eta'}{\sqrt{1 + [(y + \varepsilon\eta)']^2}} dx \right]_{\varepsilon=0} \\
 &= \int_{x_1}^{x_2} \frac{y'\eta'}{\sqrt{1 + (y')^2}} dx
 \end{aligned}$$

Later, using integration by parts (with $u = \frac{y'}{\sqrt{1+(y')^2}}$ and $dv = \eta' dx$) and equating to zero, we have

$$\delta I = \left[\frac{y'\eta}{\sqrt{1 + (y')^2}} \right]_{x_1}^{x_2} - \int_{x_1}^{x_2} \eta \frac{d}{dx} \left(\frac{y'}{\sqrt{1 + (y')^2}} \right) dx = 0$$

To satisfy the boundary conditions, the function $\eta(x)$ must be zero at x_1 and x_2 ; thus, the non-integral term in the equation above is zero.

Next, using the fundamental lemma in the integral expression leads to

$$\frac{d}{dx} \left(\frac{y'}{\sqrt{1 + (y')^2}} \right) = 0$$

Computing the derivative, we have

$$\frac{y''}{\left(\sqrt{1 + (y')^2} \right)^3} = 0$$

or

$$y'' = 0$$

This is the ordinary differential equation (ODE) whose solution represents the shortest curve between (x_1, y_1) and (x_2, y_2) .

The general solution of this ODE is given by the line

$$y = c_1 + c_2 x$$

where c_1 and c_2 are constants to be found in order to satisfy the boundary conditions. Thus, for the given problem we have:

$$c_1 = \frac{x_2 y_1 - x_1 y_2}{x_2 - x_1} \quad c_2 = \frac{y_2 - y_1}{x_2 - x_1}$$

Finally, the function $y(x)$ that represents the shortest curve connecting the points (x_1, y_1) and (x_2, y_2) is given by

$$y(x) = \frac{x_2 y_1 - x_1 y_2}{x_2 - x_1} + \left(\frac{y_2 - y_1}{x_2 - x_1} \right) x$$

8.4.5 Euler-Lagrange Equation

The Euler-Lagrange differential equation is the fundamental equation of the calculus of variations. Given a functional of the form

$$I(y) = \int_{x_1}^{x_2} L(x, y, y') \, dx \quad (8.56)$$

the Euler-Lagrange equation states that I has a stationary value if

$$\frac{\partial L}{\partial y} - \frac{d}{dx} \frac{\partial L}{\partial y'} = 0 \quad (8.57)$$

This equation can be used to directly determine stationary functions for functionals of the type given above.

The derivation of the Euler-Lagrange equation is presented below.

Considering that I has a stationary value (minimum or maximum) when $\delta I = 0$, we start by finding the first variation of I using an auxiliary function $\eta(x)$, with the boundary conditions $\eta(x_1) = \eta(x_2) = 0$

$$\begin{aligned} \delta I &= \left[\frac{d}{d\varepsilon} \int_{x_1}^{x_2} L(x, y + \varepsilon\eta, (y + \varepsilon\eta)') \, dx \right]_{\varepsilon=0} \\ &= \int_{x_1}^{x_2} \left[\frac{d}{d\varepsilon} L(x, y + \varepsilon\eta, (y + \varepsilon\eta)') \right]_{\varepsilon=0} \, dx \end{aligned} \quad (8.58)$$

After computing the derivative and setting $\varepsilon = 0$, we obtain

$$\delta I = \int_{x_1}^{x_2} \left(\frac{\partial L}{\partial y} \eta + \frac{\partial L}{\partial y'} \eta' \right) \, dx \quad (8.59)$$

Applying integration by parts to the second term leads to

$$\delta I = \int_{x_1}^{x_2} \left(\frac{\partial L}{\partial y} \eta - \eta \frac{d}{dx} \frac{\partial L}{\partial y'} \right) \, dx + \left[\frac{\partial L}{\partial y'} \eta \right]_{x_1}^{x_2} \quad (8.60)$$

where the last term vanishes because $\eta = 0$ at x_1 and x_2 . Then we have:

$$\int_{x_1}^{x_2} \eta \left(\frac{\partial L}{\partial y} - \frac{d}{dx} \frac{\partial L}{\partial y'} \right) \, dx = 0 \quad (8.61)$$

According to the fundamental lemma of the calculus of variations, the expression inside the brackets must be zero. Therefore, we recover the Euler-Lagrange equation:

$$\frac{\partial L}{\partial y} - \frac{d}{dx} \frac{\partial L}{\partial y'} = 0 \quad (8.62)$$

Exercises 8.6:

1. Determine the first variation of the following functionals:
 - a. $I(y) = \int_a^b (y^2 y' + P(x)) \, dx$
 - b. $I(y) = \int_a^b y \sqrt{1 + (y')^2} \, dx$
2. A point mass is released from point $A = (x_1, y_1)$ at time $t = 0$. It is constrained to move under the influence of constant gravity g and without friction along a curve $y = y(x)$ that ends at point $B = (x_2, y_2)$, which is lower than A . Consider $x_1 = 0, y_1 = 0$ and the y axis pointing downwards. Demonstrate that the traversal time taken by a mass to go from point A to point B is given by the functional:

$$T(y) = \int_{x_1}^{x_2} \sqrt{\frac{1 + (y')^2}{2gy}} \, dx$$

3. Find the curve that provides the shortest time for a point mass travel from point A to point B in the last exercise (brachistochrone curve).

8.5 Energy Methods

The term Energy Methods refers to a class of methods based on principles such as energy conservation, minimum potential energy, and virtual work.

8.5.1 Principle of Minimum Potential Energy

The total potential energy of a structural system is defined as the sum of the strain energy U and the potential of the external loads W :

$$\Pi = U + W \quad (8.63)$$

Strain energy corresponds to the elastic energy stored in a deformed structure. The potential of the external loads is the negative of the external work, and it accounts for the effect of applied forces in the total potential energy functional. When loads are applied to a body, deformation occurs. Assuming no energy is lost as heat, the external work done by the loads is transformed into internal strain energy.

For a conservative mechanical system, equilibrium configurations correspond to stationary values of the total potential energy. A minimum corresponds to stable equilibrium, whereas other stationary points may correspond to unstable or neutral equilibrium.

Given a functional $\Pi(u)$ in terms of the displacement field u , a stationary configuration is reached when an infinitesimal variation produces no change in energy:

$$\delta \Pi(u) = 0 \quad (8.64)$$

8.5.2 Strain energy

For a general stress state, the strain energy is

$$\begin{aligned} U &= \int_V \frac{1}{2} \boldsymbol{\sigma} : \boldsymbol{\varepsilon} \, dV \\ &= \int_V \frac{1}{2} (\sigma_{xx} \varepsilon_{xx} + \sigma_{yy} \varepsilon_{yy} + \sigma_{zz} \varepsilon_{zz} + 2\tau_{yz} \varepsilon_{yz} + 2\tau_{xz} \varepsilon_{xz} + 2\tau_{xy} \varepsilon_{xy}) \, dV \end{aligned} \quad (8.65)$$

Strain energy due to uniaxial stress

For a body under only uniaxial load, the strain energy is

$$\begin{aligned}
 U &= \int_V \frac{1}{2} \sigma \varepsilon \, dV \\
 &= \int_0^\ell \frac{1}{2} (E \varepsilon) \varepsilon A \, dx \\
 &= \int_0^\ell \frac{1}{2} EA \left(\frac{du}{dx} \right)^2 dx
 \end{aligned} \tag{8.66}$$

Strain energy due to torsional stress

Likewise, for a body under shear stress, we have

$$\begin{aligned}
 U &= \int_V \frac{1}{2} \tau \gamma \, dV \\
 &= \int_V \frac{1}{2} (G \gamma) \gamma \, dV \\
 &= \int_V \frac{1}{2} G \gamma^2 \, dV
 \end{aligned} \tag{8.67}$$

Substituting the kinematic relation $\gamma = r \frac{d\varphi}{dx}$, where φ is the angle of twist and r is the radial distance from the centroid, gives

$$\begin{aligned}
 U &= \int_V \frac{1}{2} G \left(r \frac{d\varphi}{dx} \right)^2 \, dV \\
 &= \int_0^\ell \frac{1}{2} G \left(\frac{d\varphi}{dx} \right)^2 \left(\int_A r^2 \, dA \right) dx \\
 &= \int_0^\ell \frac{1}{2} GJ \left(\frac{d\varphi}{dx} \right)^2 dx
 \end{aligned} \tag{8.68}$$

where $J = \int_A r^2 \, dA$ is the polar moment of inertia of the cross section.

Strain energy due to bending

In the case of a beam under pure bending, we have

$$\begin{aligned}
 U &= \int_V \frac{1}{2} \sigma_x \varepsilon_x \, dV \\
 &= \int_0^\ell \int_A \frac{1}{2} E \varepsilon_x^2 \, dA \, dx
 \end{aligned} \tag{8.69}$$

Under the Euler-Bernoulli assumption, the longitudinal strain is $\varepsilon_x = -y \frac{d^2 v}{dx^2}$, where y is the distance from the neutral axis. Substituting,

$$\begin{aligned}
 U &= \int_0^\ell \int_A \frac{1}{2} E y^2 \left(\frac{d^2 v}{dx^2} \right)^2 dA dx \\
 &= \int_0^\ell \frac{1}{2} E \left(\frac{d^2 v}{dx^2} \right)^2 \left(\int_A y^2 dA \right) dx \\
 &= \int_0^\ell \frac{1}{2} E I \left(\frac{d^2 v}{dx^2} \right)^2 dx
 \end{aligned} \tag{8.70}$$

where $I = \int_A y^2 dA$ is the second moment of area of the cross section.

8.5.3 Variational formulation of a 1D Bar Element

The principle of minimum potential energy can be used to derive the equilibrium equations of finite elements. This approach is also known as the Rayleigh-Ritz method. For a given element, one must define the strain energy and the potential of the external loads to form the total potential energy functional.

For a bar element, this functional is given by:

$$\Pi(u) = \int_0^\ell \frac{1}{2} E A \left(\frac{du}{dx} \right)^2 dx - F_1 u_1 - F_2 u_2 \tag{8.71}$$

The stationary condition provides the equilibrium state:

$$\delta \Pi(u) = \int_0^\ell E A \frac{du}{dx} \delta \frac{du}{dx} dx - F_1 \delta u_1 - F_2 \delta u_2 = 0 \tag{8.72}$$

As in the weighted residual method, an approximate solution is needed. We approximate $u(x)$ using two linear basis functions associated with the nodal displacements u_1 and u_2 :

$$u(x) = u_1 N_1(x) + u_2 N_2(x) \tag{8.73}$$

The first derivative of $u(x)$ with respect to x can be written as:

$$\begin{aligned}
 \frac{du}{dx} &= \frac{d}{dx} (N_1 u_1 + N_2 u_2) \\
 &= \left(\frac{dN_1}{dx} u_1 + \frac{dN_2}{dx} u_2 \right) \\
 &= \begin{pmatrix} \frac{dN_1}{dx} & \frac{dN_2}{dx} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}
 \end{aligned} \tag{8.74}$$

The variation of $\frac{du}{dx}$ follows the same pattern:

$$\begin{aligned}
 \delta \frac{du}{dx} &= \frac{d}{dx} \delta u \\
 &= \frac{d}{dx} \left(\frac{\partial u}{\partial u_1} \delta u_1 + \frac{\partial u}{\partial u_2} \delta u_2 \right) \\
 &= \frac{d}{dx} (N_1 \delta u_1 + N_2 \delta u_2) \\
 &= \left(\frac{dN_1}{dx} \delta u_1 + \frac{dN_2}{dx} \delta u_2 \right)
 \end{aligned} \tag{8.75}$$

This results can be expressed in matrix form as:

$$\begin{aligned}\delta \frac{du}{dx} &= \begin{pmatrix} \frac{dN_1}{dx} & \frac{dN_2}{dx} \end{pmatrix} \begin{pmatrix} \delta u_1 \\ \delta u_2 \end{pmatrix} \\ &= (\delta u_1 \quad \delta u_2) \begin{pmatrix} \frac{dN_1}{dx} \\ \frac{dN_2}{dx} \end{pmatrix}\end{aligned}\quad (8.76)$$

Substituting $\frac{du}{dx}$ and $\delta \frac{du}{dx}$ into the stationary condition, we obtain:

$$EA \int_0^\ell (\delta u_1 \quad \delta u_2) \begin{pmatrix} \frac{dN_1}{dx} \\ \frac{dN_2}{dx} \end{pmatrix} \begin{pmatrix} \frac{dN_1}{dx} & \frac{dN_2}{dx} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} dx - (\delta u_1 \quad \delta u_2) \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} = 0 \quad (8.77)$$

The displacement vector and its variation do not depend on x , so they can be taken outside the integral:

$$EA(\delta u_1 \quad \delta u_2) \int_0^\ell \begin{pmatrix} \frac{dN_1}{dx} \\ \frac{dN_2}{dx} \end{pmatrix} \begin{pmatrix} \frac{dN_1}{dx} & \frac{dN_2}{dx} \end{pmatrix} dx \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} - (\delta u_1 \quad \delta u_2) \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} = 0 \quad (8.78)$$

For this to hold for any admissible variation of u_1 and u_2 , it is required that:

$$EA \int_0^\ell \begin{pmatrix} \frac{dN_1}{dx} \\ \frac{dN_2}{dx} \end{pmatrix} \begin{pmatrix} \frac{dN_1}{dx} & \frac{dN_2}{dx} \end{pmatrix} dx \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} - \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (8.79)$$

Evaluating the derivatives and carrying out the integration gives the finite element equations for a bar element:

$$\frac{AE}{\ell} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} \quad (8.80)$$

These results are identical to those obtained via the weighted residual method, as both approaches solve the same differential equation with the same basis functions.

8.5.4 Variational formulation of a beam element

For a beam element, the total potential energy is given by:

$$\Pi(v) = \frac{1}{2} \int_0^\ell EI \left(\frac{d^2 v}{dx^2} \right)^2 dx - \int_0^\ell qv dx - M_1 \theta_1 - M_2 \theta_2 - V_1 v_1 - V_2 v_2 \quad (8.81)$$

To find the equilibrium state we apply the stationary condition as:

$$\delta \Pi(v) = \int_0^\ell EI \frac{d^2 v}{dx^2} \delta \frac{d^2 v}{dx^2} dx - \int_0^\ell q \delta v dx - M_1 \delta \theta_1 - M_2 \delta \theta_2 - V_1 \delta v_1 - V_2 \delta v_2 = 0 \quad (8.82)$$

The same approximate solution form with the same basis functions as the weighted residuals method are used:

$$v(x) = v_1 N_1(x) + \theta_1 N_2(x) + v_2 N_3(x) + \theta_2 N_4(x) \quad (8.83)$$

The deflection's second derivative can be written in matrix notation as:

$$\begin{aligned}
 \frac{d^2 v}{dx^2} &= \frac{d^2}{dx^2} (N_1 v_1 + N_2 \theta_1 + N_3 v_2 + N_4 \theta_2) \\
 &= \left(\frac{d^2 N_1}{dx^2} v_1 + \frac{d^2 N_2}{dx^2} \theta_1 + \frac{d^2 N_3}{dx^2} v_2 + \frac{d^2 N_4}{dx^2} \theta_2 \right) \\
 &= \begin{pmatrix} \frac{d^2 N_1}{dx^2} & \frac{d^2 N_2}{dx^2} & \frac{d^2 N_3}{dx^2} & \frac{d^2 N_4}{dx^2} \end{pmatrix} \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix}
 \end{aligned} \tag{8.84}$$

and its first variation as:

$$\begin{aligned}
 \delta \frac{d^2 v}{dx^2} &= \frac{d^2}{dx^2} \delta v \\
 &= \begin{pmatrix} \frac{d^2 N_1}{dx^2} & \frac{d^2 N_2}{dx^2} & \frac{d^2 N_3}{dx^2} & \frac{d^2 N_4}{dx^2} \end{pmatrix} \begin{pmatrix} \delta v_1 \\ \delta \theta_1 \\ \delta v_2 \\ \delta \theta_2 \end{pmatrix}
 \end{aligned} \tag{8.85}$$

Substituting the last two equations into the stationary condition we get:

$$\begin{aligned}
 EI \int_0^\ell (\delta v_1 \quad \delta \theta_1 \quad \delta v_2 \quad \delta \theta_2) \begin{pmatrix} \frac{d^2 N_1}{dx^2} \\ \frac{d^2 N_2}{dx^2} \\ \frac{d^2 N_3}{dx^2} \\ \frac{d^2 N_4}{dx^2} \end{pmatrix} \begin{pmatrix} \frac{d^2 N_1}{dx^2} & \frac{d^2 N_2}{dx^2} & \frac{d^2 N_3}{dx^2} & \frac{d^2 N_4}{dx^2} \end{pmatrix} \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix} dx \\
 - \int_0^\ell q (\delta v_1 \quad \delta \theta_1 \quad \delta v_2 \quad \delta \theta_2) \begin{pmatrix} N_1 \\ N_2 \\ N_3 \\ N_4 \end{pmatrix} dx - (\delta v_1 \quad \delta \theta_1 \quad \delta v_2 \quad \delta \theta_2) \begin{pmatrix} V_1 \\ M_1 \\ V_2 \\ M_2 \end{pmatrix} = 0
 \end{aligned} \tag{8.86}$$

For this equation to be equal to zero we need:

$$\begin{aligned}
 EI \int_0^\ell \begin{pmatrix} \frac{d^2 N_1}{dx^2} \\ \frac{d^2 N_2}{dx^2} \\ \frac{d^2 N_3}{dx^2} \\ \frac{d^2 N_4}{dx^2} \end{pmatrix} \begin{pmatrix} \frac{d^2 N_1}{dx^2} & \frac{d^2 N_2}{dx^2} & \frac{d^2 N_3}{dx^2} & \frac{d^2 N_4}{dx^2} \end{pmatrix} \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix} dx \\
 - \int_0^\ell q \begin{pmatrix} N_1 \\ N_2 \\ N_3 \\ N_4 \end{pmatrix} dx - \begin{pmatrix} V_1 \\ M_1 \\ V_2 \\ M_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}
 \end{aligned} \tag{8.87}$$

After calculating all derivatives and performing the integration, we obtain the finite element equilibrium equations for a beam element:

$$\frac{EI}{\ell^3} \begin{pmatrix} 12 & 6\ell & -12 & 6\ell \\ 6\ell & 4\ell^2 & -6\ell & 2\ell^2 \\ -12 & -6\ell & 12 & -6\ell \\ 6\ell & 2\ell^2 & -6\ell & 4\ell^2 \end{pmatrix} \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} V_1 \\ M_1 \\ V_2 \\ M_2 \end{pmatrix} + \frac{q\ell}{12} \begin{pmatrix} 6 \\ \ell \\ 6 \\ -\ell \end{pmatrix} \quad (8.88)$$

As expected, this system of equations is the same as the one obtained using the weighted residuals method.

Exercises 8.7:

1. A two node bar element that also includes the effect of a distributed force $q(x)$ along the axis is required. Determine the finite element equations using the principle of minimum potential energy.
2. A special bar element considers that the cross-sectional area varies linearly according to:

$$A(x) = A_1 N_1(x) + A_2 N_2(x)$$

where A_1 and A_2 are the cross-sectional areas at nodes 1 and 2, respectively. Find the finite element equations using the principle of minimum potential energy.

3. A one-dimensional two-node element is being designed to account for the torsional deformation of circular shafts. The degrees of freedom are given by the angles φ_1 and φ_2 which are related to twisting moments T_1 and T_2 at nodes 1 and 2, respectively. Considering that the strain energy for torsion is

$$U = \int_0^\ell \frac{1}{2} GJ \left(\frac{d\varphi}{dx} \right)^2 dx$$

where G is the material shear modulus and J is the cross-section polar moment of inertia, find the finite element equations.

8.6 Virtual Work

The principle of virtual work can be used to derive the equations for mechanical finite elements in static equilibrium. It states that stresses, body forces, and tractions are in equilibrium if and only if the internal virtual work equals the external virtual work for any admissible virtual displacement field:

$$\delta W_{\text{int}} = \delta W_{\text{ext}} \quad (8.89)$$

Note 8.6: The equality $\delta W_{\text{int}} = \delta W_{\text{ext}}$ represents the equilibrium condition. Here δW_{int} and δW_{ext} denote the first variations of the internal and external work functionals with respect to the displacement field. The principle of virtual work states that a body is in equilibrium if and only if the first variation of the total work vanishes,

$$\delta W_{\text{ext}} - \delta W_{\text{int}} = 0,$$

for every kinematically admissible virtual displacement field. For an elastic body, $\delta W_{\text{int}} = \delta U$, so this condition is equivalent to the stationary condition $\delta \Pi = \delta U - \delta W_{\text{ext}} = 0$ of the total potential energy.

For a body, the internal virtual work is defined by

$$\delta W_{\text{int}} = \int_V \boldsymbol{\sigma} : \delta \boldsymbol{\varepsilon} dV \quad (8.90)$$

where σ is the stress tensor, ε is the strain tensor, and the operator $:$ denotes the so-called double-dot product of tensors. Each component of the strain tensor is defined by:

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (8.91)$$

Thus, the virtual strain components are:

$$\delta \varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial \delta u_i}{\partial x_j} + \frac{\partial \delta u_j}{\partial x_i} \right) \quad (8.92)$$

On the other hand, the external virtual work is defined as:

$$\delta W_{\text{ext}} = \int_V \mathbf{b} \cdot \delta \mathbf{u} \, dV + \int_{\Gamma} \mathbf{t} \cdot \delta \mathbf{u} \, d\Gamma \quad (8.93)$$

where $\mathbf{b}$ is the body-force vector and $\mathbf{t}$ is the traction vector acting over the boundary surface Γ .

The application of the virtual work principle leads to the static equilibrium equations of a body. In fact, applying the divergence theorem (i.e. $\int_V \frac{\partial f}{\partial x_i} \, dV = \int_{\Gamma} f n_i \, d\Gamma$) on the internal virtual work term we have:

$$\begin{aligned} \int_V \sigma_{ij} \delta \varepsilon_{ij} \, dV &= \int_V \sigma_{ij} \frac{\partial \delta u_i}{\partial x_j} \, dV \\ &= \int_V \frac{\partial \sigma_{ij} \delta u_i}{\partial x_j} \, dV - \int_V \frac{\partial \sigma_{ij}}{\partial x_j} \delta u_i \, dV \\ &= \int_{\Gamma} \sigma_{ij} n_j \delta u_i \, d\Gamma - \int_V \frac{\partial \sigma_{ij}}{\partial x_j} \delta u_i \, dV \end{aligned} \quad (8.94)$$

Equating this result to the external virtual work, and considering $\sigma_{ij} n_j = t_i$, the principle of virtual work recovers the static equilibrium differential equation:

$$\frac{\partial \sigma_{ij}}{\partial x_j} + b_i = 0 \quad (8.95)$$

or in tensor notation:

$$\nabla \cdot \sigma + \mathbf{b} = \mathbf{0} \quad (8.96)$$

8.6.1 Virtual Work Formulation of a Bar Element

The finite element equations for mechanical problems can also be found by the use of the principle of virtual work. First we assume a virtual displacement field over a bar element. Equating the associated internal and external work done by the axial stress and external nodal forces we have:

$$\begin{aligned} \delta W_{\text{int}} &= \delta W_{\text{ext}} \\ \int_V \sigma_x \delta \varepsilon_x \, dV &= F_1 \delta u_1 + F_2 \delta u_2 \end{aligned} \quad (8.97)$$

Note that the internal virtual work is defined in terms of the virtual strain field $\delta \varepsilon_x$ that in turn can be written in terms of the virtual displacement field δu , thus $\delta \varepsilon_x = \frac{d}{dx} \delta u$.

Substituting $\sigma_x = E \varepsilon_x$ and $\varepsilon_x = \frac{du}{dx}$ into the virtual work equation leads to:

$$\int_0^\ell EA \frac{du}{dx} \delta \frac{du}{dx} dx = F_1 \delta u_1 + F_2 \delta u_2 \quad (8.98)$$

which is the same expression as the one obtained with the use of the minimum potential energy principle.

Assuming an approximate solution of the form $u(x) = N_1 u_1 + N_2 u_2$ we can write the virtual work equality as:

$$EA(\delta u_1 \quad \delta u_2) \int_0^\ell \begin{pmatrix} \frac{dN_1}{dx} \\ \frac{dN_2}{dx} \end{pmatrix} \begin{pmatrix} \frac{dN_1}{dx} & \frac{dN_2}{dx} \end{pmatrix} dx \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = (\delta u_1 \quad \delta u_2) \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} \quad (8.99)$$

Since the equation above is valid for any virtual displacement field, then:

$$EA \int_0^\ell \begin{pmatrix} \frac{dN_1}{dx} \\ \frac{dN_2}{dx} \end{pmatrix} \begin{pmatrix} \frac{dN_1}{dx} & \frac{dN_2}{dx} \end{pmatrix} dx \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} \quad (8.100)$$

Finally, after using linear basis functions, we obtain the conventional finite element equations for a bar element.

8.6.2 Virtual Work Formulation of a Beam Element

Given a virtual displacement field δv and equating the associated internal and external work we have:

$$\delta W_{\text{int}} = \delta W_{\text{ext}} \\ \int_V \sigma_x \delta \varepsilon_x dV = \int_0^\ell q \delta v dx + M_1 \delta \theta_1 + M_2 \delta \theta_2 + V_1 \delta v_1 + V_2 \delta v_2 \quad (8.101)$$

Note that the axial stress and strain are variable along the beam depth, thus in order to find the virtual internal work it is necessary to integrate along x and y directions:

$$\begin{aligned} \int_V \sigma_x \delta \varepsilon_x dV &= \int_0^\ell \int_{-\frac{h}{2}}^{\frac{h}{2}} \sigma_x \delta \varepsilon_x b dy dx \\ &= \int_0^\ell \int_{-\frac{h}{2}}^{\frac{h}{2}} E \varepsilon_x \delta \varepsilon_x b dy dx \end{aligned} \quad (8.102)$$

where b and h are the beam width and height, respectively.

Recalling the relationship between the strain ε_x and the beam's deflection we have

$$\varepsilon_x = -y v'' = -y \frac{d^2 v}{dx^2} \quad (8.103)$$

Substituting this expression into the internal virtual work term we get

$$\begin{aligned}
 \int_V \sigma_x \delta \varepsilon_x dV &= \int_0^\ell \int_{-\frac{h}{2}}^{\frac{h}{2}} E \frac{d^2 v}{dx^2} \delta \frac{d^2 v}{dx^2} y^2 b dy dx \\
 &= \int_0^\ell E \frac{d^2 v}{dx^2} \delta \frac{d^2 v}{dx^2} \int_{-\frac{h}{2}}^{\frac{h}{2}} y^2 b dy dx \\
 &= \int_0^\ell E \frac{d^2 v}{dx^2} \delta \frac{d^2 v}{dx^2} \int_{-\frac{h}{2}}^{\frac{h}{2}} y^2 dA dx \\
 &= \int_0^\ell EI \frac{d^2 v}{dx^2} \delta \frac{d^2 v}{dx^2} dx
 \end{aligned} \tag{8.104}$$

Using this expression in the virtual work equation we obtain the same equation as the one obtained by applying the principle of minimum potential energy

$$\int_0^\ell EI \frac{d^2 v}{dx^2} \delta \frac{d^2 v}{dx^2} dx = \int_0^\ell q \delta v dx + M_1 \delta \theta_1 + M_2 \delta \theta_2 + V_1 \delta v_1 + V_2 \delta v_2 \tag{8.105}$$

The approximate solution for the deflection v can be chosen as:

$$v(x) = v_1 N_1(x) + \theta_1 N_2(x) + v_2 N_3(x) + \theta_2 N_4(x) \tag{8.106}$$

By using the same approximate solution form and the same basis functions as in the weighted residual method, we obtain the conventional finite element equations for a beam element.

Exercises 8.8:

1. Find the finite element equations for a three-node bar element using the virtual work approach. The element has three nodal displacements u_1 , u_2 and u_3 , thus the approximate solution has the form:

$$u(x) = u_1 N_1(x) + u_2 N_2(x) + u_3 N_3(x)$$

The basis functions are:

$$N_1(x) = \frac{2x^2}{\ell^2} - \frac{3x}{\ell} + 1$$

$$N_2(x) = -\frac{4x^2}{\ell^2} + \frac{4x}{\ell}$$

$$N_3(x) = \frac{2x^2}{\ell^2} - \frac{x}{\ell}$$

2. Consider a one-dimensional two-node element designed to account for the torsional deformation of circular shafts. The degrees of freedom are given by the angles φ_1 and φ_2 which are related to twisting moments T_1 and T_2 at nodes 1 and 2, respectively. Use the principle of virtual work to determine the finite element equations.
3. Find the equivalent nodal forces for a beam element with a triangular distributed load $q(x)$. Consider $q(0) = 0$ and $q(\ell) = q$.